

IIT JEE

ELP Booklet

PHYSICS



Practice Booklet

Table of Contents

S.No.	Chapter Name		Pg. No.
1.	Geometrical Optics	ELP-01-10	01-23
2.	Simple Harmonic Motion	ELP-11-15	24-35
3.	Electrostatic	ELP-16-25	36-62
4.	Gravitation	ELP-26-27	63-66
5.	Current Electricity	ELP-28-34	67-85
6.	Capacitor	ELP-35-38	86-97
7.	MEC	ELP-39-45	98-118
8.	EMI & AC	ELP-46-54	119-144
9.	Wave on String and Sound Wave	ELP-55-63	145-166
10.	Wave Optics	ELP-64-66	167-173
11.	Modern Physics	ELP-67-71	174-183
	Answer key		184-193

Note: (*) → Multiple Correct Type Question

Total Marks: 32

Max. Time: 22 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.8

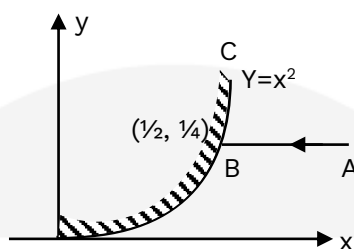
(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.9 to Q.10

(4 marks, 3 min.)

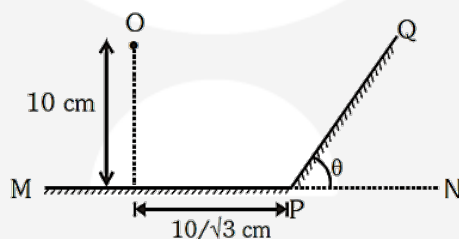
(There is partial marking in multiple correct questions)

1. In the given figure, incident ray AB is parallel to x-axis. The angle of reflection is:



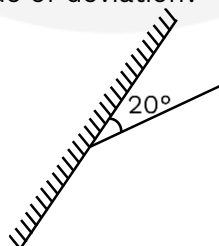
- (A) 30° (B) 60° (C) 45° (D) None of these

2. Two plane mirror MN and PQ inclined at an angle θ are shown in figure. O is a point source 10 cm above the mirror MN. For light rays striking on the mirror MN directly from O not to strike the mirror PQ, the maximum value of θ is:



- (A) 30° (B) 45° (C) 60° (D) 75°

3. In the shown figure, what is the value of deviation?



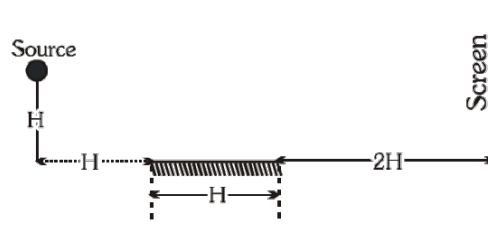
- (A) 40° (B) 140° (C) 80° (D) None of these

4. The man, whose height is 1.75 m is standing at a distance 6m from a vertical pillar of height $h = 7$ m. At what distance from the man should we put a small flat mirror on the ground horizontally to see the image of the top of the pillar in it?

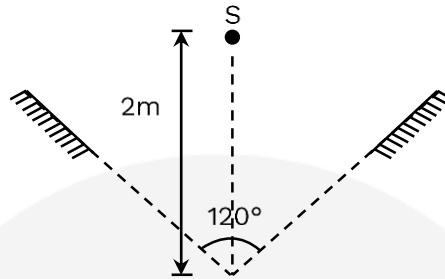
- (A) 4.8 m (B) 1.2 m (C) 5.2 m (D) 4.2 m



5. A point source of light has been placed as shown in the figure. What is the length on the screen that will receive reflected light from the mirror?



- (A) $2H$ (B) $3H$ (C) H (D) None
6. Area of triangle formed by S and its two images is:



- (A) $3\sqrt{3} \text{ m}^2$ (B) $\frac{3\sqrt{3}}{4} \text{ m}^2$ (C) $\frac{4}{3\sqrt{3}} \text{ m}^2$ (D) Can't be determined
7. A train accelerates from rest at a constant rate α for distance x_1 and time t_1 . After that, retards at constant rate β for distance x_2 in time t_2 and comes to the rest. Which of the following relation is correct?
- (A) $\frac{x_1}{x_2} = \frac{\alpha}{\beta} = \frac{t_1}{t_2}$ (B) $\frac{x_1}{x_2} = \frac{\beta}{\alpha} = \frac{t_1}{t_2}$ (C) $\frac{x_1}{x_2} = \frac{\alpha}{\beta} = \frac{t_2}{t_1}$ (D) $\frac{x_1}{x_2} = \frac{\beta}{\alpha} = \frac{t_2}{t_1}$
8. The speed v of a particle moving along the x -axis, is given by $v^2 = 10x - 5x^2$. The magnitude of its acceleration, when it is at $x = 1 \text{ m}$ is:
- (A) 0 m/s^2 (B) 2 m/s^2 (C) 5 m/s^2 (D) 10 m/s^2
- 9*. A ray of light travelling in the direction $\frac{1}{5}(\hat{3}\hat{i} + 4\hat{j})$ is incident on a plane mirror. After reflection, it travels along the direction $\frac{1}{5}(\hat{3}\hat{i} - 4\hat{j})$. Then choose the correct statement(s):
- (A) Mirror is kept in x - y plane (B) Normal of the mirror is along $-\hat{j}$
(C) Angle of incident is 37° (D) Mirror is kept in x - z plane
- 10*. A particle is moving in a straight line along positive y -axis. Its position from origin at any time t is given by $y = 5t^2 - 10t + 5$, where y is in meters and t is in seconds.
- (A) The velocity at $t = 2\text{s}$ will be 10 m/s
(B) Particle is at $y = 5\text{m}$, when its velocity is zero.
(C) Average velocity for the time interval $t = 0$ to $t = 10 \text{ s}$ is 40 m/s
(D) The particle has a uniform acceleration of 5 m/s^2



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-02

Note: (*) → Multiple Correct Type Question

Total Marks: 38

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.7 to Q.9

(4 marks, 3 min.)

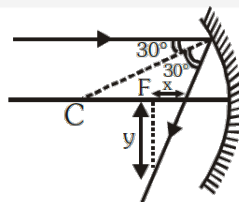
Match the Column Type Questions (No negative marking) Q.10

(8 marks, 4 min.)

(There is partial marking in multiple correct questions)

1. A plane mirror is moving with velocity $4\hat{i} + 5\hat{j} + 8\hat{k}$. A point object in front of the mirror moves with a velocity $3\hat{i} + 4\hat{j} + 5\hat{k}$. Here $\hat{k}$ is along the normal to the plane mirror and facing towards the object. The velocity of the image is:
- (A) $-3\hat{i} - 4\hat{j} + 5\hat{k}$ (B) $3\hat{i} + 4\hat{j} + 11\hat{k}$ (C) $-3\hat{i} - 4\hat{j} + 11\hat{k}$ (D) $7\hat{i} + 9\hat{j} + 11\hat{k}$

2. A marginal ray is falling on a concave spherical mirror as shown in figure. The ratio of x and y is:

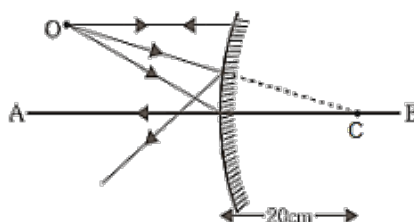


- (A) $\sqrt{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{2}{\sqrt{3}}$ (D) Can't be determined

3. A light ray travelling parallel to the principal axis of a concave mirror strikes the mirror at angle of incidence θ . If radius of curvature of the mirror is R, then after reflection, the ray meets the principle axis at distance d from the centre of curvature, then d is:

- (A) $\frac{R}{2}$ (B) $R\left(1 - \frac{1}{2\cos\theta}\right)$ (C) $\frac{R}{2\cos\theta}$ (D) $\frac{R}{2}(1 + \cos\theta)$

4. In the figure, a convex mirror of radius of curvature 20 cm is shown. An object O is placed in front of this mirror. Its ray diagram is shown. How many mistakes are there in the diagram? (AB is its principal axis)



- (A) 3 (B) 2 (C) 1 (D) 0

5. A particle starting from rest has a constant acceleration of 4 ms^{-2} for 4 s. It then retards uniformly for next 8 s and comes to rest. Average speed of the particle during the motion is:

- (A) 16 ms^{-1} (B) 8 ms^{-1} (C) 24 ms^{-1} (D) None of these



6. In question 5, magnitude of average velocity of the particle for a time interval from $t = 0$ to $t = 8$ s is:
(A) 10 ms^{-1} (B) 8 ms^{-1} (C) 12 ms^{-1} (D) None of these
- 7*. Which of the following statements about a plane mirror is (are) correct,
(A) When a plane mirror is displaced a distance L perpendicular to its surface, image point for any fixed object point moves a distance $2L$.
(B) When a plane mirror is displaced a distance L perpendicular to its surface, image point for any fixed object point moves a distance L .
(C) When a plane mirror is rotated through an angle α about an axis that lies in the plane of the mirror, the reflected ray for any particular incident ray that is perpendicular to the axis, rotates through an angle 2α .
(D) When a plane mirror is rotated through an angle α about an axis that lies in the plane of the mirror, the reflected ray for any particular incident ray that is perpendicular to the axis, rotates through an angle α .
- 8*. Two mirrors are inclined at an angle of 30° . A ray is incident parallel to one of the mirrors. Select the correct statement(s)-
(A) Ray will undergo five reflections from the system
(B) Total deviation from the system will 120°
(C) Ray will retrace its path finally
(D) Total deviation suffered by the ray will be 90°
- 9*. If displacement along x-axis is defined as $x = \frac{t^3}{3} - \frac{5t^2}{2} + 6t + 107$, then find value (s) of t for which velocity is zero:
(A) $t = 1$ (B) $t = 2$ (C) $t = 3$ (D) $t = 4$
10. The position of a mirror and a particle moving along X-axis is given by $x_m = t^3 - 6t^2$ and $x_p = 9t^2 - 60t$. Match the column I with column II.
- | Column-I | Column-II |
|---|-------------|
| (A) Velocity of image of particle is zero | (P) 2 sec |
| (B) Acceleration of image of particle is zero | (Q) 3 sec |
| (C) Acceleration of image of particle is negative | (R) 3.5 sec |
| (D) Velocity of image of particle is towards the origin | (S) 4 sec |
| | (T) 5 sec |



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-03

Note: (*) → Multiple Correct Type Question

Total Marks: 33

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.7

(3 marks, 2 min.)

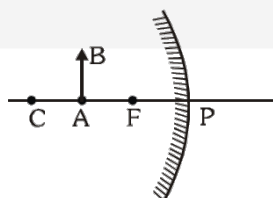
Multiple Correct Questions ('-2' negative marking) Q.8 to Q.10

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

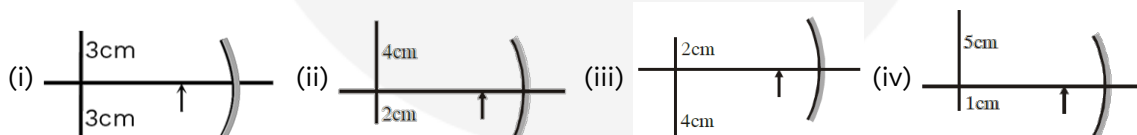
1. An object is placed in front of a convex mirror at a distance of 50 cm. A plane mirror is introduced between object and convex mirror covering the lower half of the convex mirror. If the distance between the object and the plane mirror is 30 cm, it is found that there is no gap between the images formed by the two mirrors. The radius of the convex mirror is:
- (A) 12.5 cm (B) 25 cm (C) 50 cm (D) 100 cm

2. An object is kept between centre of curvature C and focus F. Image will be formed



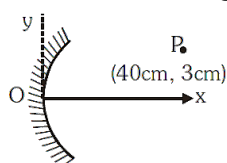
- (A) between F & C (B) between C and ∞
(C) between F and P (D) None of these

3. A candle of length 2 cm is placed perpendicular to principal axis of a concave mirror of radius of curvature 40 cm at a distance of 30 cm from the mirror. A white screen is placed perpendicular to principal axis to obtain the image. If the screen is a square sheet of side 6 cm, then in which of the set of given arrangements the image can be totally obtained?



- (A) (i) and (iii) (B) (ii) and (iv) (C) (i), (ii) & (iv) (D) None of these

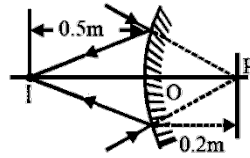
4. The co-ordinates of the image of point object P formed by a concave mirror of radius of curvature 20 cm (consider paraxial rays only) as shown in the figure is:



- (A) (13.33 cm, -1 cm) (B) (13.33 cm, +1 cm)
(C) (-13.33 cm, +1 cm) (D) (-13.33 cm, -1 cm)

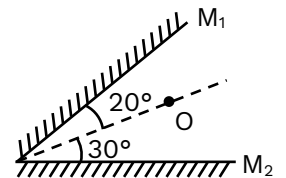


5. A beam of convergent light converges to a point 0.5 m in front of the mirror after reflection at a convex mirror but in the absence of the mirror the beam converges to a point 0.2 m behind the mirror. The radius of curvature of the mirror is:



- (A) 20 cm (B) 50 cm (C) 66.67 cm (D) 28.57 cm
6. A stone is dropped from a certain height which can reach the ground in 5 seconds. It is stopped after 3 seconds of its fall and is again released. The total time taken by the stone to reach the ground will be:
- (A) 6 s (B) 6.5 s (C) 7s (D) 7.5 s
7. From the top of a tower, a stone is thrown up. It reaches the ground in time t_1 . A second stone thrown down with the same speed reaches the ground in time t_2 . A third stone released from rest reaches the ground in time t_3 . Then:
- (A) $t_3 = \frac{t_1 + t_2}{2}$ (B) $t_3 = \sqrt{t_1 t_2}$ (C) $\frac{1}{t_3} = \frac{1}{t_1} - \frac{1}{t_2}$ (D) $t_3^2 = t_1^2 - t_2^2$

- 8*. An object O is placed between two inclined plane mirrors M_1 & M_2 as shown in the figure. Choose the correct statement(s):
- (A) Eight images are formed
- (B) Angular positions of images with respect to mirror M_2 measured anticlockwise are $70^\circ, 130^\circ, 170^\circ, 190^\circ, 330^\circ, 290^\circ, 230^\circ, 280^\circ$
- (C) Seven images are formed
- (D) Angular position of images with respect to mirror M_2 measured anticlockwise are $70^\circ, 130^\circ, 170^\circ, 190^\circ, 330^\circ, 290^\circ, 230^\circ$



- 9*. A point object is placed on the axis of a concave mirror of the focal length 24 cm such that the image is formed at a distance of 48 cm from the pole. Distance of the object from pole may be :
- (A) 48 cm (B) 16 cm (C) 32 cm (D) 20 cm
- 10*. A particle moving along a straight line with uniform acceleration has velocities 7m/s at A and 17m/s at C. B is the mid-point of AC. Then:
- (A) the velocity at B is 12m/s
- (B) the average velocity between A and B is 10m/s
- (C) the ratio of the time to go from A to B to that from B to C is 3 : 2
- (D) the average velocity between B and C is 15m/s

Note: (*) → Multiple Correct Type Question

Total Marks: 36

Max. Time: 26 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.4

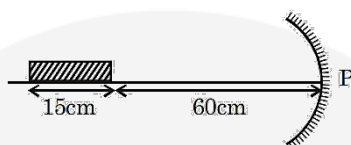
(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.5 to Q.10

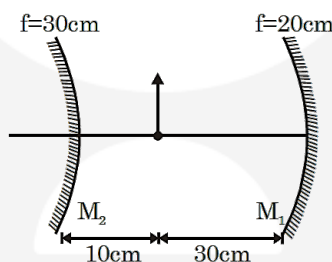
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

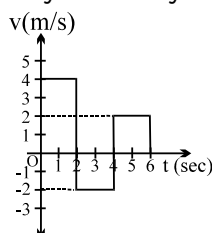
1. An object of length 15 cm is placed on principle axis of a concave mirror of focal length 30 cm. Its one end is at a distance of 60 cm from the pole of mirror as shown in diagram. Find the length of image formed by mirror.



- (A) 10 cm (B) 15 cm
(C) 20 cm (D) 30 cm
2. In the figure shown, find the total magnification after two successive reflection first on M_1 and then on M_2 . (Assume paraxial rays only)



- (A) +6 (B) -6 (C) +3 (D) -3
3. A particle in uniform acceleration in a straight line has speed v (m/s) at a position x (meter) which is given by $v = \sqrt{180 - 16x}$. The acceleration of the particle (in m/s^2) is:
- (A) 8 (B) -8 (C) 164 (D) $\frac{-8}{\sqrt{180 - 16x}}$
4. The velocity-time graph of a body moving in a straight line is shown in the figure. The ratio of displacement to the distance travelled by the body in first 6 seconds is:



- (A) 1 : 1 (B) 1 : 2 (C) 1 : 3 (D) 1 : 4

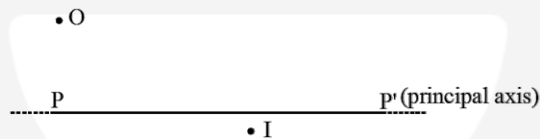


- 5*. You are required to design a little dentist's mirror to be fixed at the end of a shaft for use in someone's mouth. The image should be erect and when held at 1.5 cm from a tooth, it should produce an image twice the size of the tooth.
- (A) The mirror is convex with $|f| = 3$ cm
(B) The mirror is concave with $|f| = 3$ cm
(C) The image of this tooth is 3 cm behind the mirror.
(D) The image of this tooth is 3 cm in front of the mirror.

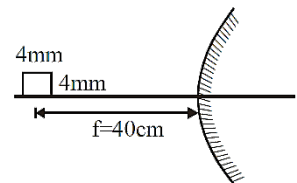
- 6*. The image formed by a concave mirror is twice the size of the block. The focal length of the mirror is 20 cm. the distance of the object from the mirror is (are):
- (A) 10 cm (B) 30 cm (C) 25 cm (D) 15 cm

- 7*. A particle is moving towards a fixed convex mirror. The image also moves. If V_i = speed of image and V_o = speed of the object, then:
- (A) $V_i < V_o$ if $|u| < |F|$ (B) $V_i > V_o$ if $|u| > |F|$
(C) $V_i < V_o$ if $|u| > |F|$ (D) $V_i = V_o$ if $|u| = |F|$

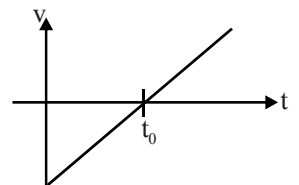
- 8*. The location I of image for object location O can be obtained by placing



- (A) Concave mirror left of O facing towards O.
(B) Concave mirror right of I facing towards I.
(C) Convex mirror between O & I facing towards O.
(D) Convex mirror right of I facing away from I.
- 9*. A small square object is placed in front of convex mirror on principle axis as shown.
- (A) The shape of image is square
(B) The shape of image is nearly rectangular
(C) the area of image is nearly 2 mm^2
(D) The area of image is nearly 4 mm^2



- 10*. The velocity time graph of a particle moving in a straight line is given in the figure. Then starting from $t = 0$, the particle:
- (A) continuously speeds up
(B) first slows down and then speeds up
(C) moves with constant acceleration
(D) moves with acceleration of constant magnitude which changes direction at $t = t_0$



Note: (*) → Multiple Correct Type Question

Total Marks: 28

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

Comprehension Type Questions ('-1' Negative marking) Q.7 to Q.9

Subjective Type Questions (No negative marking) Q.10

(There is partial marking in multiple correct questions)

Max. Time: 21 min.

(3 marks, 2 min.)

(2 marks, 2 min.)

(4 marks, 3 min.)

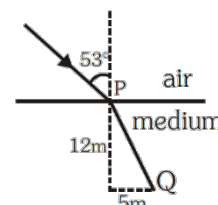
1. A light ray passes air and enters into another medium at point P. How long does it take the light ray to travel from P to Q?

(A) $9 \times 10^{-10} \text{s}$

(B) $9 \times 10^{-7} \text{s}$

(C) $9 \times 10^{-8} \text{s}$

(D) $9 \times 10^{-6} \text{s}$



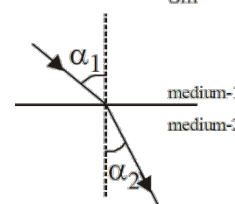
2. A beam of light propagates through a medium-1 and falls onto another medium-2, at an angle α_1 as shown. After that it propagates in medium-2 at an angle α_2 as shown. The light's wavelength in medium-1 is λ_1 . What is the wavelength of light in medium-2?

(A) $\frac{\sin \alpha_1}{\sin \alpha_2}$

(B) $\frac{\sin \alpha_2}{\sin \alpha_1} \lambda_1$

(C) $\frac{\cos \alpha_1}{\cos \alpha_2} \lambda_1$

(D) $\frac{\cos \alpha_2}{\cos \alpha_1} \lambda_1$



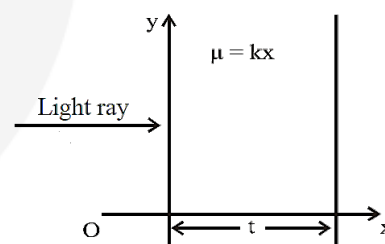
3. Refractive index of a transparent slab varies as $\mu = kx$, where x is the distance from origin. Time taken by the light to travel the slab of thickness t (as shown) is:

(A) $\frac{t^2 k}{2c}$

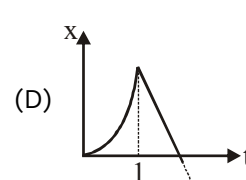
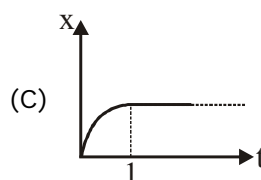
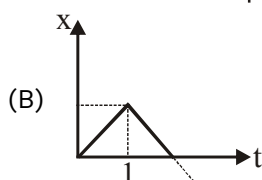
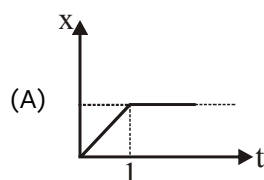
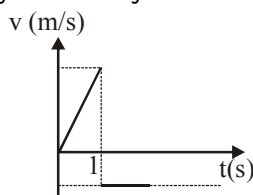
(B) $\frac{t^2 k}{c}$

(C) $\frac{tk}{c}$

(D) $\frac{2tk}{2c}$



4. The velocity time graph for a particle moving along x-axis is shown in the figure. The corresponding displacement time graph is correctly shown by:

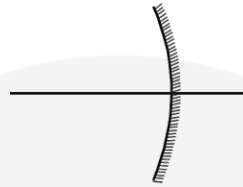




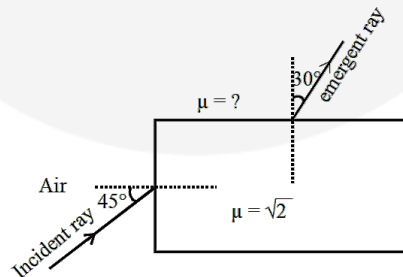
5. Two bodies are projected with the same velocity. If one is projected at an angle of 30° and the other at an angle of 60° to the horizontal, the ratio of the maximum heights reached is:
(A) 3 : 1 (B) 1 : 3 (C) 1 : 2 (D) 2 : 1
6. A particle is projected from a horizontal plane (x-z plane) such that its velocity vector at time t is given by $\vec{V} = a\hat{i} + (b - ct)\hat{j}$. Its range on the horizontal plane is given by:
(A) $\frac{ba}{c}$ (B) $\frac{2ba}{c}$ (C) $\frac{3ba}{c}$ (D) None

Paragraph for Question no. 7 to 9

In the given figure an object is kept on the principal axis of a concave mirror in front of it. Magnification of image is found $-2/3$ at this position. Now mirror is shifted towards the object keeping object fixed, by amount 10 cm. At this new position distance of object from mirror becomes same as the distance of image from the mirror before displacing the mirror.



7. Distance of object from mirror before displacing the mirror is:
(A) 10 cm (B) 20 cm (C) 30 cm (D) 40 cm
8. Modulus of focal length of the mirror is:
(A) 6 cm (B) 8 cm (C) 10 cm (D) 12 cm
9. Magnification at the new position of mirror is:
(A) $-\frac{3}{2}$ (B) -3 (C) -2 (D) $-\frac{1}{3}$
10. A light ray incident from air ($\mu = 1$) to a glass slab ($\mu = \sqrt{2}$) as shown in figure & emerges in an unknown medium of refractive index $\mu = \sqrt{k}$ as shown, what is the value of k?





SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-06

Note: (*) → Multiple Correct Type Question

Total Marks: 44

Max. Time: 32 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.4

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.5 to Q.8

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9 to Q.10

(4 marks, 3 min.)

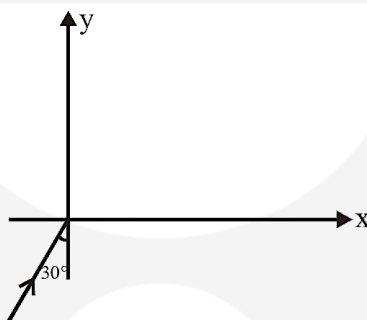
(There is partial marking in multiple correct questions)

1. Refractive indices of a medium in x-y coordinate system are -

$$\mu = \frac{2}{1+y^2} \text{ for } y > 0, \quad \text{and}$$

$$\mu = \frac{4}{3} \text{ for } y < 0.$$

A ray incident on medium as shown, then maximum y-coordinate of ray is :



- (A) $y = \sqrt{2}$ (B) $y = 0$ (C) $y = \sqrt{(\sqrt{3}-1)}$ (D) None of these

2. A beaker contains water up to a height h_1 and kerosene of height h_2 above water so that the total height of (water + kerosene) is $(h_1 + h_2)$. Refractive index of water is μ_1 and that of kerosene is μ_2 . The apparent shift in the position of the bottom of the beaker when viewed from above is :

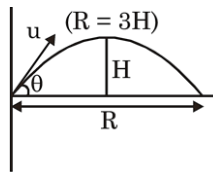
- (A) $\left(1 - \frac{1}{\mu_1}\right)h_2 - \left(1 - \frac{1}{\mu_2}\right)h_1$ (B) $\left(1 - \frac{1}{\mu_1}\right)h_1 + \left(1 - \frac{1}{\mu_2}\right)h_2$
 (C) $\left(1 + \frac{1}{\mu_1}\right)h_1 + \left(1 + \frac{1}{\mu_2}\right)h_2$ (D) $\left(1 + \frac{1}{\mu_1}\right)h_2 - \left(1 + \frac{1}{\mu_2}\right)h_1$

3. A light ray is incident at an angle 30° on a transparent surface separating two media. If the angle of refraction is 60° then critical angle is:

- (A) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (B) $\sin^{-1}(\sqrt{3})$ (C) $\sin^{-1}\left(\frac{2}{3}\right)$ (D) 45°

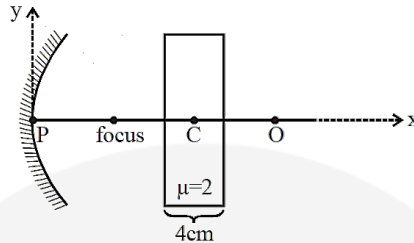


4. If range of a particle is 3 times to its maximum height, then find the angle at which particle is projected.



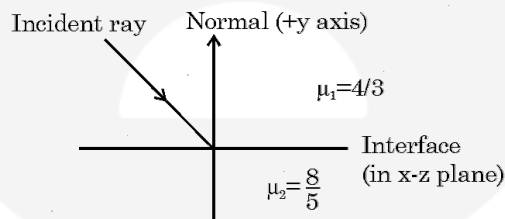
- (A) 53° (B) 37° (C) 45° (D) 60°

- 5*. A concave mirror of focal length 50 cm is placed at origin as shown in diagram. A point sized object is placed at O ($x = 152$ cm). A glass slab of thickness 4 cm, $\mu = 2$, is placed symmetrically around C. Light cannot reach from object to mirror without passing through slab. Choose the correct option(s):



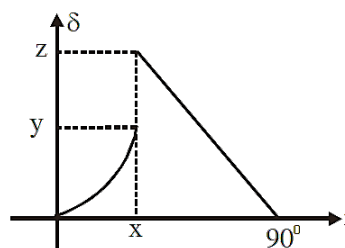
- (A) Image made by concave mirror will be at $x = 75$ cm
 (B) Final image made by entire optical system will be at $x = 77$ cm.
 (C) Final image of the system will be real.
 (D) Final image of the system will be virtual.

- 6*. A light ray incident in a medium of refractive index $\frac{4}{3}$ as shown in diagram. Interface separating the two medium lies in xz plane. Vector along incident ray is $3\hat{i} - 4\hat{j}$. Then which of the following holds true?



- (A) Angle of incident is 37°
 (B) The angle of refractive is 60°
 (C) The angle of refraction is 30°
 (D) If angle of incident of the incident ray is greater than the critical angle then total internal reflection takes place.

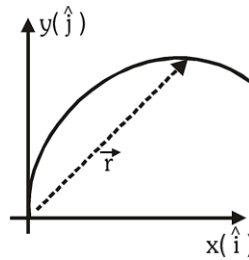
- 7*. Light is incident from glass to air. The variation of the angle of deviation δ with the angle of incident i for $0^\circ < i < 90^\circ$ is shown. The values of x, y and z , in terms of critical angle C , shown in the figure are



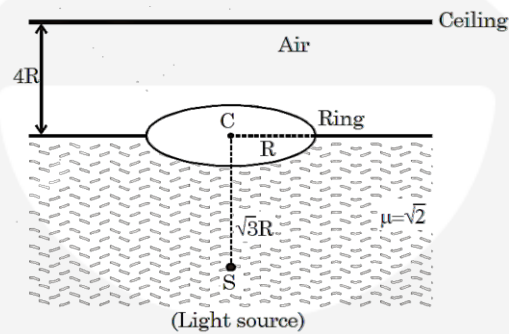
- (A) $x = C$ (B) $y = 90^\circ - C$ (C) $z = 180^\circ - 2C$ (D) $z = 180^\circ - C$



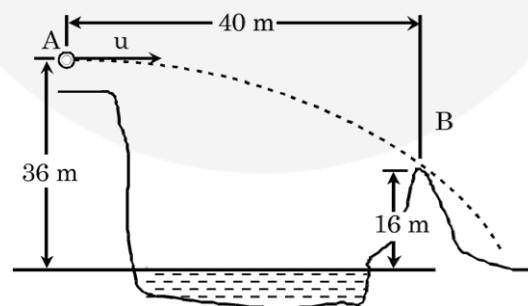
- 8*. A body is projected from origin such that its position vector varies with time as $\vec{r} = [3t\hat{i} + (4t - 5t^2)\hat{j}] \text{ m}$, where t is time in second. Then:



- (A) x-coordinate of particle is 2.4 m when y-coordinate is zero
(B) speed of projection is 5 m/s
(C) angle of projection with x-axis is $\tan^{-1}(4/3)$
(D) time when particle is again at x-axis is 0.8 s
9. On the surface of the liquid there is a circular ring of radius R . A light source is present at a depth $\sqrt{3}R$ just below the centre of ring. If the radius of shadow of the ring on the ceiling is αR , then find the value of α .



10. A boy throws a rock from A with minimum horizontal velocity 'u' so that it just clears the obstruction at B. Find the value of $\frac{u}{5}$.





SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-07

Note: (*) → Multiple Correct Type Question

Total Marks: 38

Max. Time: 28 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.2

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.3 to Q.6

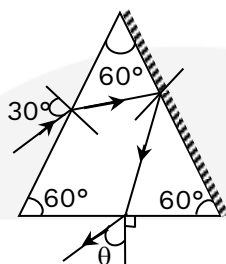
(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.7 to Q.10

(4 marks, 3 min.)

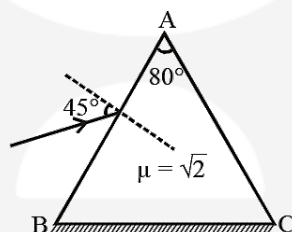
(There is partial marking in multiple correct questions)

1. A ray diagram for a prism whose one surface is reflecting is shown. The value of θ will be:



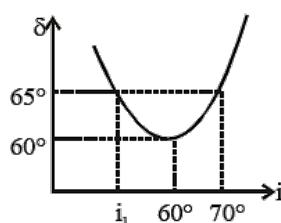
- (A) 30° (B) 60° (C) 0° (D) 90°

2. A ray of light is incident on a prism at surface AB. Side AB & AC are equal and base BC is silvered, then find net angle of deviation of emergent ray with respect to incident ray:



- (A) 45° (B) 90° (C) 60° (D) 180°

- 3*. The angle of deviation (δ) vs angle of incidence (i) is plotted for a prism. Choose the CORRECT statement(s).



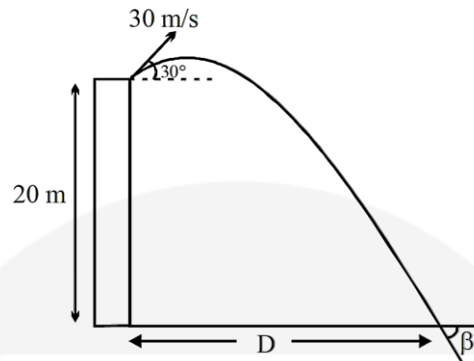
- (A) The angle of prism is 60°
 (B) The refractive index of the prism is $\mu = \sqrt{3}$
 (C) For deviation to be 65° the angle of incidence $i_1 = 55^\circ$
 (D) for minimum deviation, the angle of emergence $e = 60^\circ$



4*. If A is the angle of a prism having refractive index μ :

- (A) Light will not emerge out of the prism whatever be the angle of incidence if $A > 2 \sin^{-1} \left(\frac{1}{\mu} \right)$
- (B) If θ_c is critical angle and $\theta_c < A \leq 2\theta_c$, the ray emerges out only if $i < \sin^{-1} [\mu \sin (A - \theta_c)]$
- (C) When the ray emerges out, the deviation will be minimum when angle of incidence is $\sin^{-1} [\mu \sin (A/2)]$
- (D) When the ray emerges out, the deviation will be maximum when angle of incidence is 90°

5*. A ball is projected from a building of height 20 m at a speed of 30 m/sec making an angle of 30° with the horizontal. Then : ($g = 10 \text{ m/s}^2$)



- (A) Time after with ball strike the ground is 4 sec
- (B) ball comes to a height of 20 m again after 3 sec
- (C) value of β is $\tan^{-1} \left(\frac{5\sqrt{3}}{9} \right)$
- (D) Value of D is $60\sqrt{3} \text{ m}$.

6*. A particle is projected with a speed of 50 m/s at an angle of 37° with the horizontal. Then select the CORRECT statement (s) :

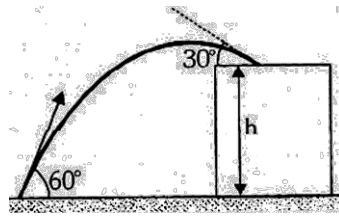
- (A) Equation of trajectory of a particle is $y = \frac{3}{4}x - \frac{x^2}{320}$
- (B) Particle moves perpendicular to initial direction at $\frac{25}{3} \text{ sec}$
- (C) Minimum speed of particle in subsequent motion is zero
- (D) Maximum height attained by the particle from the point of projection is 45 m

7. A flat glass slab of thickness 6 cm and refractive index 1.5 is placed in front of a plane mirror. An observer is standing behind the glass slab and looking at the mirror. The actual distance of the observer from the mirror is 50 cm. The observer sees the image at a distance of d_1 cm from himself. Now the slab is removed and the observer sees his image in plane mirror at a distance of d_2 cm from himself. What is the value of $d_2 - d_1$?

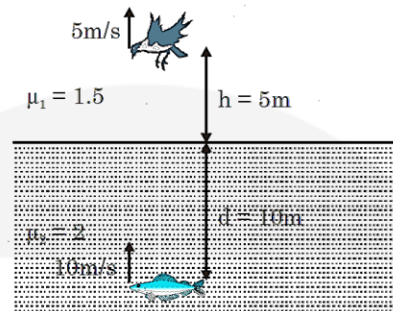
8. On the surface of a liquid reservoir of depth 3m is a circular raft of radius 6m. The surface is illuminated by light incident from all the directions. Determine the radius (in m) of the shadow of the raft at the bottom of the reservoir. The refractive index of the liquid is 1.25.



9. A stone projected at an angle of 60° from the ground level strikes at an angle of 30° on the roof of a building of height 'h'. If the speed of projection of the stone is $\sqrt{Ngh}$, find N.



10. Diagram shows a bird B and a fish F moving in two different medium of refractive index 1.5 and 2 with speed 10 m/s and 5 m/s respectively. At the given instant bird is located at height 5m above the interface while fish is located at depth of 10m. List-I indicates various parameters while list-II gives its values. Match them correctly :-



List – I

- (P) Separation of fish as observed by bird
 (Q) Separation of bird as observed by fish
 (R) Speed of fish as observed by bird
 (S) Speed of bird as observed by fish

List – II

- (1) $\frac{50}{3}$ unit
 (2) 12.5 unit
 (3) $\frac{10}{3}$ unit
 (4) 2.5 Unit



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-08

Note: (*) → Multiple Correct Type Question

Total Marks: 34

Max. Time: 24 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.7 to Q.8

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9 to Q.10

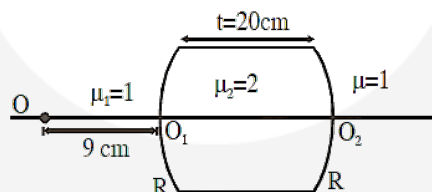
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. The ratio of angle of minimum deviation produced by a thin prism $\left(\mu = \frac{3}{2}\right)$ in air to that in liquid of refractive index $\frac{9}{7}$ is :

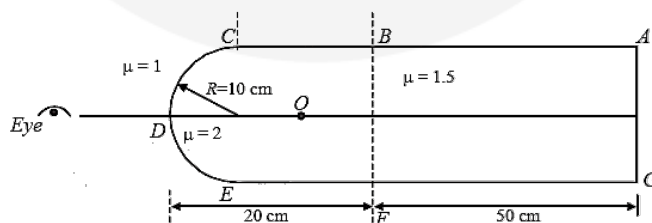
(A) $\frac{1}{3}$ (B) 3 (C) $\frac{1}{4}$ (D) 4

2. In the given figure, if object is placed at a distance 9 cm from O_1 , then find the image distance from O_2 . ($R = 9$ cm)



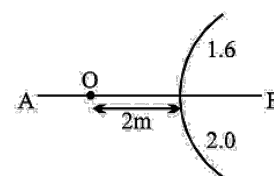
(A) 9 cm (B) 20 cm
(C) 18 cm (D) infinity

3. As shown in the figure, region BCDEF and ABFG are of refractive index 2.0 and 1.5 respectively. A particle O is kept at the mid of region BCEF. Final image of the object as seen by the eye is at a distance :



(A) 30 cm from point D (B) 22.5 cm from point D
(C) 40 cm from point D (D) 20 cm from point D

4. In the figure shown, a point object O is placed in air. A spherical boundary of radius of curvature 1.0 m separates various media. AB is principal axis. The refractive index above AB is 1.6 and below AB is 2.0. The separation between the images formed due to refraction at spherical surface is:



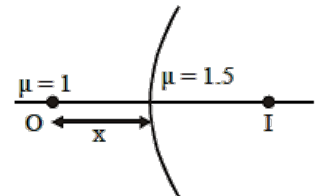
(A) 12 m (B) 20 m (C) 14 m (D) 10 m



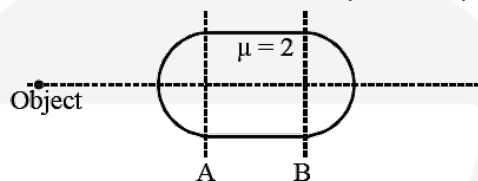
5. Two cars are moving in the same direction with a speed of 30 km h^{-1} . They are separated from each other by 5 km . Third car moving in the opposite direction meets the two cars after an interval of 4 minutes . What is the speed of the third car?
 (A) 35 km h^{-1} (B) 40 km h^{-1} (C) 45 km h^{-1} (D) 75 km h^{-1}

6. An elevator car, whose floor to ceiling distance is equal to 2.7 m , starts ascending with constant acceleration of 1.2 ms^{-2} . 2 sec after the start, a bolt begins falling from the ceiling of the car. The free fall time of the bolt is:
 (A) $\sqrt{0.54} \text{ s}$ (B) $\sqrt{6} \text{ s}$ (C) 0.7 s (D) 1 s

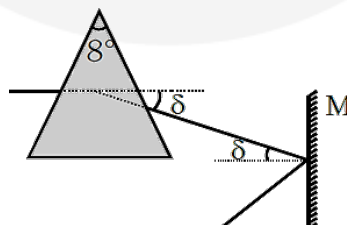
- 7*. A convex spherical surface of radius 10 cm is used to form the image of a point object as shown. What should be the object distance x so that the distance between object and the image is 1 m ?
 (A) 40 cm (B) 50 cm (C) 60 cm (D) 70 cm



- 8*. There is a uniform capsule shaped (hemispherical on sides and cylindrical in between) transparent glass, with $\mu = 2$, and radius of curvature 10 cm (for spherical portion) kept in air :



- (A) For $AB = 20 \text{ cm}$, a narrow parallel beam entering from left side will come out from other side as a parallel beam.
 (B) For $AB = 20 \text{ cm}$, and surrounding air being replaced by a medium, $\mu = 3/2$, on both sides, parallel light beam will come out as a diverging beam.
 (C) For $AB = 20 \text{ cm}$, and surrounding air being replaced by a medium, $\mu = 3/2$, on both sides, parallel light beam will come out as a parallel beam.
 (D) For surrounding $\mu = 3/2$, if emergent beam has to be parallel then AB has to be bigger than 20 cm .
9. A horizontal ray of light passes through the glass prism (Refractive index = $3/2$) whose apex is 8° & then strikes the vertical mirror M as shown in figure. For the ray to become horizontal after reflection, find angle by which mirror must be rotated.



10. Particle is projected from the ground with some velocity. This particle grazes top of the two poles at time 2 seconds and 6 second . The ratio of maximum height reached by the particle to that of height of the pole is $\frac{4}{x}$. Find the value of x .



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-09

Note: (*) → Multiple Correct Type Question

Total Marks: 28

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.4

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.5 to Q.7

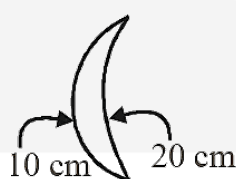
(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.8 to Q.10

(2 marks, 2 min.)

(There is partial marking in multiple correct questions)

1. Figure shows a lens placed in air having refractive index $\mu = \frac{3}{2}$ and radius of curvature as shown in figure. What is focal length of the lens?

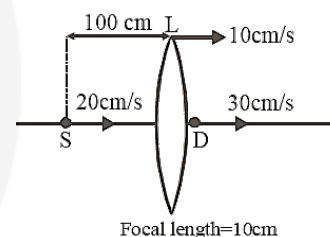


- (A) +40 cm (B) -40 cm (C) +40/3 cm (D) -40/3 cm

2. The figure shows the initial position of a point source of light S, a detector D and a lens L. Now at $t = 0$, all the three start moving towards right with different velocities as shown.

The time (s) at which the detector receives the maximum light is:

- (A) 0.56 s and 8.94 s (B) 3.8 s
(C) 8.94 s and 19.62 s (D) 0.56 s



3. For two particles A and B, given that $\vec{r}_A = 2\hat{i} + 3\hat{j}$, $\vec{r}_B = 6\hat{i} + 7\hat{j}$, $\vec{v}_A = 3\hat{i} - \hat{j}$ and $\vec{v}_B = x\hat{i} - 5\hat{j}$. What is the value of x if they collide?

- (A) 1 (B) -1 (C) 2 (D) -2

4. A swimmer swims in still water at a speed = 5 km/hr. He enters a 200 m wide river, having river flow speed = 4 km/hr at point A and proceeds to swim at an angle of 127° with the river flow direction. Another point B is located directly across A on the other side. The swimmer lands on the other bank at a point C, from which he walks the distance CB with a speed = 3 km/hr. The total time in which he reaches from A to B is:

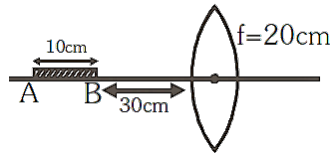
- (A) 5 minutes (B) 4 minutes (C) 3 minutes (D) None

- 5*. A convex lens made of glass ($\mu_g = 3/2$) has focal length f in air. The image of an object placed in front of it is inverted, real and magnified. Now whole arrangement is immersed in water ($\mu_w = 4/3$) without changing distance between object and lens. Then:

- (A) the new focal length will become 4f
(B) the new focal length will become f/4
(C) new image will be virtual and magnified
(D) new image will be real, inverted and smaller in size



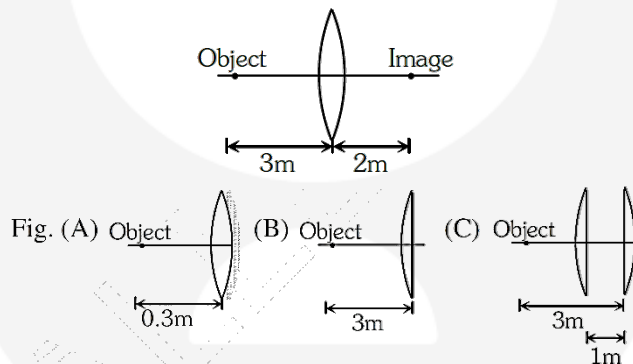
- 6*. AB is a linear object placed along optical axis as shown in figure. Tick the incorrect statement(s):



- (A) The length of image is smaller than the length of object.
(B) The length of image is larger than the length of object.
(C) The length of image is equal to the length of object.
(D) If middle portion of the lens is painted then the length of image is smaller than length of object.
- 7*. A person is standing on a truck moving with a constant velocity of 15 m/s on a horizontal road. The man throws a ball in such a way that it returns to his hand after the truck has moved 60 m. Then: ($g = 10 \text{ m/s}^2$)
(A) The speed of the ball as seen from the truck is 20 m/s
(B) The direction of initial velocity of ball is upward as seen from the truck
(C) The initial speed of the ball as seen from the ground is 25 m/s
(D) None of these

Paragraph for Q. 8 to Q. 10

An object is placed at a distance 3 meter from an equi-convex lens of refractive index $\mu = \frac{3}{2}$. Its image is formed at 2 m from the lens as shown in the figure.



8. Find the position of image if one face of the lens is silvered [Fig.(A)]
(A) 12 m from the lens towards right (B) 12 m from the lens towards left
(C) 6 m from the lens towards right (D) at infinity
9. Find the position of image if lens is cut into two symmetrical plano-convex lenses and one of the planoconvex lens is removed [Fig.(B)].
(A) 12 m from the lens towards right (B) 12 m from the lens towards left
(C) 6 m from the lens towards right (D) 6 m from the lens towards left
10. Find the position of image if plano-convex surfaces are displaced by 100 cm. [Fig.(C)]
(A) 2.94 m towards right of second lens (B) 2.94 m towards right of first lens
(C) 3.31 m towards right of second lens (D) 4.31 m towards right of first lens



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-10

Note: (*) → Multiple Correct Type Question

Total Marks: 33

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Comprehension Type Questions ('-1' Negative marking) Q.6 to Q.8

(2 marks, 2 min.)

Subjective Type Questions (No negative marking) Q.9

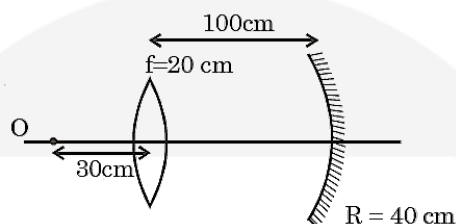
(4 marks, 3 min.)

Match the Column Type Questions (No negative marking) Q.10

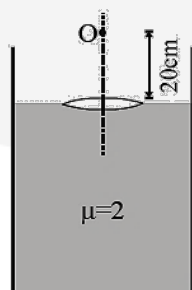
(8 marks, 4 min.)

(There is partial marking in multiple correct questions)

1. Find location of final image of O after all possible refraction & reflection:



- (A) At 60 cm right from lens.
 (B) At 40 cm left from lens.
 (C) At 30 cm right from lens.
 (D) On object.
2. A point object O is placed at a distance of 20 cm in front of a equi-convex lens ($\mu_g = 1.5$) of focal length 10 cm. The lens is placed on a liquid of refractive index 2 as shown. Image will be formed at a distance h from lens. The value of h is:



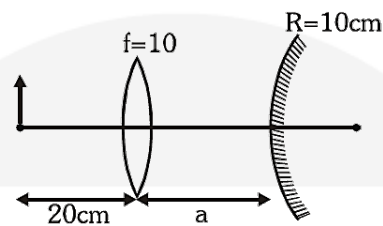
- (A) 5 cm (B) 10 cm (C) 20 cm (D) 40 cm
3. A plano-convex lens of radius of curvature 12cm and refractive index 1.5 has the plane face silvered then:
- (A) It will behave as a convex mirror of focal length 12 cm.
 (B) It will behave as a rectangular glass slab.
 (C) It will behave as a concave mirror of focal length 12 cm.
 (D) It will behave as a concave mirror of focal length 24 cm.



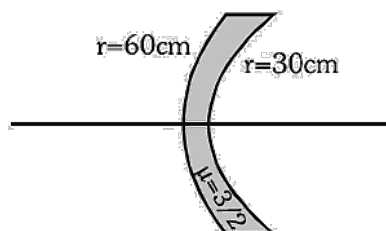
4. For four particles A, B, C & D the velocities of one with respect to other are given as $\vec{V}_{DC}$ is 20 m/s towards north, $\vec{V}_{BC}$ is 20 m/s towards east and $\vec{V}_{BA}$ is 20 m/s towards south. Then $\vec{V}_{DA}$ is:
- (A) 20 m/s towards north (B) 20 m/s towards south
(C) 20 m/s towards east (D) 20 m/s towards west
5. A man walks in rain with a velocity of 5 kmh^{-1} . The rain drops strike at him at an angle of 45° with the horizontal. The velocity of the rain if it is falling vertically downward is:
- (A) 5 kmh^{-1} (B) 4 kmh^{-1} (C) 3 kmh^{-1} (D) 1 kmh^{-1}

Paragraph for Q. 6 to Q. 8

Answer the following questions for the given setup.



6. Find nature and position of final image, if $a = 20 \text{ cm}$
- (A) At same position, real and inverted (B) At same position, virtual and inverted
(C) At same position, virtual and erect (D) At same position, real and upright
7. Find nature and position of final image, if $a = 10 \text{ cm}$
- (A) At same position, real and inverted (B) At same position, virtual and erect
(C) At same position, virtual and erect (D) At same position, real and upright
8. Find nature and position of final image, if $a = 5 \text{ cm}$
- (A) Real, inverted at 50 cm from lens towards object side
(B) Real, upright at 50 cm from lens towards mirror side
(C) At infinity
(D) None of these
9. The given lens is having radius of curvature 30 cm and 60 cm as shown in the figure respectively. The power of lens is $-5/\alpha$ diopter. Find the value of α .

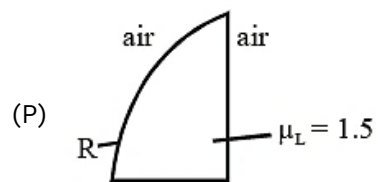




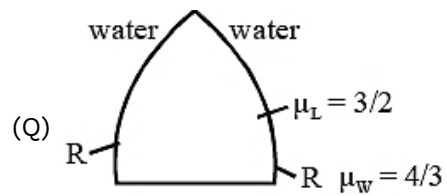
10. Using the lens maker formula, relate the focal length of thin lens in List-I with value in List-II.

List-I

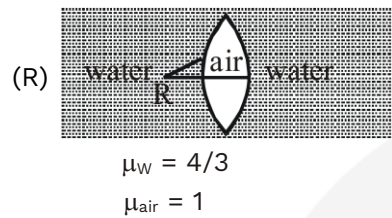
List-II



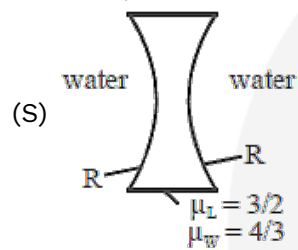
(1) $4R$



(2) $-2R$



(3) $-4R$



(4) $2R$

Codes:

	P	Q	R	S
(A)	1	3	4	2
(B)	4	2	1	3
(C)	3	2	1	4
(D)	4	1	2	3



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-11

Note: (*) → Multiple Correct Type Question

Total Marks: 33

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking)

(3 marks, 2 min.)

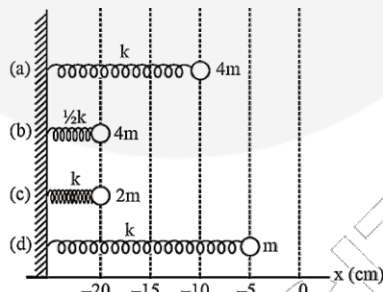
Multiple Correct Questions ('-2' negative marking)

(4 marks, 3 min.)

Subjective Type Questions (No negative marking)

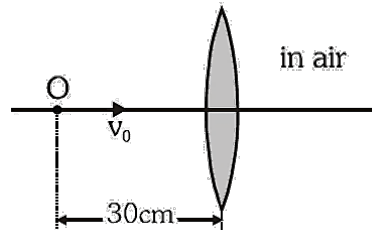
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

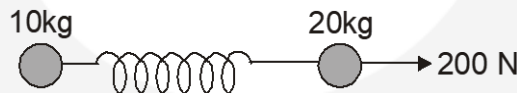
- Which of the following quantities are always positive in a simple harmonic motion?
 (A) $\vec{F} \cdot \vec{a}$ (B) $\vec{v} \cdot \vec{r}$ (C) $\vec{a} \cdot \vec{r}$ (D) $\vec{F} \cdot \vec{r}$
- The motion of a particle represented by $y = \sin \omega t - \cos \omega t$ is:
 (A) NOT S.H.M.
 (B) S.H.M. with period $\frac{2\pi}{\omega}$, amplitude $\sqrt{2}$
 (C) S.H.M. with complicated period, amplitude 1
 (D) S.H.M. with period $\frac{\sqrt{2}\pi}{\omega}$, amplitude $\sqrt{2}$
- Four springs have been compressed from their equilibrium position at $x = 0$ cm. When released, they will start to oscillate. Rank in order, from highest to lowest, the maximum speeds of the oscillators.

 (A) $c = b > a > d$ (B) $a = d > b > c$ (C) $a > d > b = c$ (D) $c > b > a = d$
- The acceleration of a particle moving rectilinearly varies with displacement as $a = -4x$. At $x = 4$ m and $t = 0$, particle is at rest. Select the INCORRECT alternative.
 (A) The maximum speed of the particle is 8 m/s
 (B) The distance travelled by the particle in first second is 20 m.
 (C) The velocity – acceleration graph of the particle is an ellipse.
 (D) The kinetic energy – displacement graph of the particle is a parabola.



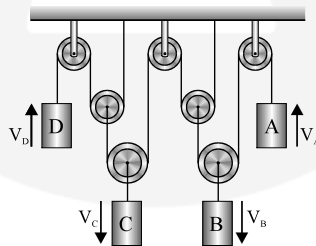
5. A bi-convex lens, made up by material of refractive index $\frac{3}{2}$ has focal length 10 cm in air. An object is placed at a distance 30 cm from it and is moving towards the lens with speed v_0 as shown. Initially the setup is in air and finally whole setup is immersed in water tank. Refractive index of water is $\frac{4}{3}$. Choose the incorrect option(s):



- (A) In air, speed of image is greater than that of the object
(B) In water, speed of image is greater than that of the object
(C) In water, nature of image is virtual
(D) In air, nature of image is real
6. Two masses of 10 kg and 20 kg respectively are connected by a massless spring as shown in figure. A force of 200 N acts on the 20 kg mass at the instant when the 10 kg mass has an acceleration of 12 ms^{-2} towards right, the acceleration of the 20 kg mass is :



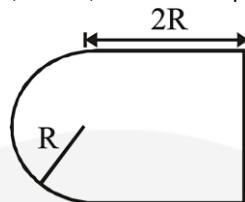
- (A) 2 ms^{-2} (B) 4 ms^{-2} (C) 10 ms^{-2} (D) 20 ms^{-2}
7. Choose the relationship which governs the velocities of the four cylinders.



- (A) $v_A + v_B - 4v_C + v_D = 0$ (B) $4v_A - 8v_B - 4v_C + v_D = 0$
(C) $4v_A - 4v_B - 4v_C + v_D = 0$ (D) $v_A + v_B - v_C + v_D = 0$
- 8*. A particle executes SHM with equation of motion given by
- $$X(t) = (20.0 \text{ cm}) \cos [(12\pi \text{ rad/s}) t - \frac{\pi}{3} \text{ rad}]$$
- (A) Magnitude of maximum velocity of particle is $240\pi \text{ cm/s}$
(B) Magnitude of maximum acceleration of particle is $2880\pi^2 \text{ cm/s}^2$
(C) Displacement of the particle from its mean position will be minimum at $t = \frac{1}{36} \text{ sec}$
(D) Net force acting on the particle will be minimum at $t = \frac{5}{72} \text{ sec}$



- 9.*** A monkey of mass 40 kg climbs on a rope which can stand maximum tension of 600 N. In which of the following case will the rope does not break?
- (A) The monkey climbs up with an acceleration of 4 m/s^2
 - (B) The monkey climbs down with an acceleration of 4 m/s^2
 - (C) The monkey climbs up with uniform speed of 5 m/s
 - (D) The monkey falls down the rope freely under gravity
- 10.** A small object placed in air at 24 cm from convex surface of a rod of glass ($n = 1.5$) of length $2R$ has a convex hemispherical surface with radius of curvature $R = 8 \text{ cm}$ and a flat surface (figure). Find the final distance of the image (in cm) from the plane surface.



Note: (*) → Multiple Correct Type Question

Total Marks: 40

Max. Time: 27 min.

Single Correct Questions ('-1' negative marking)

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

(4 marks, 3 min.)

Match the Column Type Questions (No negative marking)

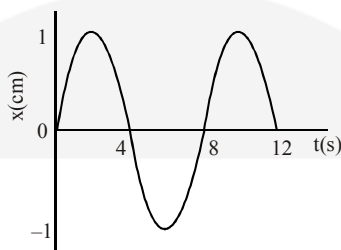
(8 marks, 4 min.)

Subjective Type Questions (No negative marking)

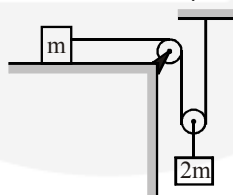
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

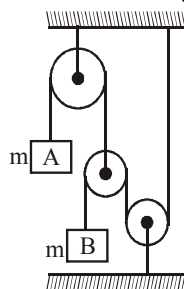
1. The x-t graph of a particle undergoing simple harmonic motion is shown below. The acceleration of the particle at $t = 4/3$ s is:



- (A) $\frac{\sqrt{3}}{32} \pi^2 \text{ cm/s}^2$ (B) $\frac{-\pi^2}{32} \text{ cm/s}^2$ (C) $\frac{\pi^2}{32} \text{ cm/s}^2$ (D) $-\frac{\sqrt{3}}{32} \pi^2 \text{ cm/s}^2$
2. The minimum angle of deviation of a prism of refractive index μ is equal to its refracting angle. The refracting angle of prism is :
- (A) $2 \cot^{-1} (\mu/2)$ (B) $2 \tan^{-1} (\mu/2)$ (C) $2 \sin^{-1} (\mu/2)$ (D) $2 \cos^{-1} (\mu/2)$
3. Find the acceleration of block of mass m . Assume pulleys are massless and frictionless.



- (A) $\frac{g}{3}$ (B) $\frac{2g}{3}$ (C) $\frac{g}{2}$ (D) None
4. In the arrangement shown, the pulleys and the strings are ideal. The acceleration of block B is:

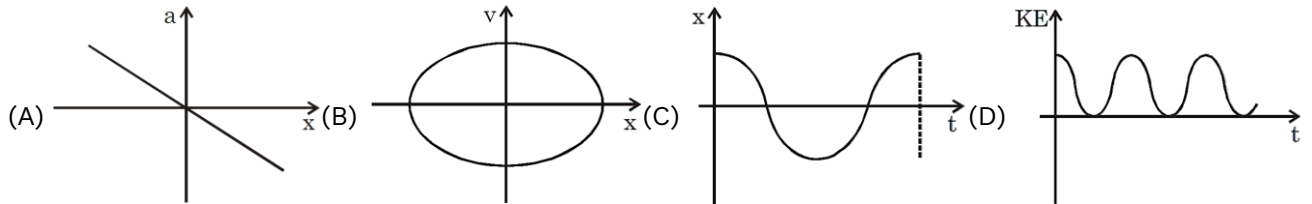


- (A) $g/5$ (B) $g/2$ (C) $2g/5$ (D) $2g/3$



- 5.* Acceleration of a particle which is at rest at $x = 0$ is $\vec{a} = (4 - 2x)\hat{i}$. Select the correct alternative(s)-
 (A) Particle further comes to rest at $x = 4$
 (B) Particle oscillates about $x = 2$
 (C) Maximum speed of particle is 4 units
 (D) All of the above

- 6.* Which of the following is/are correct graph/s for SHM $x = A \sin(\omega t)$:



- 7.* The mechanical energy of a system is given by, $E = Ax^2 + Bv^2$ where x is the displacement from ($x = 0$) and v is the velocity of the body at x then choose the correct statement(s). Assume A and B are positive constant.

- (A) The maximum x coordinate is $\sqrt{\frac{E}{A}}$
 (B) Maximum velocity of the particle during motion is $\sqrt{\frac{E}{B}}$
 (C) $x = 0$ is the equilibrium position
 (D) The acceleration of the system at any point is $\frac{-Ax}{B}$

8. Velocity of three particles A, B and C varies with time t as, $\vec{v}_A = (2t\hat{i} + 6\hat{j})$ m/s; $\vec{v}_B = (3\hat{i} + 4\hat{j})$ m/s; and $\vec{v}_C = (6\hat{i} - 4t\hat{j})$ m/s. Regarding the pseudo force match the following table:

Column I	Column II
(A) On A as observed by B	(P) Along positive x -direction
(B) On B as observed by C	(Q) Along negative x -direction
(C) On A as observed by C	(R) Along positive y -direction
(D) On C as observed by A	(S) Along negative y -direction
	(T) Zero

9. A particle performing linear SHM with time period T is at $x = +\frac{A}{2}$ moving towards positive extreme.

The time, after which it will first come to rest is $\frac{T}{N}$. Find N .

10. A ray of light enters a rectangular glass slab of refractive index $\sqrt{3}$ at an angle of incidence 60° . It travels a distance of 5 cm inside the slab and emerges out of the slab. Find out the perpendicular distance between the incident and the emergent ray.

Note: (*) → Multiple Correct Type Question

Total Marks: 35

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.8

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.9 to Q.10

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.10

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. A particle performs SHM on x-axis from $x = -2$ cm to $x = 6$ cm with time period of 0.5 sec. It was located at $x = 4$ cm and moving in positive 'x' at $t = 0$.

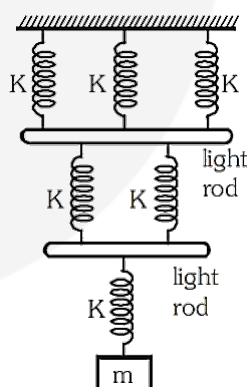
- (A) The equation of SHM of the particle is $x = 2 + 4 \sin (4\pi t + 30^\circ)$
 (B) The equation of SHM of the particle is $x = -2 + 2 \cos 4\pi t$
 (C) The velocity of the particle at $x = 4$ cm is 3 cm/s
 (D) The acceleration at $x = 4$ cm is 32π cm/s²

2. A man of mass 60 kg is standing on a platform which is executing S.H.M. in the vertical plane. The displacement from the mean position varies as $y = 0.5 \sin (2\pi ft)$. The minimum value of f , for which the man will feel weight less ness at the highest point is: (y is in meters)

- (A) $\frac{g}{4\pi}$ (B) $4\pi g$ (C) $\frac{\sqrt{2g}}{2\pi}$ (D) $2\pi\sqrt{2g}$

3. The time period of small oscillations of mass m :

- (A) $2\pi\sqrt{\frac{m}{6K}}$
 (B) $2\pi\sqrt{\frac{11m}{6K}}$
 (C) $2\pi\sqrt{\frac{6m}{11K}}$
 (D) $2\pi\sqrt{\frac{m}{K}}$



4. A body of mass 2 kg is placed on a horizontal surface having coefficient of kinetic friction 0.4 and coefficient of static friction 0.5. If a horizontal force of 25 N is applied on the body, the frictional force acting on the body will be: ($g = 10$ m/s²)

- (A) 8 N (B) 10 N (C) 20 N (D) 25 N

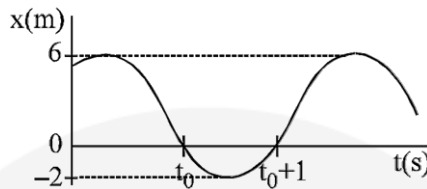
5. A particle moves on a rough horizontal ground with some initial velocity say v_0 . If $3/4^{\text{th}}$ of its kinetic energy is lost in friction in time t_0 . Then coefficient of friction between the particle and the ground is:

- (A) $\frac{v_0}{2gt_0}$ (B) $\frac{v_0}{4gt_0}$ (C) $\frac{3v_0}{4gt_0}$ (D) $\frac{v_0}{gt_0}$

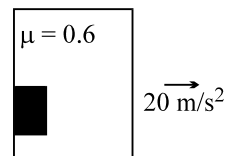


- 6.*** A particle of mass m moves rectilinearly under the action of a force $F = \alpha - \beta x$. Here x is distance from the point where it was at rest ($x = 0$). (α, β are positive constants):
- (A) Motion is simple harmonic
- (B) Amplitude of particle is $\frac{\alpha}{\beta}$
- (C) Mean position of particle may be at $\frac{\alpha}{\beta}$
- (D) Maximum acceleration of particle is $\frac{\beta}{m}$.

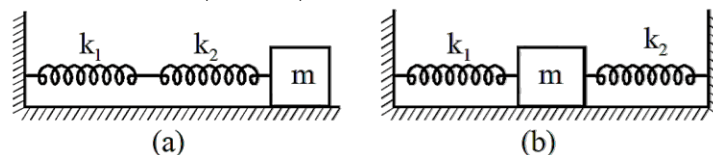
- 7.*** A particle executes SHM about a point other than $x = 0$ as shown in the graph. Then:



- (A) Amplitude is equal to 4m
- (B) Equilibrium position is at $x = 0$
- (C) Equilibrium position is at $x = 2$
- (D) Angular frequency = $\frac{2\pi}{3}$ rad/s
- 8.*** A particle on a spring executes simple harmonic motion. When the particle is found at $x = \frac{x_{\max}}{2}$, then (consider only magnitude):
- (A) Speed = $v_{\max}/2$
- (B) speed = $\sqrt{3}v_{\max}/2$
- (C) acceleration = $\sqrt{3}a_{\max}/2$
- (D) acceleration = $a_{\max}/2$
- 9.*** A box is accelerating with acceleration = 20 m/s^2 . A block of mass 10 kg is placed inside the box and is in contact with the vertical wall as shown. The friction coefficient between the block and the wall is $\mu = 0.6$ (take $g = 10 \text{ m/s}^2$)
- (A) The acceleration of the block will be 20 m/s^2
- (B) The friction force acting on the block will be 100 N
- (C) The contact force between the vertical wall and the block will be $100\sqrt{5} \text{ N}$
- (D) The contact force between the vertical wall and the block is only electromagnetic in nature



- 10.** In the figure shown, a book executes SHM under the influence of two springs on a smooth surface as shown. It is found that the time period of oscillation in case (a) is double of that in case (b). if k_1 is 10 N/m , What is the value of k_2 (in N/m)?





SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-14

Note: (*) → Multiple Correct Type Question

Total Marks: 35

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking)

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking)

(3 marks, 2 min.)

Subjective Type Questions (No negative marking)

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

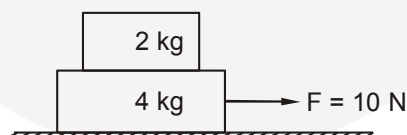
1. A wooden cube (density of wood $\frac{\rho}{3}$) of side ' ℓ ' floats in a liquid of density ' ρ ' with its upper and lower surfaces horizontal. If the cube is pushed slightly down and released, it performs simple harmonic motion of period ' T '. Then, ' T ' is equal to:

(A) $2\pi\sqrt{\frac{3\ell}{2g}}$ (B) $2\pi\sqrt{\frac{\ell}{3g}}$ (C) $2\pi\sqrt{\frac{3\ell}{g}}$ (D) $2\pi\sqrt{\frac{2\ell}{g}}$

2. A particle is placed at the lowest point of a smooth wire frame in the shape of a parabola, lying in the vertical xy-plane having equation $x^2 = 5y$ (x, y are in meter). After slight displacement, the particle is set free. Find angular frequency of oscillation (in rad/sec) (take $g = 10 \text{ m/s}^2$):

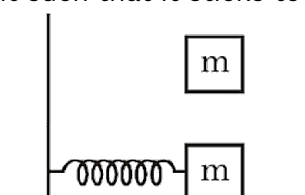
(A) 2 rad/s (B) 4 rad/s (C) 6 rad/s (D) 8 rad/s

3. In the given figure, the blocks are at rest. After that, a force of 10 N acts on the block of 4 kg mass. The co-efficient of static friction and the co-efficient of kinetic friction are $\mu_s = 0.2$ and $\mu_k = 0.15$ for both the surfaces in contact. The magnitude of friction force acting between the surface of contact between the 2 kg and 4 kg block in this situation is:



(A) 3 N (B) 4 N (C) 3.33 N (D) Zero

- 4.* A particle of mass m is performing SHM with amplitude A . When the particle is at mean position, a block of same mass was kept on it such that it sticks to it. Mark the CORRECT option(s):



(A) The new amplitude = $\frac{1}{\sqrt{2}}A$

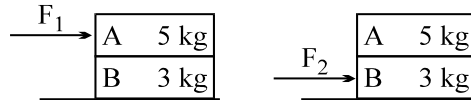
(B) The new time period remains unchanged

(C) The new amplitude = $\sqrt{\frac{7}{16}}A$

(D) The new time period increase.



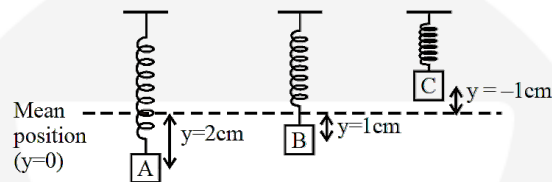
- 5*. A block A (5 kg) rests over another block B (3 kg) placed over a smooth horizontal surface. There is friction between A and B. A horizontal force F_1 gradually increasing from zero to a maximum is applied to A so that the blocks move together without having motion relative to each other. Instead of this, another horizontal force F_2 gradually increasing from zero to a maximum is applied to B so that the blocks move together without relative motion. The magnitudes of friction between the blocks in the two cases are f_1 & f_2 respectively during the variation of F_1 and F_2 . Then



- (A) $f_{1\max} > f_{2\max}$ (B) $F_{1\max} : F_{2\max} = 3 : 5$
 (C) $F_{1\max} : F_{2\max} = 5 : 3$ (D) $f_1 < F_1$

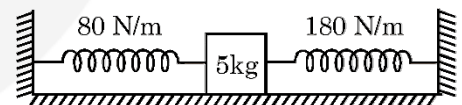
Paragraph for question No. 6 to 7

Three identical particle each of mass m are attached to three springs of identical force constant k as shown in the diagram. All masses oscillate with an amplitude 2 cm. Initially mass A is at its positive extreme ($y = +2$ cm), b is at $y = +1$ cm and going upward C is at $y = -1$ cm and going downward.



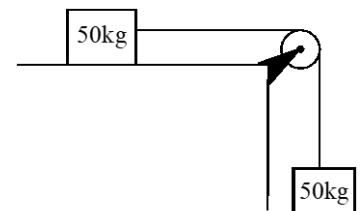
6. Maximum distance between A and B in subsequent motion:
 (A) 1 cm (B) 2 cm (C) $\sqrt{3}$ cm (4) $2\sqrt{3}$ cm
7. Time after distance between A and B is maximum:
 (A) $\frac{\pi}{3} \sqrt{\frac{m}{k}}$ (B) $\frac{\pi}{2} \sqrt{\frac{m}{k}}$ (C) $\pi \sqrt{\frac{m}{k}}$ (D) $\frac{2\pi}{3} \sqrt{\frac{m}{k}}$

8. A block of mass 5 kg is placed between two spring, just touching them and not connected to any of the spring. If the block is displaced slightly and released, then the time period of resulting oscillation is $\frac{p}{q} \pi$ (in sec). Fill the value of $(q - p)$

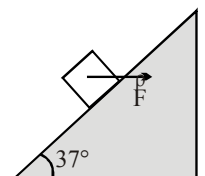


in the OMR.

9. Coefficient of friction between the 50 kg block and table is 0.2. Find the acceleration (in m/s^2) of the blocks.



10. A 5.0 kg block on an inclined plane is acted on by a horizontal force with magnitude 50 N (Figure). The coefficient of kinetic friction between block and plane is 0.20. How far (in m) up the plane does the block go if it has an initial upward speed of 4 m/s?



Note: (*) → Multiple Correct Type Question

Total Marks: 36

Max. Time: 26 min.

Single Correct Questions ('-1' negative marking)

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

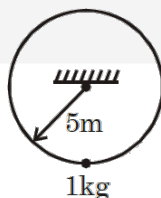
(4 marks, 3 min.)

Subjective Type Questions (No negative marking)

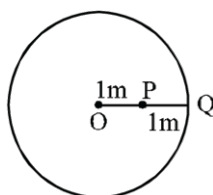
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. A disk of mass $M (= 2 \text{ kg})$ and radius $R (= 5 \text{ m})$ is pivoted at the centre. The disk lies in the vertical plane and is free to rotate about the pivot (see figure). A particle of mass $m (= 1 \text{ kg})$ is attached to the periphery of the disk as shown in the figure. The time period of small oscillation of the system is: [Taking $g = 10 \text{ m/s}^2$]



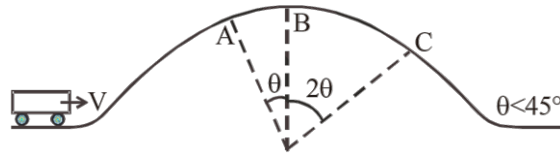
- (A) $2\pi \text{ sec}$ (B) $\pi \text{ sec}$ (C) $4\pi \text{ sec}$ (D) $(\pi/2) \text{ sec}$
2. Four simple harmonic vibrations $y_1 = 8\sin\omega t$, $y_2 = 6\sin(\omega t + \pi/2)$, $y_3 = 4\sin(\omega t + \pi)$, $y_4 = 2\sin(\omega t + 3\pi/2)$ are superimposed to each other. The resulting amplitude and initial phase are respectively:
- (A) $4\sqrt{2}$ and $\tan^{-1}(1)$ (B) $4\sqrt{2}$ and $\tan^{-1}(6/8)$
 (C) 20 and $\tan^{-1}(1/2)$ (D) 20 and $\tan^{-1}(2)$
3. Two particles of equal mass are attached to a string of length 2m as shown in figure. The string and particles are then whirled in a horizontal circle about O. The ratio of the tension in string between P and Q to tension in the string between P and O is



- (A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 2



4. A self propelled vehicle (assume it as a point mass) runs on a track with constant speed V . It passes through three positions A, B and C on the circular part of the track. Suppose N_A , N_B and N_C are the normal forces exerted by the track on the vehicle when it is passing through points A, B and C respectively then:

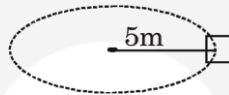


- (A) $N_A = N_B = N_C$ (B) $N_B > N_A > N_C$ (C) $N_C > N_A > N_B$ (D) $N_B > N_C > N_A$

- 5.* A 20 gm particle is subjected to two simple harmonic motion $x_1 = 2 \sin 10t$, $x_2 = 4 \sin \left(10t + \frac{\pi}{3} \right)$ along same straight line. Where x_1 and x_2 are in meter and t is in sec.

- (A) The displacement of the particle at $t = 0$ will be $2\sqrt{3}$ m.
(B) Maximum speed of the particle will be $20\sqrt{7}$ m/s
(C) Magnitude of maximum acceleration of the particle will be $200\sqrt{7}$ m/s².
(D) Energy of the resultant motion will be 28 J.

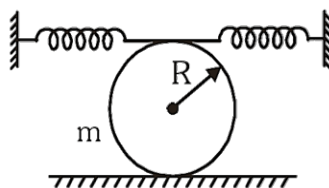
- 6.* A block of mass 25 kg rests on a horizontal floor ($\mu = 0.2$). It is attached by a 5m long horizontal rope to a peg fixed on floor. The block is pushed along the ground with an initial velocity of 10 m/s so that it moves in a circle around the peg. Choose the correct statement(s):



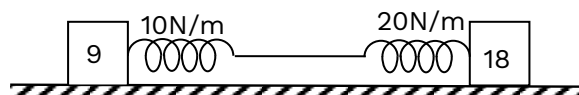
- (A) Magnitude of tangential acceleration of the block is 2 m/s²
(B) Speed of the block at time $t = 1$ sec is 8 m/s
(C) Time when tension in rope becomes zero is 5 sec
(D) Friction force acts towards centre of the circle.

7. A solid uniform cylinder of mass m performs small oscillation due to the action of two springs combined stiffness equal to k (figure). The period of these oscillation in the absence of sliding is

$T = \pi \sqrt{\frac{xm}{2k}}$. Find x :

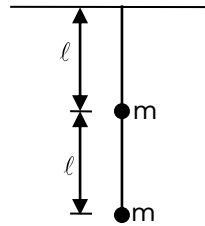


8. Two blocks of masses 9 kg & 18 kg are attached with springs as shown in figure. Find the time period of small oscillation of the two blocks (in sec.) (take $\pi^2 = 10$)

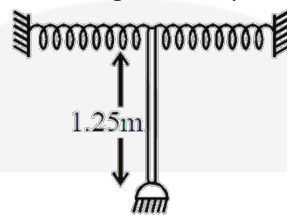




9. Two particles of mass m each are fixed to a massless rod of length 2ℓ . The rod is smoothly hinged at one end to a ceiling. It performs oscillation of small angle in vertical plane. The length of the equivalent simple pendulum is:



10. Two springs of spring constant $\frac{mg}{\ell}$ each are attached to the end of a uniform rod of mass m and length $\ell = 1.25$ m which is hinged in vertical plane as shown in figure. As the rod is rotated slightly about the hinge, it undergoes SHM with angular frequency ω . Find ω (in rad/s).



Note: (*) → Multiple Correct Type Question

Total Marks: 33

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking)

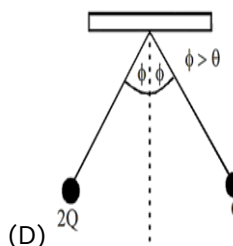
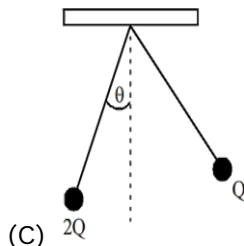
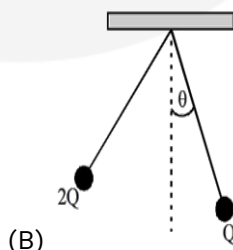
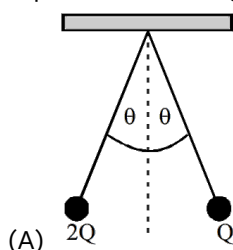
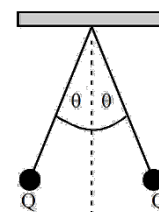
(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

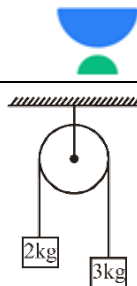
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

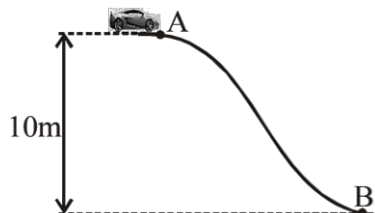
1. If charge on each of two point charges is tripled and separation between them is doubled, by what factor will the mutual force between them change?
(A) 1.50 (B) 2.25 (C) 4.50 (D) None of these
2. Charge Q is divided into two parts, which are then kept some distance apart. The force between them will be a maximum repulsive, if the two parts have charges:
(A) $Q/2$ each (B) $Q/4$ and $3Q/4$
(C) $Q/3$ and $2Q/3$ (D) e and $(Q-e)$ where e is fundamental charge
3. Two point charges q_1 and q_2 are placed at a separation r . The force on q_2 due to q_1 is F . Now the complete set-up is placed in a dielectric medium having di-electric constant $k = 4$. The force due to q_1 on q_2 is :
(A) $F/4$ (B) $\frac{3F}{4}$ (C) $\frac{F}{2}$ (D) F
4. A ring of radius R has a linear charge density $\lambda = \lambda_0 \cos^2\theta$ along its circumference. Total charge on the ring is:
(A) $3\lambda_0\pi R$ (B) $2\lambda_0\pi R$ (C) $\lambda_0\pi R$ (D) Can't be determined
5. Two pith balls with mass m are suspended from insulating threads. When the pith balls are given equal positive charge Q , they hang in equilibrium as shown. We now increase the charge on the left pith ball from Q to $2Q$ while leaving its mass essentially unchanged. Which of the following diagrams best represents the new equilibrium configuration?



6. In the figure shown, find work done by tension in string on 3 kg block in first two seconds, from beginning of motion. The system is set free at $t = 0$ ($g = 10 \text{ m/s}^2$) (Assume pulley and string are ideal)
- (A) -96 Joule (B) -48 Joule (C) +24 Joule (D) 96 Joule

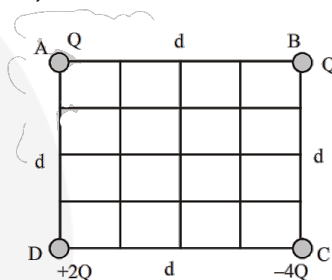


7. A car of mass 1000 kg moves from point A to B. If kinetic energy of car at point A is 100 kJ and at B is 220 kJ, then find work done by friction force on the car.

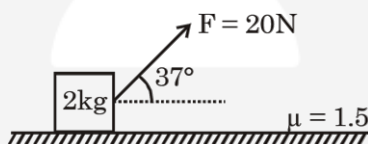


- 8*. If magnitude of force applied by charge placed at point B on A is F , then choose the CORRECT statement(s):-

- (A) Magnitude of force applied by B on C is $4F$.
 (B) Magnitude of force applied by B on D is F .
 (C) Magnitude of force applied by A on C is $2F$.
 (D) Magnitude of force applied by B on D is $\frac{F}{\sqrt{2}}$.

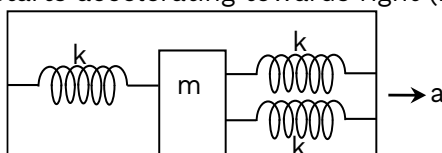


- 9*. A block of mass 2 kg is initially at rest on a rough surface having coefficient of friction $\mu = 1.5$. A force 20N is applied as shown in figure then which of the following is/are CORRECT:



- (A) Acceleration of block is 2 m/s^2 .
 (B) Work done by applied force (F) when block has moved by 2m is 32J.
 (C) Work done by kinetic friction when block has moved by 2m is -24J.
 (D) Magnitude of normal contact force applied by block on the ground is 8N.

- 10.* Figure shows a block of mass m at rest, attached with three identical ideal springs of spring constant k . if the container starts accelerating towards right (horizontal), then:



- (A) Motion of block will be SHM as seen from the frame of container and not in ground frame.
 (B) Motion of the block will be SHM from both frames.
 (C) Time period from frame of container will be $2\pi\sqrt{\frac{m}{3k}}$.
 (D) Amplitude of SHM as seen from the container frame will be $\frac{ma}{3k}$.

Note: (*) → Multiple Correct Type Question

Total Marks: 34

Max. Time: 24 min.

Single Correct Questions ('-1' negative marking)

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

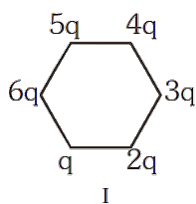
(4 marks, 3 min.)

Matrix Match Type Questions

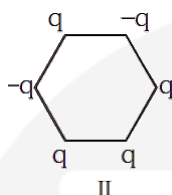
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

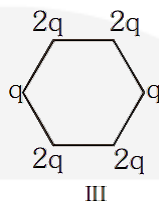
1. Figure below show regular hexagon, the charges are placed at the vertices. In which of the following cases the electric field at the centre is zero.



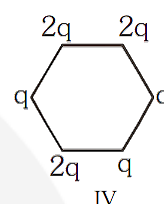
(A) IV



(B) III



(C) I



(D) II

2. Four charge q , $2q$, $-4q$ and $2q$ are placed in order at the four corners of a square of side b . The net field at the centre of the square is

(A) $\frac{q}{2\pi\epsilon_0 b^2}$ from $+q$ to $-4q$

(B) $\frac{5q}{2\pi\epsilon_0 b^2}$ from $+q$ to $-4q$

(C) $\frac{10q}{2\pi\epsilon_0 b^2}$ from $+q$ to $-4q$

(D) $\frac{20q}{2\pi\epsilon_0 b^2}$ from $-4q$ to $+q$

3. A point charge $+q$ is placed at a distance $5f$ from optical centre of converging lens of focal length f and another point charge $-q$ is placed at image position, magnitude of electric field at the optical centre due to these charge is:

(A) $\frac{16kq}{25f^2}$

(B) Zero

(C) $\frac{17kq}{25f^2}$

(D) $\frac{kq}{235f^2}$

4. Consider two identical charges placed distance $2d$ apart, along x -axis. The equilibrium of a positive test charge placed at the point O midway between them is:

(A) Neutral

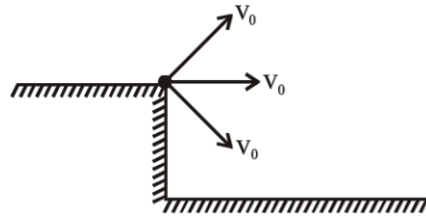
(B) Stable for displacements along the x -axis.

(C) Stable for displacement along the y -axis.

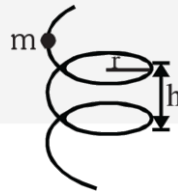
(D) Neutral for displacements along the z -axis



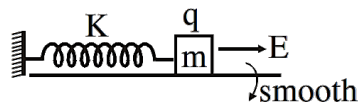
5. A ball is launched from the same height repeatedly with the same speed v_0 but in different directions A, B & C as shown. It reaches the ground with speeds v_A , v_B & v_C respectively. Which of the following is true:



- (A) $v_A = v_B = v_C$
(B) $v_A > v_B > v_C$
(C) $v_A < v_B < v_C$
(D) None of these
6. A bead of mass 'm' slide down with constant speed on a helical wire as shown, where h is the vertical displacement of the particle in one complete revolution then modulus of work done by friction in one complete revolution:



- (A) mgh
(B) will depend on speed of the particle
(C) less than mgh
(D) more than mgh
- 7*. An oil drop has a charge -9.6×10^{-19} C and has a mass 1.6×10^{-15} gm. When allowed to fall, due to air resistance it attains a constant velocity. Then if a uniform electric field is to be applied vertically to make the oil drop ascend up with the same constant speed, which of the following are correct?
- (A) the electric field is directed upward
(B) the electric field is directed downward
(C) the intensity of the electric field is $1/3 \times 10^2$ N/C
(D) the intensity of the electric field is $1/6 \times 10^5$ N/C
- 8*. In the given figure, initially an ideal sufficiently long spring is at its natural length when a constant electric field is switched on. Choose correct option/options

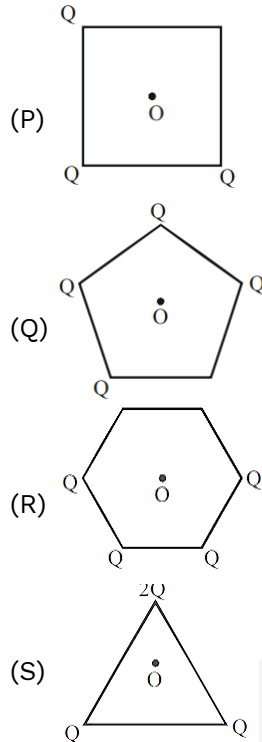


- (A) Maximum elongation of springs is $\frac{2qE}{K}$
(B) Maximum compression of springs $\frac{qE}{K}$
(C) Time period of SHM is $2\pi\sqrt{\frac{m}{K}}$ s.
(D) It will perform SHM if and only if magnitude of field is small.

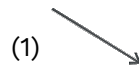


9. In the given diagrams, the direction of electric field at point O is given in list-II (O is circumcenter of the given regular polygon). Charge Q is positive charge. Match the direction of electric field for the given arrangement:

List-I

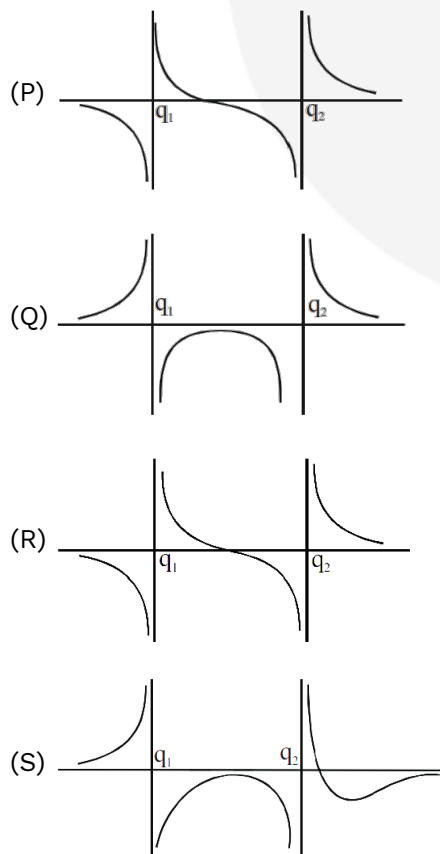


List-II



10. Variation of electric field along the line joining two charges is given in list-I. Relation between charges is given in list-II. Match the proper variation. Take electric field along right as positive.

List-I



List-II

(1) $|q_1| > |q_2|$
 $q_1 < 0; q_2 > 0$

(2) $|q_1| = |q_2|$
 $q_1 > 0; q_2 > 0$

(3) $|q_1| = |q_2|$
 $q_1 < 0; q_2 > 0$

(4) $|q_2| > |q_1|$
 $q_1 > 0; q_2 > 0$

Note: (*) → Multiple Correct Type Question

Total Marks: 35

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking)

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

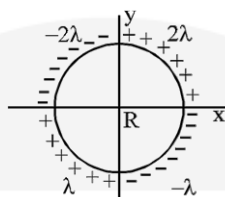
(4 marks, 3 min.)

Subjective Type Questions (No negative marking)

(4 marks, 3 min.)

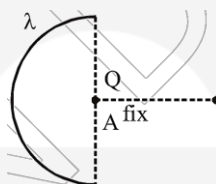
(There is partial marking in multiple correct questions)

1. The charge per unit length of the four quadrant of the ring is 2λ , -2λ , λ and $-\lambda$ respectively. The electric field at the centre is:



- (A) $-\frac{\lambda}{2\pi\epsilon_0 R} \hat{i}$ (B) $\frac{\lambda}{2\pi\epsilon_0 R} \hat{i}$ (C) $\frac{\sqrt{2}\lambda}{4\pi\epsilon_0 R} \hat{i}$ (D) None

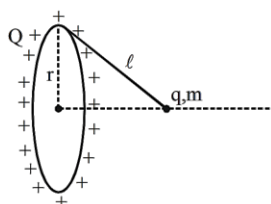
2. A charge Q is kept fixed at point A as shown in the figure in a gravity free space. A semicircular ring of linear charge density λ , mass m & radius R is released from rest. Initially centre of semicircular ring coincides with the fixed charge. The initial acceleration of the semicircular ring is:



- (A) $\frac{2K\lambda Q}{mR}$ (B) $\frac{K\lambda Q}{2mR}$ (C) $\frac{K\lambda Q}{mR}$ (D) 0

3. A uniformly charged ring (Q) of radius r is placed in a vertical plane. A thread of length (ℓ) connects the charge particle ' q ' of mass ' m ' and ring such that charge particle is in equilibrium along its axis.

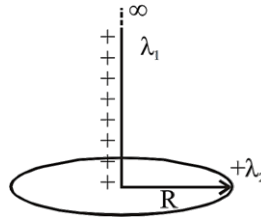
Then length of the string is: $\left(k = \frac{1}{4\pi\epsilon_0}\right)$



- (A) $\left(\frac{kQqr}{mg}\right)^{\frac{1}{3}}$ (B) $\left(\frac{kQqr}{mg}\right)^{\frac{2}{3}}$ (C) $\left(\frac{2kQqr}{mg}\right)^{\frac{1}{3}}$ (D) $\left(\frac{kQqr}{2mg}\right)^{\frac{1}{3}}$



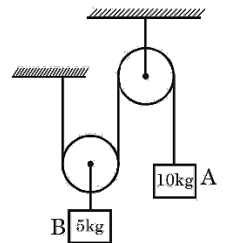
4. A semi-infinite charged wire with charge density $+\lambda_1$ is placed along the axis of a uniformly charged ring of charge density $+\lambda_2$. Find the force exerted on the wire due to the ring. The radius of ring is R.



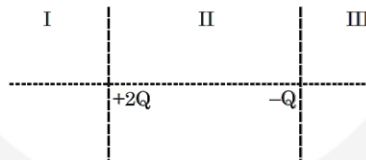
- (A) $(k\lambda_1\lambda_2) 2\pi R$ (B) $\frac{\lambda_1\lambda_2}{2\epsilon_0}$ (C) $k\lambda_1^2 2\pi R$ (D) None of these

5. Consider the situation shown in figure. The system is released from rest and the block A travels a distance 5m in downward direction, then work done by tension on the block B is:

- (A) $-\frac{250}{3}$ J (B) $\frac{500}{3}$ J (C) $\frac{250}{3}$ J (D) $-\frac{500}{3}$ J



- 6*. Two point charges $q_1 = +2Q$ and $q_2 = -Q$ are placed as shown in figure. The charges divide the line joining them in three regions I, II and III. Which of the following statement(s) is/are CORRECT:

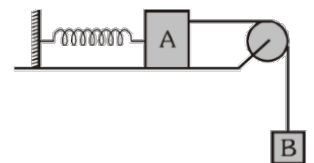


- (A) Region III has a local maxima of electric field.
 (B) Region I has a local minima of electric field.
 (C) Equilibrium position for a test charge lies in region II.
 (D) The equilibrium for constrained motion along the line joining the charges is stable for a negative charge.

- 7*. Two point charges (Q each) are placed at $(0, y)$ and $(0, -y)$. A point charge q of the same polarity can move along X-axis. Then:

- (A) the force on q is maximum at $x = \pm y / \sqrt{2}$
 (B) the charge q is in equilibrium at the origin
 (C) the charge q performs an oscillatory motion about the origin
 (D) for any position of q other than origin the force is directed away from origin

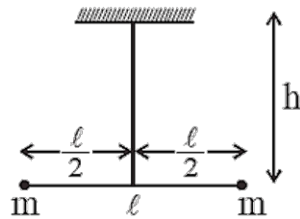
- 8*. In the figure shown, mass of block A is m and mass of block B is 2m. The spring constant is K. All the surfaces are smooth and the system is released from rest with spring initially unstretched.



- (A) The maximum extension of the spring is $\frac{4mg}{k}$
 (B) The speed of block A when extension in spring is $\frac{2mg}{k}$, is $2g\sqrt{\frac{2m}{3k}}$
 (C) Net acceleration of block B when the extension in the spring is maximum, is $\frac{2}{3}g$
 (D) Tension in the thread for extension of $\frac{2mg}{k}$ in spring, is mg

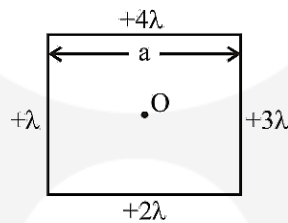


- 9.* Figure shows two particles connected with a light rod and entire arrangement hangs on an elastic wire. Neglect any gravitational effect, θ_0 = maximum angular displacement of the particle, C = torsional stiffness of wire. Mark the correct statements:



- (A) Time period of oscillation is $\pi\ell\sqrt{\frac{2m}{C}}$
- (B) Maximum tension in the rod is $\frac{C\theta_0^2}{\ell}$
- (C) The energy of oscillation is $\frac{1}{2}C\theta_0^2$
- (D) The maximum tangential acceleration of particle is $\frac{C\theta_0}{m\ell}$

- ✓ 10. Four uniformly charged wires of length a are arranged to form a square. Linear charge density of each wire is as shown. If electric field intensity at centre of square is $\frac{nk\lambda}{a}$, then value of n is:



Note: (*) → Multiple Correct Type Question

Total Marks: 33

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking)

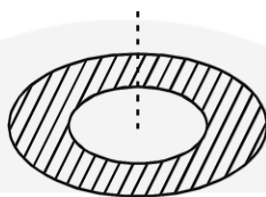
(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. A thin disc of radius $b = 2a$ has a concentric hole of radius 'a' in it (see figure). It carries uniform surface charge ' σ ' on it. If the electric field on its axis at height 'h' ($h \ll a$) from its centre is given as 'Ch' then value of 'C' is:



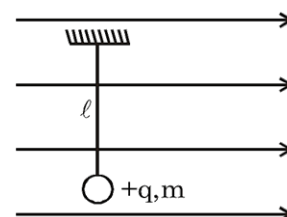
(A) $\frac{\sigma}{4a\epsilon_0}$

(B) $\frac{\sigma}{2a\epsilon_0}$

(C) $\frac{\sigma}{8a\epsilon_0}$

(D) $\frac{\sigma}{a\epsilon_0}$

2. A bob of mass m is suspended from a light string as shown. When string is vertical, then bob is imparted horizontal velocity $\sqrt{2g\ell}$ towards right. System is placed in uniform electric field $E = \frac{mg}{q}$. Find out tension in the string, when it becomes horizontal for the first time.



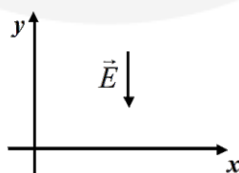
(A) $T = 0$

(B) $T = mg$

(C) $T = 3mg$

(D) $T = 2mg$

3. A particle of mass m and charge $-q$ is projected from the origin with a horizontal speed v into an electric field of intensity E directed downward. Choose the wrong statement. (Consider +y axis in upward direction and x-axis in horizontal direction. Also neglect the effect of gravity)



(A) The kinetic energy after a vertical displacement y is qEy .

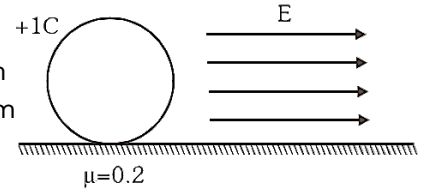
(B) The horizontal and vertical components of acceleration are $a_x = 0$, $a_y = \frac{qE}{m}$

(C) The equation of trajectory is $y = \frac{1}{2} \left(\frac{qEx^2}{mv^2} \right)$

(D) The horizontal and vertical displacement x and y after a time t are $x = vt$ and $y = \frac{1}{2} a_y t^2$



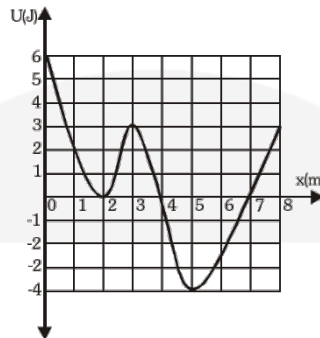
4. A non-conducting disc of mass = 2kg, total charge = +1C uniformly distributed, is placed on a rough horizontal non-conducting surface with its cross-section in vertical plane as shown. A uniform horizontal electric field E is now switched on. Find maximum magnitude of electric field E in (N/C) so that the disc rolls purely. $[g = 10 \text{ ms}^{-2}]$



(A) 10 (B) 14 (C) 12 (D) 18

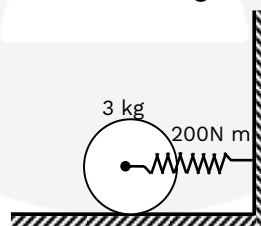
5. If potential energy in space is given as $U = x^2 + 2xy - y^3$, then find out value of force at (3, 4, 5).
 (A) $14\hat{i} + 21\hat{j}$ (B) $-14\hat{i} + 42\hat{j}$ (C) $21\hat{i} + 14\hat{j}$ (D) $7\hat{i} + 14\hat{j}$

6. Potential energy of a particle as a function of the position of the particle is shown. The particle has total mechanical energy E of 3.0 joules.



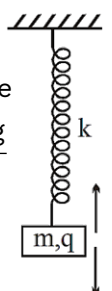
- (A) It can never be present at $x = 0 \text{ m}$
 (B) It can never be present at $x = 5 \text{ m}$
 (C) At $x = 2$ its kinetic energy is 0 J
 (D) At $x = 1$ its kinetic energy 3 J

7. A uniform solid cylinder rolls without slipping in the system shown. If the maximum compression in spring is 15 cm, the possible friction force acting on the cylinder during its motion is:



(A) 14 N (B) 10 N (C) 12 N (D) 15 N

- 8*. The following figure shows a block of mass m suspended from a fixed point by means of a vertical spring. The block is oscillating simple harmonically and carries a charge q . There also exists a uniform electric field in the region. Consider four different cases. The electric field is zero in case-1, $\frac{mg}{q}$ downward in case-2, $\frac{mg}{q}$ upward in case-3 and $\frac{2mg}{q}$ downward in case-4. The speed at mean position is same in all cases. Select the CORRECT alternative(s).



- (A) Time periods of oscillation are equal in case-1 and case-3
 (B) Amplitudes of displacement are same in case-2 and case-3
 (C) The maximum elongation (increment in length from natural length) is maximum in case-4.
 (D) Time periods of oscillation are equal in case-2 and case-4

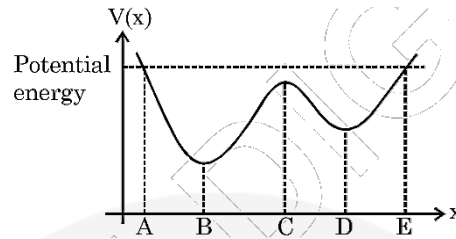


9*

A simple pendulum of length ℓ has a bob of mass m , with a charge q on it. A non-conducting thin vertical sheet, with charge σ per unit area, passes through the point of suspension of the pendulum. At equilibrium, the string makes an angle θ with the vertical. Its time period of oscillation is T . In this position,

- (A) $\tan\theta = \frac{\sigma q}{2\epsilon_0 mg}$ (B) $\tan\theta = \frac{\sigma q}{\epsilon_0 mg}$ (C) $T < 2\pi\sqrt{\frac{\ell}{g}}$ (D) $T > 2\pi\sqrt{\frac{\ell}{g}}$

- 10*. A particle moves in one dimension in a conservative force field. The potential energy is depicted in the graph below. If the particle starts to move from rest from the point A, then:



- (A) The speed is zero at the point A and E
(B) The acceleration vanishes at the points A,B,C,D,E
(C) The acceleration vanishes at the points B,C,D
(D) The speed is maximum at the point D

Note: (*) → Multiple Correct Type Question

Total Marks: 34

Single Correct Questions ('-1' negative marking)

Multiple Correct Questions ('-2' negative marking)

Subjective Type (No negative marking)

(There is partial marking in multiple correct questions)

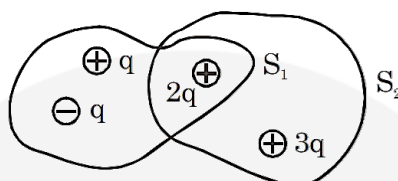
Max. Time: 24 min.

(3 marks, 2 min.)

(4 marks, 3 min.)

(4 marks, 3 min.)

1. The flux through the Gaussian surface S_1 and S_2 is:



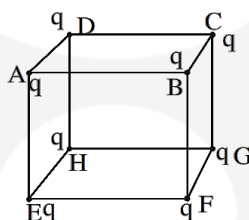
(A) $\frac{q}{\epsilon_0}, \frac{2q}{\epsilon_0}$

(B) $\frac{2q}{\epsilon_0}, \frac{5q}{\epsilon_0}$

(C) $-\frac{q}{\epsilon_0}, \frac{3q}{\epsilon_0}$

(D) $\frac{3q}{\epsilon_0}, \frac{2q}{\epsilon_0}$

2. Eight point charges having magnitude q are fixed at vertices of a cube. The electric flux through square surface ABCD of the cube is:



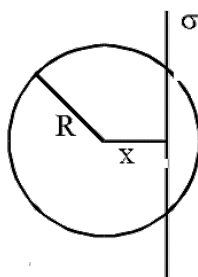
(A) $\frac{q}{24 \epsilon_0}$

(B) $\frac{q}{12 \epsilon_0}$

(C) $\frac{q}{6 \epsilon_0}$

(D) $\frac{q}{3 \epsilon_0}$

3. An infinite, uniformly charged sheet with surface charge density σ cuts through a spherical Gaussian surface of radius R at a distance x from its center, as shown in the figure. The electric flux Φ through the Gaussian surface is:



(A) $\frac{\pi R^2 \sigma}{\epsilon_0}$

(B) $\frac{2\pi(R^2 - x^2)\sigma}{\epsilon_0}$

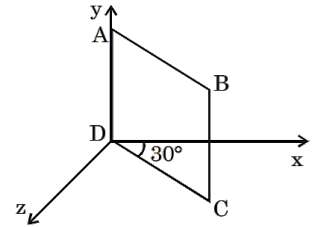
(C) $\frac{\pi(R - x)^2 \sigma}{\epsilon_0}$

(D) $\frac{\pi(R^2 - x^2)\sigma}{\epsilon_0}$



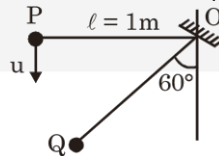
4. A square cross-section ABCD of side a is kept in the direction shown in the diagram. An electric field $E_0\hat{i} + E_0\hat{j}$ exists in the region. The flux of this field through the cross-section is:

- (A) E_0a^2 (B) $\frac{E_0a^2}{3}$
(C) $\frac{E_0a^2}{2}$ (D) $\frac{\sqrt{3}}{2}E_0a^2$



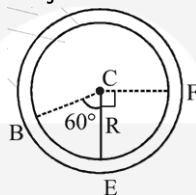
5. Which of the following statements is correct?
(A) Electric field calculated by Gauss law is the field due to only those charge which are enclosed inside the Gaussian surface.
(B) Gauss law is applicable only when there is a symmetrical distribution of charge.
(C) Electric flux through a closed surface will depends only on charge enclosed within that surface only
(D) None of these

6. A ball is given velocity u when string is at horizontal position. The string whose breaking tension is equal to three times the weight of ball breaks off at point Q as shown, value of u is:



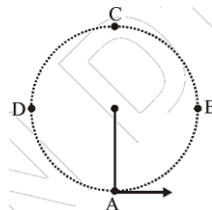
- (A) $\sqrt{\frac{g}{2}}$ (B) $\sqrt{\frac{2g}{3}}$ (C) $\sqrt{\frac{3g}{2}}$ (D) $\sqrt{\frac{g}{3}}$

- 7*. As shown in figure, BEF is a fixed vertical circular tube. A block of mass m starts moving in the tube at point B with velocity V towards E. It is just able to complete the vertical circle, then



- (A) velocity at B must be $\sqrt{3Rg}$ (B) velocity at F must be $\sqrt{2Rg}$
(C) Normal reaction at point F is $2mg$ (D) The normal reaction at point E is $6mg$

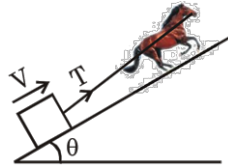
- 8*. A pendulum bob of mass m connected with a string of length ℓ lies in vertical plane. It is given just sufficient velocity so as to complete vertical circle. Mark the CORRECT statements:



- (A) Change in velocity as bob moves from A to B has magnitude $\sqrt{4g\ell}$
(B) Net work done by all the forces as bob moves from A to B is $-mg\ell$
(C) Net acceleration at B has magnitude $\sqrt{10}g$
(D) Taking top most point as reference total mechanical energy is $\frac{mg\ell}{2}$



- 9*. A horse pulls a box up a steep snow covered incline of angle θ with constant speed V . The box has a mass m & coefficient of kinetic friction between box & snow is μ . P is the power delivered by horse & T is the tension in string:



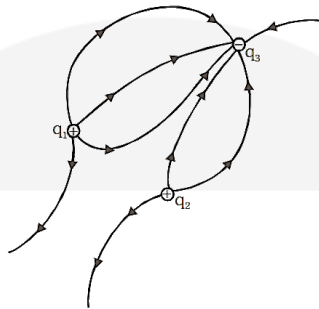
(A) $T = mg(\sin\theta + \mu\cos\theta)$

(B) $T = mg(\sin\theta - \mu\cos\theta)$

(C) $V = \frac{P}{mg(\sin\theta - \mu\cos\theta)}$

(D) $V = \frac{P}{mg(\sin\theta + \mu\cos\theta)}$

10. In the figure, electric field lines of three charges are shown. If $q_1 = +8\mu\text{C}$ and $q_3 = -12\mu\text{C}$ what is magnitude of charge q_2 in μC .



Note: (*) → Multiple Correct Type Question

Total Marks: 37

Max. Time: 27 min.

Single Correct Questions ('-1' negative marking)

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

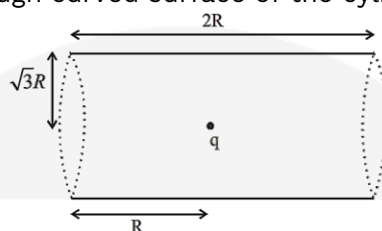
(4 marks, 3 min.)

Subjective Type Questions (No negative marking)

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. In the shown figure, a point charge q is placed as shown. An imaginary cylindrical surface is also drawn as shown. The flux through curved surface of the cylinder will be.

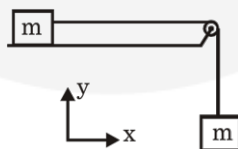


- (A) $\frac{q}{\epsilon_0}$ (B) $\frac{q}{3\epsilon_0}$ (C) $\frac{q}{2\epsilon_0}$ (D) $\frac{q}{4\epsilon_0}$

2. The centre of mass of a non-uniform rod of length L whose mass per unit length λ varies as $\lambda = \frac{k \cdot x^2}{L}$ where k is a constant & x is the distance of any point on rod from its one end, is (from the same end)

- (A) $\frac{3}{4}L$ (B) $\frac{1}{4}L$ (C) $\frac{k}{L}$ (D) $\frac{3k}{L}$

3. In the given figure, table is smooth and pulley and string are ideal then acceleration of centre of mass of the two blocks system is:



- (A) $\frac{g}{2}\hat{i} - \frac{g}{2}\hat{j}$ (B) $\frac{g}{4}\hat{i} - \frac{g}{4}\hat{j}$ (C) $-g\hat{j}$ (D) $\frac{g}{2}\hat{i} - g\hat{j}$

- 4*. Consider a region where charge density varies with radius as: (here ρ_0 is a constant)

$$\rho = \rho_0 r; \quad 0 \leq r \leq R$$

$$= 0; \quad r > R$$

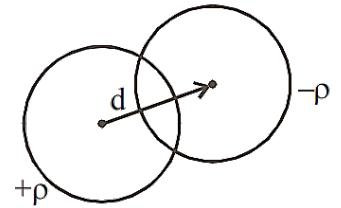
Choose the correct statement(s):

- (A) Electric field varies as r^2 for $r < R$ (B) Electric field varies as $\frac{1}{r^2}$ for $r > R$
- (C) Electric field varies as r for $r < R$ (D) Electric field varies as $\frac{1}{r}$ for $r > R$



- 5*. Two spheres, each of radius R and carrying uniform volume charge density $+\rho$ and $-\rho$, respectively, are placed so that they partially overlap. Let the vector from the positive centre to the negative centre is $\vec{d}$. Then choose the correct statement(s):

- (A) Electric field inside the overlapped region will be uniform.
 (B) Electric field inside the overlapped region is $\frac{\rho}{3\epsilon_0}\vec{d}$
 (C) Electric field inside the overlapped region is zero.
 (D) Gauss's law can not be applied if only one of the spheres is taken as Gaussian surface.

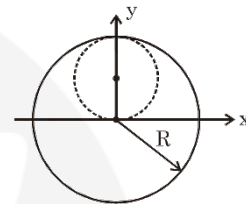


- 6*. Center of mass of a two particle system of masses 2 kg and 3 kg present at (15, 0) and (0, 20) in a x, y plane lies on lines:

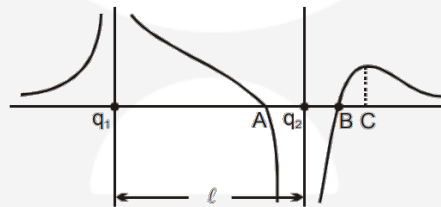
- (A) $y = 2x$ (B) $\frac{x}{12} + \frac{y}{24} = 1$ (C) $3x - y = 6$ (D) $y = x$

- 7*. A lamina is made by removing a disc of diameter R from a uniform disc of radius R and mass m .

- (A) Mass of remaining lamina is $m/2$
 (B) mass of remaining lamina is $\frac{3m}{4}$
 (C) y-coordinate of COM of lamina is $-\frac{R}{6}$
 (D) x-coordinate of COM of lamina is zero



- 8.* The curve represent distributed of potential along a line joining two charges q_1 and q_2 (separated by distance ℓ) then which of the following statements are incorrect?



- (A) $|q_1| < |q_2|$ (B) q_1 is negative in nature
 (C) A and B are equilibrium points (D) C is point of unstable equilibrium

- 9.* We have a uniformly charged ring of radius 0.1 m on the symmetric axis of the ring, the maximum electric field is found to be $8 \times 10^3 \sqrt{3}$ V/m

- (A) The charge on the ring is 4×10^{-8} C
 (B) The charge on the ring is 6×10^{-8} C
 (C) The potential at the centre of ring is 3600 V
 (D) The electric field at the centre of ring is zero

10. A sphere of radius R has charge density given by $\rho = \rho_0 \left(1 - \frac{nr}{3R}\right)$, where ρ_0 is a constant, r is distance from centre of sphere. For a spherical gaussian surface of radius R centered at the centre of sphere, the flux is zero. Find 'n',

Note: (*) → Multiple Correct Type Question

Total Marks: 37

Max. Time: 27min.

Single Correct Questions ('-1' negative marking)

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

(4 marks, 3 min.)

Matrix Match Type Questions

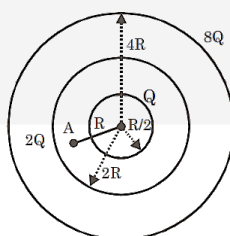
(4 marks, 3 min.)

Subjective Type Questions (No negative marking)

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. Three charged concentric nonconducting shells are given as shown in figure. Find the potential at point A:



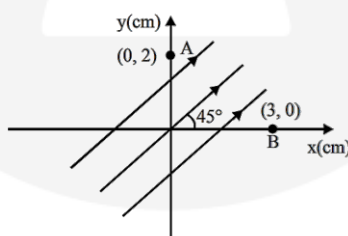
(A) $\frac{5kQ}{R}$

(B) $\frac{4kQ}{R}$

(C) $\frac{3kQ}{R}$

(D) $\frac{2kQ}{R}$

2. A uniform electric field of 400 V/m direct at 45° above the x-axis as shown in figure. The potential difference $V_A - V_B$ is given by:



(A) 0

(B) 4V

(C) 6.4V

(D) 2.8V

3. The electric potential V at a point (x, y, z) (all in meters) in space is given by the scalar function $v = (x^3 - 2y^2 + 6z)V$. (where x, y and z in m) the magnitude of electric field at the point $P(1m, -1m, 3m)$ is (in V/m)

(A) 21

(B) 13

(C) 9

(D) $\sqrt{61}$

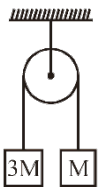


- 4.* A point charge $+Q$ is fixed at a point O . Another point charge $+q$ & mass m is projected from infinity towards $+Q$ with speed v_0 . Minimum distance of approach is r . If it is now given speed $2v_0$ at infinity then

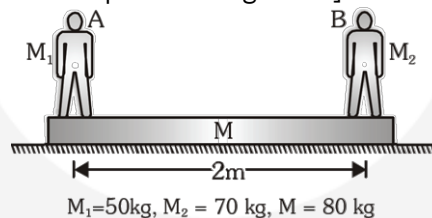
- (A) Minimum distance of approach becomes $\frac{r}{2}$
 (B) Minimum distance of approach becomes $\frac{r}{4}$
 (C) Angular momentum conservation will hold true about O
 (D) Mechanical energy will remain conserved throughout the motion

- 5.* Two blocks of mass M and $3M$ are connected by a light cord which passes over a light frictionless pulley as shown in the figure. The blocks are released from rest and are at the same height at $t = 0$.

- (A) Tension in string connecting masses is $\frac{3Mg}{4}$
 (B) The acceleration of both masses is $\frac{g}{2}$ in magnitude
 (C) The centre of mass accelerates down
 (D) The net force on system having M and $3M$ is zero



6. In the figure shown, when the persons A and B exchange their position, then match the column I and II. [There is no friction between plank and ground]



Column I

- (A) The distance (in m) moved by the centre of mass of the system is
 (B) The distance (in m) moved by the plank is
 (C) The distance (in m) moved by A with respect to ground is
 (D) The distance (in m) moved by B with respect to ground is

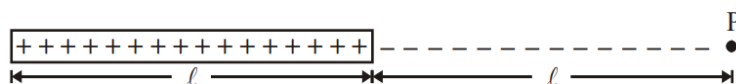
Column II

- (P) 0.2
 (Q) 1.8
 (R) 0
 (S) 2.2
 (T) 1.2

7. Two charges $+2q$ and $-3q$ are fixed at $(4m, 0, 0)$ and $(9m, 0, 0)$ respectively. The distance between 2 points on X -axis (in m) where the potential is zero is x . Then the value of $\frac{x}{2}$ is
 (Assume standard reference point)

8. Two charges q and $-2q$ are placed at $(-3, 0)$ and $(3, 0)$ in x - y plane. The locus of the point in the plane of the charges, where the field potential is zero is $(x + \alpha)^2 + y^2 = 4\beta^2$. Find the value of $(\alpha + \beta)$.

9. The long thin cylindrical glass rod shown below has length ℓ and is insulated from its surrounding. The rod has excess charge Q uniformly distributed along its length. Assume the electric potential to be zero at infinite distance from the rod. If k is the constant of Coulomb's law, the electric potential at a point P along the axis of the rod and a distance ℓ from one end is $\frac{kQ}{\ell} \ln\left(\frac{x}{3}\right)$. What is the value of x ?



10. A particle having charge 1 C is moved from a potential of 1 V to 2 V such that it gains kinetic energy of 6 J . Then find the work done by external agent in Joule.

Note: (*) → Multiple Correct Type Question

Total Marks: 37

Max. Time: 27 min.

Single Correct Questions ('-1' negative marking)

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

(4 marks, 3 min.)

Matrix Match Type Questions

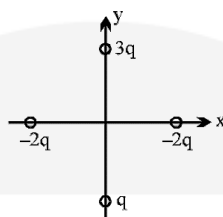
(4 marks, 3 min.)

Subjective Type Questions (No negative marking)

(4 marks, 3 min.)

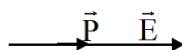
(There is partial marking in multiple correct questions)

1. Four charges are placed each at a distance 'a' from origin. The dipole moment of configuration is:



- (A) $2qa\hat{j}$ (B) $32qa\hat{j}$ (C) $2aq[\hat{i} + \hat{j}]$ (D) none

2. An electric dipole of dipole moment $\vec{P}$ is oriented parallel to a uniform electric field $\vec{E}$, as shown.



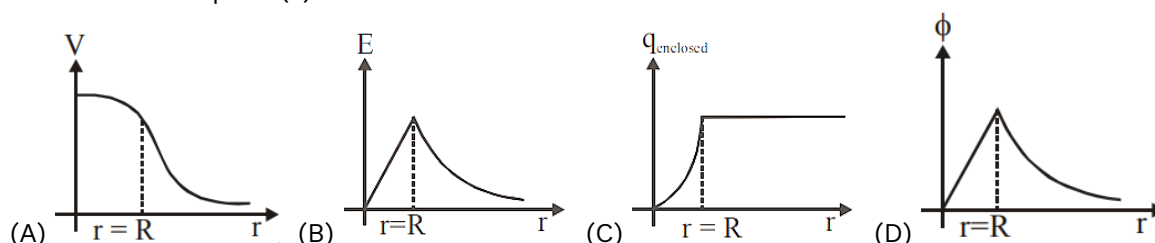
It is rotated to one of the orientations shown below. Rank the final orientations according to the change in the potential energy of the dipole-field system, most negative to most positive.

- (i) (ii) (iii) (iv)
- (A) (i), (ii), (iv), (iv) (B) (iv), (iii), (iv), (i)
(C) (i), (ii), (iii), (iv) (D) (iii), (ii) and (iv) tie, then (i)

3. A ball of mass $m = 1\text{kg}$, travelling with velocity $\hat{i} - 3\hat{j}$ receives an impulse of magnitude 2 N-s in the negative x-direction. What is the velocity of the ball immediately after the impulse?

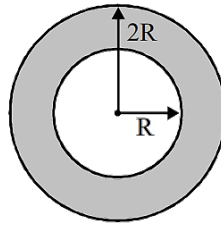
- (A) $-\hat{i} + \hat{j}$ (B) $-\hat{i} + 3\hat{j}$ (C) $-\hat{i} - 3\hat{j}$ (D) $\hat{i} - 3\hat{j}$

- 4*. Graph shows the variation of potential (V), magnitude of electric field (E), charge enclosed (q_{enclosed}) Within the concentric sphere and net flux (ϕ) through a concentric spherical Gaussian surface as a function of distance (r) from centre of a uniformly positive charged solids sphere of radius R. Choose the CORRECT option(s):





- 5*. Consider a spherical volume which has uniform charge density ρ from $x = R$ to $x = 2R$. There is no charge from $x = 0$ to $x = R$ and from $x = 2R$ to infinity. The electric potential due to this charge



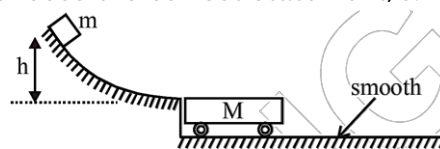
- (A) at $x = 0$ is 0
 (B) at $x = \frac{R}{2}$ is $\frac{3\rho R^2}{2\epsilon_0}$
 (C) at $x = 2R$ is $\frac{7\rho R^2}{6\epsilon_0}$
 (D) at $x = \frac{3R}{2}$ is $\frac{3\rho R^2}{2\epsilon_0}$

- 6*. Imagine a short dipole at the centre of a spherical surface. If the magnitude of the electric field at a certain point on the surface of sphere is 10 N/C, then which of the following CANNOT be the magnitude of the electric field anywhere on the surface of the sphere?
 (A) 4 N/C (B) 8 N/C (C) 16 N/C (D) 32 N/C

- 7*. A strip of wood of mass M and length ℓ is placed on a smooth horizontal surface. An insect of mass ' m ' starts at one end of the strip and walks to the other end in time t , moving with a constant speed. Select the CORRECT alternative/s.

- (A) The speed of the insect as seen from the ground is equal to $\frac{\ell}{t}$.
 (B) The speed of the insect as seen from the ground is $\frac{\ell M}{t(M+m)}$.
 (C) The speed of strip of wood as seen from the ground is equal to $\frac{m\ell}{t(M+m)}$.
 (D) total kinetic energy of the system in ground frame is equal to $\frac{1}{2}\left(\frac{\ell}{t}\right)^2 \frac{mM}{(M+m)}$.

- 8*. A cart of mass M and length L stands just at the end of a slope as shown in figure. A small block of mass m is released from rest at height h . Block slides on cart and comes to rest with respect to cart at the edge of cart. Friction exists between cart and block (coefficient of friction is μ) and all other surfaces are smooth. Choose the correct statement/s:

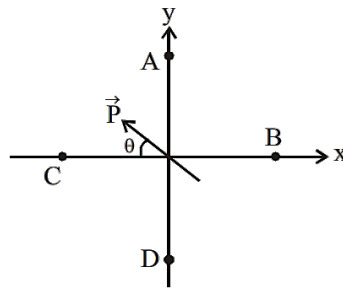


- (A) Time for which block moves on cart is $\frac{\sqrt{2gh} M}{\mu g(m+M)}$.
 (B) Net work done by kinetic friction is $-\mu mgL$.
 (C) Final velocity of centre of mass of block and cart is $\frac{m\sqrt{2gh}}{m+M}$.
 (D) Net impulse on cart is $\frac{mM\sqrt{2gh}}{m+M}$.



9

Electric dipole is placed at origin as shown. Point A,B,C & D are at distance r from origin, then match the List-I with List-II

**List-I**

- (P) Electric potential at A
 (Q) Electric potential at B
 (R) Electric potential at C
 (S) Electric potential at D

List-II

- (1) $\frac{-kP \sin \theta}{r^2}$
 (2) $\frac{kP \cos \theta}{r^2}$
 (3) $\frac{kP \sin \theta}{r^2}$
 (4) $\frac{-kP \cos \theta}{r^2}$

Code:

	P	Q	R	S
(A)	3	4	2	1
(B)	4	3	1	2
(C)	1	2	4	3
(D)	3	4	1	2

10

Find the potential difference V_{AB} in volt between A (0,0,0) and B (1 m, 1 m, 1 m) in an electric field $\vec{E} = y\hat{i} + x\hat{j}$.

Note: (*) → Multiple Correct Type Question

Total Marks: 35

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking)

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

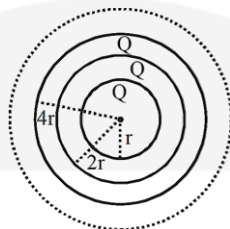
(4 marks, 3 min.)

Subjective Type Questions ('-2' negative marking)

(4 marks, 3 min.)

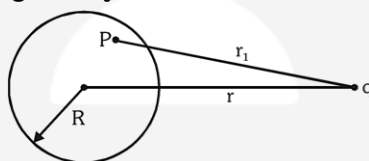
(There is partial marking in multiple correct questions)

1. A large number of concentric shells each having charge Q are arranged as shown in figure. Radii of shells are $r, 2r, 4r, \dots$. The potential due to these shells at centre is:



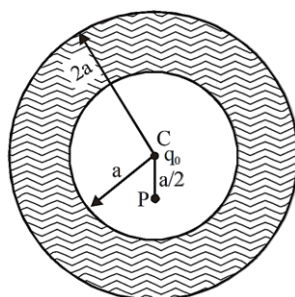
- (A) $\frac{KQ}{r}$ (B) $\frac{KQ}{2r}$ (C) $\frac{2KQ}{r}$ (D) $\frac{KQ}{r} \ln 2$

2. A point charge q is placed at a distance r from center of a conducting neutral sphere of radius R ($r > R$). The potential at any point P inside the sphere at a distance r_1 from point charge due to induced charge of the sphere is given by:



- (A) $\frac{Kq}{r_1} - \frac{Kq}{R}$ (B) $\frac{Kq}{r_1} + \frac{Kq}{R}$ (C) $\frac{Kq}{r} - \frac{Kq}{r_1}$ (D) $-\frac{Kq}{r_1} + \frac{Kq}{R}$

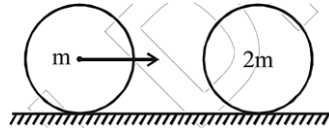
3. A point positive charge q_0 is placed at a point C inside the cavity in a neutral spherical conductor as shown in figure. Find the potential at P :



- (A) $\frac{Kq_0}{2a}$ (B) $\frac{Kq_0}{a}$ (C) $\frac{2Kq_0}{a}$ (D) $\frac{3Kq_0}{2a}$

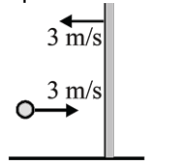


4. A uniform sphere of mass m moving along a smooth horizontal surface strikes another uniform sphere of mass $2m$ (initially at rest) head on. If loss in K.E. of first sphere is maximum, then value of coefficient of restitution is:



- (A) 0.2 (B) 0.5 (C) 0.7 (D) 1

5. A highly elastic ball moving at a speed of 3 m/s approaches a wall moving towards it with a speed of 3 m/s (figure). After the collision, the speed of the ball will be:

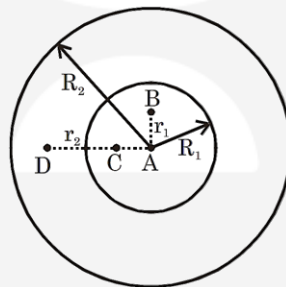


- (A) 3 m/s (B) 6 m/s (C) 9 m/s (D) zero

- 6* An electric dipole in a uniform electric field has potential U . The magnitude of the torque acting on the dipole due to the field is τ .

- (A) U is minimum and τ is zero when the dipole is perpendicular to the field.
(B) U is zero and τ is maximum when the dipole is perpendicular to the field.
(C) U is minimum and τ is maximum when the dipole is perpendicular to the field.
(D) U is minimum and τ is zero when the dipole is parallel to the field.

- 7*. Consider a thick conducting shell of inner radius R_1 & outer radius R_2 as shown in the figure. Shell is neutral and a point charge Q is kept at point A i.e. centre of shell. Point B and D lie at distance r_1 and r_2 respectively from centre, Choose CORRECT option(s) :

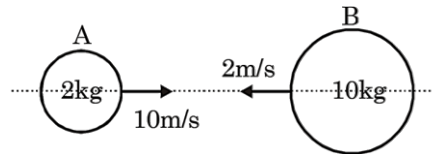


- (A) Potential at point D is $\frac{kQ}{R_2} + \frac{kQ}{r_2}$
(B) Potential at point B is $\frac{kQ}{R_2} + \frac{kQ}{r_1} - \frac{kQ}{R_1}$
(C) Potential at D will change if point charge shifted to position C
(D) Potential at B will change if point charge is shifted to position C.

- 8*. If a projectile explodes in mid air, then just before and just after explosion, choose CORRECT option(s):
- (A) Conservation of momentum is applicable.
(B) Path of the centre of mass of the projectile must be same to the original parabola till any one part of it, hits the ground.
(C) Energy released during explosion is equal to increase in kinetic energy of the projectile.
(D) Conservation of momentum is not applicable

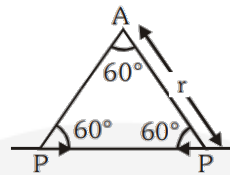


9*. If two balls are moving along same line and they collide perfectly inelastically then:



- (A) Just after collision both A & B come to rest.
- (B) Just after collision both move with speed 3 m/s.
- (C) Loss in mechanical energy due to collision is 120 J.
- (D) Linear momentum of system (A + B), does not remain conserve.

✓ 10. Potential at point A due to dipole is $\frac{3KP}{\beta r^\gamma}$ then find the value of $\beta + \gamma$?



Note: (*) → Multiple Correct Type Question

Total Marks: 35

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking)

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking)

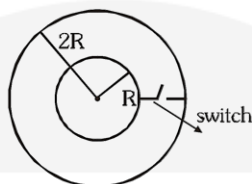
(4 marks, 3 min.)

Subjective Type Questions (No negative marking)

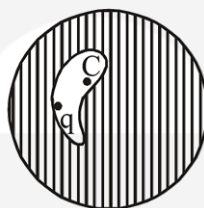
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

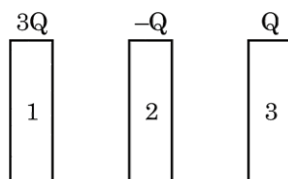
1. Net positive charge on inner spherical conductor is $3Q$ and on outer spherical conductor is Q . Switch is closed.



- (A) Charge will flow from inner shell to outer shell
 (B) Charge will flow from outer shell to inner shell
 (C) Charge will not flow
 (D) none of these
2. The figure shows a charge q placed inside a cavity in an uncharged conductor. Now if an external electric field is switched on, then:



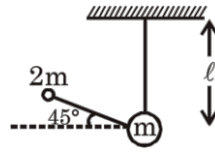
- (A) only induced charge on outer surface will redistribute
 (B) only induced charge on inner surface will redistribute
 (C) Both induced charge on outer and inner surface will redistribute
 (D) force on charge q placed inside the cavity will change
3. The charges given to three conducting plates are $3Q$, $-Q$ & Q respectively. The charges appearing on right side of plate 2 will be :



- (A) $\frac{Q}{2}$ (B) $\frac{3Q}{2}$ (C) $\frac{-Q}{2}$ (D) $\frac{-3Q}{2}$



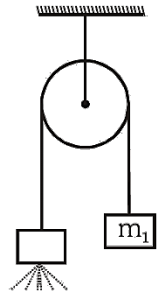
4. A ball of mass m is suspended by a massless string of length ℓ from a fixed point. A ball of mass $2m$ strikes in the directions of $\theta = 45^\circ$ from horizontal and sticks to it. What should be the velocity of mass $2m$ so that system deflects by $\phi = \frac{\pi}{2}$.



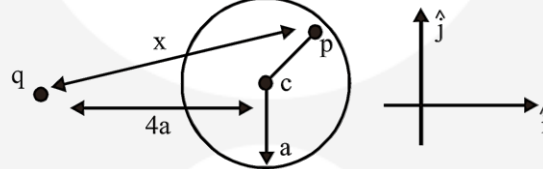
- (A) $2\sqrt{5g\ell}$ (B) $\sqrt{3g\ell}$ (C) $3\sqrt{g\ell}$ (D) $\sqrt{10g\ell}$

5. A mass m_1 is connected by a weightless cable passing over a frictionless pulley to a container of water, whose mass is m_0 at $t = 0$. If the container ejects water in downward direction at a constant rate b kg/sec, with a velocity v_0 relative to the container, then acceleration of m_1 as a function of time is

- (A) $\frac{(m_1 - m_0)g}{m_1 + (m_0 - bt)}$ (B) $\frac{m_1g - (m_0 - bt)g}{m_1 + (m_0 - bt)}$
 (C) $\frac{m_1g - (m_0 - bt)g + v_0b}{m_1 + (m_0 - bt)}$ (D) $\frac{m_1g - (m_0 - bt)g + v_0b}{m_1 - (m_0 - bt)}$

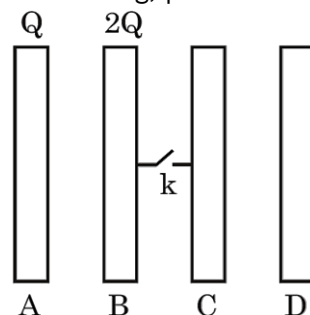


- 6*. There is a spherical conductor of radius a . A point charge is placed at a distance $4a$ from the centre of the spherical conductor as shown in figure.



- (A) The electric field at 'c' due to spherical conductor is $\frac{kq}{16a^2}(-\hat{i})$
 (B) Potential at P due to induced charge on spherical conductor is $\left(\frac{kq}{4a} - \frac{kq}{x}\right)$
 (C) Net electric field at 'P' is ZERO.
 (D) Now the conductor is earthed then charge on the spherical conductor is $\left(-\frac{q}{4}\right)$

- 7*. Initially two plates (A and B) are given charges as shown. Find the final charges on the plates after the key k is closed. All plates are conducting, parallel and of infinite length and breadth.



- (A) Final charges on plate A is Q (B) Final charge on plate B is 0
 (C) Final charge on plate C is $3Q/2$ (D) Final charge on plate D is 0



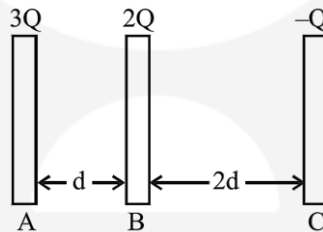
- 8*** Two concentric metallic shells of radii R and $2R$ are placed in vacuum. The inner shell is having charge Q and outer shell is uncharged. If they are connected with a conducting wire. Then
- (A) Q amount of charge will flow from inner to outer shell.
 - (B) Q/e number of electrons will flow from outer to inner shell, where 'e' is charge on an electron.
 - (C) $\frac{KQ^2}{4R}$ amount of heat is produced in the wire.
 - (D) $\frac{KQ^2}{2R}$ amount of heat is produced in the wire.

- 9*** Two bodies of mass m_A & m_B are moving with velocity $\vec{V}_A$ & $\vec{V}_B$ respectively. We know that

$$V_{cm} = \frac{m_A \vec{V}_A + m_B \vec{V}_B}{m_A + m_B}. \text{ If the two bodies collide:}$$

- (A) Maximum possible kinetic energy after collision will be $\frac{1}{2}m_A V_A^2 + \frac{1}{2}m_B V_B^2$.
- (B) Minimum possible kinetic energy after collision will be $\frac{1}{2}(m_A + m_B) V_{cm}^2$.
- (C) Minimum possible kinetic energy after collision may be zero.
- (D) Maximum loss of kinetic energy will be $\frac{1}{2} \frac{m_A m_B}{m_A + m_B} (\vec{V}_A - \vec{V}_B)^2$.

- 10.** Three large conducting plates are arranged as shown in the diagram. If potential difference between plates B & C is given by $\frac{xQd}{A \epsilon_0}$. Then find the value of x .



Note: (*) → Multiple Correct Type Question

Total Marks: 34

Max. Time: 24 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

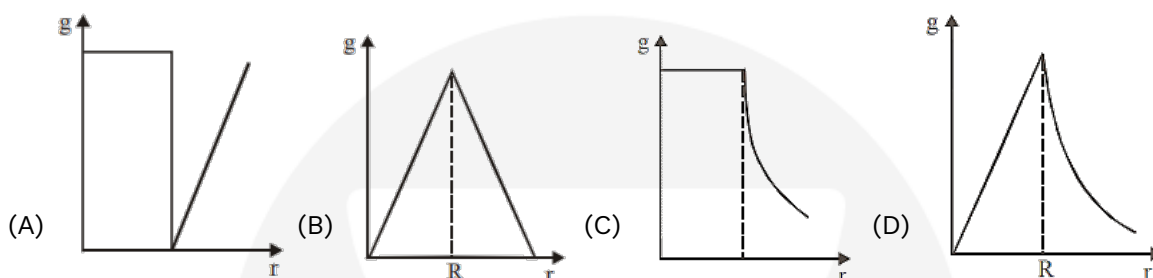
(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.7 to Q.10

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. Assuming the earth to have a constant density, point out which of the following curves show the variation of acceleration due to gravity from the centre to the points far away from the surface of earth.



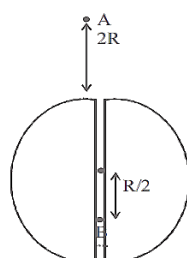
2. The fraction change in the value of free-fall acceleration g for a particle when it moves in a tunnel at depth d from the surface of the earth is:

(A) $\frac{d}{R}$ (B) $\frac{2d}{R}$ (C) $-\frac{2d}{R}$ (D) $-\frac{d}{R}$

3. A particle is projected vertically upwards the surface of the earth (radius R_e) with a speed equal to one fourth of escape velocity. What is the maximum height attained by it from the surface of the earth.

(A) $\frac{16}{15}R_e$ (B) $\frac{R_e}{15}$ (C) $\frac{4}{15}R_e$ (D) None of these

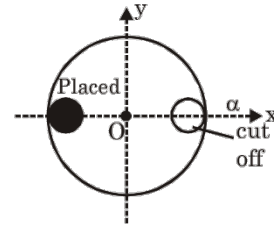
4. Suppose, if a tunnel is dug along the diameter of the earth and a body of mass m is released from a point 'A' at a distance $2R$ above earth along the line of tunnel. M is the mass of the earth & R is the radius of the earth. The velocity of the body during its fall when it crosses the point B at a distance $\frac{R}{2}$ below the earth centre as shown.



(A) $\frac{5}{2}\sqrt{\frac{GM}{3R}}$ (B) $\sqrt{\frac{GM}{3R}}$ (C) $2\sqrt{\frac{GM}{3R}}$ (D) $\frac{3}{2}\sqrt{\frac{GM}{3R}}$

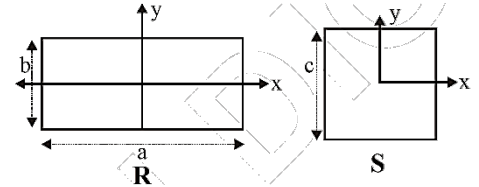


5. A small circular disc is cut off from a disc of radius r and put on the disc itself as shown in the figure. Select the correct statement:
- (A) Moment of inertia about z -axis is changed.
 (B) Moment of inertia about y -axis is changed.
 (C) Moment of inertia about x -axis is changed.
 (D) Moment of inertia about z -axis does not change.



6. A uniform square plate S (side c) and a uniform rectangular plate R (sides b, a) ($a > b$) have identical areas and masses as shown in figure. Given (a) $I_{xR}/I_{xS} < 1$; (b) $I_{yR}/I_{yS} > 1$; (c) $I_{zR}/I_{zS} > 1$ which of the following is CORRECT:

- (A) a, b
 (B) b, c
 (C) a, b, c
 (D) None of these



- 7.* If the law of gravity were inverse law instead of inverse square law, considering two satellite of same mass but in orbits of radii " r " and " $2r$ " around the earth.

- (A) The ratio of speeds of two satellite would be 1 : 1
 (B) The ratio of angular speeds of these two satellite would be 2 : 1
 (C) The ratio of the potential energy of these two satellite would be 1 : 2
 (D) The ratio of kinetic energies of these two satellite would be 2 : 1

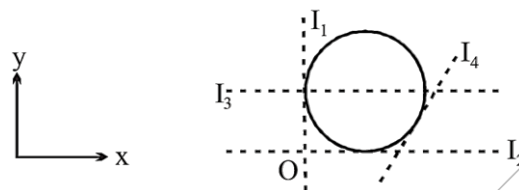
- 8.* The spherical planets have the same mass but densities in the ratio 1 : 8. For these planets, the

- (A) Acceleration due to gravity will be in the ratio 4 : 1
 (B) Acceleration due to gravity will be in the ratio 1 : 4
 (C) Escape velocities from their surfaces will be in the ratio $\sqrt{2} : 1$
 (D) Escape velocities from their surfaces will be in the ratio 1 : $\sqrt{2}$.

- 9.* Satellite A is in a circular orbit of radius r and satellite B in another circular orbit of radius $4r$, both revolving around the earth. The masses of the satellite A and B are in the ratio 3 : 1. If the symbols V , E , u and T represent the speed, total energy, escape velocity and period of revolving of satellite then.

- (A) $V_A/V_B = 2$ (B) $E_A/E_B = 12$ (C) $u_A/u_B = 2$ (D) $T_A/T_B = \frac{1}{8}$

- 10.* A disc has radius R , mass m and has moment of inertia I_1, I_2, I_3 and I_4 about axis as shown in figure. Moment of inertia of disc about z -axis passing through O will be:



- (A) $I_1 + I_2$ (B) $\frac{5}{2}mR^2$ (C) $3mR^2$ (D) $I_1 + I_4$

Note: (*) → Multiple Correct Type Question

Total Marks: 32

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.2

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.3 to Q.6

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.7 to Q.9

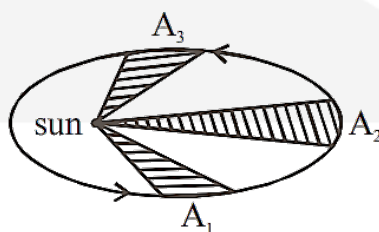
(2 marks, 2 min.)

Subjective Type Questions (No negative marking) Q.10

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. A planet moving around sun sweeps area A_1 in 2 days, A_2 in 3 days and A_3 in 6 days. Then the relation between A_1 , A_2 and A_3 is:



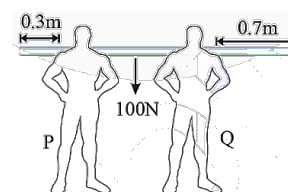
(A) $3A_1 = 2A_2 = A_3$

(B) $2A_1 = 3A_2 = 6A_3$

(C) $3A_1 = 2A_2 = 6A_3$

(D) $6A_1 = 3A_2 = 2A_3$

2. Two men P and Q support a uniform plank with their shoulders. The weight of the plank is 100 N, and its length is 2 m. If P's shoulder and Q's shoulder are 0.3 m and 0.7 m apart from the two ends respectively, then the supporting force (in N) by man P is:



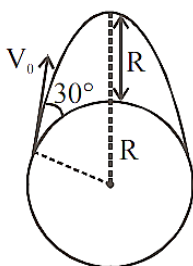
(A) 30 N

(B) 15 N

(C) 70 N

(D) 35 N

- 3*. A projectile is thrown at angle 30° with surface of earth, with velocity v_0 such that it attains a maximum height equal to radius of earth, then:



(A) Minimum velocity would be $\frac{v_0}{4}$

(B) Minimum velocity would be $\frac{\sqrt{3}v_0}{4}$

(C) At maximum height velocity will be zero (D) Trajectory would not be parabolic



- 4*. A binary star system consists of two stars having masses M and $2M$. The distance between their centres is equal to R . They revolve about their common centre of mass under their mutual gravitational force of interaction. Which of the following is/are CORRECT?

- (A) Heavier star revolves in orbit of radius $\frac{R}{3}$
 (B) Both the stars will revolve with the same time period which is equal to $\frac{2\pi}{\sqrt{3Gm}} R^{3/2}$
 (C) Kinetic energy of lighter star is twice that of the other
 (D) Kinetic energy of lighter star is 4 times that of the other

- 5*. If $\vec{F}$ be the force acting on a particle having the position vector $\vec{r}$ and $\vec{\tau}$ is the torque of this force about the origin acting on particle, then choose the CORRECT statement(s):

- (A) $\vec{\tau} \cdot \vec{r} = 0$ (B) $\vec{\tau} \cdot \vec{F} = 0$ (C) $\vec{F} \cdot \vec{r} = 0$ (D) $\vec{r} \cdot \vec{F} = 0$

- 6*. A rigid body of mass 0.5 kg at $\vec{r} = 2\hat{i} - \hat{j} + \hat{k}$ is acted on by the forces $\vec{F}_1 = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{F}_2 = 2\hat{i} + 3\hat{j} - \hat{k}$. Then:

- (A) Acceleration of body is given by $6\hat{i} + 8\hat{j} - 6\hat{k}$
 (B) Vector perpendicular to $\vec{r}$ and $\vec{F}_1$ is $\hat{i} + 5\hat{j} + 3\hat{k}$
 (C) Torque acting on the body due to forces about the origin is $-\hat{i} + 9\hat{j} + 11\hat{k}$
 (D) Force required to make the body at equilibrium is $-3\hat{i} - 4\hat{j} + 3\hat{k}$

Paragraph for Question No. 7 to 9

Consider a hypothetical planet which is very long and cylindrical. The density of the planet is ρ and its radius is R .



7. What is the possible orbital speed of the satellite moving around the planet in circular orbit in a plane which is perpendicular to the axis of planet?

- (A) $R\sqrt{\pi G \rho}$ (B) $2R\sqrt{\pi G \rho}$ (C) $R\sqrt{2\pi G \rho}$ (D) $R\sqrt{\frac{G \rho}{2\pi}}$

8. If an object is projected radially outwards from the surface such that it reaches upto a maximum distance of $3R$ from the axis then what should be the speed of projection?

- (A) $R\sqrt{\frac{2}{3}\pi \rho G}$ (B) $2R\sqrt{\pi \rho G \ln 3}$ (C) $R\sqrt{\frac{4}{3}\pi \rho G}$ (D) $R\sqrt{\frac{2}{3}\pi \rho G \ln 3}$

9. Assume that the planet is rotating about its axis with time period T . How far from the axis of the planet do the geo synchronous telecommunications satellite orbit?

- (A) $RT\sqrt{\pi G \rho}$ (B) $2RT\sqrt{\pi G \rho}$ (C) $RT\sqrt{2\pi G \rho}$ (D) $RT\sqrt{\frac{G \rho}{2\pi}}$

10. If the areal velocity of one planet is 2 times the areal velocity of other planet, then find the ratio of their radii of circular motion around the sun.



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-28

Note: (*) → Multiple Correct Type Question

Total Marks: 39

Max. Time: 26 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.6 to Q.8

(4 marks, 3 min.)

Match the column Type Questions (No negative marking) Q.9

(8 marks, 4 min.)

Subjective Type Questions (No negative marking) Q.10

(4 marks, 3 min.)

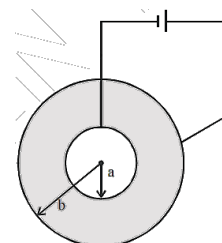
(There is partial marking in multiple correct questions)

1. If charge flow in a circuit is given by $q = \left(t + \frac{1}{t}\right) C$ then current in circuit at $t = 2$ sec will be

(A) $\frac{1}{2} A$ (B) $\frac{1}{4} A$ (C) $\frac{3}{4} A$ (D) $\frac{1}{8} A$

2. A hollow spherical conductor is having inner radius 'a' and outer radius 'b'. A constant potential difference is applied across inner and outer shell. Due to which current flows radially. Select the correct statement.

(A) Drift velocity remains constant
(B) Drift velocity increase radially outward
(C) Drift velocity decrease radially outward
(D) Drift velocity first decreases then increases radially outward.

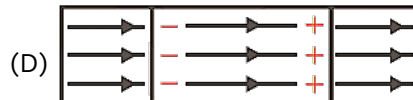
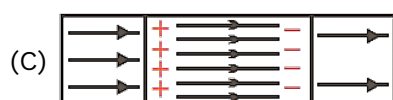
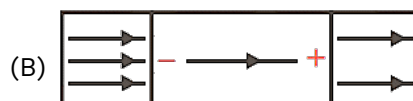
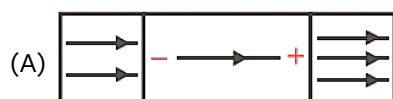
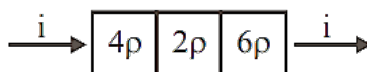


3. An insulating pipe of cross section areas 'A' contains an electrolyte which has two types of ions → their charges being $-3e$ and $+4e$. A potential difference applied between the ends if the pipe results in the drifting of the two types of ions, having drift speed $= \frac{v}{3}$ (-ve ion) and $\frac{v}{2}$ (+ve ion). Both ions

have the same number per unit volume $= n$. The current flowing through the pipe is

(A) $nev A$ (B) $nev A/4$ (C) $5 nev A/2$ (D) $3 nev A$

4. Which fit the following electric field pattern is correct inside the conductor having three materials of resistivity?



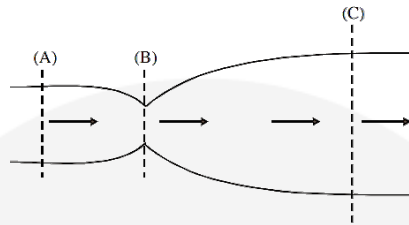


5. A rod of negligible mass and length ℓ is pivoted at its centre. A particle of mass m is fixed to its left end and another particle of mass $2m$ is fixed to the right end. If the system is released from rest and after sometime becomes vertical, the speed v of the two masses and the angular speed at that instant is:



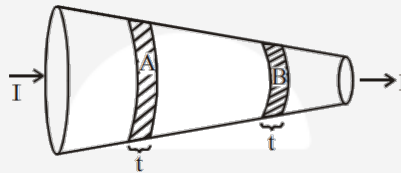
(A) $\sqrt{\frac{g\ell}{3}}, \sqrt{\frac{4g}{3\ell}}$ (B) $\sqrt{\frac{4g\ell}{3}}, \sqrt{\frac{4g}{3\ell}}$ (C) $\sqrt{\frac{4g}{3\ell}}, \sqrt{\frac{4g\ell}{3}}$ (D) $\sqrt{\frac{g\ell}{3}}, \sqrt{\frac{g}{3\ell}}$

- 6.* Figure shows a copper wire of nonuniform cross section carrying current right for sectors A, B and C.

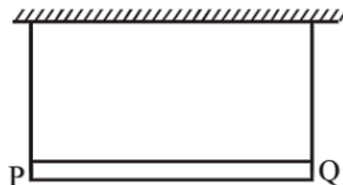


- (A) $i_B > i_A > i_C$ (i = current) (B) $E_B > E_A > E_C$ (E = electric field)
 (D) $V_B > V_A > V_C$ (V = drift velocity) (D) $J_B > J_A > J_C$ (J = current density)

- 7.* Figure shows a homogeneous conductor with decreasing cross-section area along the flow of current. Consider two elementary disks A and B of same thickness on conductor as shown. Choose the correct option(s):



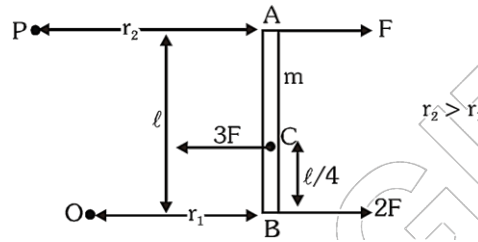
- (A) Drift velocity of electron at B is greater than at A.
 (B) No. of electrons per unit volume is same at A and B.
 (C) Resistance of segment B is bigger than that of A.
 (D) Electric field, current density and resistivity, all are bigger at B than at A.
- 8.* A horizontal rod of mass 'M' and length 'L' is tied to two vertical strings symmetrically, as shown in the figure. One of the strings at end Q is cut at $t = 0$ and the rod starts rotating about the other end P. Then:



- (A) At $t = 0$, angular acceleration of rod about P is $3g/2L$.
 (B) At $t = 0$, angular acceleration of rod about C.M. of rod is $3g/2L$.
 (C) At $t = 0$, acceleration of C.M. of rod is $3g/4$ in downward direction.
 (D) At $t = 0$, tension in the string at P is $Mg/4$.



9. In the given figure, uniform rod has mass m and length ℓ .



Column I

- (A) Torque about point O
(B) Torque about point P
(C) Torque about point C
(D) Torque about center of mass

Column II

- (P) $F\ell/4$
(Q) Fr_2
(R) $2Fr_1$
(S) More than torque about A
(T) Equal to torque about B

10. Total momentum of electrons in a straight wire of length L carrying a current I is P , if mass of electron is doubled keeping its charge constant, and length of the wire is also doubled keeping current constant. The new value of momentum will be nP . So, the value of n will be.

Note: (*) → Multiple Correct Type Question

Total Marks: 28

Max. Time: 20 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.4

(3 marks, 2 min.)

Comprehension Type Questions ('-1' Negative marking) Q.5 to Q.8

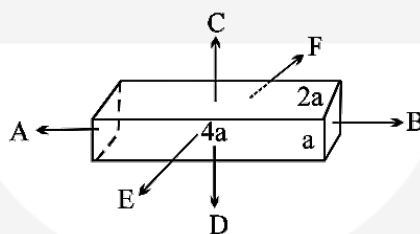
(2 marks, 2 min.)

Match the Column Type Questions (No negative marking) Q.10

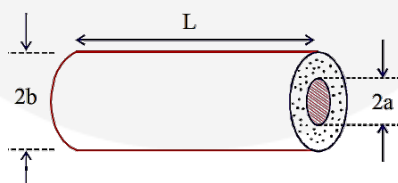
(8 marks, 4 min.)

(There is partial marking in multiple correct questions)

- A conductor wire having 10^{29} free electrons/ m^3 carries a current of 20A. If the cross-section of the wire is 1mm^2 , then the drift velocity of electrons will be ($e = 1.6 \times 10^{-19}\text{C}$)
(A) $1.26 \times 10^{-4}\text{ms}^{-1}$ (B) $1.25 \times 10^{-3}\text{ms}^{-1}$ (C) $1.25 \times 10^{-5}\text{ms}^{-1}$ (D) $6.25 \times 10^{-3}\text{ms}^{-1}$
- A conductor with rectangular cross section has dimensions ($a \times 2a \times 4a$) as shown in figure. Resistance across AB is x , across CD is y and across EF is z . Then



- (A) $x = y = z$ (B) $x > y > z$ (C) $y > x > z$ (D) $x > z > y$
- A resistor is made from a hollow cylinder of length L , inner radius a , and outer radius b . The interior of the cylinder is filled with a different material. Let r be the distance from the axis of the cylinder. The region $a < r < b$ is filled with material of resistivity ρ_1 . The region $r < a$ is filled with material of resistivity ρ_2 where $\rho_2 = \rho_1/2$. Find the resistance R across both ends of this resistor.



- (A) $R = \frac{\rho_1 L}{\pi(b^2 - a^2)}$ (B) $R = \frac{\rho_1 L}{\pi(b^2 + a^2)}$ (C) $R = \frac{\rho_1 L}{2\pi} \frac{a^2 + b^2}{(b^2 + a^2)}$ (D) $R = \frac{\rho_1 L}{2\pi} \frac{a^2 + b^2}{(b^2 - a^2)}$
- A hollow conducting sphere of inner radius R and outer radius $2R$ has resistivity ' ρ ' a function of the distance ' r ' from the centre of the sphere: $\rho = kr^2/R$. The inner and outer surface are painted with a perfectly conducting 'paint' and a potential difference ΔV is applied between the two surface. Then, as ' r ' increases from R to $2R$, the electric field inside the sphere
(A) increases (B) decreases
(C) remains constant (D) Passes through a maxima



Paragraph for Question no. 5 and 6

In a metallic resistor, there are free electrons which can move under the influence of the electric field and thus constitute current. However, in air we cannot find such charges which constitute current. Strangely, surrounding the earth is a Van allen belt where we can find that the air is conducting. How so? Actually, due to the ultraviolet radiation the air molecules split into two ions one positive and another negative. These ions move with a uniform speed on an average under the influence of electric field and constitute current. For these ions we define mobility which is their drift speed/electric field applied. To study this, let us consider that we have two parallel plates separated by a distance d . The air between them is ionized by x-ray radiation. The plates have an area S each and the voltage of V is applied across them. The ion mobility of positive charge is u^+ and ion mobility of negative charge is u^- . Assume that these ions do not recombine. And u^+ and u^- are not same. Total current is I . Modulus of charge on each ion is e .

5. Select the current statement :

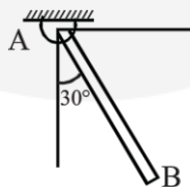
- (A) The current due to both positive and negative ions is the same but their charge density is different.
 (B) The current due to both positive and negative ions is different but their charge density is same.
 (C) The current due to both positive and negative ions as well as their charge density is same.
 (D) The current due to both positive and negative ions as well as their charge density is different.

6. The number density of positive ions is :

- (A) $\frac{Id}{eS[u^+ - u^-]V}$ (B) $\frac{Id}{eS[u^+ + u^-]V}$ (C) $\frac{Id}{eSu^+V}$ (D) $\frac{Id}{eSu^-V}$

Paragraph for Q. No. 7 to 8

A rod of length ℓ and mass m is hinged at A. It is released from horizontal position from rest. When it comes to position AB, and makes an angle of 30° with vertical, then:



7. The angular velocity in this position is :

- (A) $\left(\frac{3\sqrt{3}g}{2\ell}\right)^{1/2}$ (B) $\left(\frac{3g}{2\ell}\right)^{1/2}$ (C) $\left(\frac{9g}{2\ell}\right)^{1/2}$ (D) $\left(\frac{3\sqrt{3}g}{4\ell}\right)^{1/2}$

8. Force applied by the hinge on the rod along the rod is :

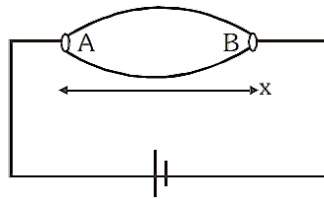
- (A) $\frac{3mg}{4}$ (B) $\frac{5\sqrt{3}mg}{4}$ (C) $\frac{3\sqrt{3}mg}{4}$ (D) $\frac{\sqrt{3}mg}{2}$

9. The angular acceleration of the rod is :

- (A) $\frac{3g}{2\ell}$ (B) $\frac{3g}{\ell}$ (C) $\frac{3g}{4\ell}$ (D) $\frac{g}{\ell}$



10. A resistor of varying cross-section is attached with a battery as shown. Variation of area from both ends are similar and areas are same at ends maximum at centre. Match the approximate graphical representation of various physical properties in column, where x axis represent position from end A of resistor and y -axis various physical properties.



Column-I

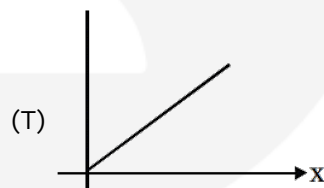
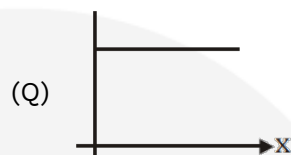
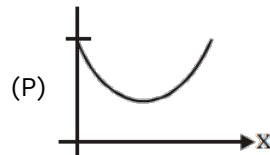
(A) i (current)

(B) j (current density)

(C) E (electric field)

(D) V_d (potential)

Column-II



Note: (*) → Multiple Correct Type Question

Total Marks: 33

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.7

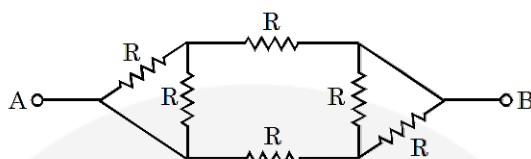
(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.8 to Q.10

(4 marks, 3 min.)

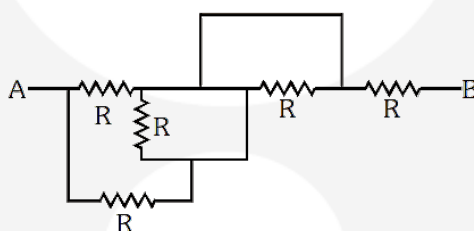
(There is partial marking in multiple correct questions)

1. The equivalent resistance between A and B is :



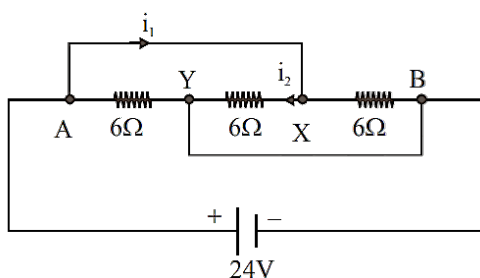
- (A) $\frac{3}{4}R$ (B) $\frac{5}{3}R$ (C) $\frac{7}{5}R$ (D) R

2. What is the equivalent resistance between A and B?



- (A) $\frac{3R}{2}$ (B) $2R$ (C) R (D) $\frac{R}{2}$

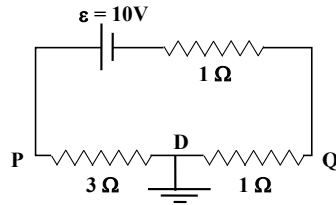
3. In the figure shown, the current in the wires which join the points A and X is i_1 and the points X and Y is i_2 . Then i_1 & i_2 are respectively.



- (A) $i_1 = 12\text{ A}, i_2 = 4\text{ A}$ (B) $i_1 = 8\text{ A}, i_2 = 4\text{ A}$ (C) $i_1 = 4\text{ A}, i_2 = 4\text{ A}$ (D) $i_1 = 3\text{ A}, i_2 = 12\text{ A}$

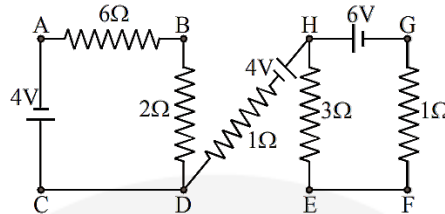


- 4 Potential of the points P and Q in the circuit shown in the figure are:



- (A) 6 V and -2 V respectively
(B) 6 V and 2 V respectively
(C) -6 V and -2 V respectively
(D) -6 V and 2 V respectively

5. In the network shown in the figure, choose the incorrect statement:

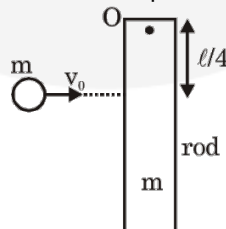


- (A) Potential difference between points H and B is 5V.
(B) Potential difference between points D and B is 1 V
(C) Potential difference between points D and E is 1 V
(D) Potential difference between points D and G is 2V

6. A Cockroach of mass m is moving on the rim of a disc (radius r) with velocity v in the anticlockwise direction with respect to ground. The moment of inertia of the disc about its own axis is I and it is rotating in the clockwise direction with angular speed ω . If the Cockroach stops moving with respect to disc, then the angular speed of the disc will be:

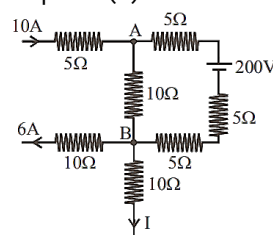
- (A) $\frac{I\omega}{I + mr^2}$ (B) $\frac{I\omega + mvr}{I + mr^2}$ (C) $\frac{I\omega - mvr}{I + mr^2}$ (D) $\frac{I\omega + mvr}{I}$

7. Rod of mass m and length ℓ is free to rotate about point 'O' in vertical plane. A particle having same mass m moving horizontally with velocity v_0 hits the rod perpendicularly at distance $\frac{\ell}{4}$ from the top end 'O' and stops. Find the magnitude of impulse due to hinge on the rod due to collision.



- (A) $\frac{mv_0}{19}$ (B) $\frac{10mv_0}{19}$ (C) $\frac{mv_0}{8}$ (D) $\frac{5mv_0}{8}$

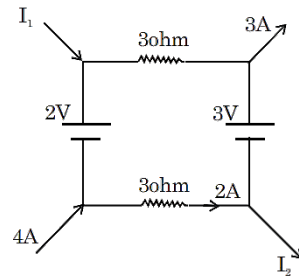
- 8.* In the given diagram choose correct option(s):



- (A) The value of I is 4A
(B) The current from the cell is 4A
(C) Voltage drop across AB is 140V
(D) The current from the cell is 10A



9.* A part of the circuit is shown in the figure. Choose the correct option(s):



- (A) $I_1 = 1/3$ (B) $I_2 = 2/3 \text{ A}$ (C) $I_1 = -1/3 \text{ A}$ (D) $I_2 = -2/3 \text{ A}$

10.* Disc of mass m is free to rotate about a vertical axis. A man of mass m stands near circumference. Radius of disc is R . Initially both are at rest. Now man moves with constant speed v relative to disc along circumference. Choose correct option(s):



- (A) Angular velocity of disc is $\frac{2v}{3R}$ (B) Angular velocity of disc is $\frac{3v}{2R}$
(C) Velocity of man is v (D) Velocity of man is $\frac{v}{3}$

Note: (*) → Multiple Correct Type Question

Total Marks: 34

Max. Time: 24 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.7 to Q.8

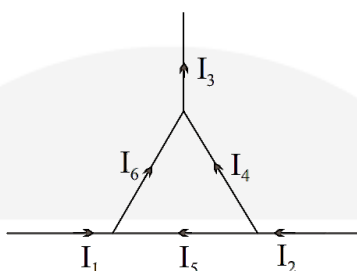
(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9 to Q.10

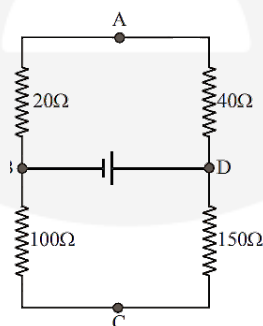
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

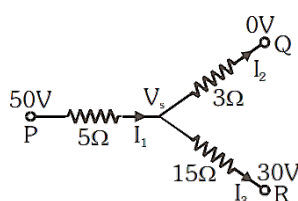
1. The diagram below shows a junction with currents labeled I_1 to I_6 . Which of the following statements is correct?



- (A) $I_1 + I_3 = I_6 + I_4$ (B) $I_1 + I_2 = I_6 + I_4$ (C) $I_4 + I_3 = I_6$ (D) $I_2 = I_6 + I_4$
2. When a 2Ω resistance is connected to a cell, then 2A current flows in it and when 9Ω resistance is connected to the same cell, then 0.5 A current flows in it. Internal resistance of cell is:
- (A) 1Ω (B) 2Ω (C) $\frac{1}{3}\Omega$ (D) $\frac{1}{4}\Omega$
3. Points, marked by A,B,C,D of the circuit are shown in the figure. Between which two points the potential difference is greater?



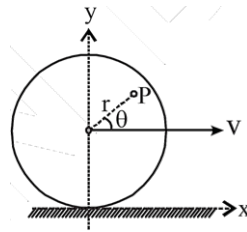
- (A) A & B (B) A & D (C) B & C (D) C & D
4. For given circuit, choose the correct statement:



- (A) Value of V_s is 20 V (B) Value of I_1 is 5A
(C) Value of I_2 is 9 A (D) Power drawn by circuit is 640W



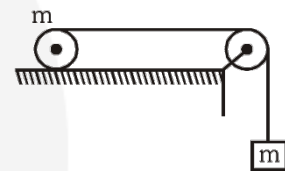
5. A disc of radius R rolls without slipping with speed v along positive x -axis. Velocity of point P at the instant shown in figure is:



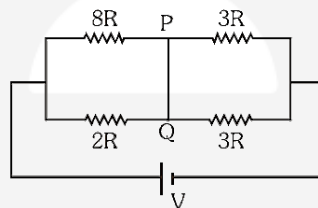
- (A) $\vec{V}_P = \left(v + \frac{vr \sin\theta}{R} \right) \hat{i} + \frac{vr \cos\theta}{R} \hat{j}$ (B) $\vec{V}_P = \left(v + \frac{vr \sin\theta}{R} \right) \hat{i} - \frac{vr \cos\theta}{R} \hat{j}$
 (C) $\vec{V}_P = v + \frac{vr \sin\theta}{R} \hat{i} + \frac{vr \cos\theta}{R} \hat{j}$ (D) $\vec{V}_P = \frac{vr \sin\theta}{R} \hat{i} - \frac{vr \cos\theta}{R} \hat{j}$

6. In the given figure, a ring of mass m is kept on a horizontal surface while a body of equal mass ' m ' is attached through a string, which is wound on the ring. When the system is released the ring rolls without slipping. Consider the following statements and choose the correct option.

- (i) acceleration of the centre of mass of ring is $\frac{g}{3}$
 (ii) acceleration of the hanging particle is $\frac{2g}{3}$
 (iii) frictional force (on the ring) acts along forward direction
 (iv) frictional force (on the ring) acts along backward direction
 (A) statement (i) and (ii) only
 (B) statement (ii) and (iii) only
 (C) statement (iii) and (iv) only
 (D) none of these

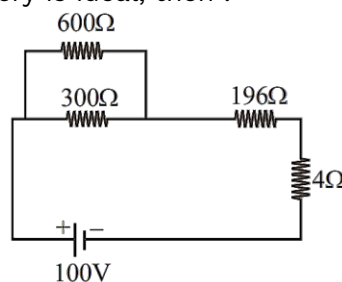


- 7.* Choose correct option(s) corresponding to the given circuit.



- (A) Current in PQ will be zero (B) Current in PQ will be $P \rightarrow Q$
 (C) Current in PQ will be $Q \rightarrow P$ (D) Equivalent resistance is $\frac{31R}{10}$

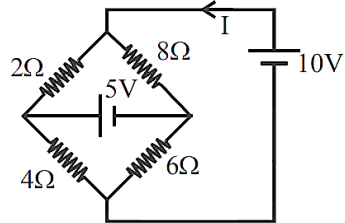
- 8.* In the given circuit diagram battery is ideal, then :



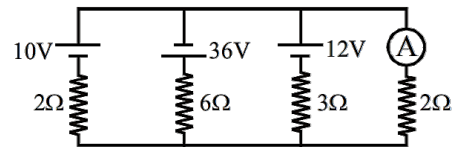
- (A) Potential difference across 300Ω is $66.67V$
 (B) Potential difference across 600Ω is $50V$
 (C) Current in 4Ω is $\frac{1}{2}A$
 (D) Current in 196Ω is $\frac{1}{4}A$



9. The current supplied to the circuit by 10 V cell is I ampere. Assuming both cells as ideal, find $4I$.



10. Find the current shown by ammeter (ideal) in ampere.



Note: (*) → Multiple Correct Type Question

Total Marks: 34

Max. Time: 24 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

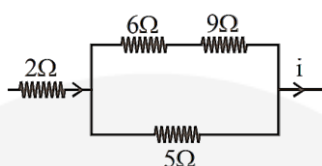
(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.7 to Q.10

(4 marks, 3 min.)

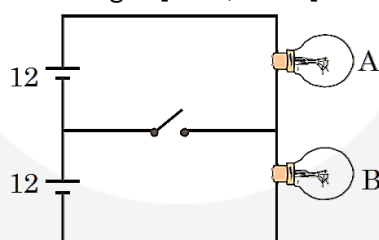
(There is partial marking in multiple correct questions)

1. In the given circuit, the heat produced in 5Ω resistance in certain time interval is 4.05 J. Then the heat produced in 2Ω resistance in the same time interval will be.



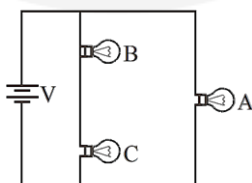
- (A) 5.76 J (B) 1.44J (C) 2.88 J (D) 2.02 J

2. Two light bulbs in the circuit have rating A [24 V, 48 W] and B [24V, 36 W] as shown



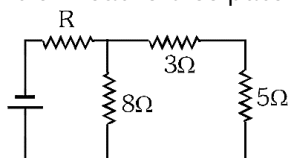
- (A) When switch is closed, bulb A delivers 24 watt & B delivers 18 Watt.
 (B) When switch is open, A delivers 9 watt & B delivers 12 watt approximately
 (C) When switch is open, A delivers 24 watt & B delivers 18 watt approximately
 (D) When switch is open, A delivers 6 watt and B delivers 9 watt approximately

3. Wattage of three bulbs A, B and C rated at same voltage are W_A , W_B and W_C respectively. Their brightness are represented by P_A , P_B and P_C . If $P_A = P_B = P_C$, then correct ratio of their wattage ($W_A : W_B : W_C$) is.



- (A) 1 : 2 : 3 (B) 1 : 2 : 2 (C) 1 : 4 : 4 (D) 2 : 2 : 1

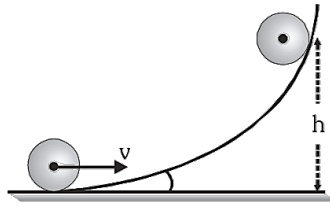
4. The value of R So that same amount of heat is dissipated in R and rest of the circuit, is:



- (A) 4Ω (B) 8Ω (C) 3Ω (D) 5Ω

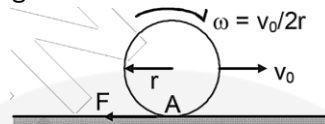


5. A disc of mass M and radius R rolls on a horizontal surface and then rolls up an inclined plane as shown in the figure. If the velocity of the disc is v , the height to which the disc will rise will be:



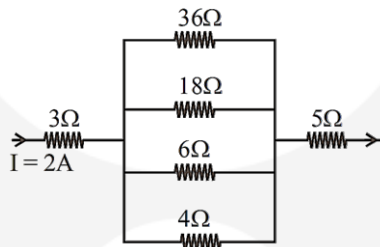
- (A) $\frac{3v^2}{2g}$ (B) $\frac{3v^2}{4g}$ (C) $\frac{v^2}{4g}$ (D) $\frac{v^2}{2g}$

6. A solid sphere of mass M and radius r slips on a rough horizontal plane. At some instant it has translational velocity v_0 and rotational velocity about the centre $v_0/2r$. The translational velocity after the sphere starts pure rolling is:



- (A) $6v_0/7$ in forward direction (B) $6v_0/7$ in backward direction
(C) $7v_0/6$ in forward direction (D) $7v_0/6$ in backward direction

- 7.* Current passing through 3Ω resistor, as shown in figure is $2A$. Then which of the following statements are correct?

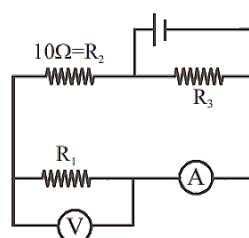


- (A) Current passing through 4Ω is $1A$ (B) Potential difference across 6Ω resistor is 4 volt
(C) Power consumption in 18Ω is $\frac{8}{9}$ watt (D) Current passing through 36Ω is $1A$

- 8.* An electric tea kettle has two heating coils. The tea kettle is connected to a constant current source to boil the tea. When one of them is switched on, the tea begins to boil in 6 min & for other in 3 min. The coils are now switched on simultaneously.

- (A) If the coils are connected in series, the boiling will begin in 2 min.
(B) If the coils are connected in series, the boiling will begin in 9 min.
(C) If the coils are connected in parallel, the boiling will begin in 2 min.
(D) If the coils are connected in parallel, the boiling will begin in 9 min.

- 9.* In the shown circuit, the power dissipated in resistor R_3 is $30W$. The reading of ammeter and voltmeter are 500 mA and $10V$ respectively. Ammeter, voltmeter and battery are ideal. Choose the correct statement(s):



- (A) The resistor R_1 is 20Ω (B) The resistor R_3 is 7.5Ω
(C) emf of battery is $15V$ (D) Power supplied by battery is $37.5W$



- 10.*** A stick with length ℓ has both translational and rotational motion, as shown in figure. Both v and ω are positive in the direction shown. Taking clockwise angular momentum L to be positive, which one of the following statements is/are true?



- (A) The L around point B is negative.
- (B) The L around point A can be positive or negative, depending on the value of v and ω .
- (C) The L around point A is positive.
- (D) The L around point B can be positive or negative, depending on the value of v and ω .



Note: (*) → Multiple Correct Type Question

Total Marks: 26

Max. Time: 22 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.2

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.3 to Q.4

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.5 to Q.10

(2 marks, 2 min.)

(There is partial marking in multiple correct questions)

1. The following four wires of length L and radius r are made of the same material. Which of these will have the largest extension, when the same tension is applied?

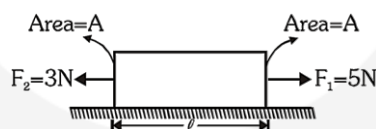
(A) $L = 100 \text{ cm}$, $r = 0.2 \text{ mm}$

(B) $L = 200 \text{ cm}$, $r = 0.4 \text{ mm}$

(C) $L = 300 \text{ cm}$, $r = 0.6 \text{ mm}$

(D) $L = 400 \text{ cm}$, $r = 0.8 \text{ mm}$

2. A block of mass M area of cross-section A and length ℓ is placed on smooth horizontal floor. Two forces $F_1 = 5 \text{ N}$ and $F_2 = 3 \text{ N}$ are applied on the block as shown in the figure. If Y is Young's modulus of elasticity of material, then total extension in the length of the block will be



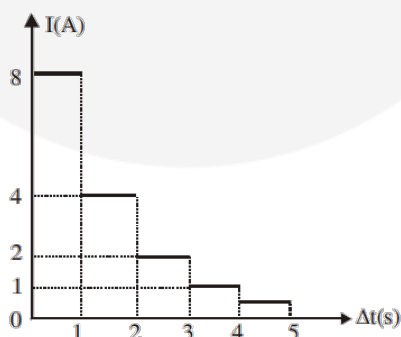
(A) $\frac{8\ell}{AY}$

(B) $\frac{2\ell}{AY}$

(C) $\frac{4\ell}{AY}$

(D) $\frac{5\ell}{2AY}$

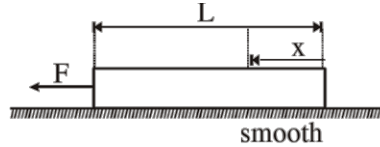
- 3.* The current in a wire of resistance 10Ω varies as shown. Select correct alternative(s).



- (A) The heat produced in the wire in 4 s is 852.5 J
 (B) The heat produced in the wire in 4 s is 850 J
 (C) The charge flows across any section of the wire in 4 s is 15 C
 (D) The charge flows across any section of the wire in 4 s is 15.5 C.



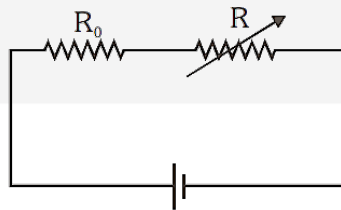
- 4.* A rod of mass m , uniform cross sectional area A and length L is accelerated by applying force F as shown in figure on a smooth surface. If Young's modulus of elasticity of the material of rod is Y . (Consider x as measured from the right end)



- (A) Tension in rod as a function of distance x is $\frac{Fx}{2L}$
 (B) Strain in rod is $\frac{F}{2AY}$
 (C) Elastic potential energy stored in the rod is $\frac{F^2L}{6AY}$
 (D) There is no stress in rod.

Paragraph for question 5 to 7

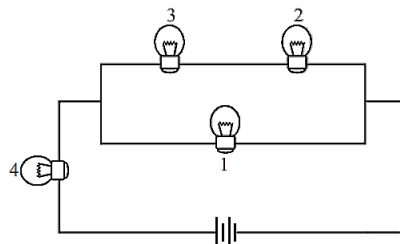
For a given circuit, with a given resistance R_0 and variable resistance R as shown:



5. For maximum power in R_0 , value of R is :
 (A) $\frac{R_0}{2}$ (B) R_0 (C) $2R_0$ (D) zero
6. For maximum power in R , value of R is :
 (A) $\frac{R_0}{2}$ (B) R_0 (C) $2R_0$ (D) zero
7. For maximum total power generated by battery, value of R is :
 (A) $\frac{R_0}{2}$ (B) R_0 (C) $2R_0$ (D) zero

Paragraph for question 8 to 10

All bulbs consume same power. The resistance of bulb 1 is 36Ω .



8. What is the resistance of bulb 3?
 (A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω
9. What is the resistance of bulb 4?
 (A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω
10. What is the voltage output of the battery if the power of each bulb is $4W$?
 (A) $12V$ (B) $16V$ (C) $24V$ (D) None of these

Note: (*) → Multiple Correct Type Question

Total Marks: 35

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.6 to Q.7

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.8 to Q.10

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. The variation of length of two metal rods A and B with change in temperature are shown in figure.

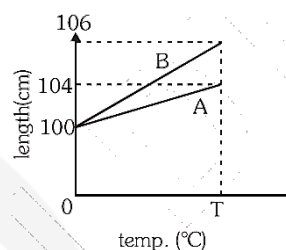
The ratio of $\frac{\alpha_A}{\alpha_B}$ is:

(A) $\frac{3}{2}$

(B) $\frac{2}{3}$

(C) $\frac{4}{3}$

(D) $\frac{3}{4}$



2. To convert galvanometer into ammeter, a shunt of 0.01Ω is used. Resistance of galvanometer coil is 50Ω and its maximum deflection current is 20 mA. Range of ammeter is :

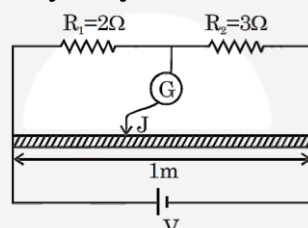
(A) 0-1 A

(B) 0-10A

(C) 0-100A

(D) 0-1000A

3. In a meter bridge experiment, initially the jockey is at null point. Now resistance R_1 & R_2 is interchanged. Shift in the position of jockey is :



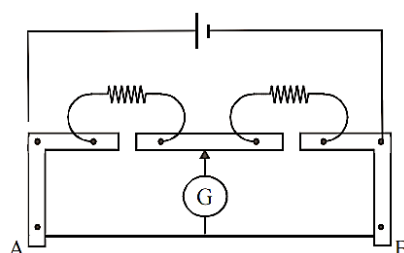
(A) 20 cm

(B) 30 cm

(C) 40 cm

(D) 50 cm

4. A meter bridge set up is shown to determine end correction at A and B. When a resistance of 15Ω is used in left gap and of 20Ω in right gap, then null point comes at a distance 42 cm from A. When these resistance are interchanged, null point comes at a distance 57 cm from A. Values of end corrections are:



(A) 1 cm, 2 cm

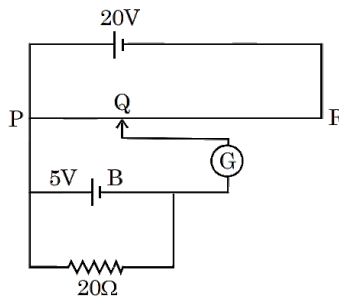
(B) 2 cm, 3 cm

(C) 3 cm, 4 cm

(D) 3 cm, 2 cm

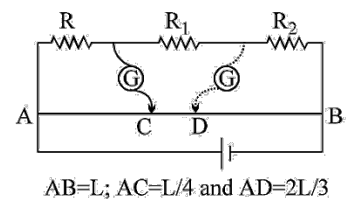


5. The figure shows a potentiometer arrangement used to measure the internal resistance of the cell B. The null point Q is located at 20 cm from point P on a 100 cm wire PR. Find the internal resistance of the cell B.



- (A) $1\ \Omega$ (B) $2\ \Omega$ (C) $5\ \Omega$ (D) $10\ \Omega$

- 6.* The diagram shows a modified meter bridge, which is used for measuring two unknown resistance at the same time. When only the first galvanometer is used, for obtaining the balance point, it is found at point C. Now the first galvanometer is removed and the second galvanometer is used, which gives balance point at D. Using the details given in the diagram, then mark the correct statement(s)?



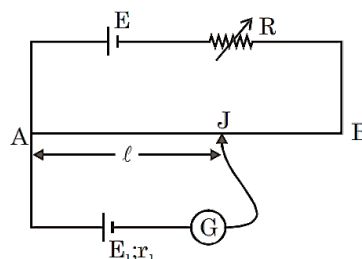
- (A) $R_1 = 5R/3$ (B) $R_2 = 7R/3$ (C) $R_1 = 4R/3$ (D) $R_2 = 4R/3$

- 7.* We wish to prepare a scale whose length does not change with temperature. Let this scale always has a length of 10 cm. For this we use two different strips of brass and iron each of different length, whose length would change in such a way that difference between their lengths remain constant and is equal to 10 cm. If $\alpha_{\text{iron}} = 1.2 \times 10^{-5}/\text{K}$ and $\alpha_{\text{Brass}} = 1.8 \times 10^{-5}/\text{K}$, then:

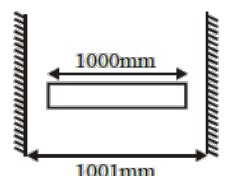
- (A) Length of Brass strip is 30 cm. (B) Length of Brass strip is 20 cm.
(C) Length of Iron strip is 30 cm. (D) Length of Iron strip is 20 cm.

8. The deflection of a moving coil galvanometer falls from 60 divisions to 12 divisions for the same value of current in the circuit, when a shunt of $12\ \Omega$ is connected. What is the resistance (in ohm) of the galvanometer?

9. In potentiometer experiment, balance length (ℓ) is x and $2x$ when variable resistor R is set at value $10\ \Omega$ and $24\ \Omega$. Resistance of potentiometer wire AB is (in Ω):



10. A rod of length 1000 mm and coefficient of linear expansion $\alpha = 10^{-4}$ per degree celsius is placed symmetrically between fixed walls separated by 1001 mm. The Young's modulus of the rod is $10^{11}\ \text{N/m}^2$. If the temperature is increased by 20°C , then the stress developed in the rod is:



Note: (*) → Multiple Correct Type Question

Total Marks: 32

Max. Time: 22 min.

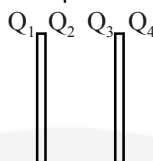
Single Correct Questions ('-1' negative marking) Q.1 to Q.8

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.9 to Q.10

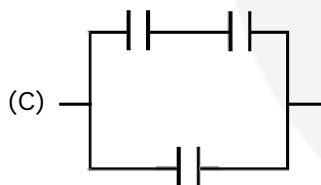
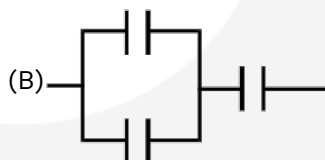
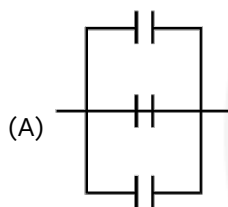
(4 marks, 3 min.)

1. In an isolated charged capacitor of capacitance 'C', the four surfaces have charges Q_1 , Q_2 , Q_3 and Q_4 as shown. Potential difference between the plates of the capacitor is:



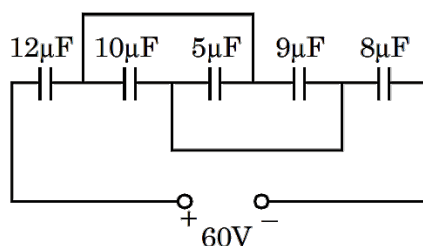
- (A) $\frac{Q_1 + Q_2 + Q_3 + Q_4}{C}$ (B) $\frac{Q_2 + Q_3}{C}$ (C) $\frac{|Q_2 - Q_3|}{2C}$ (D) $\frac{|Q_1 - Q_4|}{2C}$

2. In four options below, all the four circuits are arranged in order of equivalent capacitance. Select the correct order. Assume all capacitors are of equal capacitance.



- (A) $C_1 > C_2 > C_3 > C_4$ (B) $C_1 > C_3 > C_2 > C_4$ (C) $C_1 < C_2 < C_3 < C_4$ (D) $C_1 < C_3 < C_2 < C_4$

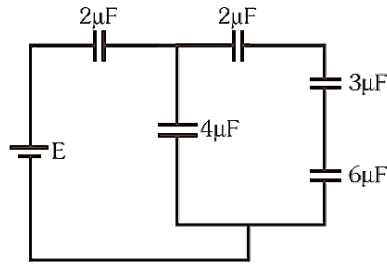
3. Find the equivalent capacitance of circuit and charge on $5\mu\text{F}$ capacitor :



- (A) $4\mu\text{F}$, $50\mu\text{C}$ (B) $8\mu\text{F}$, $25\mu\text{C}$ (C) $4\mu\text{F}$, $25\mu\text{C}$ (D) $8\mu\text{F}$, $50\mu\text{C}$

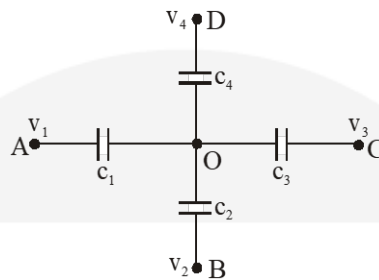


4. In the given circuit, if charge on $6\mu\text{F}$ capacitor is $10\mu\text{C}$, then charge on $4\mu\text{F}$ capacitor will be:



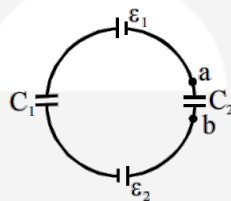
- (A) $40\mu\text{C}$ (B) $10\mu\text{C}$ (C) $20\mu\text{C}$ (D) $30\mu\text{C}$

5. Four uncharged capacitors c_1 , c_2 , c_3 and c_4 are connected as shown in the figure. The points A, B, C and D are kept at potential v_1 , v_2 , v_3 and v_4 respectively. Then the potential v_0 at point O is:



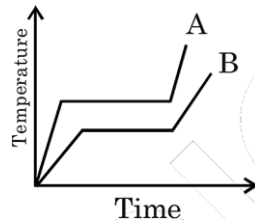
- (A) $v_1 + v_2 + v_3 + v_4$ (B) $\frac{v_1 c_1 + c_2 v_2 + c_3 v_3 + c_4 v_4}{c_1 + c_2 + c_3 + c_4}$
 (C) $\frac{1}{4} (v_1 + v_2 + v_3 + v_4)$ (D) $\frac{1}{2} \frac{c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4}{c_1 + c_2 + c_3 + c_4}$

6. Find the potential difference across points a and b in given figure:



- (A) $\left(\frac{\varepsilon_1 + \varepsilon_2}{C_1 + C_2} \right) C_1$ (B) $\left(\frac{\varepsilon_1 + \varepsilon_2}{C_1 + C_2} \right) C_2$ (C) $\left(\frac{\varepsilon_1 - \varepsilon_2}{C_1 + C_2} \right) C_1$ (D) $\left(\frac{\varepsilon_1 - \varepsilon_2}{C_1 + C_2} \right) C_2$

7. Equal masses of two liquids A and B contained in vessels of negligible heat capacity are supplied heat at the same rate. The temperature vs time graphs for the two liquids are shown in figure. If S represents specific heat and L represents latent heat of liquid, then:



- (A) $S_A > S_B$; $L_A < L_B$ (B) $S_A > S_B$; $L_A > L_B$
 (C) $S_A < S_B$; $L_A < L_B$ (D) $S_A < S_B$; $L_A > L_B$

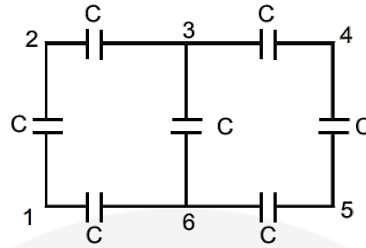


8. There are three liquids A, B and C. The mass, heat capacity and temperature for liquid A is (m_A , C_A & T_A), for liquid B is (m_B , C_B & T_B) and for liquid C is (m_C , C_C & T_C), respectively. If all the three liquids are mixed, then final temperature would be:

(A) $\frac{C_A T_A + C_B T_B + C_C T_C}{C_A + C_B + C_C}$ (B) $\frac{m_A C_A T_A + m_B C_B T_B + m_C C_C T_C}{m_A C_A + m_B C_B + m_C C_C}$

(C) $\frac{m_A T_A + m_B T_B + m_C T_C}{m_A + m_B + m_C}$ (D) $\left(\frac{T_A}{m_A} + \frac{T_B}{m_B} + \frac{T_C}{m_C} \right) (m_A + m_B + m_C)$

- 9.* For the given circuit, select the correct alternative(s)



- (A) The equivalent capacitance between points 1 & 2 is $\frac{15C}{11}$
- (B) The equivalent capacitance between points 3 & 6 is $\frac{5C}{3}$
- (C) The equivalent capacitance between points 1 & 3 is $\frac{15C}{14}$
- (D) The equivalent capacitance between points 3 & 5 is $\frac{14C}{15}$

- 10.* 5 g of steam at 100 °C is mixed with 10 g of ice at 0 °C. Choose correct alternative/s
(Given $S_{\text{water}} = 1 \text{ cal/g } ^\circ\text{C}$, $L_F = 80 \text{ cal/g}$, $L_v = 540 \text{ cal/g}$)

- (A) Equilibrium temperature of mixture is 160 °C
- (B) Equilibrium temperature of mixture is 100 °C
- (C) At equilibrium mixture contains $13\frac{1}{3}$ g of water
- (D) At equilibrium, mixture contains $1\frac{2}{3}$ g of steam

Note: (*) → Multiple Correct Type Question

Total Marks: 33

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.7

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.8 to Q.9

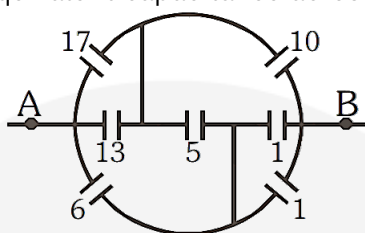
(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.10

(4 marks, 3 min.)

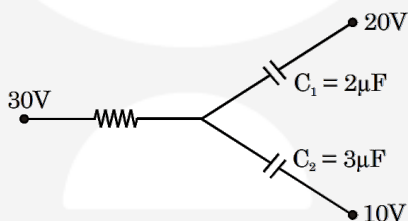
(There is partial marking in multiple correct questions)

1. In the circuit shown in figure, equivalent capacitance across AB is (all capacitance in μF):



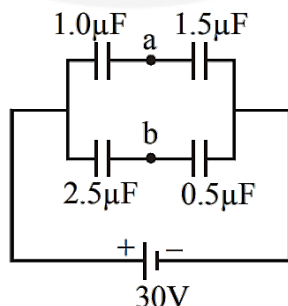
- (A) $\frac{20}{3} \mu\text{F}$ (B) $9 \mu\text{F}$ (C) $48 \mu\text{F}$ (D) None of these

2. If current passing through both the capacitors is zero, then charges on capacitor C_1 & C_2 will be respectively



- (A) 0,0 (B) $40 \mu\text{C}$, $30 \mu\text{C}$ (C) $20 \mu\text{C}$, $60 \mu\text{C}$ (D) $20 \mu\text{C}$, $30 \mu\text{C}$

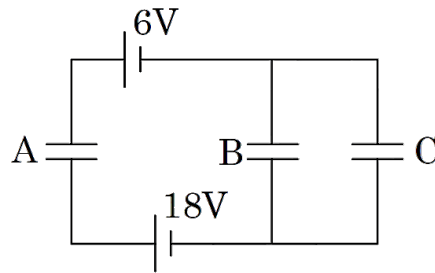
3. Four capacitors are connected as shown in figure to a 30 V battery. The potential difference between points b and a is :



- (A) 5 V (B) 9 V (C) 10 V (D) 13 V

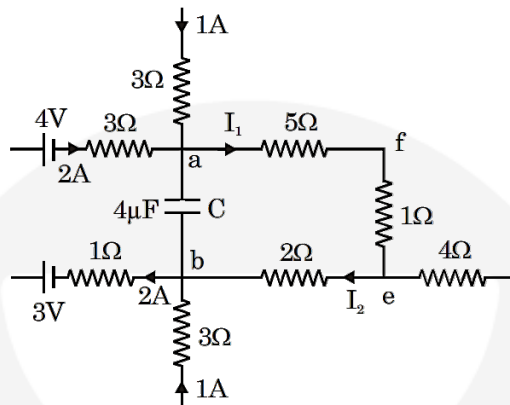


4. Three capacitors A, B and C each of capacitance $1\ \mu\text{F}$ are connected in a circuit as shown in figure. The energy stored in the capacitor C is:



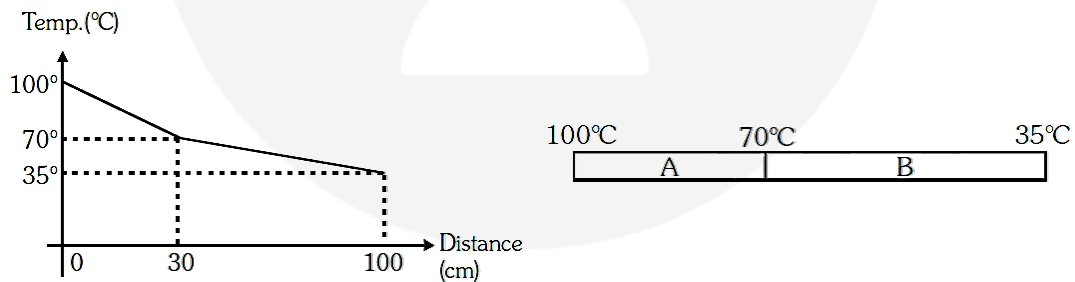
- (A) $4\ \mu\text{J}$ (B) $6\ \mu\text{J}$ (C) $8\ \mu\text{J}$ (D) $2\ \mu\text{J}$

5. The energy stored in capacitor C in steady state is



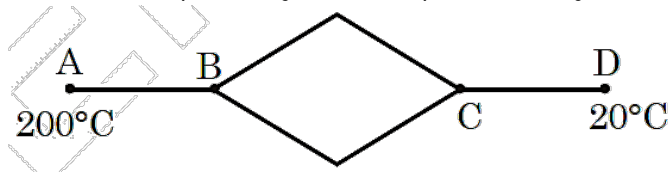
- (A) $8 \times 10^{-4}\ \text{J}$ (B) $10 \times 10^{-4}\ \text{J}$ (C) $6 \times 10^{-4}\ \text{J}$ (D) $12 \times 10^{-4}\ \text{J}$

6. Two different rods A and B are kept as shown in figure. The variation of temperature of different cross sections is plotted in a graph shown in figure. The ratio of thermal conductivities of A and B is:



- (A) 2 (B) 0.5
(C) 1 (D) $\frac{2}{3}$

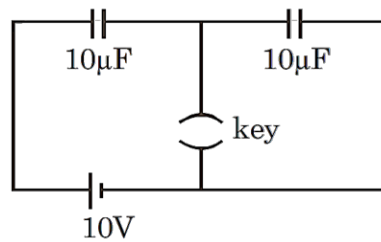
7. Six identical conducting rods are joined as shown in figure. Points A and D are maintained at temperature 200°C and 20°C , respectively. The temperature of junction 'B' will be:



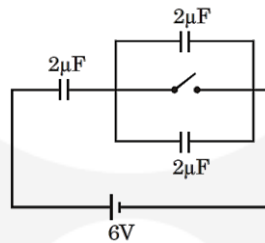
- (A) 120°C (B) 140°C
(C) 100°C (D) 80°C



- 8.*** Consider the situation shown in figure and battery is ideal. The key is open for long time and then closed. Mark the CORRECT statement(s) :



- (A) Charge flown through the battery when key is closed is $50\mu\text{C}$.
(B) Charge flown through the key when key is closed is $200\mu\text{C}$.
(C) Change in potential energy of capacitors when key is closed is $1\mu\text{J}$.
(D) Change in potential energy of capacitors when key is closed is $250\mu\text{J}$.
- 9.*** One end of a uniform rod is placed in boiling water and the other end in melting ice. After a long time: (Assume temperature of boiling water and melting ice is constant)
(A) temperature of each point of rod will become same
(B) temperature of each cross-section of rod will become constant
(C) temperature difference is same for any two equidistant cross-sections
(D) temperature of mid point of the rod is simple average of temperature of ends of rod
- 10.** Find the amount of charge (in μC) flown through the battery after switch is closed.



Note: (*) → Multiple Correct Type Questions

Total Marks: 37

Max. Time: 24 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.7

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.8 to Q.9

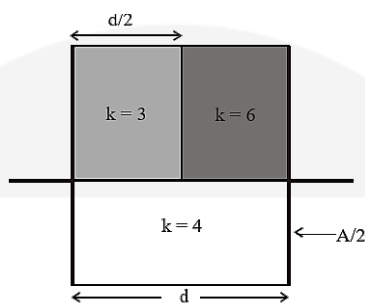
(4 marks, 3 min.)

Match the column Type Questions (No negative marking) Q.10

(8 marks, 4 min.)

(There is partial marking in multiple correct questions)

1. An air capacitor having capacity C_0 is filled with dielectric as shown in figure. The new capacitance of system will be:



- (A) C_0 (B) $2C_0$ (C) $3C_0$ (D) $4C_0$

2. A parallel plate capacitor with air between the plates has a capacitance of 9 pF. The separation between its plates is 'd'. The space between the plates is now filled with two dielectrics. One of the dielectric has dielectric constant $K_1 = 6$ and thickness $\frac{d}{3}$ while the other one has dielectric constant $K_2 = 12$ and thickness $\frac{2d}{3}$. Capacitance of the capacitor is now:

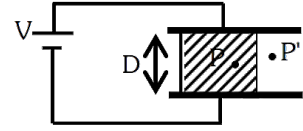
- (A) 18 pF (B) 25 pF (C) 81 pF (D) 20 pF

3. A parallel plate capacitor having a separation between the plates d, plate area A and material with dielectric constant K has capacitance C_0 . Now one-third of the material is replaced by another material with dielectric constant $2K$, so that effectively there are two capacitors one with area $\frac{1}{3}A$, dielectric constant $2K$ and another with area $\frac{2}{3}A$ and dielectric constant K. If the capacitance of this new capacitor is C, then $\frac{C_0}{C}$ is:

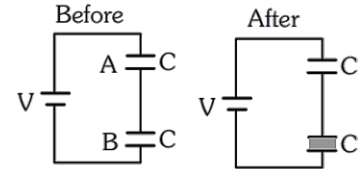
- (A) $\frac{3}{2}$ (B) $\frac{4}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$



4. A parallel plate capacitor, partially filled with a dielectric slab of dielectric constant K , is connected with a cell of emf V volt, as shown in the figure. Separation between the plates is D . Then:
- (A) electric field at point P is less than that at point P' .
(B) electric field at point P' is less than that at point P .
(C) electric fields at points P and P' are equal
(D) electric field at point P is $E = \frac{V}{KD}$.

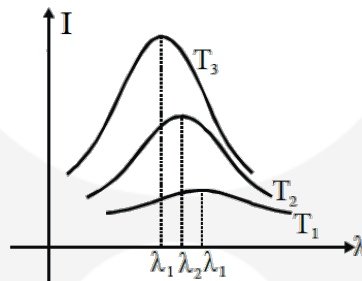


5. Two identical capacitors are connected in series as shown in the figure. A dielectric slab ($K > 1$) is placed between the plates of the capacitor B and the battery remains connected. Select correct statement(s):
- (A) The charge supplied by the battery increases.
(B) The capacitance of the system decreases.
(C) The electric field in the capacitor B increases.
(D) The electrostatic potential energy decreases.

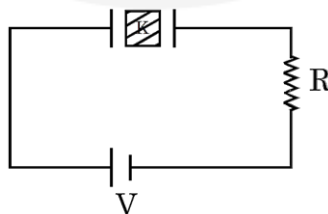


6. Two bodies A & B (with radius $R_A = 3R_B$) have maximum intensity of radiation at wavelengths $9000 \text{ \AA}$ & $3000 \text{ \AA}$ at temperature T_A & T_B , respectively. Find the ratio of rate of emission per sec of A to B:
- (A) $1/9$ (B) $9/1$ (C) $1/27$ (D) $1/3$

7. According to Wien's displacement law:



- (A) total area under the three graphs A_1, A_2, A_3 are equal
(B) areas are in the ratio: $A_1 : A_2 : A_3 = T_1^2 : T_2^2 : T_3^2$
(C) areas are in the ratio: $A_1 : A_2 : A_3 = T_1^4 : T_2^4 : T_3^4$
(D) $T_3 > T_2 > T_1$
- 8.* Consider the circuit shown in diagram. Capacitor is fully charged. Now dielectric slab is suddenly removed by an external agent. Equilibrium is again reached. During this time :



- (A) Work is done on battery.
(B) Heat is produced in resistance
(C) Work is done by battery
(D) Loss in energy of capacitor is equal to energy given to battery and heat produced in resistance R .
- 9.* Two spheres of same material are having surfaces blackened and placed in space separately. Their radii are R and $2R$ respectively and the most dominating wavelengths in their spectrum are observed to be in the ratio $1:2$. The ratio of their
- (A) Temperatures is $2:1$ (B) Emissive powers is $4:1$
(C) Rates of heat loss is $4:1$ (D) Rates of cooling is $32:1$



10. In column I, some operations are performed on capacitor kept either at constant potential or constant charge. While in column II are given one or more than one probable effects of corresponding operation performed. Match the entries of column I with the entries of column II.

Column I

- (A) A dielectric slab is inserted into the capacitor slowly, keeping the charge constant.
- (B) A dielectric slab is inserted into the capacitor slowly, keeping the voltage constant.
- (C) The plates of capacitor are moved apart keeping the charge constant.
- (D) The plates of capacitor are moved apart keeping the voltage constant.

Column II

- (P) Work done by the external agent is positive.
- (Q) Work done by external agent is negative.
- (R) Electric potential energy stored in the electric field in between capacitor plates is increasing
- (S) Electric potential energy stored in the electric field in between capacitor plates is decreasing
- (T) Charge on capacitor increases



Note: (*) → Multiple Correct Type Question

Total Marks: 29

Max. Time: 24 min.

Single Correct Questions ('-1' negative marking) Q.1

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.2 to Q.3

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.4 to Q.8

(2 marks, 2 min.)

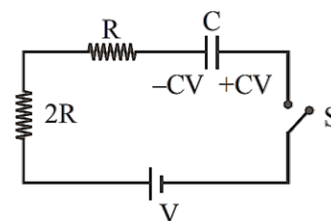
Subjective Type Questions (No negative marking) Q.9 to Q.10

(4 marks, 3 min.)

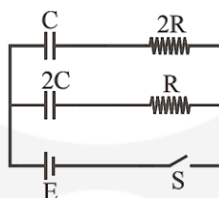
(There is partial marking in multiple correct questions)

1. A charged capacitor is connected in the circuit as shown and switch S is closed at $t = 0$, so heat loss through $2R$ will be :

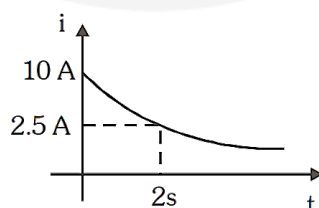
- (A) $2CV^2$ (B) $\frac{4CV^2}{3}$
(C) $\frac{2CV^2}{3}$ (D) $\frac{CV^2}{3}$



- 2.* In the circuit shown in the figure, switch S is closed at time $t = 0$. Select the correct statement(s):



- (A) Rate of increase of charge is same in both the capacitor.
(B) Ratio of charge stored in capacitor C and 2C at any time t would be 1 : 2.
(C) Time constant of both the capacitors are equal.
(D) Steady state charge in capacitors C and 2C are in the ratio 1 : 2.
- 3.* The figure shows a graph of the current in a discharging circuit of a capacitor through a resistor of resistance of 10Ω .

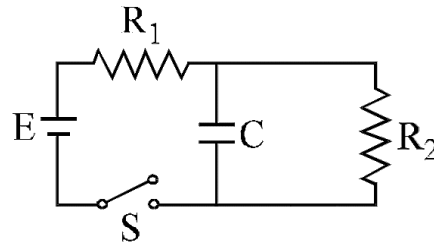


- (A) The initial potential difference across the capacitor is 100 volt.
(B) The capacitance of the capacitor is $\frac{1}{10\ln 2}$ F.
(C) The total heat produced in the circuit will be $\frac{500}{\ln 2}$ J
(D) The thermal power in the resistor will decrease with a time constant $\frac{1}{2\ln 2}$ second



Paragraph for Question No. 4 to 5

In the circuit as shown in figure the switch is closed at $t = 0$

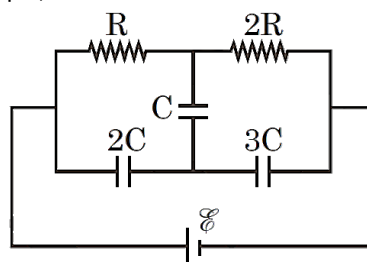


- 4.*** At the instant of closing the switch:
- (A) the battery delivers maximum current.
 - (B) no current flows through C
 - (C) Voltage drop across R_2 is zero.
 - (D) the current through the battery decreases with time and finally becomes zero.
- 5.*** A long time after closing the switch:
- (A) voltage drop across the capacitor is E.
 - (B) current through the battery is $\frac{E}{R_1 + R_2}$
 - (C) energy stored in the capacitor is $\frac{1}{2} C \left(\frac{R_2 E}{R_1 + R_2} \right)^2$
 - (D) current through the capacitor becomes zero.

Paragraph for Question 6 to 8

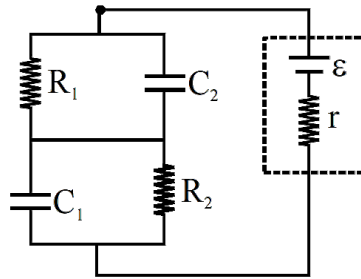
A metal block is placed in a room which is at 10°C for long time. Now it is heated by an electric heater of power 500 W till its temperature becomes 50°C . Its initial rate of rise of temperature is 2.5°C/sec . The heater is switched off and now a heater of 100 W is required to maintain the temperature of the block at 50°C . (Assume Newtons Law of cooling to be valid)

- 6.** What is the heat capacity of the block?
- (A) 50 J/ $^\circ\text{C}$ (B) 100 J/ $^\circ\text{C}$ (C) 150 J/ $^\circ\text{C}$ (D) 200 J/ $^\circ\text{C}$
- 7.** What is the rate of cooling of block at 50°C if the 100W heater is also switched off?
- (A) 5°C/s (B) 0.5°C/s (C) 1°C/s (D) 0.1°C/s
- 8.** What is the heat radiated per second when the block was 30°C ?
- (A) 100 W (B) 80 W (C) 50 W (D) 30 W
- 9.** Find the charge on the capacitor C in the figure shown. Internal resistance of a source is to be neglected. (Take : $E = 9 \text{ V}$, $C = 2\mu\text{F}$)





10. In the given circuit, $R_1 = 2\Omega$, $R_2 = 3\Omega$, $r = 1\Omega$, $\varepsilon = 6V$, $C_1 = 1\mu F$ and $C_2 = 2\mu F$. In steady state, the ratio of energy stored in the capacitors C_2 and C_1 is $\frac{x}{9}$. Then x will be.



Note: (*) → Multiple Correct Type Question

Total Marks: 38

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.7 to Q.8

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9

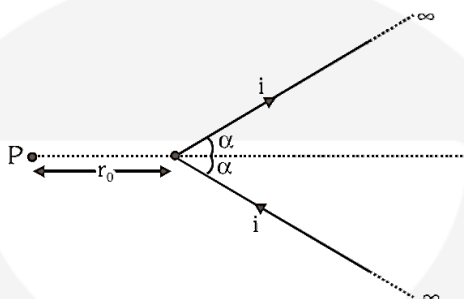
(4 marks, 3 min.)

Match the Column Type Questions (No negative marking) Q.10

(8 marks, 4 min.)

(There is partial marking in multiple correct questions)

1. A current carrying wire in the form of 'V' alphabet is kept as shown in the figure. Magnetic field intensity at point P which lies on the angular bisector of V is:



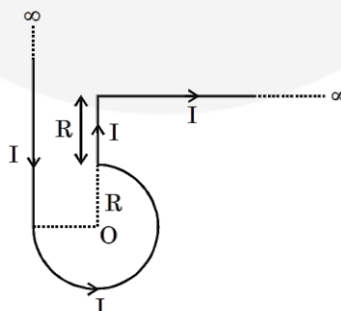
(A) $\frac{\mu_0 i}{4\pi r_0} [1 - \cos \alpha]$

(B) $\frac{\mu_0 i}{2\pi r_0} [1 - \cos \alpha]$

(C) $\frac{\mu_0 i}{4\pi r_0} \frac{[1 - \cos \alpha]}{\sin \alpha}$

(D) $\frac{\mu_0 i}{2\pi r_0} \frac{[1 - \cos \alpha]}{\sin \alpha}$

2. A long conducting wire, carrying a current I is bent into the shape as shown. The net magnetic field at the centre 'O' of the circular arc of radius 'R' is :



(A) $\frac{\mu_0 I}{2R} \left(\frac{1 + 3\pi}{\pi} \right)$

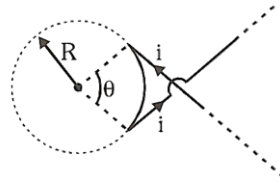
(B) $\frac{\mu_0 I}{4R} \left(\frac{2 + 3\pi}{\pi} \right)$

(C) $\frac{\mu_0 I}{8R} \left(\frac{1 + 3\pi}{\pi} \right)$

(D) $\frac{\mu_0 I}{8R} \left(\frac{2 + 3\pi}{\pi} \right)$

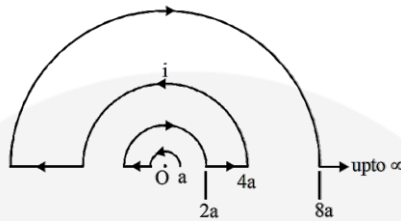


3. A current carrying wire has the configuration as shown in the figure. Two semi-infinite straight sections, each tangent to the same circle, are connected by a circular arc, of angle θ along the circumference of the circle, with all sections lying in the same plane. What must be θ in order for B (magnetic field) to be zero at the centre of the circle?



- (A) $\theta = 1$ rad (B) $\theta = 2$ rad (C) $\theta = 3$ rad (D) $\theta = 3$ rad

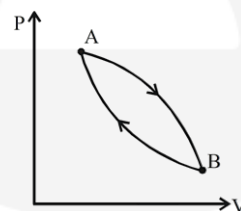
4. A conductor carrying current 'i' is bent in the form of concentric semicircles as shown in the figure. The magnetic field at the centre O is:



- (A) zero (B) $\frac{\mu_0 i}{6a}$ (C) $\frac{\mu_0 i}{a}$ (D) $\frac{\mu_0 i \ln 2}{4a}$

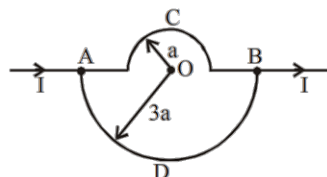
5. A mixture of gas contains 6 moles of oxygen gas and 4 moles of He gas at temperature T (kelvin). The internal energy of the system is :
- (A) $15 RT$ (B) $21 RT$ (C) $25 RT$ (D) $13 RT$

6. Figure shows a cyclic process. Heat absorbed in process $A \rightarrow B$ is 100 J. $B \rightarrow A$ is adiabatic compression. Work done on the gas in $B \rightarrow A$ is 70 J. Find change in internal energy in process $A \rightarrow B$:



- (A) $-30J$ (B) $100J$ (C) $30J$ (D) $-70J$

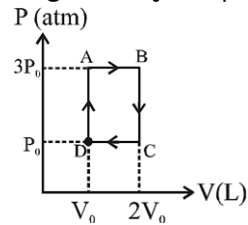
- 7*. In the given diagram, there are two semicircular parts one having radius 'a' and another having radius '3a' as shown. Resistance of ACB part is R and resistance of ADB part is $3R$. Select the correct option(s):



- (A) Current through ACB is $I/4$
 (B) Current through ADB is $I/4$
 (C) Magnitude of $\vec{B}$ at 'O' due to ACB part is $\frac{3\mu_0 I}{16a}$
 (D) Magnitude $\vec{B}$ at 'O' due to ADB part is $\frac{\mu_0 I}{48a}$



- 8*. One mole of an ideal gas is taken through the cyclic process as shown below.

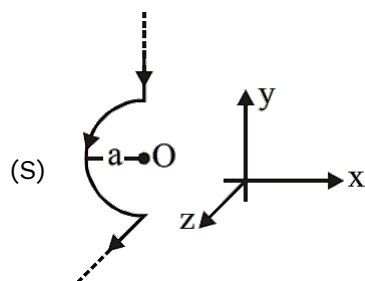
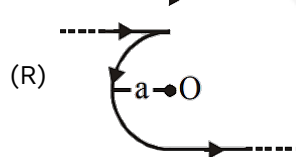
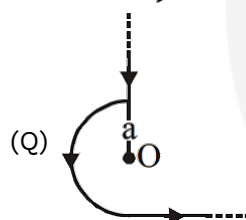
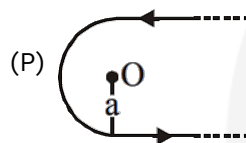


Which of the following statement(s) is/are CORRECT ?

- (A) net work done by the gas is $2P_0V_0$ (B) net work done by the gas is P_0V_0
 (C) temperature at B is $\frac{6P_0V_0}{R}$ (D) temperature at C is $\frac{P_0V_0}{R}$
9. Find the magnetic field (in μT) at point A. (Given : $i = \frac{20}{\pi}$ Amp, $R = 1\text{m}$)



10. **List-I**
Current Carrying Wires



- List-II**
 B_0

(1) $\frac{\mu_0 i}{4a} \left(1 + \frac{1}{\pi} \right)$

(2) $\frac{\mu_0 i}{4a} \sqrt{1 + \frac{1}{\pi^2}}$

(3) $\frac{\mu_0 i}{2a} \left(\frac{1}{2} + \frac{1}{\pi} \right)$

(4) $\frac{\mu_0 i}{4a}$

Note: (*) → Multiple Correct Type Question

Total Marks: 40

Max. Time: 27 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.4

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.5 to Q.8

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9

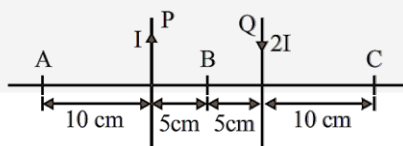
(4 marks, 3 min.)

Match the Column Type Questions (No negative marking) Q.10

(8 marks, 4 min.)

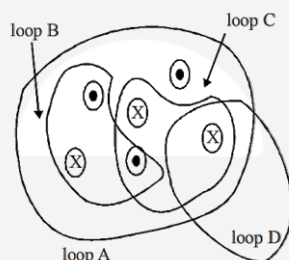
(There is partial marking in multiple correct questions)

1. Two infinitely long wires P and Q separated by a distance of 10 cm in air, carry current I and $2I$ in opposite directions as shown in figure. The resultant magnetic field due to current in these wires will be zero at:



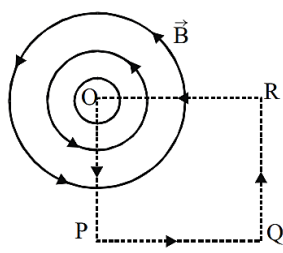
- (A) Point A (B) Point B (C) Point C (D) None of the point A, B, C

2. Consider six wires coming into or out of the page, all with the same current. Rank the line integral of the magnetic field (from most positive to most negative). Take counter clockwise as positive around each loop shown.



- (A) $B > C > D > A$ (B) $B > C = D > A$ (C) $B > A > C = D$ (D) $C > B = D > A$

3. A current carrying wire (current = i) perpendicular to the plane of the paper produces a magnetic field, as shown in the figure. A square of side a is drawn with one of its vertices on the centre of the wire. The integral $\int \vec{B} \cdot d\vec{r}$ along OPQRO has the value:



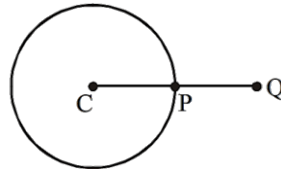
- (A) $\mu_0 i$ (B) $\frac{\mu_0 i}{8}$ (C) $\frac{\mu_0 i}{4}$ (D) $\frac{\mu_0 i}{2}$



4. An ideal gas of adiabatic exponent ' γ ' is supplied heat at a constant pressure. Then ratio $dQ : dW : dU$ is equal to :

(A) $\gamma : \gamma - 1 : \frac{1}{\gamma}$ (B) $\gamma : \gamma - 1 : 1$ (C) $1 : 1 : \gamma - 1$ (D) $\gamma : 1 : \gamma - 1$

- 5.* An electron is at P at $t = 0$. It is circulating in anti clockwise direction with a constant angular speed ω along the shown circular path. Magnetic field at Q ($CQ = 2R$, where R is radius of the circle) will be recorded as zero at times.

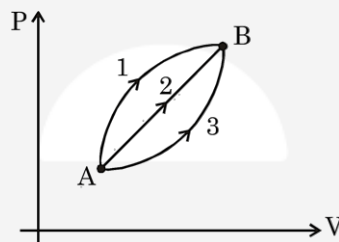


(A) $\frac{\pi}{3\omega}$ (B) $\frac{\pi}{\omega}$ (C) $\frac{5\pi}{3\omega}$ (D) $\frac{7\pi}{3\omega}$

- 6.* Two straight infinitely long and thin parallel wires are spaced 0.1 m apart and carry a current of 10 ampere each. Find the magnetic field at a point which is at a distance of 0.1 m from both wires in the two cases when the currents are in the (a) same and (b) opposite directions.

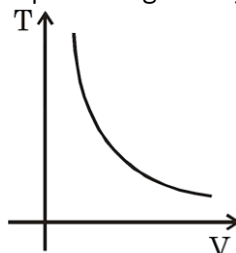
(A) When currents are in same direction, $B = 2\sqrt{3} \times 10^{-5} \text{ T}$
(B) When current are in opposite direction $B = 2 \times 10^{-5} \text{ T}$
(C) When currents are in same direction, $B = 2 \times 10^{-5} \text{ T}$
(D) When currents are in opposite direction $B = 2\sqrt{3} \times 10^{-5} \text{ T}$

- 7.* An ideal gas is taken from state A to B along three different paths as shown in figure. Select the CORRECT statement(s):



(A) Change in internal energy is same in all three paths.
(B) Temperature of gas is continuously increasing in all three paths.
(C) Heat absorbed is maximum in path-1.
(D) Work done by the gas in all paths is same.

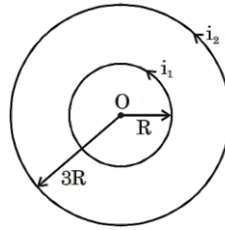
- 8.* An ideal monoatomic gas undergoes a process given by a rectangular hyperbola on T - V graph.



(A) Bulk modulus for this process will be $2P$ (where P is pressure).
(B) Molar specific heat for the process is $\frac{R}{2}$.
(C) Bulk modulus for this process will be $\frac{P}{2}$ (where P is pressure).
(D) Molar specific heat for the process is R .



9. Diagram shows two concentric current carrying loop of radius R & $3R$ carrying current i_1 & i_2 respectively as shown in the diagram. It was found that magnitude of magnetic field at the centre of loops is 3 times the field due to inner loop alone. The ratio $\left(\frac{i_2}{i_1}\right)$ is:



10. Match the following (symbols have usual meaning)

List-I

(P) $\frac{\oint \vec{B} \cdot d\vec{\ell}}{\mu_0}$

(Q) $\oint \vec{B} \cdot d\vec{s}$

(R) $\frac{\mu_0}{4\pi} \frac{id\vec{\ell} \times \vec{r}}{|\vec{r}|^3}$

(S) $\frac{\mu_0 i}{2R}$

List-II

(1) Zero

(2) Magnetic field due to current loop at its centre

(3) Current enclosed

(4) Magnetic field of small current element

Note: (*) → Multiple Correct Type Question

Total Marks: 37

Max. Time: 28 min.

Single Correct Questions ('-1' negative marking) Q.1

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.2 to Q.6

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.7 to Q.8

(2 marks, 2 min.)

Subjective Type Questions (No negative marking) Q.9

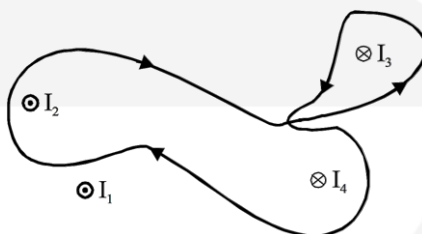
(4 marks, 3 min.)

Match the Column Type Questions (No negative marking) Q.10

(8 marks, 4 min.)

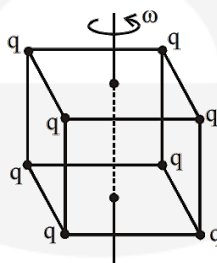
(There is partial marking in multiple correct questions)

1. Four wires carrying current $I_1 = 2A$, $I_2 = 4A$, $I_3 = 6A$ and $I_4 = 8A$ respectively cut the page perpendicularly as shown in figure. The value of $\int \vec{B} \cdot d\vec{\ell}$ for the loop shown would be :



- (A) $+2\mu_0 \text{ wb/m}$ (B) $-2\mu_0 \text{ wb/m}$ (C) $+10\mu_0 \text{ wb/m}$ (D) $-10\mu_0 \text{ wb/m}$

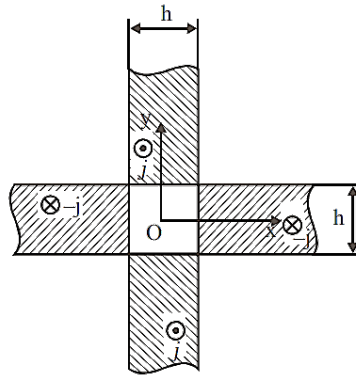
- 2.* Consider a cube of side 'a' as shown. Eight point charges are placed at the corners. The cube is rotated about the axis with constant angular velocity ' ω ':



- (A) Net magnetic field at the centre of cube is zero
 (B) Net magnetic field at the centre of cube is $\frac{\sqrt{2}\mu_0 q \omega}{\pi a}$
 (C) Net magnetic field at the centre of cube is $\frac{8}{3\sqrt{3}} \frac{\mu_0 q \omega}{\pi a}$
 (D) If polarity of any four charges are reversed, then magnetic field at the centre of cube will be zero.

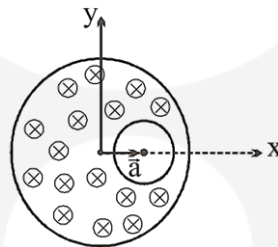


- 3.*** Figure shows two infinite sheets of thickness h each, carrying current of density (current/area) ' J '. The direction of current is indicated in the diagram. Point 'O' is at the centre of cuboid formed due to intersection.



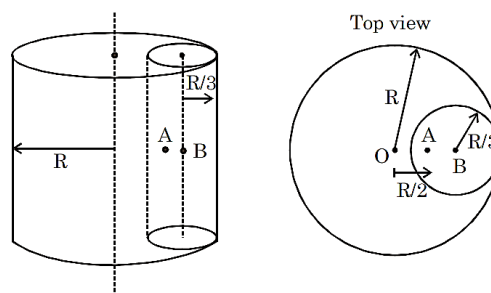
- (A) The magnitude of magnetic field at point O is $\sqrt{2}\mu_0 Jh$.
 (B) The magnitude of magnetic field at point O is zero.
 (C) The magnitude of magnetic field at any finite distance outside the sheet is zero.
 (D) The magnitude of magnetic field at any finite distance outside the sheet is $\frac{\mu_0 Jh}{\sqrt{2}}$.

- 4.*** Figure shows cross-section view of a infinite cylindrical wire with a cavity. Current density is uniform $\vec{j} = -j_0 \hat{k}$ as shown in figure.



- (A) Magnetic field inside cavity is uniform.
 (B) Magnetic field inside cavity is along $\vec{a}$.
 (C) magnetic field inside cavity is perpendicular to $\vec{a}$.
 (D) If an electron is projected with velocity $v_0 \hat{j}$ inside the cavity it will move undeviated.

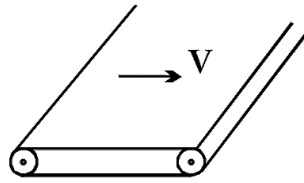
- 5.*** From a cylinder of radius R , a cylinder of radius $R/3$ is removed as shown. Current flowing in the remaining cylinder is I . The distance OA is equal to $\frac{R}{2}$. Magnetic field strength at:



- (A) Point B is $\frac{3\mu_0 I}{8\pi R}$ (B) Point B is $\frac{\mu_0 I}{8\pi R}$ (C) Point A is $\frac{3\mu_0 I}{8\pi R}$ (D) Point A is $\frac{\mu_0 I}{8\pi R}$



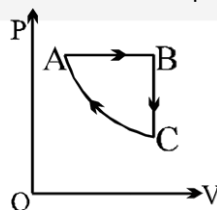
- 6.* Charge is sprayed onto a large non conducting belt above the left hand roller. The belt carries charge with a uniform surface charge density σ , as it moves with a speed v between the rollers as shown. The charge is removed by a wiper at right hand roller. For a point just above the sheet, mark the correct option(s).



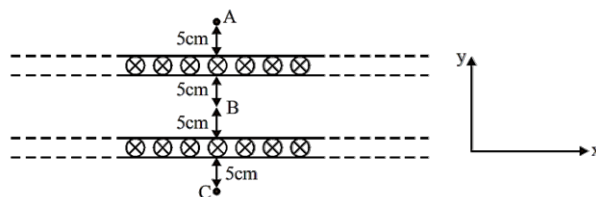
- (A) magnetic field is $\frac{\mu_0 \sigma v}{2}$, out of the plane of the page, parallel to axis of roller.
 (B) magnetic field is $\mu_0 \sigma$, out the of the plane of the page, perpendicular to axis.
 (C) Electric field is $\frac{\epsilon_0 \sigma}{2}$ perpendicular to the plane of sheet.
 (D) If an electron moves parallel to V just above the sheet it will experience an upward magnetic force.

Paragraph for Q. 7 to Q. 8

The figure represents a cyclic process in which the process CA is adiabatic



7. Choose the INCORRECT statement:
 (A) Positive work is done during the isobaric process.
 (B) Negative work is done during the adiabatic process.
 (C) The system cools down during the process BC.
 (D) The system rejects heat during the process CA.
8. If T_A , T_B and T_C are the temperatures at A, B and C respectively, then identify the correct statement:
 (A) $T_A < T_B < T_C$ (B) $T_A = T_C$ (C) $T_A > T_C$ (D) $T_A > T_B > T_C$
9. A diatomic gas ($\gamma = 1.4$) does 2000 J of work when it is expanded isobarically. Find the heat given to the gas in the above process (in kJ).
10. Figure shows cross-section of two large parallel metal sheets carrying electric currents along their surfaces. The current/length in each sheet is $\frac{10}{\pi}$ A/m along the width. Consider three points A, B and C, as shown in the figure with their positions.



List-I

- (P) Towards right
 (Q) Towards left
 (R) Zero
 (S) Towards $\perp$ to plane of paper

List-II

- (1) Magnetic field at A
 (2) Magnetic field at B
 (3) Magnetic field at C
 (4) Magnetic field at no point

Note: (*) → Multiple Correct Type Question

Total Marks: 35

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

Multiple Correct Questions ('-2' negative marking) Q.6 to Q.8

Subjective Type Questions (No negative marking) Q.9 to Q.10

(There is partial marking in multiple correct questions)

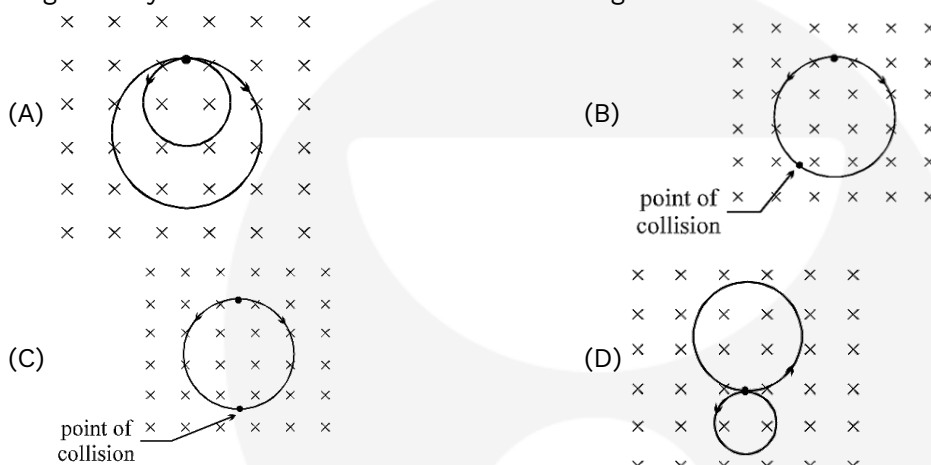
Max. Time: 25 min.

(3 marks, 2 min.)

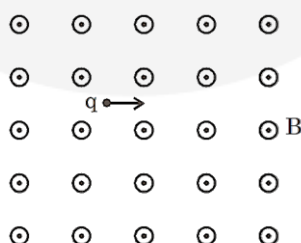
(4 marks, 3 min.)

(4 marks, 3 min.)

1. A neutral particle at rest in a magnetic field decays into two charged particles of different mass. The energy released goes into their kinetic energy. Then what can be the path of the two particles? Neglect any interaction between the two charges.



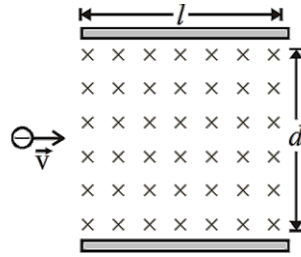
2. A charge 'q' of mass 'm' is accelerated by a potential difference 'V' and then enters a uniform magnetic field 'B' perpendicular to the velocity as shown. Choose the CORRECT option(s) :



- (A) Radius of the path followed by the particle will be $\sqrt{\frac{2mV}{qB^2}}$
- (B) If charge on the particle is doubled, then product of time period & speed remains unchanged.
- (C) If magnetic field is suddenly switched off, momentum of the particle will be $\sqrt{\frac{mqV}{2}}$.
- (D) Radius of the path followed after magnetic field is switched off will be $\sqrt{\frac{mV}{2qB^2}}$.



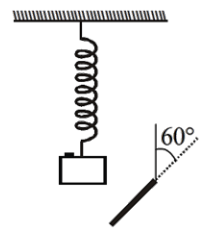
3. Particle of charge $-e$ is moving with an initial velocity $\vec{v}$ when it enters midway between two plates where exists a uniform magnetic field pointing into the page, as shown in the figure. You may ignore effects of the gravitational force. The magnitude of the velocity of the particle if it just strikes the end of the plate:



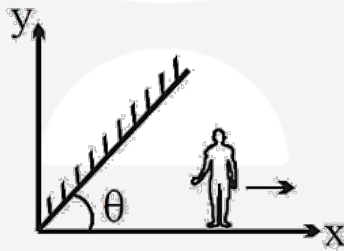
- (A) $\frac{eB}{m} \left(\frac{d}{2} + \frac{l^2}{d} \right)$ (B) $\frac{eB}{m} \left(\frac{d}{4} + \frac{l^2}{d} \right)$ (C) $\frac{eB}{m} \left(\frac{d}{4} + \frac{l^2}{2d} \right)$ (D) $\frac{eB}{m} \left(d + \frac{l^2}{d} \right)$

4. An insect of negligible mass is sitting on a block of mass M , tied with a spring of force constant k . The block performs simple harmonic motion with amplitude A in front of a plane mirror placed as shown. The maximum speed of insect relative to its image will be.

- (A) $A\sqrt{\frac{k}{M}}$ (B) $\frac{A\sqrt{3}}{2} \sqrt{\frac{k}{M}}$
(C) $A\sqrt{3} \sqrt{\frac{k}{M}}$ (D) $\frac{A}{2} \frac{k}{2\sqrt{M}}$



5. A man is walking under an inclined mirror at a constant velocity V m/s along the X axis. If the mirror is inclined at an angle θ with the horizontal, then what is the velocity of the image?



- (A) $V \sin \theta \hat{i} + V \cos \theta \hat{j}$ (B) $V \cos \theta \hat{i} + V \sin \theta \hat{j}$
(C) $V \sin 2\theta \hat{i} + V \cos 2\theta \hat{j}$ (D) $V \cos 2\theta \hat{i} + V \sin 2\theta \hat{j}$

- 6.* A particle of mass m and charge q moving with velocity $\vec{v}$ enters a region of uniform magnetic field of intensity $\vec{B}$. Then :

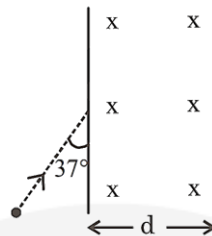
- (A) its path in the region of the field is always circular.
(B) its path in the region of the field is circular if $\vec{v} \cdot \vec{B} = 0$
(C) its path in the region of the field is a straight line if $\vec{v} \times \vec{B} = 0$
(D) distance travelled by the particle in time T does not depend on the angle between $\vec{v}$ and $\vec{B}$.

- 7.* A proton is fired from origin with velocity $\vec{v} = v_0(\hat{j} + \hat{k})$ in a uniform magnetic field $\vec{B} = B_0\hat{j}$. In the subsequent motion of the proton

- (A) the path of the proton is a helix
(B) the x -coordinate can never be zero
(C) the x and z -components will becomes zero simultaneously after every pitch
(D) the y -coordinate will be proportional to its time of flight



- 8.*** Two mirrors are inclined at an angle of 30° . A ray is incident parallel to the one of the mirror. Then select the correct statement(s):
(A) Ray will undergo five reflections from the system.
(B) Total deviation from the system will be 120°
(C) Ray will retrace its path finally.
(D) Total deviation suffered by the ray will be 90°
- 9.** A particle of mass 1 mg and charge $1\mu\text{C}$ is projected with speed 50 m/s towards a region containing a uniform magnetic field 10 T . What should be the minimum thickness (d) of this region (in m) so that the particle returns back to left side.



- 10.** Magnetic field in space is given as $\vec{B} = \sqrt{3}B_0\hat{i} + B_0\hat{j}$. A charged particle is projected along the line $y = \sqrt{3}x$ with velocity 2 m/s . Find the distance (in m) travelled by particle in 3 s :

Note: (*) → Multiple Correct Type Question

Total Marks: 26

Single Correct Questions ('-1' negative marking) Q.1 to Q.4

Multiple Correct Questions ('-2' negative marking) Q.5

Comprehension Type Questions ('-1' Negative marking) Q.6 to Q.10

(There is partial marking in multiple correct questions)

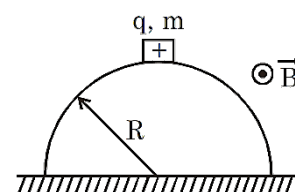
Max. Time: 21 min.

(3 marks, 2 min.)

(4 marks, 3 min.)

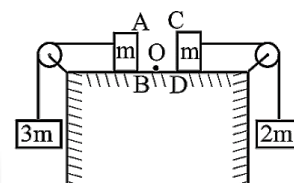
(2 marks, 2 min.)

1. A block of mass 'm' has charge 'q'. It is placed on the top of a hemisphere in external magnetic field 'B'. Find the minimum speed that should be given to the block to the left, so that it leaves the contact from the hemisphere immediately. Given $\frac{qB}{m} = 3 \text{ rad/sec}$ and $R = 1 \text{ m}$.



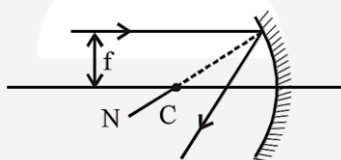
- (A) 1 m/s (B) 2 m/s (C) 3 m/s (D) 5 m/s

2. Two blocks each of mass m lie on a smooth table. They are attached to two other masses as shown in the figure. The pulleys and strings are light. An object O is kept at rest on the table. The sides AB & CD of the two blocks are made reflecting. The acceleration of two images formed in those two reflecting surfaces w.r.t. each other is:



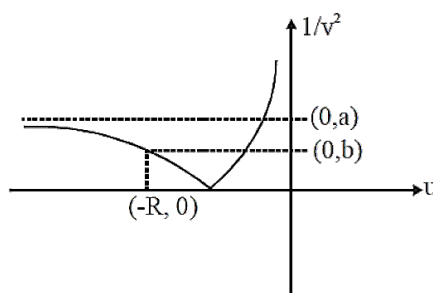
- (A) $5g/6$ (B) $5g/3$ (C) $g/3$ (D) $17g/6$

3. A light ray is coming parallel to principal axis, the distance between ray and axis is equal to focal length (as shown). Find the angle of deviation after reflection:



- (A) 60° (B) 90° (C) 120° (D) Small deviation

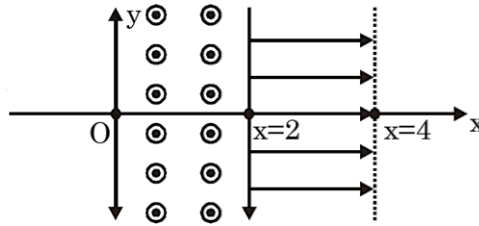
4. For a concave mirror, graph of $\frac{1}{v^2}$ and u for real object is drawn. What is the value of $\frac{b}{a}$



- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{8}$



- 5.* There exist uniform magnetic field $\vec{B} = 5T(\hat{k})$ from $x = 0$ to $x = 2m$. From $x = 2m$ to $x = 4m$ a uniform electric field $\vec{E} = 50N/C(\hat{i})$. A point charge ($q = 2C$, $m = 1kg$) is projected from origin with velocity $\vec{V} = 20 m/s(\hat{j})$. Which of the following is/are correct :



- (A) Time spend by the charge particle in magnetic field is $\frac{\pi}{20} s$
 (B) Time spend by the charge particle in electric field is $\frac{1}{5} s$
 (C) Charge particle will leave the electric field with speed $20\sqrt{2} m/s$
 (D) Charge particle will leave the electric field with velocity $20\hat{i} + 20\hat{j}$

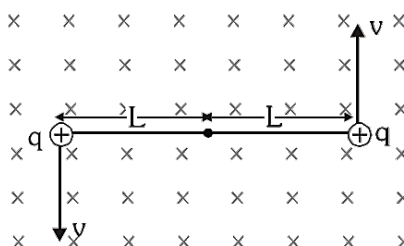
Paragraph for Question no. 6 and 7

Electric field and magnetic field in region of space are given by $\vec{E} = E_0\hat{j}$ and $\vec{B} = B_0\hat{j}$. A particle of specific charge α (charge per unit mass) is released from origin with velocity $\vec{V} = V_0\hat{i}$.

6. y coordinate of particle when it crosses y-axis for n^{th} time is :
 (A) $\frac{2n^2\pi^2E_0}{B_0^2\alpha}$ (B) $\frac{n^2\pi E_0}{B_0^2\alpha}$ (C) $\frac{2\pi^2E_0}{n^2B_0\alpha}$ (D) 0
7. Angle between particle's velocity and y-axis at that moment is :
 (A) $\tan^{-1}\left(\frac{V_0B_0}{2\pi E_0}\right)$ (B) $\tan^{-1}\left(\frac{V_0B_0}{2\pi nE_0}\right)$ (C) $\tan^{-1}\left(\frac{V_0B_0}{E_0}\right)$ (D) $\tan^{-1}\left(\frac{B_0}{E_0}\right)$

Paragraph for question no. 8 and 10

Two charge particles each of mass ' m ', carrying charge $+q$ and connected with each other by a massless inextensible string of length $2L$ are describing circular path in the plane of paper, each with speed $v = \frac{qB_0L}{m}$ (where B_0 is constant) about their centre of mass in the region in which an uniform magnetic field $\vec{B}$ exists into the plane of paper as shown in figure. Neglect any effect of electrical & gravitational forces.



8. The magnitude of the magnetic field such that no tension is developed in the string will be :
 (A) $\frac{B_0}{2}$ (B) B_0 (C) $2B_0$ (D) 0



9. If the actual magnitude of magnetic field is half to that of calculated in part (i) then tension in the string will be :
- (A) $\frac{3 q^2 B_0^2 L}{4 m}$ (B) zero (C) $\frac{q^2 B_0^2 L}{2m}$ (D) $\frac{2q^2 B_0^2 L}{m}$
10. Given that the string breaks when the tension is $T = \frac{3 q^2 B_0^2 L}{4 m}$. Now if the magnetic field is reduced to such a value that the string just breaks then find the maximum separation between the two particles during their motion.
- (A) 16 L (B) 4 L (C) 14 L (D) 2L



Note: (*) → Multiple Correct Type Question

Total Marks: 35

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.6 to Q.8

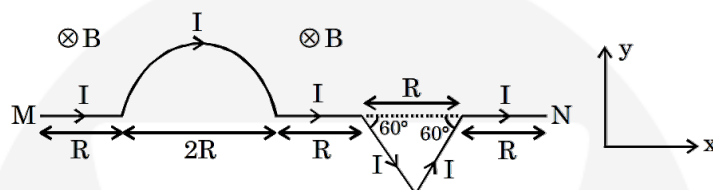
(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9 to Q.10

(4 marks, 3 min.)

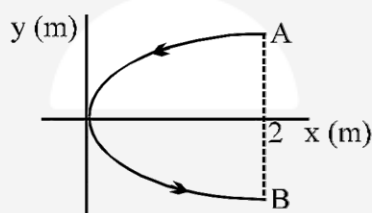
(There is partial marking in multiple correct questions)

1. A conducting wire MN carrying a current I is bent into the shape as shown and placed in xy plane. A uniform magnetic field $\vec{B} = -B\hat{k}$ is existing in the region. The net magnetic force experienced by the conducting wire MN is:



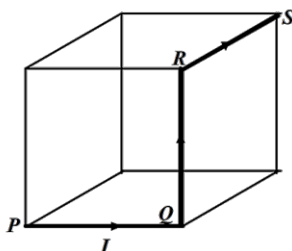
- (A) $2BIR$ (B) $3BIR$ (C) $4BIR$ (D) $6BIR$

2. A conducting wire bent in the form of a parabola $y^2 = 2x$ carries a current $i = 2$ A as shown in figure. This wire is placed in a uniform magnetic field $\vec{B} = -4\hat{k}$ Tesla. The magnetic force (in newton) on the wire is:



- (A) $-16\hat{i}$ (B) $32\hat{i}$ (C) $-32\hat{i}$ (D) $16\hat{i}$

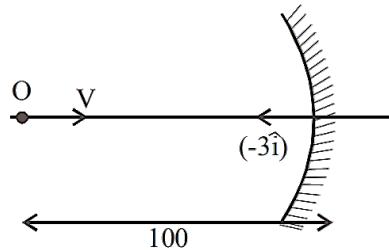
3. A wire PQRS carrying a current I runs along three edges of a cube of side l as shown. There exists a uniform magnetic field of magnitude B along one of the sides of the cube. The magnitude of the force acting on the wire is:



- (A) 0 (B) IlB (C) $\sqrt{2} IlB$ (D) $2 IlB$

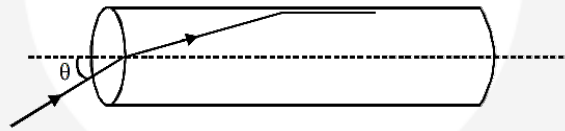


4. Radius of curvature of a concave mirror varies as $R = \left| \frac{6}{t^2 - 5t + 6} \right|$ cm. Initially the object was at a distance 100 cm from the pole as shown. At $t = 0$ the object started moving with velocity $v = 2t\hat{i}$ (cm/s). Find the speed of image at time $t = 3$ sec. Mirror is moving with constant velocity $(-3\hat{i})$ (cm/s).



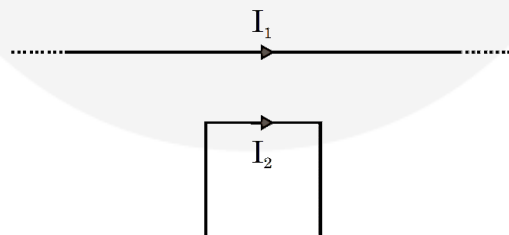
- (A) 12 cm/s (B) 20 cm/s (C) 5 cm/s (D) 24 cm/s

5. A transparent solid cylindrical rod has a refractive index of $\frac{4}{\sqrt{3}}$. It is surrounded by a medium of refractive index 2. A light ray is incident at the mid-point of one end of the rod as shown in the figure. The incident angle θ for which the light ray grazes along the wall of the rod is :-



- (A) $\sin^{-1}\left(\frac{1}{2}\right)$ (B) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (C) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (D) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

- 6*. A long straight wire carries a steady current I_1 . Nearby is a rectangular loop that carries a steady current I_2 . The directions of the two currents are shown in the figure. Which statement(s) is/are FALSE.



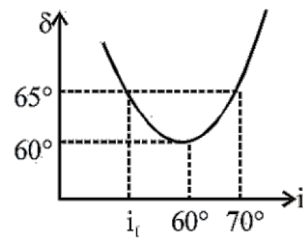
- (A) The loop experienced zero torque.
(B) There is no net force on the loop from the wire.
(C) The loop is attracted to the wire if $I_1 > I_2$; otherwise it is repelled
(D) The loop is repelled from the wire if $I_1 > I_2$; otherwise it is attracted

- 7*. A concave spherical mirror has radius of curvature of 50 cm. Find two positions of an object for which the image is four times as large as the object.

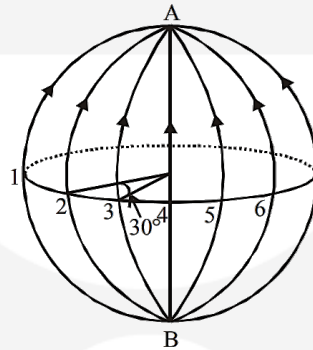
- (A) $\frac{75}{4}$ cm from mirror (B) 125 cm from mirror
(C) $\frac{125}{4}$ cm from mirror (D) 32 cm from mirror



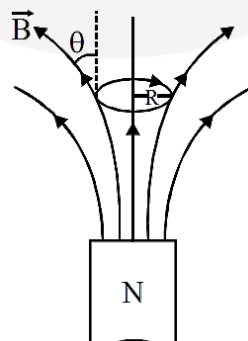
- 8*. The angle of deviation (δ) vs angle of incidence (i) is plotted for a prism. Choose the CORRECT statement(s).



- (A) The angle of prism is 60°
(B) The refractive index of the prism is $n = \sqrt{3}$
(C) For deviation to be 65° the angle of incidence $i_1 = 55^\circ$
(D) For minimum deviation, the angle of emergence $e = 60^\circ$.
9. Figure shows 12 current carrying wires wound around the sphere of radius R . Each wire carries the same current ' I ' and is separated by an angle of 30° . None of the wires are touching each other and current in all the wires is moving up as shown in figure. A constant magnetic field B exists horizontally through out the region of space. The net force on the wires is:



10. A superconducting current carrying ring of mass $m = 40 \text{ gm}$ and $R = 2 \text{ cm}$ floats in the magnetic field of a magnet. If the field lines make an angle $\theta = 30^\circ$ with vertical and the ring is placed coaxially, assuming $I = 10 \text{ A}$, find the magnetic field. If answer is $\frac{n}{3\pi}$. Write the value of n .



Note: (*) → Multiple Correct Type Question

Total Marks: 32

Max. Time: 31 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.2

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.3 to Q.5

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.6 to Q.8

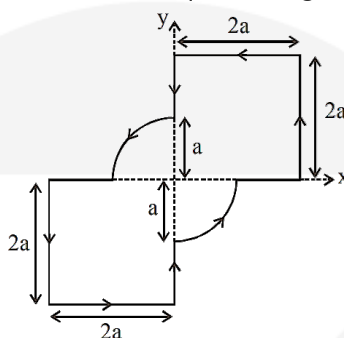
(2 marks, 2 min.)

Subjective Type Questions (No negative marking) Q.9 to 10

(4 marks, 3 min.)

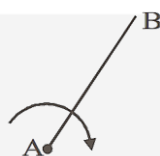
(There is partial marking in multiple correct questions)

1. A current 'I' flows through a thin wire as shown in the figure. If there exists an external magnetic field B in the same plane of the wire. Then torque acting on the coil is :



- (A) $I \left(\frac{\pi a^2}{2} + 8a^2 \right) B$ (B) $I \left(\frac{\pi a^2}{2} + 4a^2 \right) B$ (C) $I (\pi a^2 + 8a^2) B$ (D) 0

2. A non conducting rod AB of length 'l' has a positive linear charge density ' λ '. The rod is rotated about point 'A' with an angular velocity ' ω ' in plane of paper. What is the magnetic moment of rod?



- (A) $\frac{\lambda \omega l^2}{2}$ (B) $\frac{2\lambda \omega l^2}{3}$ (C) $\frac{3\lambda \omega l^3}{2}$ (D) $\frac{\lambda \omega l^3}{6}$

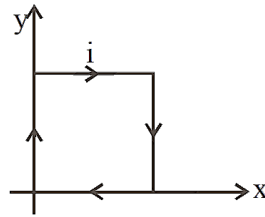
- 3.* A disk of mass m and radius R has a charge Q uniformly distributed on its surface. The disc is rotated about an axis passing through the centre of disc and perpendicular to the plane of the disc.

- (A) The magnetic moment of the disc is $\frac{q\omega R^2}{2}$
 (B) The magnetic moment of the disc is $\frac{q\omega R^2}{4}$
 (C) The magnetic field at the centre of disc is $\frac{\mu_0 Q \omega}{2\pi R}$
 (D) The magnetic field at the centre of disc is $\frac{\mu_0 Q \omega}{\pi R}$



- 4.* A square current carrying coil of edge length L is shown in figure. Magnetic field on coil is given by

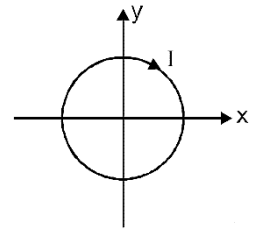
$$\vec{B} = \frac{B_0 y}{L} \hat{i} + \frac{B_0 x}{L} \hat{j} \text{ where } B_0 \text{ is positive constant.}$$



- (A) Torque on coil is $\frac{1}{2} iL^2 B_0 \hat{i}$ if coil is free to rotate about x-axis
 (B) Resultant force on coil is zero.
 (C) Equation of torque $\vec{M} \times \vec{B}$ is not valid on coil, where M is magnetic moment of coil.
 (D) Equation of torque $\vec{M} \times \vec{B}$ is valid on coil, where M is magnetic moment of coil.

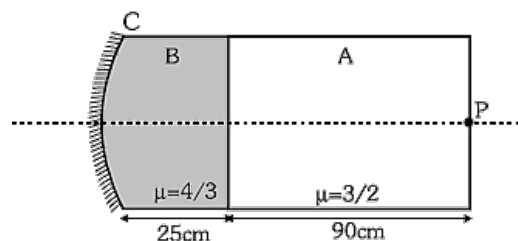
- 5.* A uniform circular loop of mass ' m ' and radius R is lying in space where a uniform magnetic field $\vec{B} = B_0 (\hat{i} + \sqrt{3}\hat{j})$ is present. The loop carries constant current I in clockwise direction and lies in a gravity free region at rest in xy -plane with its centre at origin. Then choose the CORRECT statement(s) :

- (A) The initial angular acceleration of the loop has magnitude $\frac{4I\pi B_0}{m}$
 (B) The initial angular acceleration of the loop has magnitude $\frac{2I\pi B_0}{m}$
 (C) The unit vector in the direction of initial angular acceleration is $\left(+\frac{\sqrt{3}}{2} \hat{i} - \frac{1}{2} \hat{j} \right)$
 (D) The initial acceleration of the point with coordinate $\left(\frac{\sqrt{3}}{2} R, -\frac{R}{2} \right)$ is zero.



Paragraph for Question no. 6 -8

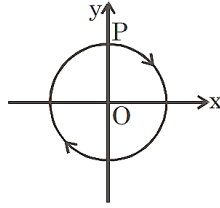
An object P is placed on the glass slab A, which is 90 cm thick with refractive index $3/2$. A container B of thickness 25 cm having water of refractive index $4/3$ is sandwiched between glass slab A and a concave mirror C of radius of curvature 70 cm.



6. As a result of first refraction, find the distance of image from glass-water interface:
 (A) 50 cm (B) 60 cm (C) 75 cm (D) 80 cm
7. Second image will be formed after reflection from the concave mirror, the distance of second image from the mirror will be:
 (A) 57.50 cm (B) 52.50 cm (C) 27.5 cm (D) 32.5 cm
8. The distance of final image from the object P, is:
 (A) 70.2 cm (B) 68.25 cm (C) 62.25 cm (D) 39.37 cm



9. Consider a metallic ring of radius 1 m, mass 1 kg and carrying a current of 1A in a gravity free space in the x-y plane with its centre O at the origin as shown in the figure. If a uniform magnetic field $(3\hat{i} + 4\hat{j})\text{T}$ is applied, then the instantaneous acceleration of the point P(which is on the y-axis at the moment) will be $C \times \pi\text{m/s}^2$. Find the value of C.



10. A moving-coil galvanometer has 100 turns and each turn has an area 2.0 cm^2 . The magnetic field produced by the magnet is 0.01 T. The deflection in the coil is 2 radian when a current of 10 mA is passed through it. If the torsional constant of the suspension wire is $N \times 10^{-6}\text{N-m/rad}$, then find N.





SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-46

Note: (*) → Multiple Correct Type Question

Total Marks: 38

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.7 to Q.8

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9

(4 marks, 3 min.)

Match the Column Type Questions (No negative marking) Q.10

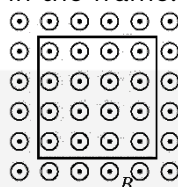
(8 marks, 4 min.)

(There is partial marking in multiple correct questions)

1. A rectangular loop of sides of length ℓ and b is placed in x-y plane. A uniform but time varying magnetic field of strength $\vec{B} = 20 t \hat{i} + 10 t^2 \hat{j} + 50 \hat{k}$ where t is time elapsed. The magnitude of induced e.m.f. at time t is:
 (A) $20 + 20 t$ (B) 20 (C) $20 t$ (D) Zero

2. A flexible wire loop in the shape of a circle has a radius that grows linearly with time. There is a magnetic field perpendicular to the plane of the loop that has a magnitude inversely proportional to the distance from the centre of loop i.e. $B(r) \propto \frac{1}{r}$. How does the emf E vary with time?
 (A) $E \propto t^2$ (B) $E \propto t$ (C) $E \propto \sqrt{t}$ (D) E is constant

3. A uniform magnetic field B is directed out of the page. A metallic wire has the shape of a square frame and is placed in the field as shown. While the shape of the wire is steadily transformed into a circle in the same plane, the current in the frame:



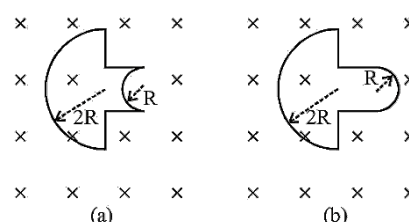
- (A) is directed clockwise (B) does not appear
 (C) is directed counterclockwise (D) is alternating
4. A conducting loop with shape (a) is placed in uniform magnetic field B . Its one curved portion is turned so that the shape becomes (b). Find the correct statement(s). Given Ω be the resistance.

(A) Change in flux is given by $2B\pi R^2$.

(B) Change in flux is given by $\frac{B\pi R^2}{2}$.

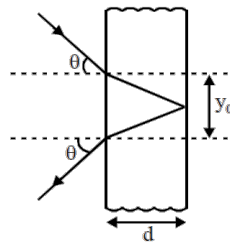
(C) Total charge crossing any point of loop is $\frac{B\pi R^2}{2\Omega}$.

(D) Total charge crossing any point of loop is $\frac{B\pi R^2}{\Omega}$.





5. A ray of light incident from air on a glass plate of refractive index n is partly reflected and partly refracted at the two surfaces of the glass. The displacement y_0 in the figure is:



- (A) $\frac{2d \sin \theta}{\sqrt{n^2 - \sin^2 \theta}}$ (B) $\frac{2d \sin \theta}{\sqrt{\sin^2 \theta - \frac{1}{n^2}}}$ (C) $\frac{2d \sqrt{n^2 - \sin^2 \theta}}{\sin \theta}$ (D) None of these

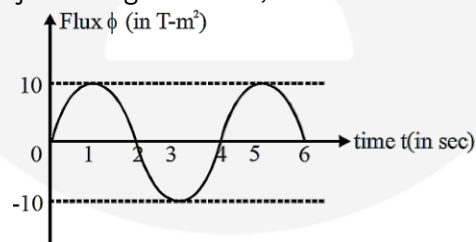
6. When a thin convergent glass lens ($\mu_g = 1.5$) of power +5 D is immersed in a liquid of refractive index μ_l it acts as a divergent lens of focal length 100 cm. Then μ_l is:

- (A) 4/3 (B) 5/3 (C) 5/4 (D) 6/5

- 7.* A ring of radius 20 cm has a total resistance of 0.04Ω . A uniform magnetic field varying with time $B = (0.4t)$ T is perpendicular to the plane of the ring.

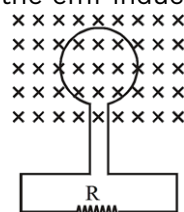
- (A) Induced current in the ring is $\frac{2\pi}{5}$ A .
(B) The ring will be in tension.
(C) The ring will be in compression.
(D) The magnetic field due to ring at the center of the ring is in direction opposite to applied magnetic field.

- 8.* A closed conducting loop, having resistance R , is being rotated about an axis perpendicular to the magnetic field. Magnetic flux through the closed conducting loop is continuously changing according to the graph shown in the adjacent figure. Then, which of the following statement(s) is/are correct?



- (A) The electric current through the loop is minimum (zero) at $t = 1$ s, 3s and 5s.
(B) The electric current through the loop is minimum (zero) at $t = 0$ s, 2s and 6s.
(C) Total charge flown through any cross-section of a closed conducting loop between 0 and 6 s is zero
(D) Total work done in rotating the loop in the magnetic field is zero

9. In the figure, the flux due to magnetic field through the loop perpendicular to the plane of the coil and directed into the paper is varying according to the relation, $\phi = t^2 + 7t + 1$ where ϕ is in weber and t is in seconds. The magnitude of the emf induced in the loop at $t = 1$ s is (in volt).

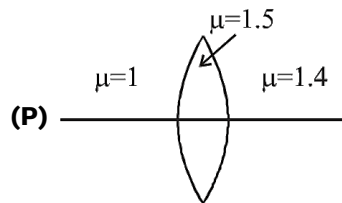




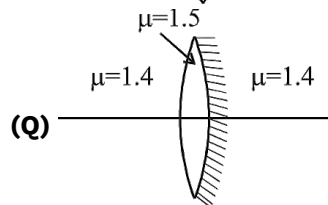
10. List-I gives different lens configurations. The radius of curvature of each curved surface is R . paraxial rays of light, parallel to the axis of lens traversing through the lens get focused at distance f from the lens. List-II gives corresponding values of f . (μ represent refractive index of medium)

List-I

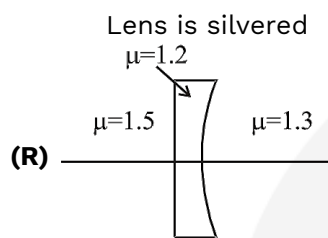
List-II



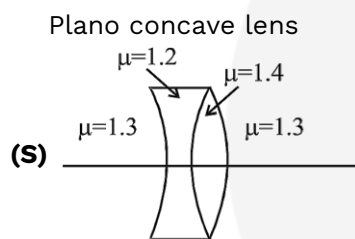
(1) $13 R$



(2) $\frac{13R}{4}$



(3) $\frac{7R}{3}$



(4) $\frac{7R}{16}$

(5) $\frac{16R}{7}$

Note: (*) → Multiple Correct Type Question

Total Marks: 25

Max. Time: 20 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

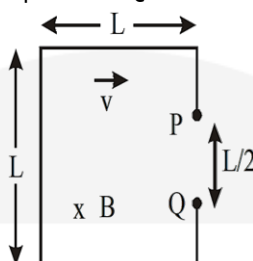
(3 marks, 2 min.)

Comprehension Type Questions ('-1' Negative marking) Q.6 to Q.10

(2 marks, 2 min.)

(There is partial marking in multiple correct questions)

1. The loop shown moves with a velocity v in a uniform magnetic field of magnitude B , directed into the paper. The potential of P with respect to Q is :



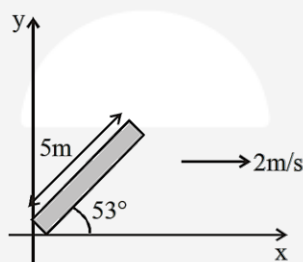
(A) $e = \frac{7}{2}Blv$

(B) $e = -\frac{7}{2}Blv$

(C) $\frac{Blv}{2}$

(D) $-\frac{Blv}{2}$

2. A conducting rod PQ of $l = 5\text{m}$ oriented as shown is moving with $V = (2\text{m/s}) \hat{i}$ without any rotation in a uniform magnetic field $(3\hat{j} + 4\hat{k})$ Tesla. Emf induced in the rod is:



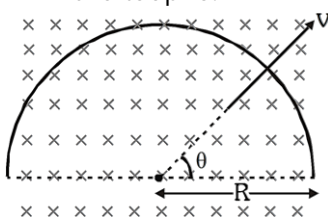
(A) 32 V

(B) 40 V

(C) 50 V

(D) None

3. A semi circular loop of radius R is placed in a uniform magnetic field as shown. It is pulled with a constant velocity. The induced emf in the loop is:



(A) $Bv (\pi R) \cos\theta$

(B) $Bv (\pi R) \sin\theta$

(C) $Bv (2R) \cos\theta$

(D) $Bv (2R) \sin\theta$

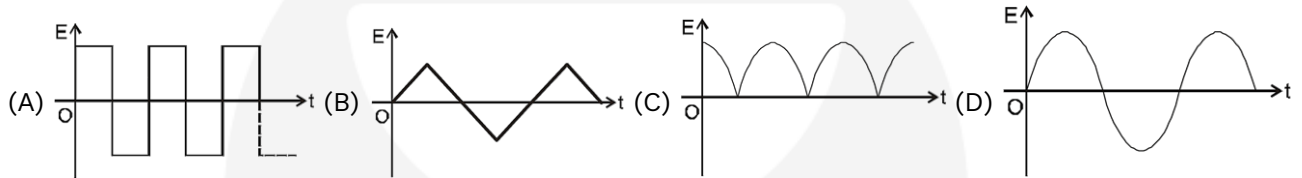
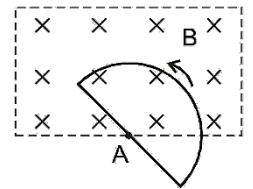


4. In figure shown, wire P_1Q_1 and P_2Q_2 , both are moving towards right with speed 5 cm/sec. Resistance of each wire is 2Ω . Then current through 19Ω resistor is:



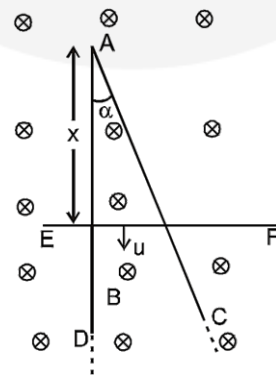
- (A) 0 (B) 0.1 mA (C) 0.2 mA (D) 0.3 mA

5. A uniform and constant magnetic field B is directed perpendicularly into the plane of the page everywhere within a rectangular region as shown in figure. A wire circuit in the shape of a semicircle is uniformly (that is with uniform angular speed) rotated counter clockwise in the plane of the page about an axis A . The axis A is perpendicular to the page at the edge of the field and directed through the centre of the straight line portion of the circuit. Which of the following graphs best approximates the emf E induced in the circuit as a function of time t ?

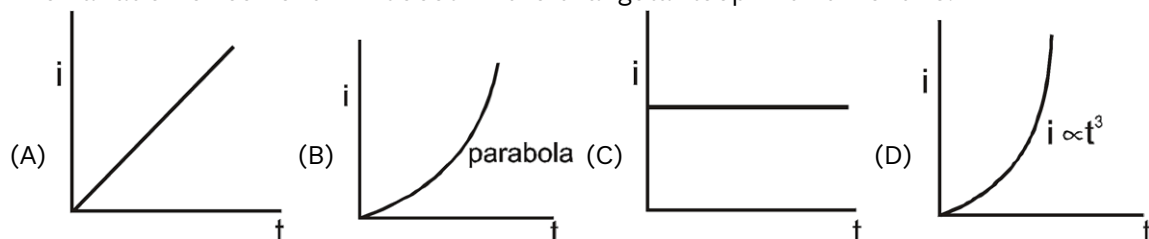


Paragraph for Questions 6 and 7

A long straight wire having uniform resistance per unit length λ (ohms/metre) is bent into V- shape as shown (angle $DAC = \alpha$). A uniform and constant magnetic field B (tesla) exists in space which is perpendicular to the plane of V-shaped wire. Another straight wire EF of same uniform resistance per unit length λ (ohms/metre) is pulled with constant velocity u (m/s) such that the wire EF is always perpendicular to side AD of V-shape (the wire EF is always in conducting contact with the V-shaped wire at two points). At $t = 0$ (t is time in seconds) the value of $x = 0$ (x = distance of wire EF from end A , in metres)



6. The variation of current ' i ' induced in the triangular loop with time ' t ' is:



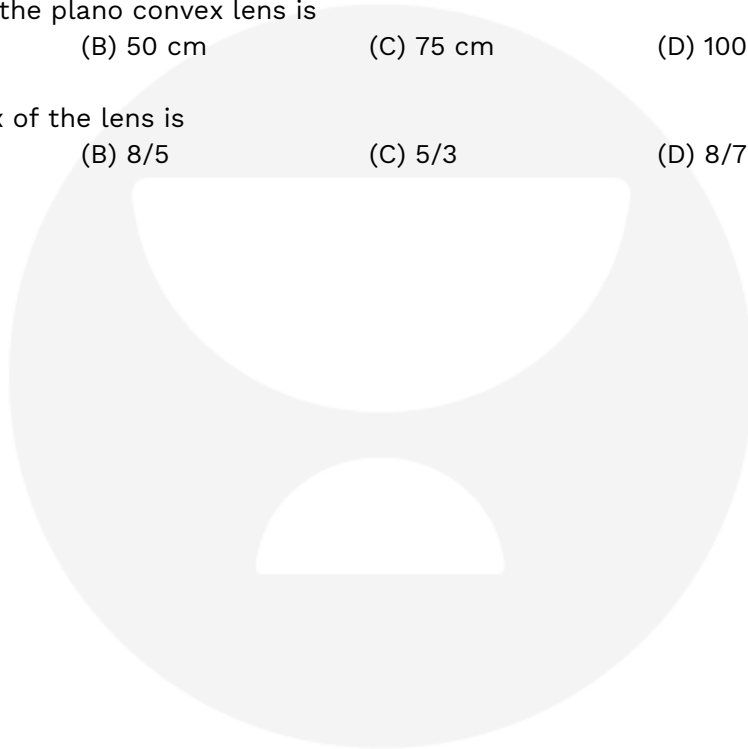


7. The magnetic force acting on the wire EF due to uniform magnetic field B at any time t (in N) is:
- (A) $\frac{B^2 t u^2 \tan^2 \alpha}{\lambda(1 + \tan \alpha + \sec \alpha)}$ (B) $\frac{B^2 u^2 \tan^2 \alpha}{\lambda(1 + \tan \alpha + \sec \alpha)}$
- (C) $\frac{B^2 t u^2}{\lambda(2 + \tan \alpha)}$ (D) None of these

Paragraph for questions 8 to 10

A plano convex lens has thickness 4 cm, when placed on a horizontal table with the curved surface in contact with it, the apparent depth of the bottom most point of the lens is found to be 3 cm. If the lens is inverted such that the plane is in contact with the table, the apparent depth of the centre of the plane face of the lens is found to be $25/8$ cm.

8. radius of curvature of curved surface is
(A) 25 cm (B) 50 cm (C) 79 cm (D) 100 cm
9. Focal length of the plano convex lens is
(A) 25 cm (B) 50 cm (C) 75 cm (D) 100 cm
10. Refractive index of the lens is
(A) $4/3$ (B) $8/5$ (C) $5/3$ (D) $8/7$



Note: (*) → Multiple Correct Type Question

Total Marks: 32

Max. Time: 24 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.4

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.5 to Q.7

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.8 to Q.9

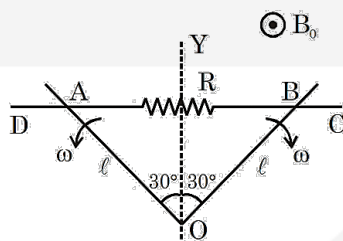
(2 marks, 2 min.)

Subjective Type Questions (No negative marking) Q.10

(4 marks, 3 min.)

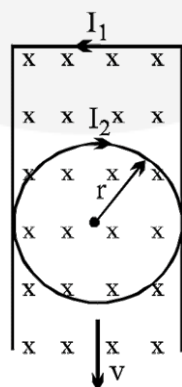
(There is partial marking in multiple correct questions)

1. In the shown figure, uniform magnetic field B_0 is pointing out of the plane in the region. Wire CD is fixed and has resistance R , while OA and OB are conducting wires rotating with angular velocity ω about O as shown. If at some instant $OA = OB = \ell$ and each wire makes angle $\theta = 30^\circ$ with y-axis, then current through resistance R is:



- (A) $\frac{B_0 \omega \ell^2}{R}$ (B) $\frac{\sqrt{3} B_0 \omega \ell^2}{2R}$ (C) $\frac{\sqrt{3} B_0 \omega \ell^2}{4R}$ (D) Zero

2. Figure shows a uniform cross-sectional conducting ring of radius r and total resistance R in contact with two vertical conducting rails, which are joined at the top. The rails have no friction and resistance. There is a uniform magnetic field B perpendicular to the plane of rails. If ring falls with the velocity v , then



- (A) $I_1 = 0; I_2 = \frac{4Bvr}{R}$ (B) $I_1 = \frac{4Bvr}{R}; I_2 = 0$
 (C) $I_1 = \frac{8Bvr}{R}; I_2 = \frac{4Bvr}{R}$ (D) $I_1 = \frac{4Bvr}{R}; I_2 = \frac{2Bvr}{R}$



3. A wire of mass m and length ℓ can slide freely on a pair of smooth, vertical rails (figure). A magnetic field B exists in the region in the direction perpendicular to the plane of the rails. The rails are connected at the top end by a capacitor of capacitance C . The acceleration of the wire neglecting any electric resistance is:

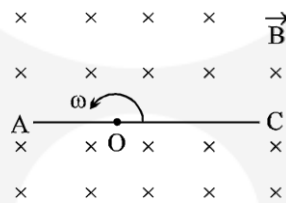


- (A) $\frac{mg}{m + CB^2\ell^2}$ (B) $\frac{2mg}{m + CB^2\ell^2}$ (C) $\frac{mg}{CB^2\ell^2}$ (D) $\frac{mg}{2(m + CB^2\ell^2)}$

4. Two thin symmetrical lenses of different nature have equal radii of curvature $R = 20$ cm. The lenses are put close together and immersed in water. The converging focal length of the system is 24 cm. Find the difference between refractive indices of the two lenses. (Refractive index for water = $\frac{4}{3}$)

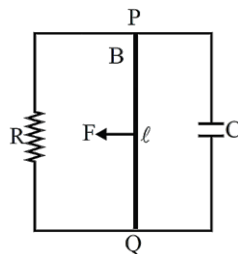
- (A) $(\mu_1 - \mu_2) = \frac{1}{24}$ (B) $(\mu_1 - \mu_2) = \frac{1}{20}$ (C) $(\mu_1 - \mu_2) = \frac{5}{9}$ (D) $(\mu_1 - \mu_2) = \frac{4}{3}$

- 5*. conducting rod AC of length $4l$ is rotated about a point O in a uniform magnetic field $\vec{B}$ directed into the paper. $AO = l$ and $OC = 3l$. Then:



- (A) $V_A - V_O = \frac{B\omega l^2}{2}$ (B) $V_O - V_C = \frac{9}{2}B\omega l^2$ (C) $V_A - V_C = 4B\omega l^2$ (D) $V_C - V_O = \frac{9}{2}B\omega l^2$

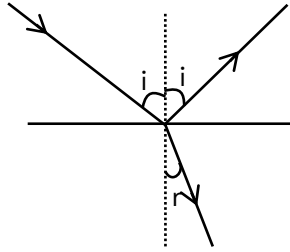
- 6*. conducting rod PQ of length ℓ is pulled with a constant force F along two smooth parallel rails separated by a distance ℓ as shown in the adjacent figure. Then choose the correct statement(s):



- (A) terminal velocity of the rod, $v_t = \frac{2FR}{B^2\ell^2}$
 (B) terminal velocity of the rod, $v_t = \frac{FR}{B^2\ell^2}$
 (C) maximum charge on the capacitor, $q_{\max} = \frac{FCR}{B\ell}$
 (D) maximum charge on the capacitor, $q_{\max} = \frac{2FCR}{B\ell}$



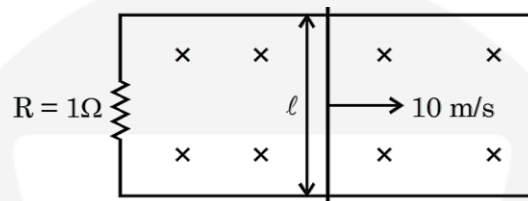
- 7*. A ray of light from a denser medium strikes a rarer medium at an angle of incidence i . The reflected and refracted rays make an angle of 90° with each other. The angles of reflection and refraction are i and r . The critical angle is:



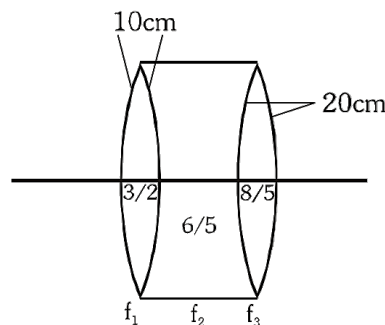
- (A) $\sin^{-1}(\cot r)$ (B) $\sin^{-1}(\tan i)$ (C) $\sin^{-1}(\tan r)$ (D) $\tan^{-1}(\sin i)$

Paragraph for Questions No. 8 and 9

A conducting rod of negligible resistance can slide on smooth U shaped rail as shown. At $t = 0$, conducting rod is given a velocity of 10 m/s towards right. A uniform magnetic field $B = 1$ T exists in the region. [Mass of conducting rod is 1 kg and $\ell = 1$ m]



8. The distance travelled by the conducting rod when its speed is 5 m/s is:
 (A) $\frac{5}{2}$ m (B) 5 m
 (C) 10 m (D) 20 m
9. Heat loss in the resistor during the time interval $t = 0$ to time t when the speed of the conducting rod is 6 m/s is:
 (A) 64 J (B) 36 J
 (C) 32 J (D) 50 J
10. In the shown figure the focal length (in cm) of equivalent system is in the form of $\left(\frac{50x}{13}\right)$. Find the value of x .



Note: (*) → Multiple Correct Type Question

Total Marks: 31

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.6 to Q.8

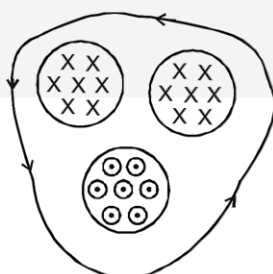
(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.9 to Q.10

(2 marks, 2 min.)

(There is partial marking in multiple correct questions)

1. Figure shows three regions of magnetic field, each of area A , and in each region magnitude of magnetic field decreases at a constant rate α . If $\vec{E}$ is induced electric field then value of line integral $\oint \vec{E} \cdot d\vec{r}$ along the given loop is equal to:



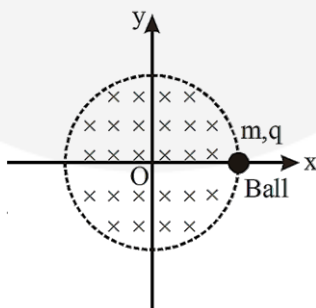
(A) αA

(B) $-\alpha A$

(C) $3\alpha A$

(D) $-3\alpha A$

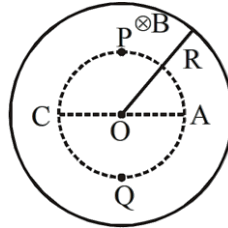
2. A small ball having charge q and mass m lies on a horizontal smooth surface (take it as xy -plane) at distance b from origin. A magnetic field exists within the region enclosed by the circle $x^2 + y^2 = b^2$ and directed towards negative z -axis as shown. If magnetic field starts increasing with time as $B = B_0(10 + t)$, then:



- (A) The ball will start to move towards +ve x -axis with acceleration $\frac{bqB_0}{2m}$
- (B) The ball will start to move towards -ve x -axis with acceleration $\frac{bqB_0}{2m}$
- (C) The ball will start to move towards +ve y -axis with acceleration $\frac{bqB_0}{2m}$
- (D) The ball will start to move towards -ve y -axis with acceleration $\frac{bqB_0}{2m}$



3. A uniform magnetic field B increasing with time exists in a cylindrical region of centre O and radius R . The direction of magnetic field is inwards the paper as shown. The work done by external agent in taking a unit positive charge slowly from A to C via paths APC , AOC and AQC be W_{APC} , W_{AOC} and W_{AQC} respectively. Then



- (A) $W_{APC} = W_{AOC} = W_{AQC}$
 (B) $W_{APC} > W_{AOC} > W_{AQC}$
 (C) $W_{APC} < W_{AOC} < W_{AQC}$
 (D) $W_{APC} = W_{AQC} < W_{AOC}$
4. Given a ring of mass M , radius R resting on a rough horizontal plane, with coefficient of friction ' μ '. A cylindrical, vertical magnetic field passing through ring is switched on at $t = 0$ which varies with time as $B = Kt^2$. Ring has charge Q distributed uniformly over it. Find the time after which ring starts rotating.

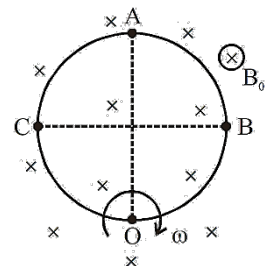
- (A) $\frac{\mu Mg}{QKR}$ (B) $\frac{2\mu Mg}{QKR}$ (C) $\frac{\mu Mg}{2QKR}$ (D) $\frac{\mu Mg}{4QKR}$

5. The string of a simple pendulum is replaced by a thin uniform rod of mass M and length L . The mass of the bob is m . If it is allowed to oscillate with small amplitude, the period of oscillation is:

- (A) $2\pi\sqrt{\frac{2(M+3m)L}{3(M+2m)g}}$ (B) $2\pi\sqrt{\frac{3(M+2m)L}{2(M+3m)g}}$ (C) $2\pi\sqrt{\frac{(M+2m)L}{(M+3m)g}}$ (D) $2\pi\sqrt{\frac{2(M+m)L}{3(M+2m)g}}$

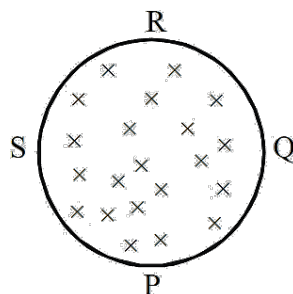
- 6*. In the figure, there is a conducting ring having resistance R placed in the plane of paper in a uniform magnetic field B_0 . If the ring is rotating about an axis passing through point O and perpendicular to the plane of paper with constant angular speed ω in clockwise direction, then:

- (A) point A will be at higher potential than O .
 (B) the potential of point B and C will be same.
 (C) the current in ring will be zero.



- (D) the current in the ring will be $\frac{2B_0\omega r^2}{R}$.

- 7*. A conducting ring is placed in a uniform external magnetic field present in the space within the ring and perpendicular to plane of ring. The magnetic field changes at a constant rate due to which a current of magnitude 4 amp flows in the ring. The resistance of parts PQR and PSR are 4Ω and 8Ω respectively. Then:



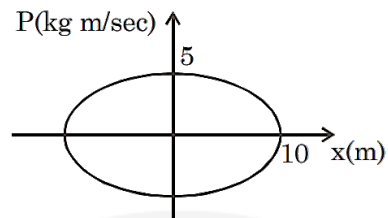
- (A) The emf induced in the ring 48 V
 (B) Potential difference between P & R is 24 V
 (C) Potential difference between P & R is 8V
 (D) Potential difference between P & R is zero as conservative electric field in the ring wire is zero



- 8*.** x-t equation of a particle moving along x-axis is given as:
 $x = A + A(1 - \cos \omega t)$
(A) particle oscillates simple harmonically between points $x = 2A$ and $x = A$.
(B) Velocity of particle is maximum at $x = 2A$.
(C) Time taken by the particle in travelling from $x = A$ to $x = 3A$ is π/ω .
(D) Time taken by particle in travelling from $x = A$ to $x = 2A$ is $\pi/2\omega$.

Paragraph for Questions 9 and 10

Momentum (P) displacement (x) graph of a particle of unit mass performing SHM along x-axis is shown in the figure.



- 9.** Angular frequency of the SHM is:
(A) 1 rad/sec
(B) $\frac{1}{2}$ rad/sec
(C) 2 rad/sec
(D) 4 rad/sec
- 10.** Total energy of oscillation is (potential energy at mean position = 0):
(A) 500 Joule
(B) 25 Joule
(C) 12.5 Joule
(D) 100 Joule

Note: (*) → Multiple Correct Type Question

Total Marks: 30

Max. Time: 22 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.7 to Q.8

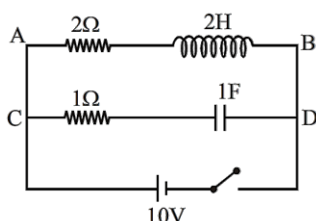
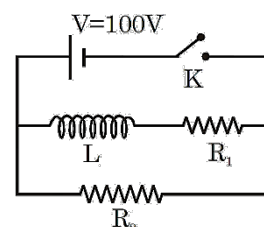
(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.9 to Q.10

(2 marks, 2 min.)

(There is partial marking in multiple correct questions)

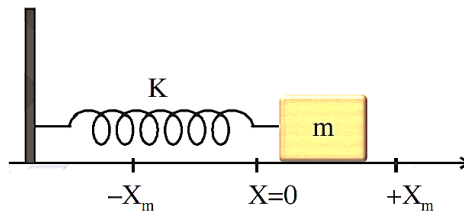
- A magnetic flux of $5 \times 10^{-4} \text{ wb}$ is associated with every 10 turns of a 500 turns coil. The electric current flowing through the wire is 5A. The self inductance of the coil will be:
(A) 0.5 H (B) $5 \times 10^{-3} \text{ H}$ (C) 5 H (D) $5 \times 10^{-2} \text{ H}$
- A long solenoid is made of a superconducting wire carrying a current i_0 . The solenoid is slowly stretched so that its cross section does not change but its length changes from L_0 to L_1 . What is the new current in the solenoid assuming the solenoid is still tightly wound?
(A) $\frac{L_0 i_0}{L_1}$ (B) $\frac{L_1 i_0}{L_0}$ (C) $\frac{L_0^2 i_0}{L_0}$ (D) $\frac{L_1^2 i_0}{L_0^2}$
- Self inductance of a coil is 8 H. The power (in watt) consumed by coil (purely inductive) is given by $P = 8i^2$ where 'i' is current in ampere. The time for the current to change from i_0 to $2i_0$ will be
(A) $\ln 2$ (B) $5 i_0$ (C) $\frac{i_0^2}{\ln 2}$ (D) None of these
- In the circuit shown below, the key K is closed at $t = 0$. If current through battery at $t = 0$ is 20 A and at $t = \infty$ is 40 A, then value of R_1 and R_2 is:
(A) $R_1 = 5\Omega, R_2 = 4\Omega$ (B) $R_1 = 4\Omega, R_2 = 5\Omega$
(C) $R_1 = 5\Omega, R_2 = 5\Omega$ (D) $R_1 = 4\Omega, R_2 = 4\Omega$
- In the given circuit shown, the switch is closed at $t = 0$. The battery is ideal. Then,



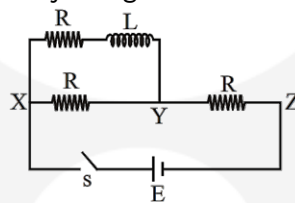
- Initial current through battery is 5A.
- Current through battery after a long time is 10 A.
- The time when current in both branches (AB & CD) are equal is $\ln 2$ sec.
- The time when current in both branches (AB & CD) are equal is $\ln 3$ sec.



6. At $t = 0$, the displacement of the block in a linear oscillator as shown is -0.08 m. At the same moment $t = 0$, its velocity is -1.6 m/s and acceleration is 32 m/s². Choose the INCORRECT statement.



- (A) Angular frequency of the motion is 20 rad/s
 (B) Amplitude of the motion is 11.3 cm
 (C) Phase constant ϕ of the motion if the equation of motion is expressed as $x = A \sin(\omega t + \phi)$ is $\frac{\pi}{4}$
 (D) Phase constant ϕ of the motion if the equation of motion is expressed as $x = A \sin(\omega t + \phi)$ is $\frac{5\pi}{4}$
- 7*. A time varying voltage $V = 10 - 2i$ (volt) is applied across an inductor of inductance $L = 2$ H where i is current at any time t with $i = 0$ at $t = 0$. Then
 (A) Current increases with time
 (B) Energy stored in magnetic field at $t = \ln 2$ sec is 6.25 J
 (C) At time $t = \ln 2$ sec energy stored in magnetic field is increasing at a rate of 12.5 J/sec
 (D) Energy stored in magnetic field is zero all the time
- 8*. The switch is closed at $t = 0$ in the adjoining circuit. Which of the following is/are CORRECT?



- (A) The potential difference across YZ at $t = 0$ is $2E/3$
 (B) The potential difference across XY at $t = \infty$ is $E/2$
 (C) The potential difference across YZ at $t = 0$ is $E/2$
 (D) The potential difference across XY at $t = \infty$ is $2E/3$

Paragraph for Questions No. 9 and 10

A variable force given by $F = -8x + 16$ N is applied on a block of mass 2 kg. At $t = 0$ particle is at rest at $x = 5$ m.

9. Equation of motion of block is given by :
 (A) $x = 3 \cos 2\pi t$ (B) $x = 2 + 3 \cos 2t$ (C) $x = 2 - 3 \cos 2t$ (D) None of these
10. Speed of block at $t = \pi/4$ sec is :
 (A) 6 m/s (B) 2 m/s (C) 3 m/s (D) None of these



aSUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-51

Note: (*) → Multiple Correct Type Question

Total Marks: 35

Max. Time: 27 min.

Single Correct Questions ('-1' negative marking) Q.1

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.2 to Q.7

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.8 to Q.9

(2 marks, 2 min.)

Subjective Type Questions (No negative marking) Q.10

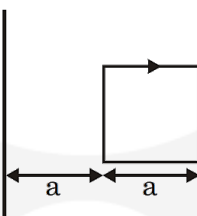
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

- 1.** The KE of particle executing SHM is 16 J when it is at mean position. If amplitude is 25 cm and mass of particle is 5.12 Kg, the time period is :

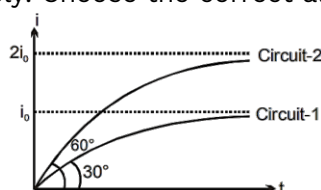
(A) $\frac{\pi}{5}$ sec (B) 2π sec (C) 20π sec (D) 5π sec

- 2*.** A very long straight conductor and a square conducting frame lie in a plane and are separated from each other as shown in the figure ($a = 10$ cm). If current in the square frame is increasing at a rate of $2A/s$ in clockwise direction.



- (A) Mutual inductance of the two loops is $2 \ell n 2 \times 10^{-8} H$.
 (B) The current induced in the wire is upwards.
 (C) The current induced in the wire is downwards.
 (D) The current induced in the wire is zero.

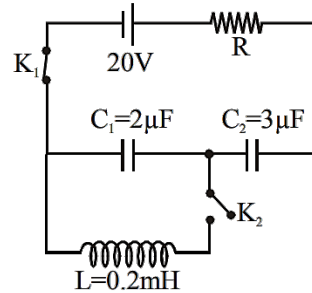
- 3*.** Growth of current in two different L-R circuit are depicted by the i-t graph shown. Angle subtended by the curves with time axis at time $t = 0$ are also shown in the graph. τ_1 and τ_2 are time constants for the circuits 1 and 2 respectively. Choose the correct alternative.



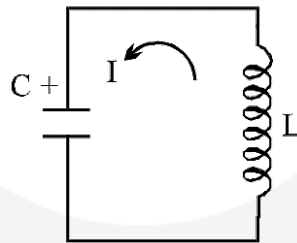
- (A) $\frac{\tau_1}{\tau_2} = \frac{2}{3}$
 (B) $\frac{\tau_1}{\tau_2} = \frac{3}{2}$
 (C) Initial rate of change of current for circuit-2 is 2 times that of circuit-1.
 (D) Initial rate of change of current for circuit-2 is 3 times that of circuit-1.



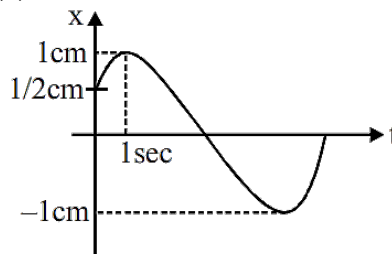
- 4*. A circuit containing capacitors C_1 and C_2 , shown in the figure is in the steady state with the key K_1 closed. At the instant $t = 0$, K_1 is opened and K_2 is closed. The ratio of energy in the inductor and that of capacitor C_1 becomes one third at time 't' for the first time. If charge on capacitor C_1 at that instant is q_1 and that on capacitor C_2 is q_2 then choose the CORRECT statement(s):



- (A) Angular frequency of oscillations of the L.C. circuit will be 5×10^4 rad/s
 (B) Time 't' will be 21 μ s.
 (C) Charge on capacitor C_1 at that instant will be, $q_1 = 12\sqrt{3}\mu$ C.
 (D) Charge on capacitor C_2 at that instant will be, $q_2 = 24 \mu$ C
- 5*. At time $t = 0$, the LC circuit shown in the figure has equal amount of energy stored in the capacitor $C = 20 \mu$ F and in the inductor, each equal to 200 μ J. The current amplitude in the circuit is $1/\sqrt{10}$ A



- (A) The natural frequency of oscillation is 398 Hz.
 (B) The natural frequency of oscillation is 199 Hz.
 (C) The equation for current in terms of t is $\frac{1}{\sqrt{10}} \cos\left(2500t + \frac{\pi}{4}\right)$
 (D) The equation for current in terms of t is $\frac{1}{\sqrt{10}} \cos\left(2500t + \frac{3\pi}{4}\right)$
- 6*. The displacement time graph of a block of mass 1 kg executing SHM is given by following graph. Choose the correct statement(s)



- (A) The equation of motion can be $x = 1 \text{ cm} \sin\left(2\pi t + \frac{\pi}{6}\right)$
 (B) The force constant is $\frac{\pi^2}{9}$ N/m
 (C) After $t = 0$, the block first crosses mean position at $t = 2$ sec.
 (D) The energy of SHM is $\frac{\pi^2}{18} \times 10^{-4}$ J



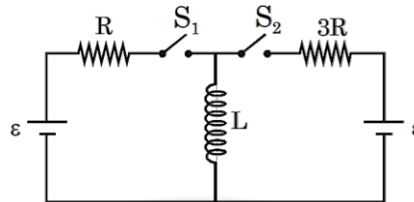
7*. Two simple harmonic motions are represented by the equation

$$y_1 = 3 (\sqrt{3} \cos 3\pi t + \sin 3\pi t) \text{ and } y_2 = 6 \sin (6\pi t + \pi/6)$$

- (A) The ratio A_1/A_2 of their amplitude is $1/2$
- (B) The ratio A_1/A_2 of their amplitude is 1
- (C) The ratio V_1/V_2 of their maximum velocities is $1/2$
- (D) The ratio V_1/V_2 of their maximum velocities is 2

Paragraph for question no. 8 & 9

In the given switch S_2 is kept closed and S_1 is kept open for a very long time such that the system is in steady-state. Now S_1 is also closed keeping S_2 closed.



8. Current passing through R just after S_1 closed is

- (A) $\frac{\varepsilon}{R}$
- (B) $\frac{\varepsilon}{3R}$
- (C) $\frac{\varepsilon}{4R}$
- (D) $\frac{\varepsilon}{2R}$

9. Current passing through inductor L long time after S_1 is closed is :

- (A) $\frac{\varepsilon}{3R}$
- (B) $\frac{4\varepsilon}{3R}$
- (C) $\frac{\varepsilon}{2R}$
- (D) $\frac{\varepsilon}{R}$

10. Two coils, 1 and 2, have mutual inductance = M and resistance R , each. A current flows in coil 1, which varies with time as : $i_1 = kt^2$, where k is a constant and 't' is time. The total change that has flown through coil 2, between $t = 0$ and $t = T$ is $\left(\frac{\lambda}{-\lambda + 1} \right) \frac{kMT^2}{R}$. Find $\left(\frac{1}{\lambda} \right)$.

Note: (*) → Multiple Correct Type Question

Total Marks: 25

Max. Time: 20 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Comprehension Type Questions ('-1' Negative marking) Q.6 to Q.10

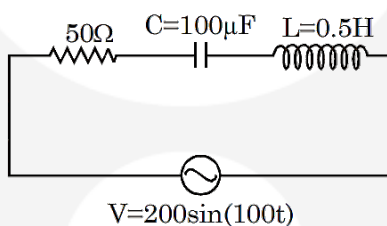
(2 marks, 2 min.)

(There is partial marking in multiple correct questions)

1. In an ac circuit, the instantaneous voltage $\varepsilon(t)$ and current $I(t)$ are given by $\varepsilon(t) = 5 [\cos \omega t + \sqrt{3} \sin \omega t]$ volt ; $I(t) = 5 \left[\sin \left(\omega t + \frac{\pi}{4} \right) \right]$ amp then:

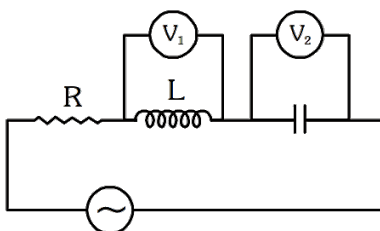
- (A) Current leads voltage by $\frac{\pi}{4}$ (D) Voltage leads current by $\frac{\pi}{3}$
 (C) Voltage leads current by $\frac{\pi}{6}$ (D) Current leads voltage by $\frac{\pi}{12}$

2. In the given LCR AC circuit, the effective current flowing through the circuit will be :



- (A) 2A (B) $2\sqrt{2}$ A (C) 4A (D) $4\sqrt{2}$ A

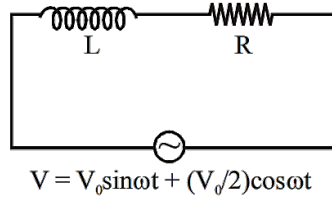
3. In the circuit shown, resistance $R = 100\Omega$, inductance $L = \frac{2}{\pi}$ H and capacitance $C = \frac{8}{\pi} \mu\text{F}$ are connected in series with an ac source of 200 volt and frequency 'f'. If the readings of the hot wire voltmeters V_1 and V_2 are same then which of the following is INCORRECT:



- (A) $f = 125$ Hz (B) $V_1 = V_2 = 1000$ V
 (C) Current through R is 1A (D) Current through R is 2A.

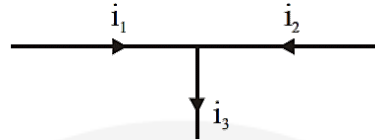


4. In the given figure, the current amplitude is :



- (A) $\sqrt{\frac{5V_0^2}{4(\omega^2 L^2 + R^2)}}$ (B) $\frac{V_0}{\sqrt{4(\omega^2 L^2 + R^2)}}$ (C) $\frac{\sqrt{5}V_0}{2\sqrt{\omega^2 L^2 - R^2}}$ (D) $\frac{\sqrt{5}V_0}{2\sqrt{R - \omega^2 L^2}}$

5. In the given figure, if $i_1 = 3 \sin \omega t$ and $i_2 = 4 \cos \omega t$, then i_3 is :



- (A) $5 \sin (\omega t + 53^\circ)$ (B) $5 \sin (\omega t + 37^\circ)$ (C) $5 \sin (\omega t + 45^\circ)$ (D) $5 \cos (\omega t + 53^\circ)$

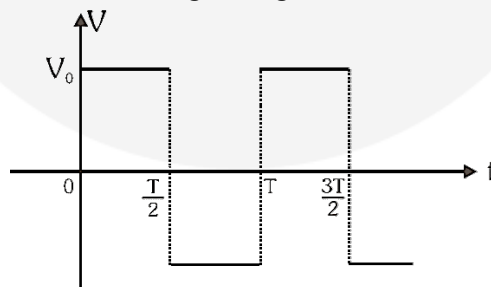
Paragraph for question 6 and 7

An electric current that reverses its direction with a constant frequency is called alternating current.

The average or mean value of ac is defined for half cycle as $I_{\text{mean}} = \frac{\int_0^{T/2} I dt}{\int_0^{T/2} dt}$. The effective, virtual or

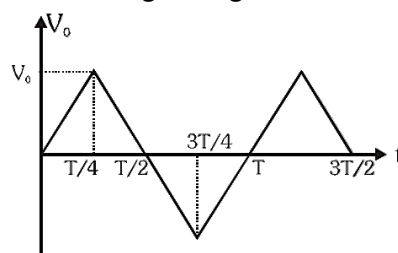
rms value of ac is defined as $I_{\text{rms}} = \sqrt{\frac{\int_0^T I^2 dt}{\int_0^T dt}}$

6. The mean and rms value of an alternating voltage as shown in figure is :



- (A) V_0, V_0 (B) $\frac{V_0}{2}, V_0$ (C) $\frac{3V_0}{2}, \frac{V_0}{2}$ (D) $\frac{V_0}{4}, \frac{V_0}{2}$

7. The mean and rms value of an alternating voltage as shown in figure is :

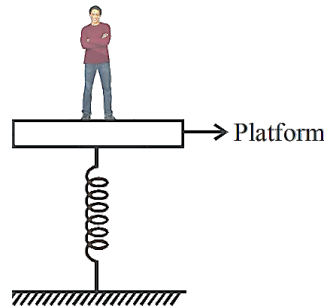


- (A) $\frac{V_0}{3}, \frac{V_0}{2}$ (B) $\frac{V_0}{\sqrt{2}}, \frac{V_0}{\sqrt{3}}$ (C) $\frac{V_0}{2}, \frac{V_0}{3}$ (D) $\frac{V_0}{2}, \frac{V_0}{\sqrt{3}}$



Paragraph for question 8 to 10

Mr. Saransh having mass 50 kg is standing on a platform of mass 10 kg oscillating up & down (doing SHM) of frequency 0.5 sec^{-1} & amplitude 0.4 m. Platform has a weighing machine fitted in it (which is massless) on which person is standing. (Take $\pi^2 = 10$ & $g = 10 \text{ m/s}^2$). Assume that at $t = 0$ (platform + man) is at their highest point of oscillation.



- 8.** Maximum reading of weighing machine is :
(A) 60 kg (B) 70 kg (C) 84 kg (D) 50 kg
- 9.** Find compression in spring when weighing machine reads 55 kg :
(A) 1 m (B) 1.1 m (C) 1.4 m (D) 0.1 m
- 10.** Find minimum time after which speed of person is $0.2 \pi \text{ m/s}$:
(A) $\frac{1}{6} \text{ sec}$ (B) $\frac{1}{3} \text{ sec}$ (C) 1 sec (D) $\frac{1}{4} \text{ sec}$

Note: (*) → Multiple Correct Type Question

Total Marks: 38

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.7 to Q.9

(4 marks, 3 min.)

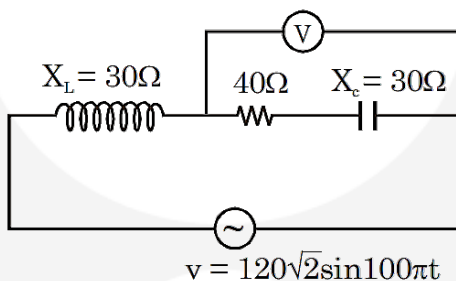
Match the Column Type Questions (No negative marking) Q.10

(8 marks, 4 min.)

(There is partial marking in multiple correct questions)

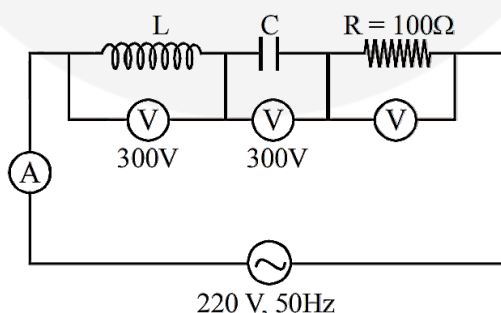
- When resonance is produced in a series LCR circuit, then which of the following is NOT CORRECT?
 (A) Current in the circuit is in phase with the applied voltage.
 (B) Inductive and capacitive reactance are equal.
 (C) If R is reduced, the voltage across capacitor will increase.
 (D) Impedance of the circuit is maximum

- In the following figure, the reading of ideal AC voltmeter will be.



- (A) 90 V (B) 150 V (C) 300 V (D) 120 V

- In the circuit shown below, what will be the readings of the voltmeter connected across R and ammeter?



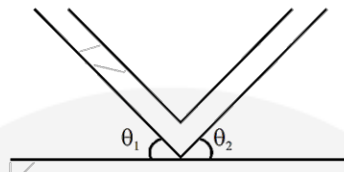
- (A) 800 V, 2A (B) 300 V, 2A (C) 220 V, 2.2A (D) 100 V, 2A

- An inductor of reactance $1\ \Omega$ and a resistor of $2\ \Omega$ are connected in a series to the terminals of a 6 V (rms) a.c. source. The power dissipated in the circuit is :
 (A) 8 W (B) 12 W (C) 14.4 W (D) 18 W



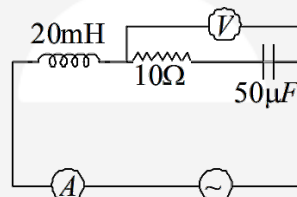
5. A series RLC circuit is driven by a generator at frequency 1000 Hz. The inductance is 90.0 mH; capacitance is 0.500 μF ; and the phase constant has a magnitude of 60.0° (Take $\pi^2 = 10$)
Choose the incorrect statement:
(A) Here currents leads the voltage in phase
(B) Here voltage leads the current in phase
(C) Resistance of circuit is $\frac{80\pi}{\sqrt{3}} \Omega$
(D) At resonance $\omega = \frac{\sqrt{2}}{3} \times 10^4 \text{ rad/sec}$

6. A non-viscous liquid of density ρ is filled in a tube with A as the area of cross-section, as shown in the figure. If the liquid is slightly displaced in one of the arms, the liquid column oscillates with a frequency:

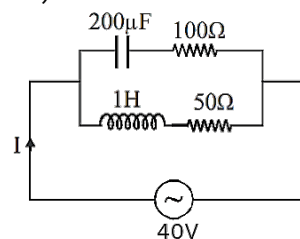


- (A) $\frac{1}{2\pi} \sqrt{\frac{\rho g A \sin\left(\frac{\theta_1 + \theta_2}{2}\right)}{m}}$
(B) $\frac{1}{2\pi} \sqrt{\frac{\rho g A (\sin\theta_1 - \sin\theta_2)}{m}}$
(C) $\frac{1}{2\pi} \sqrt{\frac{\rho g A (\sin\theta_1 + \sin\theta_2)}{m}}$
(D) $\frac{1}{2\pi} \sqrt{\frac{\rho g A \sin\left(\frac{\theta_1 - \theta_2}{2}\right)}{m}}$

- 7*. For the AC circuit shown, the reading of ammeter and voltmeter are 5A and $50\sqrt{5}$ volts respectively, then



- (A) Average power delivered by the source is 250W.
(B) rms value of AC source is 50 volts
(C) rms value of AC source is 100 volts.
(D) frequency of AC source is $100/2\pi \text{ Hz}$
- 8*. In the given circuit, the AC source has $\omega = 50 \text{ rad/s}$. Considering the inductor and capacitor to be ideal, the CORRECT choice (s) is (are) :



- (A) The voltage across 100 Ω resistor = $20\sqrt{2} \text{ V}$
(B) The voltage across 50 Ω resistor = $20\sqrt{2} \text{ V}$
(C) The current through the circuit I is $\frac{2}{\sqrt{10}} \text{ A}$.
(D) The current through the circuit, I is 1.2A



- 9*. Speed of a particle moving along x-axis is given by $v = \sqrt{16x - x^2 - 48}$. Which of the following is/are correct? (x is in m)
- (A) Amplitude of oscillation is 4 m.
 - (B) Maximum speed of particle is 4 m/s.
 - (C) Mean position of particle is (8,0)
 - (D) Acceleration of particle at $x = 9$ m is 2 m/s^2 .

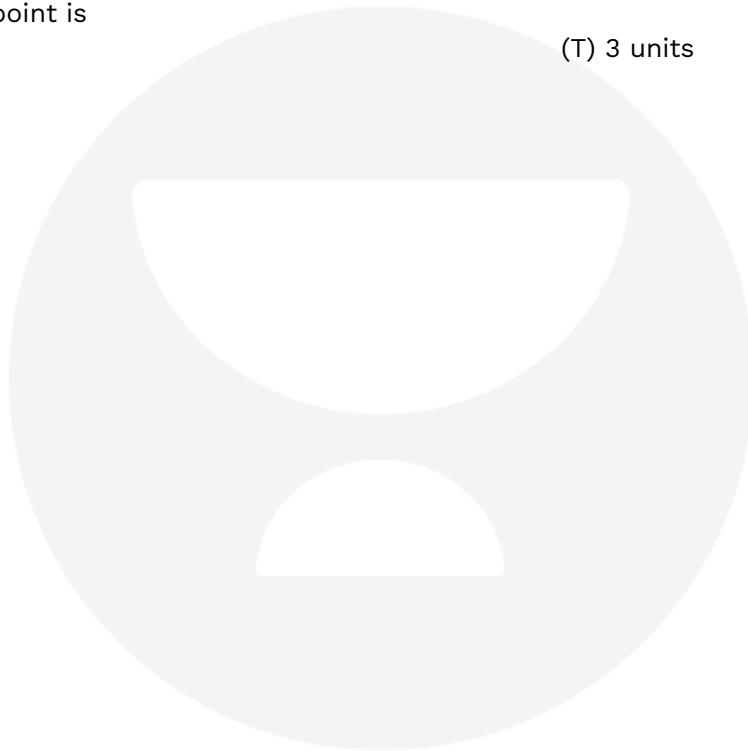
10. The speed (v) of a particle moving along a straight line, when it is at a distance (x) from a fixed point on the line is given by : $v^2 = 144 - 9x^2$

Column-I

- (A) Displacement of the particle in $\frac{4\pi}{3}$ sec is
- (B) Time period of simple harmonic motion is
- (C) Maximum displacement from the fixed point is
- (D) Magnitude of acceleration at a distance 3 cm from the fixed point is

Column-II

- (P) $T = 2\pi/3$ units
- (Q) Less than distance moved by it
- (R) 27 units
- (S) 4 units
- (T) 3 units



Note: (*) → Multiple Correct Type Question

Total Marks: 37

Single Correct Questions ('-1' negative marking) Q.1 to Q.7

Multiple Correct Questions ('-2' negative marking) Q.8

Subjective Type Questions (No negative marking) Q.9

Match the Column Type Questions (No negative marking) Q.10

(There is partial marking in multiple correct questions)

Max. Time: 24 min.

(3 marks, 2 min.)

(4 marks, 3 min.)

(4 marks, 3 min.)

(8 marks, 4 min.)

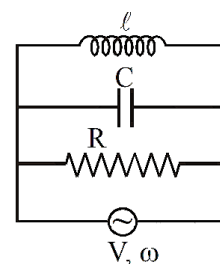
1. Consider L, C, R circuit as shown in figure, with a.c. source of peak value V and angular frequency ω . Then the peak value of current through the ac source is :

(A) $\frac{V}{\sqrt{\frac{1}{R^2} + \left(\omega L - \frac{1}{\omega C}\right)^2}}$

(B) $V \sqrt{\frac{1}{R^2} + \left(\frac{1}{\omega L} - \omega C\right)^2}$

(C) $\frac{V}{\sqrt{R^2 + \left(\omega C - \frac{1}{\omega C}\right)^2}}$

(D) $\frac{VR\omega}{\sqrt{\omega^2 C^2 + R(\omega^2 C^2 - 1)^2}}$



2. A series LR circuit is connected to an ac source of frequency ω and the inductive reactance is equal to $2R$. A capacitance of capacitive reactance equal to R is added in series with L and R . The ratio of the old power factor to the new one is:

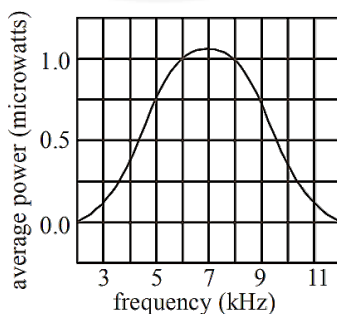
(A) $\sqrt{\frac{2}{3}}$

(B) $\sqrt{\frac{2}{5}}$

(C) $\sqrt{\frac{3}{2}}$

(D) $\sqrt{\frac{5}{2}}$

3. The plot given below is of the average power delivered to an LCR circuit versus frequency. The quality factor of the circuit is nearly :



(A) 5

(B) 2.4

(C) 2.8

(D) 1.4

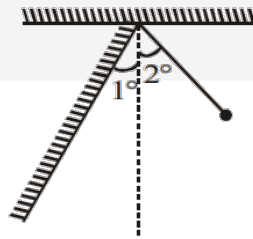


4. Mark the INCORRECT statement regarding an ideal transformer:
- (A) In a ideal transformer primary has negligible resistance and all the flux in the core links both primary and secondary windings.
 - (B) The alternating magnetic flux induces eddy currents in the iron core and causes heating.
 - (C) Soft iron core is used in transformer to minimize hysteresis loss.
 - (D) Primary current and secondary current are in same phase.

5. If a 2 kg mass is attached to a spring of force constant $k = 1250 \text{ N/m}$, the period of oscillation is $T = \frac{\pi}{12} \text{ sec}$. The damping constant 'b' has the value.

(A) 9.8 kg/sec (B) 98 kg/sec (C) 2.8 kg/sec (D) 28 kg/sec

6. A simple pendulum of length 1m is allowed to oscillate with amplitude 2° . It collides elastically with a wall inclined at 1° to the vertical. Its time period will be : (use $g = \pi^2$)



(A) $2/3 \text{ sec}$ (B) $4/3 \text{ sec}$ (C) 2 sec (D) None of these

7. Frequency of oscillation of a body is 6 Hz when force F_1 is applied and 8 Hz when F_2 is applied. If both forces F_1 & F_2 are applied together then the frequency of oscillation, is :-

(A) 14 Hz (B) 2 Hz (C) 10 Hz (D) $10\sqrt{2} \text{ Hz}$

- 8.* A circuit, containing an inductance and a resistance connected in series, has an AC source of 200 V, 50 Hz connected across it. An AC current of 10A rms flows through the circuit and the power loss is measured to be 1 kW.

(A) The inductance of the circuit is $\frac{\sqrt{3}}{10\pi} \text{ H}$.

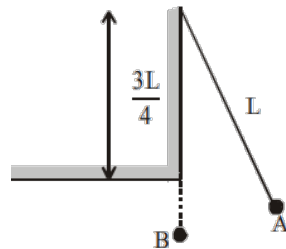
(B) The frequency of the AC when the phase difference between the current and emf becomes $\pi/4$, with the above components is $\frac{50}{\sqrt{3}} \text{ Hz}$.

(C) The frequency of the AC when the phase difference between the current and emf becomes $\pi/3$, with the above components is $\frac{25}{\sqrt{3}} \text{ Hz}$.

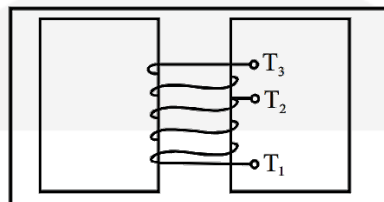
(D) The frequency of the AC when the phase difference between the current and emf becomes $\pi/4$, with the above components is $\frac{25}{\sqrt{3}} \text{ Hz}$



9. A pendulum has period T for small oscillations. An obstacle is placed directly beneath the pivot, so that only the lowest one quarter of the string can follow the pendulum bob when it swings in the left of its resting position as shown in the figure. The pendulum is released from rest at a certain point A. The time taken by it to return to that point is $nT/4$. Find the value of n ?



10. An autotransformer consists of a single coil with a ferromagnetic core. Three connections are provided. Between connections T_1 and T_2 there are 200 turns and between T_2 and T_3 there are 800 turns. Any two connections can be used as primary and any two can be used as secondary.



List-I

List-II

- | | |
|---|----------|
| (A) For step up transformer, maximum value of $\frac{V_{\text{secondary}}}{V_{\text{primary}}}$ | (1) 0.8 |
| (B) For step down transformer, maximum value of $\frac{V_{\text{secondary}}}{V_{\text{primary}}}$ | (2) 0.2 |
| (C) For step up transformer, minimum value of $\frac{V_{\text{secondary}}}{V_{\text{primary}}}$ | (3) 1.25 |
| (D) For step down transformer, minimum value of $\frac{V_{\text{secondary}}}{V_{\text{primary}}}$ | (4) 5 |

Note: (*) → Multiple Correct Type Question

Total Marks: 37

Max. Time: 24 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.7

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.8 to Q.9

(4 marks, 3 min.)

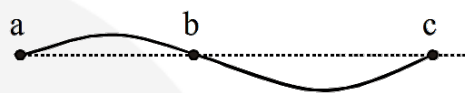
Match the Column Type Questions (No negative marking) Q.10

(8 marks, 4 min.)

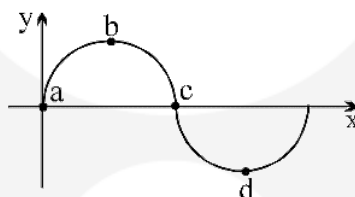
(There is partial marking in multiple correct questions)

1. The diagram below shows an instantaneous position of a string as a transverse progressive wave travels along it from left to right. Which one of the following correctly shows the direction of the velocity of the points a, b and c on the string?

- | | a | b | c |
|-----|---|---|---|
| (A) | → | → | → |
| (B) | → | ← | → |
| (C) | ↓ | ↓ | ↓ |
| (D) | ↓ | ↑ | ↓ |



2. The figure shows a simple harmonic progressive transverse wave, progressing towards the right. The particle having the maximum acceleration directed downwards is represented by:

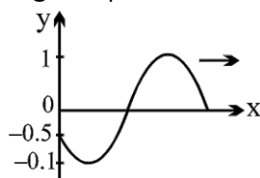


- (A) a (B) b (C) c (D) d

3. A wave propagates in a string in the positive x-direction with velocity v. The shape of the string at $t = t_0$ is given by $f(x, t_0) = A \sin \left(\frac{x^2}{a^2} \right)$: then, the wave equation at any instant t is given by

- | | |
|---|---|
| (A) $g(x, t) = A \sin \frac{[x - v(t - t_0)]^2}{a^2}$ | (B) $g(x, t) = A \sin \frac{[x + v(t - t_0)]^2}{a^2}$ |
| (C) $g(x, t) = A \sin \frac{[x - v(t + t_0)]^2}{a^2}$ | (D) $g(x, t) = A \sin \frac{[x + v(t + t_0)]^2}{a^2}$ |

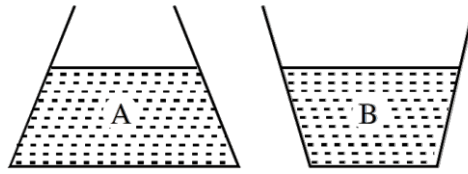
4. The equation of a wave travelling along the positive x-axis, as shown in figure at $t = 0$ is given by



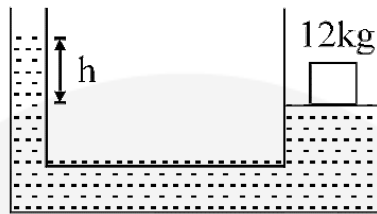
- (A) $\sin \left(kx - \omega t + \frac{\pi}{6} \right)$ (B) $\sin \left(kx - \omega t - \frac{\pi}{6} \right)$ (C) $\sin \left(\omega t - kx + \frac{\pi}{6} \right)$ (D) $\sin \left(\omega t - kx - \frac{\pi}{6} \right)$



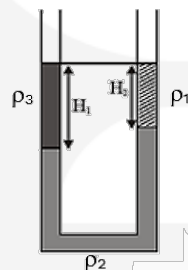
5. Two vessels A and B of same cross-sections area of base; contains same liquid, up to the same height. Choose the correct option(s) :



- (A) Pressure at bottom of vessels, A is more than pressure at bottom of vessel B
 (B) Pressure at bottom of vessel B is more than pressure at bottom of vessel A
 (C) If kept on weighing machine, reading is more for vessel A
 (D) If kept on weighing machine, reading is more for vessel B
6. The area of cross-section of the wider tube shown in figure is 800 cm^2 . If a mass of 12 kg is placed on the massless piston, the difference in height h in the level of water tubes is :



- (A) 10 cm (B) 6 cm (C) 15 cm (D) 2 cm
7. There are three different liquids, with densities ρ_1 , ρ_2 and ρ_3 , in a U-shaped container as shown in picture. The lengths shown are $H_1 = 15 \text{ cm}$ and $H_2 = 10 \text{ cm}$. Which of the following equations gives the correct relation between the densities of the fluids in the container?



- (A) $3\rho_3 = 2\rho_1 + \rho_2$ (B) $\rho_3 = 2\rho_1 + 3\rho_2$
 (C) $2\rho_3 = 3\rho_1 + \rho_2$ (D) $\rho_3 = 3\rho_1 + 2\rho_2$

- 8.* A wave is represented by the equation :

$$y = A \sin \left[20\pi x + 30\pi t + \frac{\pi}{6} \right]$$

Where x is in meters and t is in seconds. The expression represents :

- (A) Wave is travelling in the positive x -direction with velocity of 1.5 m/s
 (B) Wavelength of the wave is 10 cm .
 (C) Frequency of the wave is 15 Hz .
 (D) Wave is travelling along positive y -direction with velocity of $\frac{2}{3} \text{ m/s}$.

- 9.* Equation of a travelling wave is $y = 5e^{(-ax^2 - bt^2 - 2\sqrt{ab}xt)}$, where x & y are in meter and t is in sec.
 (Given : $a = 25 \text{ m}^2$ and $b = 9 \text{ sec}^{-2}$)

- (A) Travelling wave propagates along (+) x direction
 (B) Travelling wave propagates along (–) x direction
 (C) Speed of wave is $\frac{3}{5} \text{ m/sec}$.
 (D) Maximum displacement of particle is 5 m .



✓ 10

Following is the equation of a travelling wave (all in SI units) :

$$y = (0.02) \sin 2\pi(10t - 5x)$$

match the followings.

List-I

- (P) Speed of wave
- (Q) Frequency of wave
- (R) Wavelength of wave
- (S) Maximum particle speed

List-II (only magnitude in SI units)

- (i) 10
- (ii) 0.4π
- (iii) 2
- (iv) 0.2



Note: (*) → Multiple Correct Type Question

Total Marks: 26

Max. Time: 21 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.4

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.5

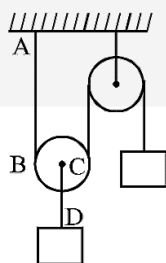
(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.6 to Q.10

(2 marks, 2 min.)

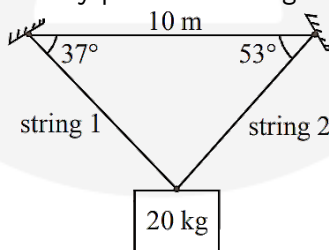
(There is partial marking in multiple correct questions)

1. Both the strings, shown in figure are made of same material and diameter of CD is double that of AB. The pulleys are light. The speed of a transverse wave in the string AB is v_1 and in CD it is v_2 , then $\frac{v_1}{v_2}$ is.



- (A) 1 (B) 2 (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

2. A mass of 20 kg is hanging with support of two strings of same linear mass density. Now pulses are generated in both strings at same time near the joint at mass. Ratio of time, taken by a pulse to travel through string 1 to that taken by pulse on string 2 is :



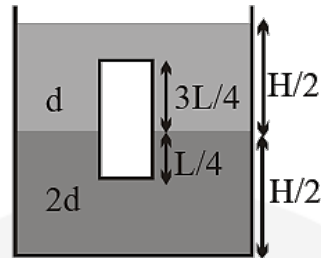
- (A) $\frac{4}{3}$ (B) $\frac{\sqrt{4}}{\sqrt{3}}$ (C) $\frac{4\sqrt{4}}{3\sqrt{3}}$ (D) $\frac{3\sqrt{4}}{4\sqrt{3}}$

3. A heavy but uniform rope of length L is suspended from a ceiling. A particle is dropped from the ceiling at the instant when the bottom end is given a transverse wave pulse. Where will the particle meet the pulse.

- (A) At a distance $\frac{2L}{3}$ from the bottom (B) At a distance $\frac{L}{3}$ from the bottom
(C) At a distance $\frac{3L}{4}$ from the bottom (D) None of the above



4. A container of a large uniform cross-sectional area A resting on a horizontal surface holds two immiscible, non viscous and incompressible liquids of densities d and $2d$ each of height $\frac{H}{2}$ as shown. The lower density liquid is open to atmosphere. A homogenous solid cylinder of length L $\left(< \frac{H}{2} \right)$, cross-sectional area $\frac{A}{5}$ is immersed such that it floats with its axis vertical to the liquid- liquid interface with length $\frac{L}{4}$ in denser liquid. If D is the density of the solid cylinder then.



- (A) $D = \frac{3d}{2}$ (B) $D = \frac{d}{2}$ (C) $\frac{2d}{3}$ (D) $\frac{5d}{4}$

5.* A long string of mass per unit length 0.2 kg/m is stretched to a tension of 500 N . Choose the correct option(s):

- (A) Speed of transverse wave on string is 50 ms^{-1}
 (B) Speed of transverse wave on string is 25 ms^{-1}
 (C) Mean power required to maintain a travelling wave of amplitude 10 mm and wavelength 0.5 m is 197 W .
 (D) Mean power required to maintain a travelling wave of amplitude 10 mm and wavelength 0.5 m is 150 W .

Paragraph for question no 6 to 8

Sinusoidal waves travel on five identical strings. Use the mathematical forms of the waves given below. In the expressions given below x is in m , y is in millimeters and t is in seconds.

- (a) $y = (2\text{mm}) \sin(2x - 4t)$
 (b) $y = (3\text{mm}) \sin(4x - 10t)$
 (c) $y = (2.5\text{mm}) \sin(6x - 12t)$
 (d) $y = (1\text{mm}) \sin(8x - 16t)$
 (e) $y = (4\text{mm}) \sin(10x - 20t)$

6 Four of the strings have the same tension, but the fifth has a different tension. Identify it.

- (A) a (B) b (C) d (D) e

7. Two of the strings have the same energy in one wavelength, identify the pair.

- (A) b and c (B) a and e (C) a and d (D) b and d

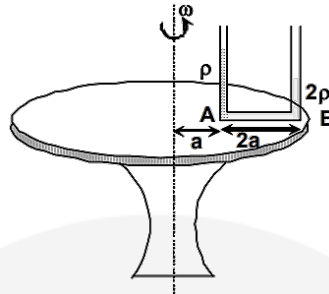
8. Which string has maximum slope at $x = 8 \text{ m}$ and at $t = 4 \text{ sec}$.

- (A) a (B) c (C) d (D) e



Paragraph for question No. 9 to 10

A U tube containing two different liquids of density ρ and 2ρ is fixed vertically on a rotating table as shown in the figure. Initially table is at rest and table can rotate about a vertical axis passing through the centre of the table. The interface of two liquids of densities ρ and 2ρ respectively lies at the point A in the U tube kept on the table at rest. The height of liquid column above A is $\frac{8}{3}a$, where $AB = 2a$. The cross-sectional area of the tube is S . Now the table is whirled with angular velocity ω about a vertical axis as shown in figure such that the interface of the liquids shifts towards B by $\frac{2}{3}a$, then at this instant.



9. Find $P_B - P_A$:
(A) ρga (B) $2\rho ga$ (C) $3\rho ga$ (D) $4\rho ga$
10. Find the value of ω .
(A) $\sqrt{\frac{9g}{4a}}$ (B) $\sqrt{\frac{9g}{32a}}$ (C) $\sqrt{\frac{9g}{8a}}$ (D) $\sqrt{\frac{9g}{16a}}$

Note: (*) → Multiple Correct Type Question

Total Marks: 34

Max. Time: 24 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.7 to Q.8

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9 to Q.10

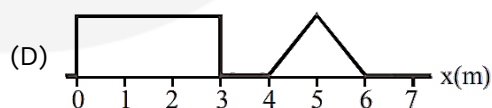
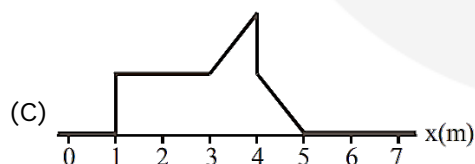
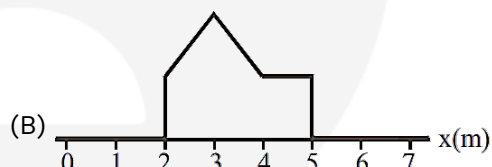
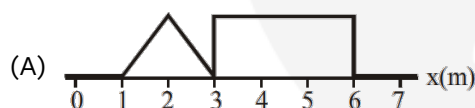
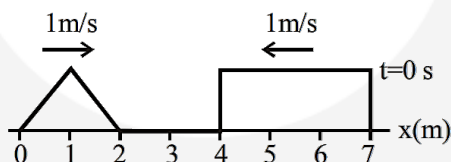
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. In which case is the power being delivered by a given progressive sinusoidal wave on a given string is doubled?

- (A) The wave amplitude is doubled (keeping the frequency the same)
- (B) The wave frequency is cut in half (keeping the amplitude the same)
- (C) the string is made two time as taut (keeping its linear density the same)
- (D) The diameter of the string is doubled (keeping the tension the same)

2. Two pulses on a string approach each other at speeds of 1 m/s. What is the shape of the string at $t = 3$ s.



3. Standing waves are set on a stretched sonometer wire. Choose the correct statement:

- (A) Both kinetic and potential energy minimises together in the wire at antinode.
- (B) Both kinetic and potential energy maximises together in the wire at antinode.
- (C) Both kinetic and potential energy maximises together at all points in the wire.
- (D) Total energy of one loop (minimum distance two nodes) varies with time.

4. A steel string of area 10 mm^2 and length 60 cm is fixed at both ends. It vibrates at 600 Hz in the third harmonic. What is the tension in the string? The density of steel is 8000 kg/m^3 .

- (A) 4608 N
- (B) 4508 N
- (C) 4550 N
- (D) 4600 N



5. Two solids P and Q float in water. It is observed that P floats with half of its volume immersed and Q floats with $\left(\frac{2}{3}\right)^{\text{rd}}$ of its volume immersed. The ratio of densities of P and Q is:
(A) 4/3 (B) 3/4 (C) 2/3 (D) 3/2
6. A block has weight 15 N in air. It weighs 12 N when immersed in water. When immersed in another liquid, it weighs 13 N. Then find the ratio of density of block and the other liquid. ($\rho_{\text{block}}/\rho_{\text{liquid}}$)
(A) $\frac{15}{2}$ (B) 15 (C) $\frac{15}{4}$ (D) $\frac{13}{12}$
- ✓ 7* A wave disturbance in a medium is described by -
 $y(x, t) = 0.02 \cos(50\pi t + \pi/2) \cos(10\pi x)$,
Where x and y are in meters and t in seconds.
(A) A node occurs at x = 0.15 m (B) An antinode occurs at x = 0.3 m
(C) The speed of the wave is 5.0 m/s (D) The wavelength is 0.2 m
- ✓ 8* String of length L whose one end is at x = 0, vibrates according to the relations given by different equations. Choose the CORRECT statement(s).
(A) $y = A \sin \frac{\pi x}{L} \sin \omega t$ has 1 antinodes, 2 nodes
(B) $y = A \cos \frac{\pi x}{L} \sin \omega t$ has 2 antinodes, 1 nodes
(C) $y = A \sin \frac{2\pi x}{L} \sin \omega t$ has 3 nodes, 2 antinodes
(D) $y = A \cos \frac{2\pi x}{L} \sin \omega t$ has 3 antinodes, 2 nodes
- ✓ 9 Two identical wires, one made of iron and the other of aluminium are stretched along-side on a sonometer board by equal stretched forces.
[Density of iron = 7.5 gm/cc, density aluminium = 2.7 g/cc]
If the frequency of lowest harmonic for which both wires vibrate in unison is 100α Hz, fill the value of α . (Given that the length of the wires is 1 m, their diameters 1 mm and tension 75π Newton)
- ✓ 10 A solid floats with 1/3 of its volume immersed in liquid-A and with 3/5 of its volume immersed in liquid-B. A homogenous mixture is created by mixing equal volumes of both liquid-A and B, then fraction of solid immersed in mixture is a/b, fill the value of $\frac{a+b}{2}$ in

Note: (*) → Multiple Correct Type Question

Total Marks: 35

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.6 to Q.8

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9 to Q.10

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

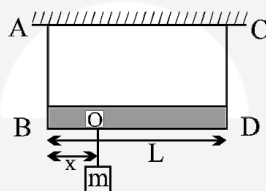
1. A string of length 1 m fixed at both ends is vibrating in 3rd overtone. Tension in string is 200 N and linear mass density is 5 gm/m. frequency of these vibrations is :

- (A) 200 Hz (B) 250 Hz (C) 300 Hz (D) 400 Hz

2. A steel wire of length L, cross-sectional area A and density ρ is fixed between two rigid supports with the wire just taut. Y = young's modulus and α = coefficient of thermal expansion of steel. If the wire temperature is now lowered by ΔT , then the frequency of the fundamental note produced by plucking the wire in the middle is :

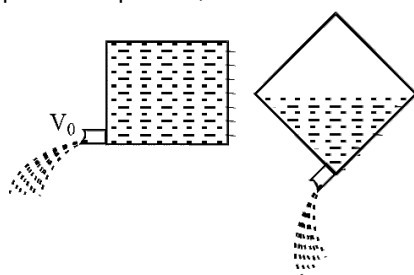
- (A) $\frac{1}{L} \left[\frac{\alpha Y L \Delta T}{\rho} \right]^{1/2}$ (B) $\frac{1}{2L} \left[\frac{\alpha Y \Delta T}{\rho} \right]^{1/2}$ (C) $\frac{1}{2L} \left[\frac{\alpha Y \Delta T}{\rho L} \right]^{1/2}$ (D) $\frac{1}{L} \left[\frac{\alpha Y \Delta T}{\rho} \right]^{1/2}$

3. A massless rod is suspended by two identical strings AB and CD of equal length. A block of mass m is suspended from point O such that BO is equal to 'x'. Further, it is observed that the frequency of 1st harmonic (fundamental frequency) in AB is equal to 2nd harmonic frequency in CD. Then, length of BO is



- (A) $\frac{L}{5}$ (B) $\frac{L}{4}$ (C) $\frac{4L}{5}$ (D) $\frac{3L}{4}$

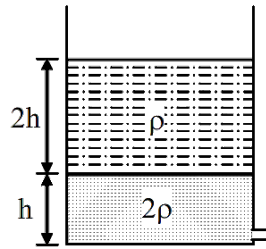
4. A cubical box of oil has a small spout located in one of the bottom corners. When the box is full and placed on a level surface, opening the spout results in a flow of oil with a initial speed of v_0 (see figure). When the box is half empty, someone tilts it at 45° so that the spout is at the lowest point (see figure). When the spout is opened, the oil will flow out with a speed of :



- (A) v_0 (B) $v_0/2$ (C) $v_0/\sqrt{2}$ (D) $v_0/\sqrt[4]{2}$

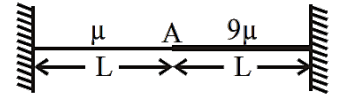


5. The velocity of the liquid coming out of a small hole of a vessel containing two different liquids of densities 2ρ and ρ as shown in figure is:



- (A) $\sqrt{6gh}$ (B) $2\sqrt{gh}$ (C) $2\sqrt{2gh}$ (D) $\sqrt{gh}$

- 6*. A light string is tied at one end to fixed support and to a heavy string of equal length L at the other end as shown in figure. Mass per unit length of the strings are μ and 9μ and the tension is T . Find the possible values of frequencies such that point A is an antinode.



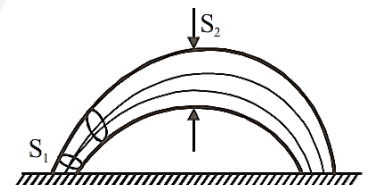
- (A) $\frac{1}{4L}\sqrt{\frac{T}{\mu}}$ (B) $\frac{3}{4L}\sqrt{\frac{T}{\mu}}$ (C) $\frac{3}{2L}\sqrt{\frac{T}{\mu}}$ (D) $\frac{7}{6L}\sqrt{\frac{T}{\mu}}$

- 7*. The equation of a plane progressive wave is $y = 0.02 \sin 8\pi \left[t - \frac{x}{20} \right]$. When it is reflected at a rarer medium (medium with higher velocity) at $x = 0$, its amplitude becomes 75% of its previous value. Incorrect equation(s) of the reflected wave is (are):

- (A) $y = 0.02 \sin 8\pi \left[t - \frac{x}{20} \right]$ (B) $y = 0.02 \sin 8\pi \left[t + \frac{x}{20} \right]$
 (C) $y = +0.015 \sin 8\pi \left[t + \frac{x}{20} \right]$ (D) $y = -0.015 \sin 8\pi \left[t + \frac{x}{20} \right]$

- 8*. Water jet is projected at an angle to the horizontal. At the point of projection, the area of the jet is S_1 and at the highest point, the area of the jet is S_2 . The initial velocity of projection is u .

- (A) The angle of projection is $\cos^{-1} \left(\frac{S_1}{S_2} \right)$
 (B) The range on the level ground is $\frac{2u^2}{g} \frac{S_1}{S_2} \sqrt{1 - \frac{S_1^2}{S_2^2}}$
 (C) The maximum height reacted from the ground is $\frac{2u^2}{g} \left(1 - \frac{S_1^2}{S_2^2} \right)$
 (D) The rate of volume flow is $S_2 u$



9. A standing wave is created in a string and is given by,

$$y = 6 \sin (100\pi t) \cos (5\pi x)$$

If mass per unit length of string is 10 gm/m then find tension in string (in N)

10. A string AB of length 5 m and linear mass density 1 kg m^{-1} is clamped at both the ends with a tension of 144 N . Find minimum frequency (in Hz) of transverse standing wave in string so that a node appears at a distance 3 m from end A.

Note: (*) → Multiple Correct Type Question

Total Marks: 27

Max. Time: 22 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.3

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.4 to Q.5

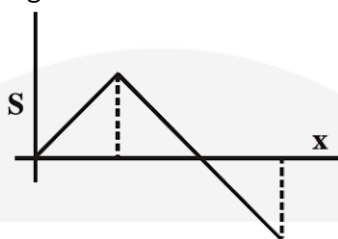
(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.6 to Q.10

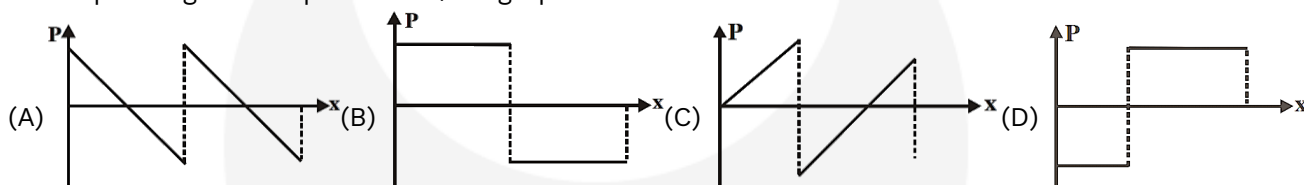
(2 marks, 2 min.)

(There is partial marking in multiple correct questions)

1. A sound wave is propagating through a medium such that displacement y vs x graph is



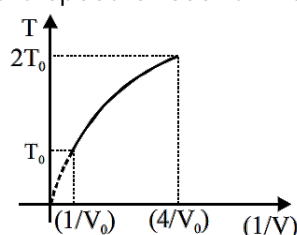
Corresponding excess pressure p vs x graph will be



2. Mark the INCORRECT statement about longitudinal pressure wave.

- (A) The density variation and the longitudinal velocity of the particles have their maxima and minima at the position of equilibrium of the particles.
- (B) At the position of maximum displacement, the density is at the ambient (environment) value.
- (C) Greatest compression occurs at the same time and place where the velocity of the particles is maximum and in direction of wave propagation.
- (D) Greatest rarefaction occurs at the same time and place where the velocity of the particles is maximum and in direction of wave propagation.

3. Figure shows a parabolic graph between T and $\frac{1}{V}$ for a mixture of a gas undergoing an adiabatic process. What is the ratio of v_{rms} and speed of sound in the mixture?



(A) $\sqrt{\frac{3}{2}}$

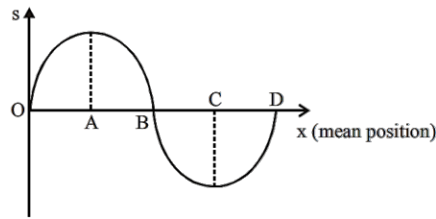
(B) $\sqrt{2}$

(C) $\sqrt{\frac{2}{3}}$

(D) $\sqrt{3}$

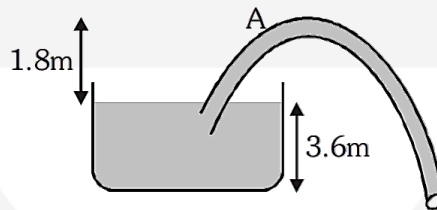


- 4.* A sound wave is travelling in air along positive x-direction. Displacement(s) of particles from their mean positions at a particular time t is shown in the figure. Choose the correct option(s) for that instant only.



- (A) Particle located at C has zero velocity
 (B) Particle located at D has its velocity in the negative direction.
 (C) The pressure at C is equal to normal atmospheric pressure.
 (D) Particles located near B are under compression.

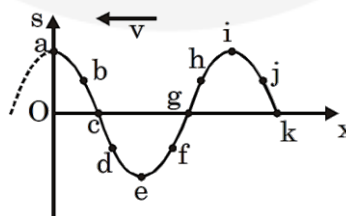
- 5.* A siphon has a uniform circular base of diameter $\frac{8}{\sqrt{\pi}}$ cm with its crest A, 1.8 m above water level as in figure. Then [use $P_0 = 10^5 \text{ N/m}^2$ & $g = 10 \text{ m/s}^2$]



- (A) Velocity of flow = $6\sqrt{2} \text{ m/sec}$
 (B) Discharge rate of flow is = $9.6\sqrt{2} \times 10^{-3} \text{ m}^3/\text{sec}$.
 (C) Absolute pressure at the crest level A is = $4.6 \times 10^4 \text{ N/m}^2$.
 (D) Speed of flow will decrease with time.

Paragraph for questions no. 6 to 8

The figure represents the instantaneous displacement-position graph of a longitudinal harmonic wave travelling along the negative x-axis. Identify the correct statement(s) related to the movement of the points shown in the figure

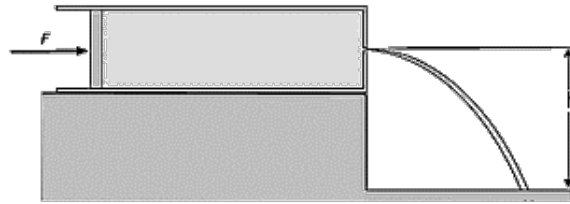


- 6.* The points moving opposite to the direction of propagation are
 (A) a (B) d (C) f (D) j
- 7.* The stationary points are
 (A) a (B) c (C) g (D) k
- 8.* The points of maximum compression are
 (A) c (B) g (C) e (D) k



Paragraph for question 9 & 10

An ideal liquid of density ρ is filled in a horizontally fixed syringe fitted with piston. There is no friction between the piston and the inner surface of the syringe. Cross-section area of the syringe is A . At one end of the syringe, an orifice of negligible cross-section area is made. When the piston is pushed into the syringe, the liquid comes out of the orifice following parabolic path and falls on the ground.



9. With what velocity the liquid comes out of the orifice?

(A) $\sqrt{\frac{F}{\rho A}}$ (B) $\sqrt{\frac{2F}{\rho A}}$ (C) $\sqrt{\frac{F + 2\rho ghA}{\rho A}}$ (D) $\sqrt{\frac{F + \rho ghA}{\rho A}}$

10. With what speed the liquid strike the ground? Neglect the air drag.

(A) $\sqrt{\frac{F + \rho ghA}{\rho A}}$ (B) $\sqrt{\frac{F + 2\rho ghA}{\rho A}}$ (C) $\sqrt{\frac{2F + \rho ghA}{\rho A}}$ (D) $\sqrt{\frac{2(F + \rho ghA)}{\rho A}}$

Note: (*) → Multiple Correct Type Question

Total Marks: 31

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.6 to Q.8

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.9 to Q.10

(2 marks, 2 min.)

(There is partial marking in multiple correct questions)

1. A point source of power 50π watts is producing sound waves. The velocity of sound is 330 m/s, density of air is 1.0 kgm^{-3} . Then at $r = \sqrt{330}$ m from the point source. [using $\pi = 22/7$]
 (A) Pressure amplitude is 10 Nm^{-2} (B) Pressure amplitude is 5 Nm^{-2}
 (C) Pressure amplitude is 6 Nm^{-2} (D) Pressure amplitude is 12 Nm^{-2}
2. The average density of Earth's crust 10 km beneath the surface is 2.7 gm/cm^3 . The speed of longitudinal seismic waves at that depth is 5.4 km/s. The bulk modulus of Earth's crust considering its behavior as fluid at that depth is :
 (A) $7.9 \times 10^{10} \text{ Pa}$ (B) $5.6 \times 10^{10} \text{ Pa}$ (C) $7.9 \times 10^7 \text{ Pa}$ (D) $1.46 \times 10^7 \text{ Pa}$
3. A sound wave of wavelength λ and angular frequency ω travels with a speed v in a medium of density ρ and bulk modulus B . Let $k = \frac{2\pi}{\lambda}$ be the propagating constant. If p and A are the pressure amplitude and displacement amplitude respectively, then the intensity of sound wave is (Choose the INCORRECT option)
 (A) $I = \frac{Bk\omega A^2}{2}$ (B) $I = \frac{v\rho^2}{2B}$ (C) $I = \frac{3p^2}{2\rho v}$ (D) $I = \frac{p^2}{2\sqrt{\rho B}}$
4. A person hears a single-frequency sound from a speaker that emits sound uniformly in all directions. The person records the decibel level reading. Which of the following processes would increase the decibel reading the most.
 (A) The power output of the speaker is doubled
 (B) The distance of the person to the speaker is halved
 (C) The frequency of the sound is doubled
 (D) The temperature of the air is doubled on the Kelvin scale.
5. Eight equal drops of water each of radius $r = 2 \text{ mm}$ are falling through air with a terminal velocity of 16 cm/s. The eight drops combine to form a big drop. The terminal velocity of drop will be.
 (A) 16 cm/s (B) 32 cm/s (C) 64 cm/s (D) 30 cm/s
6. A small source of sound operates at 1kHz, $24\pi \text{ W}$ emitting sound uniformly in all directions. The speed of sound is 360 m/s in air and density of air is 1.2 kg/m^3
 (A) The intensity of sound at a distance 10 m from the source is 0.06 W/m^2
 (B) The intensity of sound at a distance 10 m from the source is 0.24 W/m^2
 (C) The pressure amplitude at distance 10 m is 7.2 Pa.
 (D) The pressure amplitude at distance 10 m is 28.8 Pa.



7.* Two mechanical waves, $y_1 = 2 \sin 2\pi(50t - 2x)$ & $y_2 = 4 \sin 2\pi(ax + 100t)$ propagate in a medium with same speed.

- (A) The ratio of their intensities is 1 : 16 (B) The ratio of their intensities is 1 : 4
(C) The value of 'a' is 4 units (D) The value of 'a' is 2 units

8.* A solid sphere of radius r and density ρ_s is dropped in a liquid of density ρ_ℓ and coefficient of viscosity η then choose the CORRECT option(s) : (Depth of liquid is large enough & $\rho_s > \rho_\ell$)

- (A) Buoyant force applied by liquid on ball is $\frac{4}{3}\pi r^3 \rho_\ell g$.
(B) Viscous force applied by liquid on sphere when velocity of sphere is v , is $6\pi\eta r v$.
(C) For some time, speed of sphere increases and then becomes constant.
(D) Terminal velocity of sphere is $\frac{2}{9} \frac{r^2(\rho_s - \rho_\ell)g}{\eta}$

Paragraph for questions 9 and 10

A small sphere of radius R is arranged to pulsate so that its radius varies in simple harmonic motion between a minimum of $R - \Delta R$ and a maximum of $R + \Delta R$ with frequency f . This produces sound waves in the surrounding air of density ρ and bulk modulus B . (Assume the amplitude of oscillation of the sphere is the same as that of the air at the surface of the sphere)

9. The total acoustic power radiated by the sphere is

- (A) $P = 8\pi^3 \sqrt{\rho B} f^2 R^2 (\Delta R)^2$ (B) $P = \pi^3 \sqrt{\rho B} f^2 R^2 (\Delta R)^2$
(C) $P = 2\pi^3 \sqrt{\rho B} f^2 R^2 (\Delta R)^2$ (D) $P = 4\pi^3 \sqrt{\rho B} f^2 R^2 (\Delta R)^2$

10. At a distance $d \gg R$ from the centre of the sphere, the pressure amplitude is :

- (A) $P_{\max} = \pi \sqrt{\rho B} \left(\frac{Rf}{d} \right) \Delta R$ (B) $P_{\max} = 2\pi \rho B \left(\frac{Rf}{d} \right) \Delta R$
(C) $P_{\max} = 2\pi \sqrt{\rho B} \left(\frac{Rf}{d} \right) \Delta R$ (D) $P_{\max} = 2\pi \sqrt{\rho B} \left(\frac{R^2 f}{d^2} \right) \Delta R$

Note: (*) → Multiple Correct Type Question

Total Marks: 35

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.6 to Q.8

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9 to Q.10

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. An open pipe is suddenly closed at one end with the result that the frequency of third harmonic of the closed pipe is found to be higher by 100 Hz than the fundamental frequency of the open pipe. The fundamental frequency of the open pipe is :

(A) 200 Hz (B) 300 Hz (C) 240 Hz (D) 480 Hz

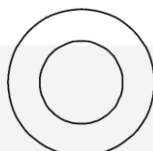
2. Two sound waves of slightly different frequencies have amplitude ratio $\frac{11}{9}$. What is the difference of sound levels in decibels of maximum and minimum intensities heard at a point ?

(A) 100 (B) 10 (C) 16 (D) 20

3. Two soap bubbles with radii r_1 and r_2 ($r_1 > r_2$) come in contact. Their common surface has a radius of curvature r . Choose the correct option:

(A) $r = \frac{r_1 + r_2}{2}$ (B) $r = \frac{r_1 r_2}{r_1 - r_2}$ (C) $r = \frac{r_1 r_2}{r_1 + r_2}$ (D) $r = \sqrt{r_1 r_2}$

4. A soap bubble of radius R is surrounded by another soap bubble of radius $2R$, as shown. Take surface tension = S . Then, the pressure inside the smaller soap bubble, in excess of the atmospheric pressure will be



Atmosphere

(A) $4S/R$ (B) $3S/R$ (C) $6S/R$ (D) None of these

5. An open capillary tube is lowered in a vessel filled with mercury. The difference between the levels of the mercury in the vessel and in the capillary tube $\Delta h = 4.6$ mm. What is the radius of curvature of the mercury meniscus in the capillary tube? Surface tension of mercury is 0.46 N/m, density of mercury is 13.6 gm/cc.

(A) $\frac{1}{340}$ m (B) $\frac{1}{680}$ m (C) $\frac{1}{1020}$ m (D) Information insufficient

- 6.* Which of the following statements is /are INCORRECT ?

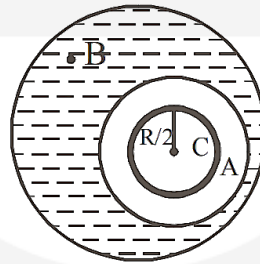
(A) In wave motion, particle speed depends on tension and linear mass density of the string.
 (B) In wave motion particle speed can be equal to the wave speed.
 (C) The speed of sound in air at 20°C is twice that at 5°C .
 (D) A 60 dB sound has twice the intensity of a 30 dB sound.



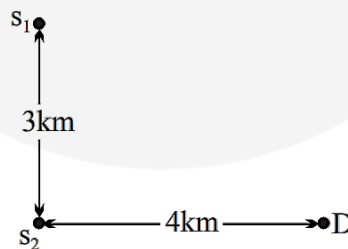
- 7.* Two coherent waves represented by $y_1 = A \sin\left(\frac{2\pi x_1}{\lambda} - \omega t + \frac{\pi}{4}\right)$ and $y_2 = A \sin\left(2\pi \frac{x_2}{\lambda} - \omega t + \frac{\pi}{6}\right)$ are superposed. The two waves will produce:
- (A) Constructive interference at $(x_1 - x_2) = 2\lambda$
(B) Constructive interference at $(x_1 - x_2) = \frac{23}{24}\lambda$
(C) Destructive interference at $(x_1 - x_2) = 1.5\lambda$
(D) Destructive interference at $(x_1 - x_2) = \frac{11}{24}\lambda$.

- 8.* The second overtone of an open pipe A and a closed pipe B have the same frequency at a given temperature. It follows that the ratio of the
- (A) length of A and B is 4 : 3
(B) Fundamental frequencies of A and B is 5 : 6
(C) length of B to that of A is 5 : 6
(D) Frequencies of first overtone of A and B is 10 : 9

9. Suppose there is an air bubble of radius $\frac{R}{2}$ inside the air cavity of radius R inside a water drop of radius 3R as shown in figure. Find the ratio of gauge pressure at point C to gauge pressure at point B.



10. Two point sound sources S_1 and S_2 are both of same power and send out sound waves in the same phase. The wavelength of both the waves is $\frac{48}{5}$ m. The intensity due to S_2 alone at D is 25 W/m^2 . The resultant intensity at D is I. Find I (in W/m^2)



Note: (*) → Multiple Correct Type Question

Total Marks: 31

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.6

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.7 to Q.8

(2 marks, 2 min.)

Subjective Type Questions (No negative marking) Q.9 to Q.10

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

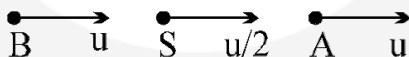
1. Two identical tuning forks are vibrating at a frequency of 535 Hz. A mischievous Physics professor shaves the forked end of one of the tuning forks so that its mass is smaller than that of the other fork. When the two forks are set into vibration, 5 beat/sec. are heard. What is the frequency of the damaged fork?

- (A) 535 Hz (B) 540 Hz (C) 530 Hz (D) 545 Hz

2. A sound source of frequency 512 Hz approaches a stationary observer with a velocity of 36 km/hr. The speed of sound is 330 m/s. What is the frequency heard by the observer?

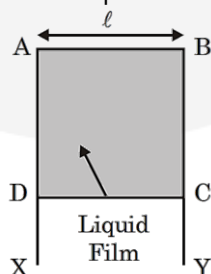
- (A) 522 Hz (B) 528 Hz (C) 524 Hz (D) 532 Hz

3. A source of sound is moving with velocity $u/2$ and two observers A and B are moving with velocity u as shown. Find ratio of wavelength received by A and B. Given that velocity of sound is $10u$.



- (A) $\frac{19}{21}$ (B) $\frac{17}{21}$ (C) $\frac{21}{23}$ (D) $\frac{17}{23}$

4. A liquid film is formed over a frame ABCD as shown in figure. A massless wire CD can slide without friction. The mass to be hung from CD to keep it in equilibrium is (Surface tension of liquid is T):



- (A) Tl/g (B) $2Tl/g$ (C) $g/2Tl$ (D) $Tl/2g$

5. 'n' drops of a liquid, each with surface energy E , join to form a single drop.

(A) Some energy will be released in the process.

(B) Some energy will be absorbed in the process

(C) The energy released or absorbed will be $\frac{E}{n}(n - n^{2/3})$

(D) The energy released or absorbed will be $nE(2^{2/3} - 1)$



- 6.* Consider a spherical bubble of radius R at a depth of h from the surface of water. At the instant when force becomes zero (atmospheric pressure is P_0 , surface tension, viscosity & density of the water is T , η & ρ), then (neglect density of air in compare to water):

- (A) Pressure inside the bubble is $P_0 + \rho gh + \frac{2T}{R}$
(B) Velocity of the sphere cannot remain constant
(C) Velocity of the sphere is given by $\frac{2}{9} \frac{\rho g r^2}{\eta}$
(D) As sphere moves up its radius increases

Paragraph for Question No. 7 and 8

A car traveling at 25 m/s is approaching a stationary wall. Wind is blowing in a direction opposite to car at 5 m/s. At an instant, the car driver briefly blows the horn. He heard an echo exactly $\frac{11}{3}$ seconds later after sounding the horn and notes that its frequency is 156 Hz.

(Take speed of sound 330 m/s)

7. The actual frequency of horn is:
(A) 134 Hz (B) 136 Hz (C) 138 Hz (D) 140 Hz
8. At what distance from the wall the driver blows the horn?
(A) 600 m (B) 650 m (C) 700 m (D) 750 m
9. ✓ An open pipe 40 cm long and a closed pipe 31 cm long, both having same diameter, are producing their first overtone, and these are in unison. Determine the end correction in cm of these pipe.
10. Two trains move towards each other with a speed of 72 km/hr each. Sound speed is 340 m/s. One of the train emits a signal of frequency $f_0 = 64$ Hz. What will be the frequency 'f' (in Hz) heard by the other train passengers?



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-63

Note: (*) → Multiple Correct Type Question

Total Marks: 42

Max. Time: 32 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.8

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.9 to Q.10

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.11 to Q.15

(2 marks, 2 min.)

(There is partial marking in multiple correct questions)

1. Three coherent waves of equal frequencies having amplitude $10\ \mu\text{m}$, $4\ \mu\text{m}$ and $7\ \mu\text{m}$ respectively, arrive at a given point with successive phase difference of $\pi/2$. The amplitude of the resulting wave in μm is given by:

- (A) 5 (B) 6 (C) 3 (D) 4

2. Two waves of sound having intensities I and $4I$ interfere to produce interference pattern. The phase difference between the waves is $\pi/2$ at point A and π at point B. Then the difference between the resultant intensities at A and B is:

- (A) $2I$ (B) $4I$ (C) $5I$ (D) $7I$

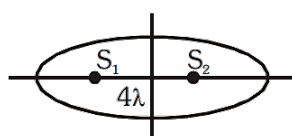
3. The ratio of maximum to minimum intensity due to superposition of two waves is $\frac{49}{9}$. Then the ratio of the intensity of component waves is :

- (A) $\frac{25}{4}$ (B) $\frac{16}{25}$ (C) $\frac{4}{49}$ (D) $\frac{9}{49}$

4. The ratio of intensities between two coherent sound sources is $4 : 1$. The difference of loudness in dB between maximum and minimum intensities when they interfere in space is :

- (A) $10 \log 2$ (B) $20 \log 3$ (C) $10 \log 3$ (D) $20 \log 2$

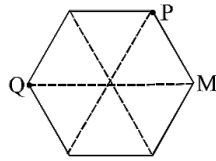
5. S_1 , S_2 are two coherent sources of sound located along x -axis separated by 4λ , where λ is wavelength of sound emitted by them. Number of maxima located on the elliptical boundary around it will be:



- (A) 16 (B) 12 (C) 8 (D) 4

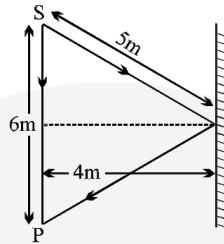


6. Two sound sources emitting sound of wavelength 1 m are located at points P and Q shown in figure. All sides of the polygon are equal and of length 1 m. The intensity of sound at M due to both the individual sources is I_0 . What will be the intensity of sound at point M when both the sources are on?



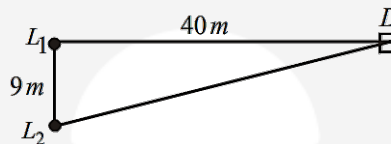
- (A) $4I_0$ (B) $2I_0$ (C) I_0 (D) $(1/2)I_0$

7. A person standing at a distance of 6 m from a source of sound receives sound wave in two ways, one directly from the source and other after reflection from a rigid boundary as shown in the figure. The maximum wavelength for which, the person will receive maximum sound intensity, is:



- (A) 4 m (B) $\frac{16}{3}$ m (C) 2 m (D) $\frac{8}{3}$ m

8. Two loudspeakers L_1 and L_2 , driven by a common oscillator and amplifier, are arranged as shown. The frequency of the oscillator is gradually increased from zero and the detector at D records a series of maxima and minima. If speed of sound is 330 m/s, then the frequency at which the first maximum is observed is:

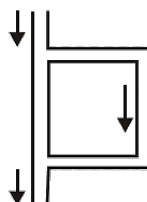


- (A) 165 Hz (B) 330 Hz (C) 495 Hz (D) 660 Hz

- 9.* Three coherent sources kept along the same line produce intensity I_0 each at point P on this line. When S_1 and S_2 are switched on simultaneously, intensity at point P is $2I_0$. When S_2 and S_3 are switched on simultaneously, intensity at point P is $2I_0$. Then

- (A) When S_1 and S_3 are switched on simultaneously, intensity at point P can be $2I_0$.
 (B) When S_1 and S_3 are switched on simultaneously, intensity at point P can be 0
 (C) When all 3 sources are switched on simultaneously, intensity at point P can be I_0 .
 (D) When all 3 sources are switched simultaneously, intensity at point P can be $3I_0$.

- 10.* A sound consists of four frequencies $\rightarrow$ 300 Hz, 900 Hz, 2400 Hz and 4500 Hz. A sound 'filter' is made by passing this sound through a bifurcated pipe as shown. The sound waves have to travel a distance of 50 cm more in the right branch-pipe than in the straight pipe. The speed of sound in air is 300 m/s. Then, which of the following frequencies will be almost completely muffled or "silenced" at the outlet?

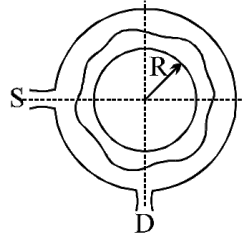


- (A) 300 Hz (B) 900 Hz (C) 2400 Hz (D) 4500 Hz



Paragraph for question no. 11 to 15

A narrow tube is bent in the form of a circle of radius R , as shown in the figure. Two small holes S and D are made in the tube at the positions right angle to each other. A source placed at S generated a wave of intensity I_0 which is equally divided into two parts : One part travels along the longer path, while the other travels along the shorter path. Both the waves meet at the point D where a detector is placed.



- 11.*** If a maxima is formed at the detector then, the magnitude of wavelength λ of the wave produced is given by :
- (A) πR (B) $\frac{\pi R}{2}$ (C) $\frac{\pi R}{4}$ (D) $\frac{2\pi R}{3}$
- 12.*** If the minima formed at the detector then, the magnitude of wavelength λ of the wave produced is given by :
- (A) $2\pi R$ (B) $\frac{3\pi R}{2}$ (C) $\frac{2\pi R}{3}$ (D) $\frac{2\pi R}{5}$
- 13.** The maximum intensity produced at D is given by :
- (A) $4I_0$ (B) $2I_0$ (C) I_0 (D) $3I_0$
- 14.** The maximum value of λ to produce a maxima at D is given by :
- (A) πR (B) $2\pi R$ (C) $\frac{\pi R}{2}$ (D) $\frac{3\pi R}{2}$
- 15.** The maximum value of λ to produce a minima at D is given by :
- (A) πR (B) $2\pi R$ (C) $\frac{\pi R}{2}$ (D) $\frac{3\pi R}{2}$

Note: (*) → Multiple Correct Type Question

Total Marks: 38

Max. Time: 25 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.6

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.7 to Q.9

(4 marks, 3 min.)

Match the Column Type Questions (No negative marking) Q.10

(8 marks, 4 min.)

(There is partial marking in multiple correct questions)

- In Young's double slit experiment, if the widths of the slit are in the ratio 4 : 9, ratio of intensity of maxima to intensity of minima will be :
 (A) 25 : 1 (B) 9 : 4 (C) 3 : 2 (D) 81 : 16
- Consider an YDSE that has different slit widths, as a result, amplitude of waves from two slits are A and 2A, respectively. If I_0 be the maximum intensity of the interference pattern, then intensity of the pattern at a point where phase difference between waves is ϕ is :
 (A) $I_0 \cos^2 \phi$ (B) $\frac{I_0}{3} \sin^2 \frac{\phi}{2}$ (C) $\frac{I_0}{9} [5 + 4 \cos \phi]$ (D) $\frac{I_0}{9} [5 + 8 \cos \phi]$
- In a young's double slit experiment, the intensity at a point where the path difference is $\frac{\lambda}{6}$ (λ being the wavelength of the light used) is I. If I_0 denotes the maximum intensity, then (I/I_0) is equal to:
 (A) $\frac{1}{\sqrt{2}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$
- In a Young's double slit interference pattern, the intensity of the central fringe at $P = I_0$, When one slit width is reduced to one forth, the intensity at P will be:
 (A) $\frac{I_0}{2}$ (B) $\frac{I_0}{4}$ (C) $\frac{9}{16} I_0$ (D) $\frac{3}{4} I_0$
- In a young's double slit experiment, 12 fringes are observed to be formed in a certain segment of the screen when light of wavelength 600 nm is used. If the wavelength of light is changed to 400 nm, number of fringes observed in the same segment of the screen is given by:
 (A) 12 (B) 18 (C) 24 (D) 30
- Intensities of light due to the two slits of YDSE are I and 4I. How far from the central maxima will the intensity be equal to the average intensity on the screen? (β is the fringe width)
 (A) $\frac{\beta}{2}$ (B) $\frac{\beta}{3}$ (C) $\frac{\beta}{6}$ (D) $\frac{\beta}{4}$



7.* Coherent light consisting of two wavelengths $\lambda_1 = 4500 \text{ \AA}$ & $\lambda_2 = 6000 \text{ \AA}$ is sent through both slits of a young's double slit apparatus (Source is placed symmetrically for both the slits). Choose the correct option/s :

- (A) Central maxima for both wavelengths will coincide
- (B) The third order bright fringe of λ_1 will coincide with the fourth order bright fringe of λ_2 .
- (C) The third order dark fringe of λ_2 will coincide with fourth order dark fringe of λ_1 .
- (D) The fringes of wavelength λ_1 will be wider than the fringes of wavelength λ_2 .

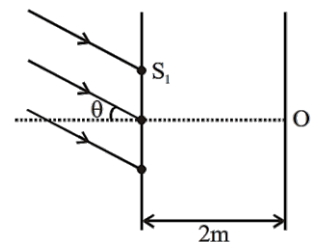
8.* A double-slit apparatus is set up and green light is used to form the interference pattern. The green light is then replaced by red light. Which of the following changes may allow the spacing of the bright bands on the screen to remain unaffected by the change of light colour?

(Given that the distance between the slits is d and that between the plane of the slits and the screen is D):

- (A) Decrease ' D '
- (B) Increase ' d '
- (C) Increase ' d ' and decrease ' D '
- (D) Increase the brightness of the light source

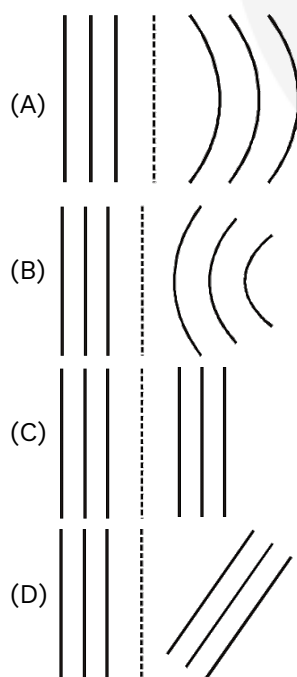
9.* A parallel beam of light ($\lambda = 5000 \text{ \AA}$) is incident at an angle $\theta = 30^\circ$ with the normal to the slit plane in a Young's double slit experiment. The intensity due to each slit is I_0 . Point O is equidistant from S_1 and S_2 . The distance between slits is 1 mm, then :

- (A) The intensity at O is $4 I_0$.
- (B) The intensity at O is zero.
- (C) The intensity at a point on the screen 4 mm above O is $4 I_0$.
- (D) The intensity at a point on the screen 4 mm above O is zero.



10. Wave front shown as dark lines in column-I, travelling from left to right is incident on a optical device marked by dotted line. Match possible wave front with the optical device shown in column-II.

Column-I



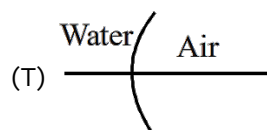
Column-II

(P) Prism

(Q) Converging lens

(R) Diverging lens

(S) parallel sided glass slab



Note: (*) → Multiple Correct Type Question

Total Marks: 39

Max. Time: 26 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.6 to Q.8

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9

(4 marks, 3 min.)

Match the Column Type Questions (No negative marking) Q.10

(8 marks, 4 min.)

(There is partial marking in multiple correct questions)

- A glass slab of thickness 4 cm contains the same number of waves as 5 cm of water when both are traversed by the same monochromatic light. If the refractive index of water is $4/3$, what is that of glass.

(A) $5/3$ (B) $5/4$ (C) $16/15$ (D) $3/2$
- In Young's double slit experiment with light of wavelength $\lambda = 600\text{nm}$, intensity of central fringe is I_0 . Now one of the slit is covered by glass plate of refraction index 1.4 and thickness $t = 5\text{ }\mu\text{m}$, the new intensity at the same point on screen will be:

(A) $\frac{I_0}{4}$ (B) $\frac{3I_0}{4}$ (C) I_0 (D) $\frac{I_0}{2}$
- In Young's double slit experiment, the figures are displaced by a distance x when a glass plate of refractive index 1.5 is introduced in the path of one of the beams. When this plate is replaced by another plate of the same thickness, the shift of fringes is $(3/2)x$. The refractive index of the second plate is :

(A) 1.75 (B) 1.50 (C) 1.25 (D) 1.00
- A transparent slab of thickness t and refractive index μ is inserted in front of upper slit of YDSE apparatus. The wavelength of light used is λ . Assume that there is no absorption of light by the slab. Mark the INCORRECT statement.

(A) The intensity of dark fringes will be zero, if slits are identical.

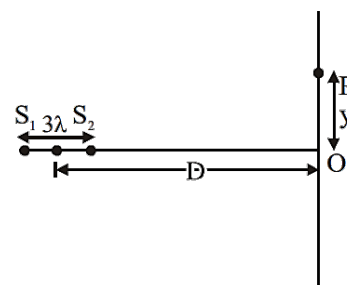
(B) The change in optical path due to insertion of plate is μt .

(C) For making intensity maximum at the centre of screen, the thickness could be $\frac{5\lambda}{(\mu - 1)}$

(D) For making intensity zero at the centre of screen, the thickness could be $\frac{5\lambda}{2(\mu - 1)}$
- Figure shows two coherent microwave source S_1 and S_2 emitting waves of wavelength λ and separated by a distance 3λ . For $\lambda \ll D$ and $y \neq 0$, the minimum value of y for point P to be an intensity maximum is

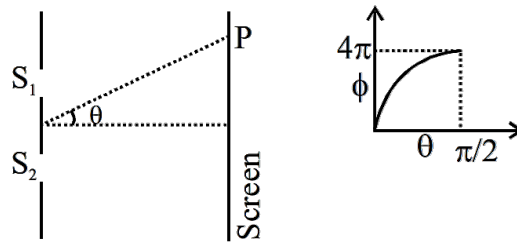
(A) $\frac{D\sqrt{5}}{2}$ (B) $\frac{D\sqrt{7}}{2}$

(C) $\frac{D\sqrt{3}}{2}$ (D) $\frac{D\sqrt{15}}{2}$

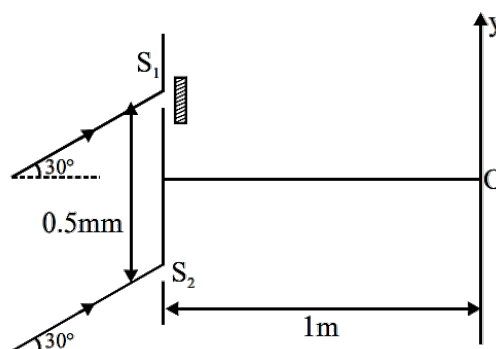




- 6.* In a YDSE setup, we plot the phase difference (ϕ) between both waves at point P on the screen against the angular position (θ) of point P on the screen. The graph is as shown below.

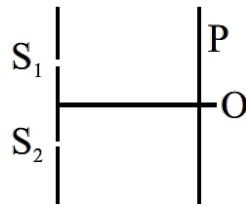


- (A) The distance $S_1S_2 = 2\lambda$
(B) There are a total of 4 minima on the screen
(C) The first maxima above the centre is at $\theta = \frac{\pi}{4}$
(D) At $\theta = \frac{\pi}{6}$, intensity is maximum
- 7.* Bright and dark fringes are seen on a screen when light of wavelength λ , from a single source reaches at two narrow slits of width w and a short distance d apart. The number of fringes per unit length on the screen can be halved
- (A) If the distance between the slits is changed to $d' = \frac{d}{2}$
(B) If the wavelength is changed to $\lambda' = 2\lambda$
(C) If the distance between the slits is changed to $d' = \frac{d}{2}$ and the wavelength is changed to $\lambda' = 4\lambda$
(D) Only if the width of the slits is changed to $w' = 2w$
- 8.* A thin paper of thickness 0.02 mm having refractive index 1.45 is pasted across one of the slits in a Young's double slit experiment. The paper transmits $\frac{4}{9}$ fraction of light falling on it ($\lambda_{\text{light}} = 600\text{nm}$).
- (A) Amplitude of light wave transmitted through the paper will be $\frac{2}{3}$ times that of incident wave
(B) The ratio of maximum and minimum intensity in the fringe pattern will be 25
(C) The total number of fringe crossing the centre if an identical paper is pasted on the other slit is 15.
(D) The ratio of maximum and minimum intensity in the fringe pattern will be 5.
9. In Young's double slits experiment, parallel monochromatic electromagnetic waves of wavelength $3 \times 10^{-5} \text{ m}$ fall on the slits at an angle of 30° with the normal to the plane of slits as shown in the figure. A transparent sheet of thickness $5 \times 10^{-5} \text{ m}$ and refractive index 1.5 is introduced near the slit S_1 . The distance between the two slits is 0.5 mm and the distance between the plane of slits and the screen is 1 m. Find the y-coordinate (in cm) of the closest maxima to the origin on the positive y-axis.





10. In the standard YDSE apparatus shown in figure; Δx is the path difference between S_2P & S_1P . Now a glass slab is introduced in front of S_2 . Then match the following.



Column-I

- (A) Δx at P will
- (B) Fringe width will
- (C) Fringe pattern will
- (D) Number of fringes between O & P will

Column-II

- (P) Increase
- (Q) Decrease
- (R) Remain same
- (S) Shift upward
- (T) Shift downward





SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-66

Note: (*) → Multiple Correct Type Question

Total Marks: 30

Max. Time: 20 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.10

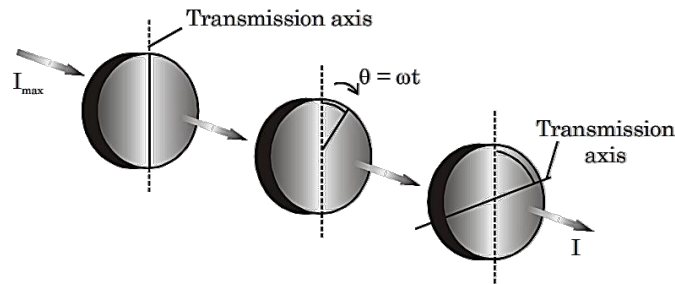
(3 marks, 2 min.)

(There is partial marking in multiple correct questions)

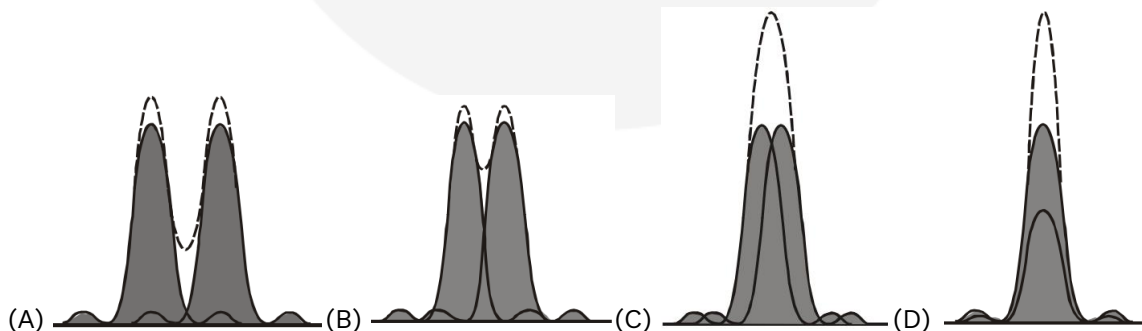
- An electromagnetic wave of frequency 1×10^{14} hertz is propagating along z-axis. The amplitude of electric field is 4 V/m. If $\epsilon_0 = 8.8 \times 10^{-12} \frac{C^2}{N-m^2}$, then average energy density of electric field will be:
 (A) $35.2 \times 10^{-10} \text{ J/m}^3$ (B) $35.2 \times 10^{-11} \text{ J/m}^3$
 (C) $35.2 \times 10^{-12} \text{ J/m}^3$ (D) $35.2 \times 10^{-13} \text{ J/m}^3$
- Choose the correct equation of maxwell -ampere circular law :
 (A) $\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$ (B) $\oint \vec{B} \cdot d\vec{l} = \mu_0 + \epsilon_0 \frac{d\phi_E}{dt}$
 (C) $\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_c + \epsilon_0 \frac{d\phi_E}{dt} \right)$ (D) $\oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \epsilon_0 \frac{d\phi_E}{dt}$
- An unpolarized light beam is incident on a surface at an angle of incidence equal to Brewster's angle. Then :
 (A) The reflected and the refracted beams are both partially polarized.
 (B) The reflected beam is partially polarized and the refracted beam is completely polarized and are at right angle to each other.
 (C) The reflected beam is completely polarized and the refracted beam is partially polarized and are at right angle to each other.
 (D) Both the reflected and the refracted beam are completely polarized and are at right angle to each other.
- Two polaroid sheets are placed one over the other with their axes inclined to each other at 30° . The ratio of the intensity of the unpolarized incident light and the polarized emergent light is :
 (A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{8}{3}$ (D) $\frac{3}{8}$
- The critical angle of a certain medium is $\sin^{-1}\left(\frac{3}{5}\right)$. The polarizing angle of medium is :
 (A) $\sin^{-1}\left(\frac{4}{5}\right)$ (B) $\tan^{-1}\left(\frac{5}{3}\right)$ (C) $\tan^{-1}\left(\frac{3}{4}\right)$ (D) $\tan^{-1}\left(\frac{4}{3}\right)$



6. In figure, suppose the transmission axes of the left and right polarizing disks are perpendicular to each other. Also, let the center disk be rotated about the common axis with an angular speed ω . If unpolarized light is incident on the incident on the left disk with an intensity $I_{\max}$ the intensity of the beam emerging from the right disk is given by.



- (A) $I = \frac{1}{16} I_{\max} (1 - \cos 4\omega t)$ (B) $I = \frac{1}{16} I_{\max} (1 - \cos 2\omega t)$
 (C) $I = \frac{1}{8} I_{\max} (1 - \cos 4\omega t)$ (D) $I = \frac{1}{8} I_{\max} (1 - \cos 2\omega t)$
7. Consider a single slit diffraction using light of wavelength $\lambda = 500 \text{ nm}$ and a lens of diameter 0.1 mm , then width of central maxima obtained on screen at a distance of 1 m will be:
 (A) 5 mm (B) 1 mm (C) 10 mm (D) 2.5 mm
8. You are driving in your car listening to music on the radio. Your car is equipped with radio that can receive radio signals of frequency $f_1 \approx 3 \text{ MHz}$, $f_2 \approx 30 \text{ MHz}$ and $f_3 = 3 \text{ GHz}$. You enter a tunnel with a circular opening of diameter 10 m . Which kind of radio signal will you be able to receive the longest as you continue to travel in the tunnel?
 (A) Radio signal of frequency f_1
 (B) Radio signal of frequency f_2
 (C) Radio signal of frequency f_3
 (D) Distance upto which signal can penetrate in tunnel is independent of frequency.
9. The diagrams below show the intensity distribution in diffraction of light of two sources. In which of the following cases the sources are just resolved?



10. Two luminous point sources separated by a certain distance are at 10 km from an observer. If the aperture of his eye is $2.5 \times 10^{-3} \text{ m}$ and the wavelength of light used is 500 nm , the distance of separation between the point sources just seen to be resolved is :
 (A) 12.2 m (B) 24.2 m (C) 2.44 m (D) 2.22 m



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-67

Note: (*) → Multiple Correct Type Question

Total Marks: 32

Max. Time: 22 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.8

(3 marks, 2 min.)

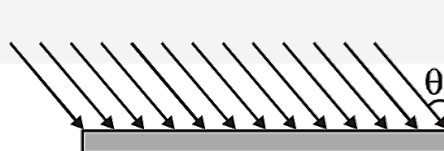
Multiple Correct Questions ('-2' negative marking) Q.9 to Q.10

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. Find the number of photons emitted per second from a source of light which results in a photocurrent with drift velocity of 1.5 m/s in a conductor with cross-section area 0.25 m^2 , volume density of electrons 10^{20} per m^3 , (Assume that 60% of photons emitted results in electron emission).
 (A) 6.25×10^{19} (B) 6.25×10^{20} (C) 6×10^{219} (D) 6×10^{20}

2. The radiation of intensity I is falling obliquely on a surface. Area of surface exposed to radiation is A . Select correct alternative.

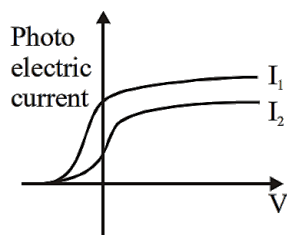


- (A) If surface is perfectly reflecting surface, then force experienced due to radiation is $\frac{IA \cos^2 \theta}{c}$
 (B) If surface is perfectly absorbing surface, then force experienced due to radiation is $\frac{IA \cos^2 \theta}{c}$
 (C) If surface is perfectly absorbing surface, then force experienced due to radiation is $\frac{IA \cos \theta}{c}$
 (D) If surface is partially reflecting surface, then force experienced due to radiation is $\frac{\beta IA \cos^2 \theta}{c}$
 where β is a constant and depends upon nature of surface.

3. A photo cell is illuminated by a small bright source placed 1 m away. When the same source of light is placed 2 m away, the electrons emitted by photo cathode,
 (A) Each carry one quarter of their previous energy.
 (B) Each carry one quarter of their previous momenta.
 (C) Are half in numbers.
 (D) Are one-quarter in numbers.
4. In a photoelectric experiment, if the potential difference across the electrodes is equal to the stopping potential and the emitter plate is held at negative potential with respect to the collector plate, mark the correct statement :
 (A) No electron is emitted from the emitter plate.
 (B) Electron is emitted from the emitter plate but it stops at half the length of the tube.
 (C) Electron is emitted from the emitter plate but it stops just before reaching the collector plate.
 (D) All the emitted electrons will reach the collector plate.



5. In an experiment of photoelectric effect, the variation of photoelectric current is plotted against potential difference across emitter and collector plates for different intensities. On the basis of graph, choose the incorrect option:



- (A) All emitted photoelectrons do not have same kinetic energy.
(B) If frequency of incident light is kept constant, the maximum kinetic energy of emitted photoelectrons is independent of intensity.
(C) Maximum value of photoelectric current is independent of intensity of radiation if its frequency is constant.
(D) From the graph, we can conclude radiation intensity $I_1 > I_2$
6. Photons with energy 5 eV are incident on a cathode C, on a photoelectric cell. The maximum energy of the emitted photoelectrons is 2 eV. When photons of energy 6 eV are incident on C, no photoelectrons will reach the anode A if the stopping potential of A relative to C is:
(A) V (B) -3V (C) -1V (D) 4 V
7. When a metallic surface is illuminated with monochromatic light of wavelength λ , the stopping potential is $5V_0$. When the same surface is illuminated with light of wavelength 3λ , the stopping potential is V_0 . Then the work function of the metallic surface is :
(A) $\frac{hc}{6\lambda}$ (B) $\frac{hc}{5\lambda}$ (C) $\frac{hc}{4\lambda}$ (D) $\frac{2hc}{4\lambda}$
8. Fastest electrons ejected from a surface by incident light of wavelength 330 nm are bent into circle of radius 1 m as they pass through magnetic field of 2.5×10^{-6} T. Find the work function of the surface.
(A) $\phi = 4.3$ eV (B) $\phi = 2.2$ eV (C) $\phi = 2.2$ eV (D) $\phi = 3.2$ eV
- 9*. Moonlight has an intensity of 0.84×10^{-3} W/m². The average wavelength of moonlight is 5000 Å.
(A) The force exerted by moonlight on a perfectly absorbing surface of area 100 m² is 0.28×10^{-9} N.
(B) The force exerted by moonlight on a perfectly absorbing surface of area 100 m² is 0.84×10^{-2} N.
(C) The number of photons/m³ in the beam is 7×10^6
(D) The number of photons/m³ in the beam is 7×10^3
- 10*. In a photoelectric-effect experiment, if 'f' is the frequency of incident radiations on the metal surface and I is the intensity of the incident radiations, then:
(A) If 'f' is increases, keeping I and work function constant, then stopping potential and maximum kinetic energy of photoelectron increases.
(B) If distance between cathode and anode increases, stopping potential remains same.
(C) If 'I' is increased keeping 'f' and work function constant, saturation current increases but stopping potential remains same.
(D) Work function is decreased keeping f and I constant. Then, stopping potential and maximum kinetic energy of photoelectrons increases.



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-68

Note: (*) → Multiple Correct Type Question

Total Marks: 29

Max. Time: 22 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Comprehension Type Questions ('-1' Negative marking) Q.6 to Q.8

(2 marks, 2 min.)

Subjective Type Questions (No negative marking) Q.9 to Q.10

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

- Radiation of wavelength $\frac{\lambda_0}{3}$ (λ_0 being the threshold wavelength) is incident on a photosensitive metallic sphere of radius R. The charge developed on the sphere when electrons cease to be emitted will be:

(A) $\frac{4\pi\epsilon_0 R h c}{e \lambda_0}$ (B) $\frac{6\pi\epsilon_0 R h c}{e \lambda_0}$ (C) $\frac{8\pi\epsilon_0 R h c}{e \lambda_0}$ (D) $\frac{12\pi\epsilon_0 R h c}{e \lambda_0}$
- The time taken to prepare a print by exposing a small sheet to a point light source of 60 W placed at a distance of 25 cm from the sheet is 5 sec. If the distance is increased to 40 cm, the time taken to prepare a similar print will be.

(A) 16 sec (B) 12.8 sec (C) 3.1 sec (D) 1 sec
- A proton and α particle have equal kinetic energies. If λ_α and λ_p are the De-Broglie wavelengths of α particle and proton, then the ratio of $\frac{\lambda_\alpha}{\lambda_p}$ is:

(A) 2 (B) 1 (C) 1/2 (D) 1/4
- An electron (mass m) with an initial velocity $\vec{v} = v_0 \hat{i}$ is in an electric field $\vec{E} = E_0 \hat{j}$. If $\lambda_0 = h/mv_0$, its de-Broglie wavelength at time t is given by :

(A) λ_0 (B) $\lambda_0 \sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}$ (C) $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$ (D) $\frac{\lambda_0}{\left(1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}\right)}$
- The additional energy that should be given to an electron to reduce its de-Broglie wavelength from 1 nm to 0.5 nm is

(A) 2 times the initial kinetic energy (B) 3 times the initial kinetic energy
(C) 0.5 times the initial kinetic energy (D) 4 times the initial kinetic energy



Paragraph for question 6 to 8

Some phenomenon of light can be explained by wave theory and other some phenomenon show the particle nature of light. De-Broglie proposed a hypothesis based on the dual nature of light. According to De-Broglie each moving particle consists of a wave. This wave is known as De-Broglie wave or matter wave and its wavelength is given by $\lambda = \frac{h}{p}$. Actually this expression correlate wave and particle nature. (λ) represent the wave nature and momentum (p) represent the particle nature.

6. Proton, deuteron and α particles are accelerated through the same potential difference. Then the ratio of their wavelength as :
- (A) $1 : \sqrt{2} : 1$ (B) $1 : 1 : 1$ (C) $1 : 2 : 2 : \sqrt{2}$ (D) $2\sqrt{2} : 2 : 1$
7. When an electron is accelerated through the 150 volt potential difference, the corresponding wavelength associated with it will be :
- (A) $1 \text{ \AA}$ (B) $2 \text{ \AA}$ (C) $1/2 \text{ \AA}$ (D) $4 \text{ \AA}$
8. A proton and α particle have equal kinetic energies. If λ_α and λ_p are the De-Broglie wavelength of α particle and proton, then the ratio of $\frac{\lambda_\alpha}{\lambda_p}$ is
- (A) 2 (B) 1 (C) $1/2$ (D) $1/4$
9. A small plate of a metal is placed at a distance of $\frac{2}{\sqrt{\pi}}$ meter from a monochromatic light source of wavelength 660 nm and power 48 watt. The light falls normally on the plate. The number of photons striking the metal plate per square cm per sec is 10×10^{2y} then the value y will be (take $h = 6.6 \times 10^{-34} \text{ J sec}$)
10. The potential energy of a particle of mass m is given by $V(x) = \begin{cases} E_0 & 0 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$. λ_1 and λ_2 are the de-Broglie wavelengths of the particle, when $0 \leq x \leq 1$ and $x > 1$, respectively. If the total energy of particle is $2E_0$, find $(\lambda_1/\lambda_2)^2$.

Note: (*) → Multiple Correct Type Question

Total Marks: 39

Max. Time: 26 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.5

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.6 to Q.8

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.9

(4 marks, 3 min.)

Match the Column Type Questions (No negative marking) Q.10

(8 marks, 4 min.)

(There is partial marking in multiple correct questions)

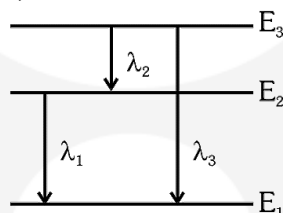
1. Energy of electron in first orbit of He^+ is y joule. Then the potential energy of electron in second orbit of He^+ is :

(A) $\frac{y}{5}$ J (B) $\frac{y}{2}$ J (C) $\frac{-y}{2}$ J (D) $\frac{-y}{9}$ J

2. The angular momentum of an electron in the hydrogen atom is $\frac{3h}{2\pi}$. Here h is Planck's constant. The kinetic energy of this electron is :

(A) 4.53 eV (B) 1.51 eV (C) 3.4 eV (D) 6.8 eV

3. For the given transitions of electron, the relation between λ_1 , λ_2 & λ_3 , is :



(A) $\lambda_3 = \sqrt{\lambda_1 \lambda_2}$ (B) $\lambda_3 = \frac{\lambda_1 + \lambda_2}{2}$ (C) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$ (D) $\lambda_3 = \lambda_1 + \lambda_2$

4. The shortest wavelength in the Lyman series is 911.6 Å. Then the longest wavelength in the Lyman series is :

(A) 1215 Å (B) ∞ (C) 2430 Å (D) 600 Å

5. A neutron having kinetic energy 13.6 eV collides with a hydrogen atom in ground state at rest. Assume that the mass of neutron and hydrogen atoms are same and the neutron does not leave its line of motion. Then which of the following is the possible KE of the neutron after the collision?

(A) zero (B) 3.4 eV (C) 1.5 eV (D) 6.8 eV

- 6.* The total energy of an electron in the first excited state of the hydrogen atom is -3.4 eV. Select correct alternative(s):

(A) The kinetic energy of the electron in this state is 3.4 eV.

(B) The potential energy of the electron in this state is -6.8 eV.

(C) Answer (A) would change if the choice of zero of potential energy is changed.

(D) Answer (B) would change if the choice of zero of potential energy is changed.



- 7.*** Whenever a hydrogen atom emits a photon in the Balmer series,
(A) It may emit photon in the Balmer series.
(B) it must emit another photon in the Lyman series.
(C) The second photon, if emitted, will have a wavelength of about 122 nm.
(D) it may emit a second photon, but wavelength of this photon cannot be predicted.
- 8.*** If electron in H-atom would be as heavy as proton, then :
(A) Entire Paschen series would lie in UV-region.
(B) Orbital radius corresponding to any quantum number would reduce drastically.
(C) Speed of electron would remain unchanged.
(D) Angular momentum of electron corresponding to any quantum state would decrease.
- 9.** In a transition to a state of ionization potential 30.6 V, a hydrogen like atom emits a 541.17 Å photon. Determine the principal quantum number of the initial state,
- 10.** Suppose the potential energy between electron and proton at a distance r is given by $\frac{-ke^2}{r^3}$. Column-

I gives the various parameter and in column-II its dependency on n & z . match the parameters with its proportionality. (Assume Bohr model to be valid)

Column-I

- (A) radius of n^{th} orbit
(B) Velocity of e^- in n^{th} orbit
(C) Time period of e^- in n^{th} orbit
(D) Energy in n^{th} orbit (∞ to be at zero potential energy)

Column-II

- (P) $n^3 Z^{-1}$
(Q) $n^{-2} Z$
(R) $n^{-2} Z^{-1}$
(S) $n^6 Z^{-2}$
(T) $n^{-5} Z^2$



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-70

Note: (*) → Multiple Correct Type Question

Total Marks: 33

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.7

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.8 to Q.9

(4 marks, 3 min.)

Subjective Type Questions (No negative marking) Q.10

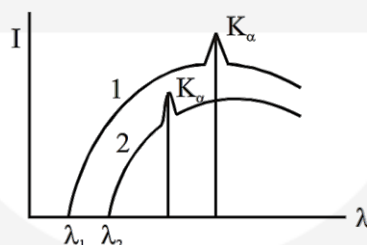
(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. Frequency of the K_{α} X-ray for the element calcium (Ca, $Z = 20$) is 8.95×10^{17} Hz. Frequency of K_{α} X-ray for element cadmium (Cd, $Z = 48$) will be:

(A) 5.46×10^{18} Hz (B) 6.31×10^{18} Hz (C) 5.71×10^{17} Hz (D) 6.31×10^{17} Hz

2. When two different materials A and B having atomic number Z_1 and Z_2 are used as the target in Coolidge X-ray tube at different operating voltage V_1 and V_2 respectively their spectrums are found as below.



- (A) $V_1 > V_2$ and $Z_1 > Z_2$ (B) $V_1 < V_2$ and $Z_1 < Z_2$
(C) $V_1 < V_2$ and $Z_1 > Z_2$ (D) $V_1 > V_2$ and $Z_1 < Z_2$

3. In a Coolidge tube experiment, the minimum wavelength of the continuous X-ray spectrum is equal to 66.3 pm.

- (A) Electrons accelerate through a potential difference of 12.75 kV in the Coolidge tube.
(B) Electrons accelerate through a potential difference of 18.75 kV in the Coolidge tube.
(C) De-Broglie wavelength of the electrons reaching the anticathode is of the order of $10 \mu\text{m}$
(D) De-Broglie wavelength of the electrons reaching the anticathode is $0.01 \text{ \AA}$.

4. A nucleus (of nuclear density ρ) disintegrates into two daughter nuclei with masses in the ratio 8 : 27. Density of the smaller nucleus is :-

- (A) $\frac{2}{3}\rho$ (B) $\frac{2}{5}\rho$ (C) $\frac{8}{27}\rho$ (D) ρ

5. The binding energy of deuteron ${}^2_1\text{H}$ is 1.112 MeV per nucleon and an α -particle ${}^4_2\text{He}$ has a binding energy of 7.047 MeV per nucleon. Then in the fusion reaction, ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^4_2\text{He} + Q$, the energy Q released is :

- (A) 1 MeV (B) 11.9 MeV (C) 23.8 MeV (D) 931 MeV



6. A nucleus of mass $M + \Delta m$ is at rest and decay into two daughter nuclei of mass $\frac{M}{4}$ & $\frac{3M}{4}$ each. Speed of light is C . The speed of daughter nucleus of mass $\frac{M}{4}$ is :
- (A) $C\sqrt{\frac{\Delta m}{M + \Delta m}}$ (B) $C\frac{\Delta m}{M + \Delta m}$ (C) $C\sqrt{\frac{2\Delta m}{M}}$ (D) $C\sqrt{\frac{6\Delta m}{M}}$
7. The equation $x \rightarrow y + \alpha$ represents a decay process. If x is at rest just before decay and k_1 and k_2 represent the kinetic energy of y and α having mass m_2 and m_1 respectively, then
- (A) $\frac{k_1}{k_2} = \frac{m_1}{m_2}$ (B) $\frac{k_1}{k_2} = \frac{m_1^2}{m_2^2}$ (C) $\frac{k_1}{k_2} = \sqrt{\frac{m_2}{m_1}}$ (D) None of these
- 8.* Which of the following statements about X-rays is/are TRUE?
- (A) $E(K_\alpha) + E(L_\beta) = E(K_\beta) + E(M_\alpha) = E(K_\gamma)$
(B) For the harder X-rays, the intensity is higher than soft X-rays
(C) The continuous and the characteristics X-rays of same wavelengths differ only in the method of creation
(D) The cut-off wavelength $\lambda_{\min}$ depends only on the accelerating voltage applied between the target and the filament.
- 9.* Choose the correct statement(s):
- (A) The stability of nucleus is directly proportional to total binding energy.
(B) As the accelerating potential in Coolidge apparatus increases, difference between $\lambda_{\text{cut-off}}$ and wavelength of K_α characteristics X-ray increases due to change in both $\lambda_{\text{cut-off}}$ & λ_{K_α} .
(C) A neutron with kinetic energy of 24.2 eV collides with stationary H-atom in ground state then collision may be elastic, inelastic & perfectly inelastic.
(D) Presence of K_α in characteristics X-ray ensures the availability of all types of characteristics X-rays.
10. In a fission reaction ${}^{236}_{92}\text{U} \rightarrow {}^{117}\text{X} + {}^{117}\text{Y} + n + n$, the binding energy per nucleon of X and Y is 8.5 MeV whereas of ${}^{236}\text{U}$ is 7.58 MeV. The total energy liberated is 25 N MeV. Find the value of N.



SUBJECT: IIT-PHYSICS

COURSE: EXCEL (XII)

ELP No.-71

Note: (*) → Multiple Correct Type Question

Total Marks: 29

Max. Time: 23 min.

Single Correct Questions ('-1' negative marking) Q.1 to Q.3

(3 marks, 2 min.)

Multiple Correct Questions ('-2' negative marking) Q.4

(4 marks, 3 min.)

Comprehension Type Questions ('-1' Negative marking) Q.5 to Q.8

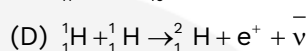
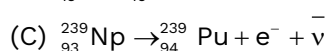
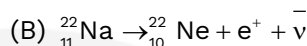
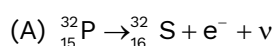
(2 marks, 2 min.)

Subjective Type Questions (No negative marking) Q.9 to Q.10

(4 marks, 3 min.)

(There is partial marking in multiple correct questions)

1. Which of the following reaction is correct :



2. A radioactive nuclei X converts into stable nuclei Y. half life of X is 50 years. The age of radioactive sample when the ratio of X and Y is 1 : 15, is :

(A) 100 years

(B) 200 years

(C) 300 years

(D) 400 years

3. Two radioactive materials A and B have decay constants 5λ and λ respectively. Initially both A and B have the same number of nuclei. The ratio of the number of nuclei of A to that of B will be $\frac{1}{e}$ after a time:

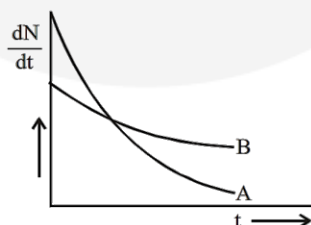
(A) $\frac{1}{5\lambda}$

(B) $\frac{1}{4\lambda}$

(C) $\frac{5}{4\lambda}$

(D) $\frac{4}{5\lambda}$

4.* The variation of decay rate of two radioactive samples A and B with time is shown in figure. Which of the following statements is/are true?



(A) Decay constant of A is greater than that of B, hence A always decay faster than B.

(B) Decay constant of B is greater than that of A but its decay rate is always smaller than that of A.

(C) Decay constant of A is greater than that of B but it does not always decay faster than B.

(D) Decay constant of B is smaller than that of A but still its decay rate at some instant becomes equal to that of A.



Paragraph for question no 5 and 6

Energy is often released or absorbed in a nuclear reaction. A reaction releases energy when kinetic energy of product nuclei after reaction is greater than the kinetic energy of interacting nuclei before reaction, the increase results from the transformation of rest mass into kinetic energy. The Q value of nuclear reaction (amount of energy released)

$$Q \text{ Value} = \Sigma E_{kf} - \Sigma E_{ki}$$

ΣE_{kf} = final kinetic energy

ΣE_{ki} = initial kinetic energy

$$Q \text{ value} = \Sigma B_i - \Sigma B_f$$

ΣB_i = initial binding energy

ΣB_f = Final binding energy

For an exoergic reaction $Q > 0$, such reaction can proceed even if the reaction particles are at rest. For endoergic reaction $Q < 0$, such reaction require threshold kinetic energy. The nuclear reaction follows certain conservation of

(i) Charge (ii) Linear and angular momentum (iii) mass and energy

5. The binding energy per nucleon of ${}^7_3\text{Li}$ and ${}^4_2\text{He}$ nuclei are 5.6 MeV and 7.06 MeV, then in the reaction $p + {}^7_3\text{Li} \rightarrow 2({}^4_2\text{He})$, the Q value of the reaction is :
- (A) 28.24 MeV (B) 17.28 MeV (C) 1.46 MeV (D) 39.2 MeV
6. A nucleus with mass no. 220 initially at rest emits an α particle. If the Q value of reaction is 5.5 MeV, then kinetic energy of α particle is :
- (A) 4.4 MeV (B) 5.4 MeV (C) 5.6 MeV (D) 6.5 MeV

Paragraph for Question Nos. 7 & 8

A capacitor of capacitance 1 F and charge 10 C starts discharging through a resistance of 100 Ω , at that instant a radio-active element has N_0 undecayed nuclei. The ratio of number of undecayed nuclei and charge on capacitor remains equal to 100 nuclei per coulomb throughout time.

7. The half life of radio-active element is:
- (A) 138.6 sec (B) 6.93 sec (C) 1.386 sec (D) 69.3 sec.
8. The initial radioactivity of element is:
- (A) 100 decays/sec (B) 10 decays/sec (C) 1 decays/sec (D) 5 decays/sec
9. In an α - decay the KE of α -particle is 48 MeV and Q-value of the reaction is 50 MeV. If the mass number of the mother nucleus is x. Then the value of $\left\{ \frac{2x}{100} \right\}$ will be (Assume that daughter nucleus is in ground state)
10. Two radioactive materials A and B have initial activity A_0 and $2A_0$. After 1 half life of A, the total combined activity of A and B is $\frac{1}{4}$ times of their initial total activity. Their half lives are T_A and T_B then $\frac{T_A}{T_B}$



ANSWER KEY

GEOMETRICAL OPTICS (ELP-1 to 10)

ELP No.-01

- | | | | | | | | | | | | | | |
|-----------|-----|-----------|-------|------------|------|-----------|-----|-----------|-----|-----------|-----|-----------|-----|
| 1. | (C) | 2. | (C) | 3. | (A) | 4. | (B) | 5. | (A) | 6. | (A) | 7. | (B) |
| 8. | (A) | 9. | (BCD) | 10. | (AC) | | | | | | | | |

ELP No.-02

- | | | | | | | | | | | | | | |
|-----------|------|-----------|------|------------|------------------------------------|-----------|-----|-----------|-----|-----------|-----|-----------|------|
| 1. | (B) | 2. | (B) | 3. | (C) | 4. | (A) | 5. | (B) | 6. | (A) | 7. | (AC) |
| 8. | (AC) | 9. | (BC) | 10. | (A)-P,T; (B)-R; (C)-P,Q; (D)-Q,R,S | | | | | | | | |

ELP No.-03

- | | | | | | | | | | | | | | |
|-----------|------|-----------|------|------------|-------|-----------|-----|-----------|-----|-----------|-----|-----------|-----|
| 1. | (B) | 2. | (D) | 3. | (B) | 4. | (A) | 5. | (C) | 6. | (C) | 7. | (B) |
| 8. | (CD) | 9. | (AB) | 10. | (BCD) | | | | | | | | |

ELP No.-04

- | | | | | | | | | | | | | | |
|-----------|------|-----------|------|------------|------|-----------|-----|-----------|------|-----------|------|-----------|------|
| 1. | (A) | 2. | (B) | 3. | (B) | 4. | (B) | 5. | (BC) | 6. | (AB) | 7. | (AC) |
| 8. | (BD) | 9. | (BC) | 10. | (BC) | | | | | | | | |

ELP No.-05

- | | | | | | | | | | | | | | |
|-----------|-----|-----------|-----|------------|-----|-----------|-----|-----------|-----|-----------|-----|-----------|-----|
| 1. | (C) | 2. | (B) | 3. | (A) | 4. | (D) | 5. | (B) | 6. | (B) | 7. | (C) |
| 8. | (D) | 9. | (A) | 10. | 6 | | | | | | | | |

ELP No.-06

- | | | | | | | | | | | | | | |
|-----------|--------|-----------|-----|------------|-----|-----------|-----|-----------|-------|-----------|------|-----------|-------|
| 1. | (A) | 2. | (B) | 3. | (A) | 4. | (A) | 5. | (ABD) | 6. | (AC) | 7. | (ABC) |
| 8. | (ABCD) | 9. | 5 | 10. | 4 | | | | | | | | |

ELP No.-07

- | | | | | | | | | | | | | | |
|-----------|------|-----------|-----|------------|----------------------------|-----------|-------|-----------|--------|-----------|-------|-----------|---|
| 1. | (AD) | 2. | (D) | 3. | (ABCD) | 4. | (ACD) | 5. | (ABCD) | 6. | (ABD) | 7. | 4 |
| 8. | 2 | 9. | 3 | 10. | P - 2; Q - 1; R - 4, S - 3 | | | | | | | | |

**ELP No.-08**

- | | | | | | | | | | | | | | |
|----|-------|----|--------------|-----|-----|----|-----|----|-----|----|-----|----|------|
| 1. | (B) | 2. | (A) | 3. | (A) | 4. | (A) | 5. | (C) | 6. | (C) | 7. | (AB) |
| 8. | (ABD) | 9. | 2° clockwise | 10. | 3 | | | | | | | | |

ELP No.-09

- | | | | | | | | | | | | | | |
|----|-----|----|-----|-----|-----|----|-----|----|------|----|-------|----|-------|
| 1. | (A) | 2. | (A) | 3. | (B) | 4. | (B) | 5. | (AC) | 6. | (ACD) | 7. | (ABC) |
| 8. | (D) | 9. | (A) | 10. | (A) | | | | | | | | |

ELP No.-10

- | | | | | | | | | | | | | | |
|----|-----|----|-----|-----|-----|----|-----|----|-----|----|-----|----|-----|
| 1. | (D) | 2. | (D) | 3. | (C) | 4. | (D) | 5. | (A) | 6. | (D) | 7. | (D) |
| 8. | (D) | 9. | 6 | 10. | (D) | | | | | | | | |

SHM(ELP-11-15)**ELP No.-11**

- | | | | | | | | | | | | | | |
|----|-------|----|--------|-----|-----|----|-----|----|-----|----|-----|----|-----|
| 1. | (A) | 2. | (B) | 3. | (D) | 4. | (B) | 5. | (A) | 6. | (B) | 7. | (B) |
| 8. | (ABD) | 9. | (ABCD) | 10. | 32 | | | | | | | | |

ELP No.-12

- | | | | | | | | | | | | | | |
|----|------------------------------------|----|-----|-----|-----|----|-----|----|------|----|-------|----|--------|
| 1. | (D) | 2. | (D) | 3. | (B) | 4. | (C) | 5. | (AB) | 6. | (ABD) | 7. | (ABCD) |
| 8. | (A)→(T); (B)→(R); (C)→(R); (D)→(Q) | 9. | 6 | 10. | 2.5 | | | | | | | | |

ELP No.-13

- | | | | | | | | | | | | | | |
|----|------|----|--------|-----|-----|----|-----|----|-----|----|-------|----|-------|
| 1. | (A) | 2. | (C) | 3. | (B) | 4. | (A) | 5. | (A) | 6. | (ABC) | 7. | (ACD) |
| 8. | (BD) | 9. | (ABCD) | 10. | 10 | | | | | | | | |

ELP No.-14

- | | | | | | | | | | | | | | |
|----|-----|----|-----|-----|-----|----|------|----|------|----|-----|----|-----|
| 1. | (B) | 2. | (A) | 3. | (D) | 4. | (AD) | 5. | (CD) | 6. | (B) | 7. | (A) |
| 8. | 7 | 9. | 4 | 10. | 10 | | | | | | | | |

ELP No.-15

- | | | | | | | | | | | | | | |
|----|-----|----|----------------|-----|-----|----|-----|----|--------|----|-------|----|---|
| 1. | (A) | 2. | (A) | 3. | (B) | 4. | (B) | 5. | (ABCD) | 6. | (ABC) | 7. | 3 |
| 8. | 6 | 9. | $\frac{5l}{3}$ | 10. | 6 | | | | | | | | |



3. Electrostatics (ELP-16-25)

ELP No.-16

1. (B) 2. (A) 3. (D) 4. (C) 5. (D) 6. (A) 7. (D)
8. (ABC) 9. (ABCD) 10. (ACD)

ELP No.-17

1. (B) 2. (B) 3. (C) 4. (B) 5. (A) 6. (A) 7. (BC)
8. (AC) 9. $(P) \rightarrow (3); (Q) \rightarrow (1); (R) \rightarrow (2); (S) \rightarrow (4)$ 10. $(P) \rightarrow (4); (Q) \rightarrow (3); (R) \rightarrow (2); (S) \rightarrow (1)$

ELP No.-18

1. (A) 2. (A) 3. (A) 4. (B) 5. (B) 6. (AD) 7. (ABD)
8. (AC) 9. (ABCD) 10. 8

ELP No.-19

1. (A) 2. (C) 3. (A) 4. (C) 5. (B) 6. (A) 7. (B)
8. (ABCD) 9. (AC) 10. (AC)

ELP No.-20

1. (B) 2. (C) 3. (D) 4. (C) 5. (C) 6. (C) 7. (ABC)
8. (BCD) 9. (AD) 10. 6

ELP No.-21

1. (C) 2. (A) 3. (B) 4. (AB) 5. (AB) 6. (ABC) 7. (BCD)
8. (ABC) 9. (ACD) 10. 4

ELP No.-22

1. (B) 2. (D) 3. (D) 4. (BCD) 5. (BC) 6. (A)-R; (B)-P; (C)-S; (D)-Q
7. 6 8. 7 9. 6 10. 7

**ELP No.-23**

1. (A) 2. (C) 3. (C) 4. (ABC) 5. (BC) 6. (AD) 7. (BCD)
8. (ABCD) 9. (A) 10. 1

ELP No.-24

1. (C) 2. (C) 3. (D) 4. (B) 5. (C) 6. (BD) 7. (BD)
8. (ABC) 9. (AC) 10. 5

ELP No.-25

1. (A) 2. (A) 3. (A) 4. (C) 5. (C) 6. (ABCD) 7. (ACD)
8. (ABC) 9. (ABCD) 10. 6

4. Gravitation (ELP26-27)**ELP No.-26**

1. (D) 2. (D) 3. (B) 4. (A) 5. (D) 6. (C) 7. (AB)
8. (BD) 9. (ABCD) 10. (ABD)

ELP No.-27

1. (A) 2. (A) 3. (BD) 4. (ABC) 5. (AB) 6. (ABCD) 7. (C)
8. (B) 9. (D) 10. 4

5. Current Electricity (ELP-28-34)**ELP No.-28**

1. (C) 2. (C) 3. (D) 4. (A) 5. (A) 6. (BCD) 7. (ABC)
8. (ABCD) 9. $(A) \rightarrow (P,T) : (B) \rightarrow (P,T) ; (C) \rightarrow (P,T) (D) \rightarrow (P,T)$ 10. 4

ELP No.-29

1. (B) 2. (D) 3. (B) 4. (C) 5. (B) 6. (B) 7. (A)
8. (B) 9. (C) 10. $(A) \rightarrow (Q); (B) \rightarrow (P); (C) \rightarrow (P); (D) \rightarrow (R)$

**ELP No.-30**

1. (A) 2. (A) 3. (B) 4. (A) 5. (C) 6. (C) 7. (D)
8. (ABC) 9. (BC) 10. (AD)

ELP No.-31

1. (B) 2. (C) 3. (B) 4. (A) 5. (B) 6. (A) 7. (CD)
8. (BD) 9. 9 10. 1

ELP No.-32

1. (C) 2. (B) 3. (C) 4. (A) 5. (B) 6. (A) 7. (ABC)
8. (AD) 9. (ABCD) 10. (CD)

ELP No.-33

1. (A) 2. (C) 3. (BC) 4. (BC) 5. (D) 6. (B) 7. (D)
8. (B) 9. (A) 10. (B)

ELP No.-34

1. (B) 2. (C) 3. (A) 4. (D) 5. (C) 6. (AD) 7. (BC)
8. 48 9. 4 10. 10^8 Pa

6. Capacitor (ELP-35-38)**ELP No.-35**

1. (C) 2. (B) 3. (A) 4. (A) 5. (B) 6. (A) 7. (D)
8. (A) 9. (ABC) 10. (BCD)

ELP No.-36

1. (B) 2. (C) 3. (D) 4. (C) 5. (A) 6. (B) 7. (B)
8. (AD) 9. (BCD) 10. 4

**ELP No.-37**

1. (D) 2. (C) 3. (D) 4. (C) 5. (A) 6. (A) 7. (D)
8. (AB) 9. (ACD) 10. $(A) \rightarrow (Q,S); (B) \rightarrow (P,R,T); (C) \rightarrow (P,R); (D) \rightarrow (Q,S)$

ELP No.-38

1. (B) 2. (BCD) 3. (ABCD) 4. (AC) 5. (BCD) 6. (D) 7. (B)
8. (C) 9. 4 10. 8

7. MEC (ELP-39-45)**ELP No.-39**

1. (D) 2. (C) 3. (B) 4. (B) 5. (B) 6. (D) 7. (BCD)
8. (AC) 9. 3 10. $(P) - 3; (Q) - 1; (R) - 4; (S) - 2.$

ELP No.-40

1. (A) 2. (C) 3. (C) 4. (B) 5. (ACD) 6. (AB) 7. (ABC)
8. (AB) 9. 6 10. $(P)-3; (Q)-1; (R)-4; (S)-2$

ELP No.-41

1. (B) 2. (CD) 3. (BD) 4. (ACD) 5. (AC) 6. (AD) 7. (D)
8. (C) 9. 7 10. $(P) \rightarrow 1; (Q) \rightarrow 3; (R) \rightarrow 2; (S) \rightarrow 4$

ELP No.-42

1. (B) 2. (A) 3. (B) 4. (C) 5. (D) 6. (BCD) 7. (ACD)
8. (AC) 9. 1 10. 6

ELP No.-43

1. (B) 2. (D) 3. (C) 4. (A) 5. (AC) 6. (A) 7. (B)
8. (B) 9. (C) 10. (C)

**ELP No.-44**

1. (D) 2. (B) 3. (C) 4. (A) 5. (B) 6. (BCD) 7. (AC)
8. (ABCD) 9. 24 BiR 10. 6

ELP No.-45

1. (A) 2. (D) 3. (BC) 4. (ABC) 5. (ACD) 6. (D) 7. (B)
8. (D) 9. 8 10. 1

08. EMI And AC (ELP 46-54)**ELP No.-46**

1. (D) 2. (D) 3. (A) 4. (D) 5. (A) 6. (B) 7. (ACD)
8. (AC) 9. 9 10. $(P) \rightarrow 3; (Q) \rightarrow 4; (R) \rightarrow 1; (S) \rightarrow 2$

ELP No.-47

1. (C) 2. (A) 3. (D) 4. (B) 5. (A) 6. (C) 7. (A)
8. (A) 9. (C) 10. (A)

ELP No.-48

1. (A) 2. (C) 3. (A) 4. (C) 5. (BC) 6. (BC) 7. (AB)
8. (B) 9. (C) 10. 2

ELP No.-49

1. (B) 2. (C) 3. (C) 4. (A) 5. (A) 6. (ABC) 7. (AC)
8. (BCD) 9. (B) 10. (C)

ELP No.-50

1. (B) 2. (B) 3. (A) 4. (C) 5. (D) 6. (C) 7. (ABC)
8. (C) 9. (B) 10. (A)

**ELP No.-51**

1. (A) 2. (AC) 3. (BD) 4. (ACD) 5. (AC) 6. (BD) 7. (BC)
8. (C) 9. (B) 10. 2

ELP No.-52

1. (D) 2. (A) 3. (C) 4. (A) 5. (A) 6. (A) 7. (D)
8. (B) 9. (B) 10. (A)

ELP No.-53

1. (D) 2. (B) 3. (C) 4. (C) 5. (A) 6. (C) 7. (AB)
8. (ABC) 9. (ABC) 10. (A)→(Q); (B)→(P); (C)→(S); (D)→(R)

ELP No.-54

1. (B) 2. (B) 3. (D) 4. (D) 5. (D) 6. (B) 7. (C)
8. (AB) 9. 3 10. (A)-4; (B)-1; (C)-3; (D)-2

09-Wave on String and Sound Wave_(ELP-55-63)**ELP No.-55**

1. (D) 2. (B) 3. (A) 4. (D) 5. (D) 6. (C) 7. (A)
8. (BC) 9. (BCD) 10. $P \rightarrow \text{(iii)}; Q \rightarrow \text{(i)}; R \rightarrow \text{(iv)}; S \rightarrow \text{(ii)}$

ELP No.-56

1. (C) 2. (C) 3. (B) 4. (D) 5. (AC) 6. (B) 7. (C)
8. (D) 9. (B) 10. (B)

ELP No.-57

1. (D) 2. (C) 3. (A) 4. (A) 5. (B) 6. (A) 7. (ABD)
8. (ABCD) 9. 5 10. 5

**ELP No.-58**

1. (D) 2. (B) 3. (A) 4. (D) 5. (C) 6. (AB) 7. (ABD)
8. (AB) 9. 4 10. 6

ELP No.-59

1. (D) 2. (D) 3. (B) 4. (ABCD) 5. (ABCD) 6. (C) 7. (A)
8. (AD) 9. (B) 10. (D)

ELP No.-60

1. (B) 2. (A) 3. (C) 4. (B) 5. (C) 6. (AC) 7. (AC)
8. (ABCD) 9. (A) 10. (C)

ELP No.-61

1. (A) 2. (D) 3. (B) 4. (C) 5. (B) 6. (ACD) 7. (BD)
8. (CD) 9. 16 10. 61

ELP No.-62

1. (B) 2. (B) 3. (A) 4. (B) 5. (A) 6. (ABCD) 7. (A)
8. (B) 9. 2 10. 72

ELP No.-63

1. (A) 2. (B) 3. (A) 4. (B) 5. (A) 6. (A) 7. (A)
8. (B) 9. (BC) 10. (ABD) 11. (ABC) 12. (ACD) 13. (B) 14. (A)
15. (B)

10-Wave Optics_(ELP-64-66)**ELP No.-64**

1. (A) 2. (C) 3. (D) 4. (C) 5. (B) 6. (D) 7. (A)
8. (ABC) 9. (AC) 10. (A) R,T (B) Q, (C) P,S (D) P

**ELP No.-65**

1. (A) 2. (A) 3. (A) 4. (B) 5. (A) 6. (ABD) 7. (AB)
8. (ABC) 9. 1 10. $(A \rightarrow (P); (B \rightarrow (R); (C \rightarrow (T); (D \rightarrow (R)$

ELP No.-66

1. (C) 2. (C) 3. (C) 4. (C) 5. (B) 6. (A) 7. (C)
8. (B) 9. (B) 10. (C)

11-Modern Physics (ELP -67-71)**ELP No.-67**

1. (A) 2. (C) 3. (D) 4. (D) 5. (C) 6. (B) 7. (A)
8. (D) 9. (AC) 10. (ABCD)

ELP No.-68

1. (C) 2. (B) 3. (C) 4. (C) 5. (B) 6. (D) 7. (A)
8. (C) 9. 7 10. 2

ELP No.-69

1. (B) 2. (B) 3. (C) 4. (A) 5. (A) 6. (ABD) 7. (BC)
8. (ABC) 9. 4 10. $(A \rightarrow (Q); (B \rightarrow (P); (C \rightarrow (T); (D \rightarrow (S)$

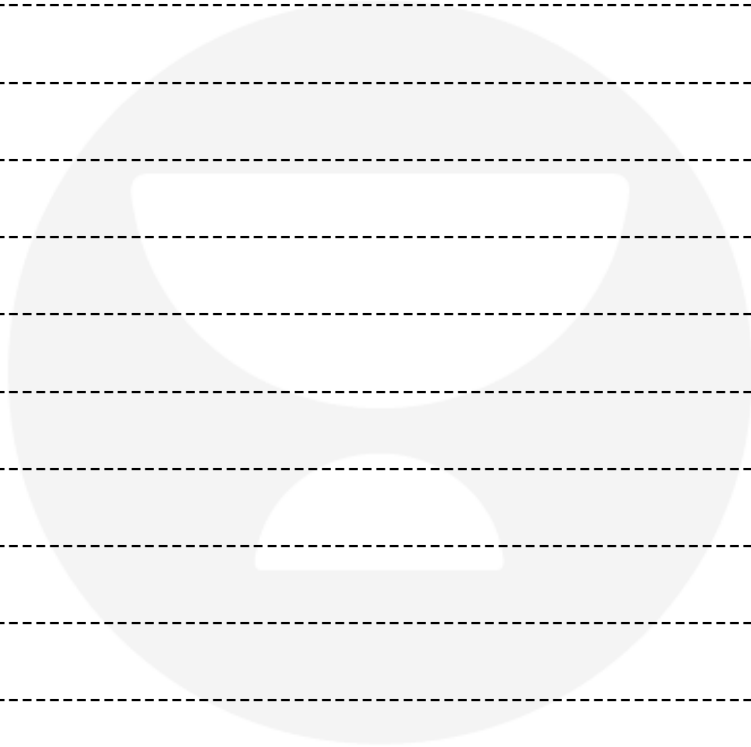
ELP No.-70

1. (A) 2. (D) 3. (B) 4. (D) 5. (C) 6. (D) 7. (A)
8. (ACD) 9. (CD) 10. 8

ELP No.-71

1. (C) 2. (B) 3. (B) 4. (CD) 5. (B) 6. (B) 7. (D)
8. (B) 9. 2 10. 3

Quick Revision Notes



The content provided herein is created and owned by third party licensors and licensed to the Company and reviewed and edited/modified by third party service providers. The Company disclaims all rights and liabilities in relation to the content. The author of the content or the editors or the reviewers shall be solely responsible, towards their respective works, for any liability that may arise with respect to the same.

